

Final run ledger and IUPAC results

This document lists the dates, data partitions, proteins, and exact-assignment IUPAC residue-atom results for the completed final shielding and chemical-shift runs.

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Time zone Europe/London (BST; UTC+1)

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1 Run provenance

1.1 Completed calculations

Calculation	Date (BST)	Fit	Selector	Evaluation	Conditions
Shielding fit	2026-09-09–10	521	25	—	Structure; AIMNet; MOPAC; Joint
Shielding holdout	2026-09-10	—	—	61 + 18	Structure; AIMNet; MOPAC; Joint
Shielding designations	2026-09-10	—	—	61 + 18; all 79	Structure; AIMNet; MOPAC; Joint
Shielding trajectories	2026-09-10	—	—	1L2Y; 5AWL	Structure; AIMNet; MOPAC; Joint
Static-shift fit	2026-09-10	151	22	—	Structure; AIMNet; MOPAC; no weight transfer / weight transfer
Static-shift external	2026-09-10	—	—	386; 1,742	Structure; AIMNet; MOPAC; no weight transfer / weight transfer
MD-shift fit: Single Frame at 150 ps	2026-09-10	151	22	—	Structure; AIMNet; MOPAC; no weight transfer / weight transfer
MD-shift evaluation	2026-09-11	151	22	all 173	Single Frame at 150 ps; 100-Frame Mean
Matched tleap-shift fit	2026-09-11	106	22	—	Structure; AIMNet2; MOPAC PM7; no weight transfer / weight transfer
Matched GROMACS-shift fit: Single Frame at 150 ps	2026-09-12	106	22	—	Structure; AIMNet2; MOPAC PM7; no weight transfer / weight transfer
Matched preparation and motion evaluation	2026-09-12	106	22	45	tleap; GROMACS Single Frame at 150 ps; GROMACS 100-Frame Mean

2 Protein sets

2.1 Shielding fitting and selector set

Member	Split
A0A075FR84	fitting
A0A075FUC0	fitting
A0A075FUI5	fitting
A0A075G604	fitting
A0A075G9H6	fitting
A0A075GBG4	fitting
A0A075GBP5	fitting
A0A075GJJ0	fitting
A0A075GL14	fitting
A0A075GNQ9	fitting
A0A075GT79	fitting
A0A075H126	fitting
A0A075I1Q2	selector
A0A075I3Q2	selector
A0A075I8Y7	selector
A0A075I9Q6	fitting
A0A087S4K0	fitting
A0A0U3H9J4	fitting
A0A101DYY5	fitting
A0A101DZD3	fitting
A0A101E0G0	fitting
A0A101GND4	fitting
A0A101IR95	fitting
A0A101ISX8	fitting
A0A117LQ45	fitting
A0A117MF79	fitting
A0A124F7Z6	fitting
A0A139CG29	fitting
A0A1C8ZW37	fitting
A0A1G6L254	fitting
A0A1G7QZ38	fitting
A0A1H1EH39	fitting
A0A1I0QAW8	fitting
A0A1L3Q2M2	fitting
A0A1L9C4R6	fitting
A0A1Q9NRY5	selector
A0A1S6H833	fitting
A0A1S6H9M5	fitting
A0A1V4YX59	fitting
A0A219ANU8	fitting
A0A284VNA8	fitting
A0A285EYT8	fitting
A0A2E0QIE5	fitting
A0A2G1WNH3	fitting
A0A2G9PKM8	fitting
A0A2H9PSD1	fitting
A0A2R4ZQL9	selector
A0A2R6C3Z5	fitting
A0A2R6C5B4	fitting
A0A2R6C8T4	fitting
A0A2R6CA67	fitting
A0A2R6CAA9	fitting
A0A2R6D1W6	fitting
A0A2R6LXP5	fitting
A0A2R6MFM1	fitting
A0A2R6MFZ7	fitting
A0A2V4GT18	fitting
A0A2V4H8E0	fitting
A0A345DZ50	fitting

Member	Split
A0A371MR54	fitting
A0A371MRP4	fitting
A0A371MS07	fitting
A0A371MTC3	fitting
A0A3A5VK63	fitting
A0A3A5VSI2	fitting
A0A3A5W4U3	fitting
A0A3A5WVH7	fitting
A0A3M0CXD9	fitting
A0A3M1LH11	fitting
A0A3M1LHW0	fitting
A0A3M1LK61	fitting
A0A3M1LLL7	fitting
A0A3M1LPS3	fitting
A0A3M9LMY0	fitting
A0A3M9ZV07	fitting
A0A3M9ZZ00	fitting
A0A3R7DRJ2	fitting
A0A3R7VPW5	fitting
A0A421DU79	fitting
A0A432I0P7	fitting
A0A495R914	fitting
A0A495R9V0	fitting
A0A497IH14	fitting
A0A4U7F0R4	fitting
A0A4U7F4A3	fitting
A0A519BR66	fitting
A0A519BS09	fitting
A0A519BYY9	fitting
A0A519CKS0	fitting
A0A519CU77	fitting
A0A520KLZ7	selector
A0A523ASN7	selector
A0A523AYL0	fitting
A0A523AYZ6	fitting
A0A523BEK8	fitting
A0A523SSL6	fitting
A0A533U6D3	fitting
A0A533UJM1	fitting
A0A533UWQ2	fitting
A0A533V157	fitting
A0A533V2Q5	fitting
A0A533V4X6	fitting
A0A533VIP4	fitting
A0A533VJR1	fitting
A0A533VSR0	fitting
A0A533W8U5	fitting
A0A533WKQ8	fitting
A0A533WM15	fitting
A0A533WMV9	fitting
A0A533WY04	fitting
A0A533WZX1	fitting
A0A533X1P5	fitting
A0A533XGD3	fitting
A0A533XLT4	fitting
A0A538P2N2	fitting
A0A538P8P9	fitting
A0A538P9J5	fitting
A0A550D5K8	fitting
A0A558GAE9	fitting
A0A5C9ESF5	fitting
A0A5E4HI60	fitting
A0A5E4HJN0	fitting
A0A5E4HP61	fitting

Member	Split
A0A5E4I7H3	selector
A0A5E4IJB0	fitting
A0A5E4ISJ4	fitting
A0A5E4J1K7	fitting
A0A5E4J7J9	fitting
A0A5E4JHA9	fitting
A0A5E4K421	fitting
A0A5E4K5U7	fitting
A0A5E4K9E0	fitting
A0A5E4KHE1	fitting
A0A5E4KT12	fitting
A0A5E4L400	fitting
A0A5E4LE60	fitting
A0A5N5UDU8	fitting
A0A660HR30	fitting
A0A662PMX3	fitting
A0A662PVT5	fitting
A0A662PX72	fitting
A0A662Q049	fitting
A0A662Q584	fitting
A0A662Q601	fitting
A0A662QA52	fitting
A0A662QCF6	fitting
A0A662QCK9	fitting
A0A662QEP8	fitting
A0A662R097	fitting
A0A662R2Z3	fitting
A0A662R555	fitting
A0A662RAM6	fitting
A0A662RTJ4	fitting
A0A662T3H2	fitting
A0A662T947	fitting
A0A662UDV1	selector
A0A6A7LI67	fitting
A0A6A7LJ60	fitting
A0A6A7LLL2	fitting
A0A6A7LMN5	fitting
A0A6A8AGZ0	fitting
A0A6A8AP17	fitting
A0A6A9T401	fitting
A0A6B0HK18	fitting
A0A6G3LQ52	fitting
A0A6V8CBP4	fitting
A0A6V8CC87	fitting
A0A6V8CDD8	fitting
A0A6V8CLY9	fitting
A0A6V8CMT5	fitting
A0A6V8CMX5	selector
A0A6V8CNG8	fitting
A0A6V8CR43	fitting
A0A6V8CTA7	fitting
A0A6V8CUM7	fitting
A0A6V8CZK2	fitting
A0A6V8D2T6	fitting
A0A6V8D831	fitting
A0A6V8DL76	fitting
A0A6V8DQK0	fitting
A0A6V8E5D1	fitting
A0A6V8E729	fitting
A0A6V8ESH0	fitting
A0A6V8ESU9	fitting
A0A7C0XQX0	fitting
A0A7C1M065	fitting
A0A7C1M999	fitting

Member	Split
A0A7C1R072	fitting
A0A7C2AFH7	fitting
A0A7C2DAU6	fitting
A0A7C2GI08	fitting
A0A7C2L465	fitting
A0A7C2QK30	fitting
A0A7C2QRH1	fitting
A0A7C2VVD2	fitting
A0A7C3M2E6	fitting
A0A7C3QTI1	fitting
A0A7C3S3R7	fitting
A0A7C3S9L1	fitting
A0A7C3SKL2	fitting
A0A7C3SL62	fitting
A0A7C3SRL8	fitting
A0A7C3V221	fitting
A0A7C3V662	fitting
A0A7C3VBE0	fitting
A0A7C3VLX4	fitting
A0A7C3WCV6	fitting
A0A7C3YB50	selector
A0A7C4BA98	fitting
A0A7C4BAF9	fitting
A0A7C4BEL2	fitting
A0A7C4C6T0	fitting
A0A7C4C740	fitting
A0A7C4CKX8	fitting
A0A7C4CNR5	fitting
A0A7C4CNU0	fitting
A0A7C4CRC4	fitting
A0A7C4CV75	fitting
A0A7C4CVP5	fitting
A0A7C4D0I2	fitting
A0A7C4D0Z0	fitting
A0A7C4D2B3	fitting
A0A7C4D2X1	fitting
A0A7C4D481	fitting
A0A7C4D4U5	fitting
A0A7C4DB51	fitting
A0A7C4DZ89	fitting
A0A7C4F165	fitting
A0A7C4FBN7	fitting
A0A7C4FEM6	fitting
A0A7C4GES3	fitting
A0A7C4GGW0	fitting
A0A7C4GN29	fitting
A0A7C4GR21	fitting
A0A7C4GSI3	fitting
A0A7C4GW05	fitting
A0A7C4H515	fitting
A0A7C4H7N1	fitting
A0A7C4HAL7	fitting
A0A7C4HDC8	fitting
A0A7C4HL71	fitting
A0A7C4HU33	fitting
A0A7C4HZJ5	fitting
A0A7C4JB99	fitting
A0A7C4JMT3	selector
A0A7C4JNN4	fitting
A0A7C4M5D4	fitting
A0A7C4M8L5	fitting
A0A7C4M8N8	fitting
A0A7C4MCQ5	fitting
A0A7C4MFI0	selector

Member	Split
A0A7C4NC93	fitting
A0A7C4NEZ2	fitting
A0A7C4NIQ4	fitting
A0A7C4NIY2	fitting
A0A7C4NQE1	fitting
A0A7C4PVJ1	fitting
A0A7C4PZ72	selector
A0A7C4Q0N0	fitting
A0A7C4S1G2	fitting
A0A7C4S653	fitting
A0A7C4S8S6	fitting
A0A7C4TUE3	selector
A0A7C4VDL7	fitting
A0A7C4VJN0	fitting
A0A7C4VQX9	selector
A0A7C4W7X4	fitting
A0A7C4WIF3	fitting
A0A7C4XD95	fitting
A0A7C4XSG8	fitting
A0A7C4ZS34	fitting
A0A7C5A9M3	fitting
A0A7C5IX38	fitting
A0A7C5K1Z9	fitting
A0A7C5N6U4	fitting
A0A7C5T4J2	fitting
A0A7C5T7W6	fitting
A0A7C5TFR7	fitting
A0A7C5TK25	fitting
A0A7C5U1D8	fitting
A0A7C5U2B7	fitting
A0A7C5U5Q9	fitting
A0A7C5VQ90	fitting
A0A7C5X1V8	fitting
A0A7C5X2V4	fitting
A0A7C5XHQ0	fitting
A0A7C5XHS9	fitting
A0A7C5XJ97	fitting
A0A7C5XTQ5	fitting
A0A7C5Y7Y0	fitting
A0A7C5YCQ1	fitting
A0A7C5YSG7	fitting
A0A7C5ZT79	fitting
A0A7C6AIW3	selector
A0A7C6DX77	fitting
A0A7C6E4A0	fitting
A0A7C6FRR6	fitting
A0A7C7CNG0	fitting
A0A7C7GIG1	fitting
A0A7C7MPZ4	fitting
A0A7C7PIY6	fitting
A0A7C7PPS5	fitting
A0A7C7PU94	fitting
A0A7C7Y7G9	fitting
A0A7G9Y4P4	fitting
A0A7G9Z2P5	fitting
A0A7G9Z8V8	fitting
A0A7H1KP24	fitting
A0A7J2IPL3	fitting
A0A7J2IQR4	fitting
A0A7J2ISG1	fitting
A0A7J2LST6	fitting
A0A7J2MF60	fitting
A0A7J2MGK5	fitting
A0A7J2MGV4	fitting

Member	Split
A0A7J2MHH4	fitting
A0A7J2PVB2	fitting
A0A7J2PZK0	fitting
A0A7J2QZC8	fitting
A0A7J2R6L5	fitting
A0A7J2R8L4	fitting
A0A7J2T743	fitting
A0A7J2TJN1	fitting
A0A7J2TN26	fitting
A0A7J2UQV7	fitting
A0A7J2V0Z7	fitting
A0A7J2V9U9	fitting
A0A7J2VTQ0	fitting
A0A7J2VWF2	fitting
A0A7J2WFC1	fitting
A0A7J2Y502	fitting
A0A7J2Y6X9	fitting
A0A7J2YMB9	fitting
A0A7J2Z0G0	fitting
A0A7J2ZCD7	fitting
A0A7J2ZM04	fitting
A0A7J3AC04	selector
A0A7J3ADC8	fitting
A0A7J3B0X2	fitting
A0A7J3B843	fitting
A0A7J3BD52	fitting
A0A7J3BEY8	fitting
A0A7J3C7I1	fitting
A0A7J3CJ29	fitting
A0A7J3D6Q8	fitting
A0A7J3D8F9	fitting
A0A7J3D9G8	selector
A0A7J3DAY3	fitting
A0A7J3EGB7	fitting
A0A7J3EMJ0	fitting
A0A7J3F9X2	fitting
A0A7J3FAQ0	fitting
A0A7J3FC68	fitting
A0A7J3FC71	fitting
A0A7J3FCW3	fitting
A0A7J3GV54	fitting
A0A7J3HB40	fitting
A0A7J3HPJ8	fitting
A0A7J3I3M4	fitting
A0A7J3I437	fitting
A0A7J3I6Q7	fitting
A0A7J3IA32	fitting
A0A7J3IFT0	fitting
A0A7J3IY93	fitting
A0A7J3J0Q3	fitting
A0A7J3J3L3	fitting
A0A7J3JCS0	fitting
A0A7J3KAV9	fitting
A0A7J3LXD3	fitting
A0A7J3LYC8	fitting
A0A7J3M1M9	fitting
A0A7J3M345	fitting
A0A7J3M4E2	fitting
A0A7J3M654	fitting
A0A7J3MZ99	fitting
A0A7J3P4N5	fitting
A0A7J3PA89	fitting
A0A7J3QFR6	fitting
A0A7J3QSA6	selector

Member	Split
A0A7J3QSZ4	fitting
A0A7J3QT34	fitting
A0A7J3QTT3	fitting
A0A7J3R097	fitting
A0A7J3S713	fitting
A0A7J3SZL7	fitting
A0A7J3T0Q9	fitting
A0A7J3TGH2	fitting
A0A7J3TX92	fitting
A0A7J3UYK2	fitting
A0A7J3W6K5	fitting
A0A7J3W6T1	fitting
A0A7J3W7Q6	fitting
A0A7J3W909	fitting
A0A7J3WCY0	fitting
A0A7J3WZ86	fitting
A0A7J3XGR4	fitting
A0A7J3XUT2	fitting
A0A7J3ZAE1	fitting
A0A7J3ZM45	fitting
A0A7J3ZQ93	fitting
A0A7J4BA39	fitting
A0A7J4BAQ2	fitting
A0A7J4FTH4	fitting
A0A7J4G0F4	fitting
A0A7J4NWE9	fitting
A0A7J9SMA8	fitting
A0A7K3Y6W1	fitting
A0A7K3YIN1	fitting
A0A7K4NEG8	fitting
A0A7K4NF07	fitting
A0A7M3SGS7	fitting
A0A7M3SKX7	fitting
A0A7M3WGX1	fitting
A0A7M3WHN5	fitting
A0A7M3WSQ7	fitting
A0A7M3X120	fitting
A0A7M3X1A0	fitting
A0A7M3XQK6	fitting
A0A7M3XRS6	fitting
A0A7M3XTS2	fitting
A0A7M3XVG2	fitting
A0A7M3XVL2	fitting
A0A7M3XYR1	fitting
A0A7M3XZU0	fitting
A0A7M3YDZ0	fitting
A0A7M3YIS4	fitting
A0A7M3YK94	fitting
A0A7M3YPB3	fitting
A0A7M3YUC1	fitting
A0A7M3Z7R6	fitting
A0A7M3Z7W4	fitting
A0A7M3ZAB9	fitting
A0A7M3ZI84	fitting
A0A7M3ZZV4	fitting
A0A7M4A1E0	fitting
A0A7M4A5Q8	fitting
A0A7M4AAC8	fitting
A0A811TBK2	fitting
A0A812A2F1	fitting
A0A822IIB9	selector
A0A822IKI2	selector
A0A822ILJ2	fitting
A0A822IR84	fitting

Member	Split
A0A822IUL2	selector
A0A831V7D1	fitting
A0A831YJC5	fitting
A0A832DFU9	fitting
A0A832Y0B8	fitting
A0A832YEZ8	fitting
A0A842Q616	fitting
A0A842Y7E2	fitting
A0A843CAZ7	fitting
A0A849PJP1	fitting
A0A8J7XH87	fitting
A0A8J8C3Q5	fitting
A0A8J8DHJ8	fitting
A0A8S8Y833	fitting
A0A8S8YVN2	fitting
A0A8T4DH60	fitting
A0A8T5JI94	fitting
A0A8T6SHV5	fitting
A0A8T6TUW3	fitting
A0A8T6U953	fitting
A0A8T6UP66	fitting
A0A8T6WDJ4	fitting
A0A8T6XLB1	selector
A1TJT8	fitting
A1TYL8	fitting
A1V881	fitting
A1WVA1	fitting
A2S7J8	fitting
A3MRX6	fitting
A3P091	fitting
A4VHQ1	fitting
A4WFA7	fitting
A5FMZ9	fitting
A5FRW8	fitting
A6QCS0	fitting
A6W371	fitting
A7MPF5	fitting
A7MWH9	fitting
A7ZSI8	fitting
A8A5A4	fitting
A8ETK2	fitting
A9ADL5	fitting
B1GZA6	fitting
B1IQ00	fitting
B1JIY2	fitting
B1JU44	fitting
B1LHB3	fitting
B1X6F1	fitting
B3CQF1	fitting
B5FGD7	fitting
B8GV37	fitting
C0HLA1	fitting
C6C1A7	fitting
H0AEA3	fitting
I7CSI0	fitting
M0B978	fitting
M0BL78	fitting
O96797	fitting
P0A404	fitting
P0A497	fitting
P0A7R0	fitting
P17684	fitting
P80502	selector
P83867	fitting

Member	Split
P84477	fitting
P84807	fitting
P86498	fitting
Q0ABF4	fitting
Q0T003	fitting
Q0TCG2	fitting
Q0VSI2	fitting
Q1BRX0	fitting
Q1R632	fitting
Q1XDA2	fitting
Q21M37	fitting
Q21QP5	fitting
Q2JI37	fitting
Q2JT51	fitting
Q30TS7	fitting
Q318K4	fitting
Q31VX7	fitting
Q32B52	fitting
Q3YWW0	fitting
Q3Z957	fitting
Q3ZZQ0	fitting
Q46501	fitting
Q47J81	fitting
Q4A8J4	fitting
Q4AAG3	fitting
Q5E893	fitting
Q5L8D2	fitting
Q62GM7	fitting
Q661C1	fitting
Q664U2	fitting
Q6B8L0	fitting
Q6KI32	fitting
Q6YQZ8	fitting
Q73PL1	fitting
Q7MTN6	fitting
Q8ZJ91	fitting
Q98Q05	fitting
Q9TLX5	fitting
Q9UR66	fitting

2.2 Shielding holdout set

Member	Split
A0A062V9G2	previous_paper
A0A075G8S5	historical_primary
A0A075I9N7	historical_primary
A0A087RU44	historical_primary
A0A0F2L3J9	historical_primary
A0A285ESG6	previous_paper
A0A2E4JTY0	historical_primary
A0A2V3JMT2	historical_primary
A0A371MSL8	historical_primary
A0A3M1LFP0	historical_primary
A0A432I6S4	historical_primary
A0A510DZD2	previous_paper
A0A519CLK2	previous_paper
A0A523AT74	historical_primary
A0A523AVT4	previous_paper
A0A5E4IG34	historical_primary
A0A5E4IH91	historical_primary
A0A662Q1I9	previous_paper
A0A662Q1V9	historical_primary

Member	Split
A0A662R080	historical_primary
A0A662R0I2	previous_paper
A0A662R243	historical_primary
A0A662RX65	historical_primary
A0A6A7LI12	historical_primary
A0A6B9QZA6	historical_primary
A0A6V8CN64	historical_primary
A0A6V8CNU4	previous_paper
A0A6V8CQS9	previous_paper
A0A6V8DTT4	historical_primary
A0A6V8E1N8	historical_primary
A0A6V8ERQ8	historical_primary
A0A7C0XZN5	previous_paper
A0A7C0Y320	historical_primary
A0A7C1BUZ8	historical_primary
A0A7C3TX69	historical_primary
A0A7C4CQ25	historical_primary
A0A7C4D3B7	historical_primary
A0A7C4GSQ4	historical_primary
A0A7C4R5B6	previous_paper
A0A7C4R964	historical_primary
A0A7C4RCS8	historical_primary
A0A7C5BHP9	historical_primary
A0A7C5T765	historical_primary
A0A7C5U7X1	historical_primary
A0A7C5ZRH2	previous_paper
A0A7C7GE12	historical_primary
A0A7J2H756	historical_primary
A0A7J3DNK5	historical_primary
A0A7J3ELT1	previous_paper
A0A7J3I0Z5	historical_primary
A0A7J3J2N7	previous_paper
A0A7J3LWS6	historical_primary
A0A7J3MGX5	historical_primary
A0A7J3MYD0	historical_primary
A0A7J3QT08	previous_paper
A0A7J3QYA8	historical_primary
A0A7J3XUF5	historical_primary
A0A7M3W8C7	historical_primary
A0A7M3X6H7	historical_primary
A0A7M4AP21	historical_primary
A0A822IN67	previous_paper
A0A831VDP7	historical_primary
A0A843BFT9	previous_paper
A0A8S8YJI6	previous_paper
A0A949JHM8	historical_primary
A1VES0	historical_primary
A5GN09	historical_primary
A5GRN4	historical_primary
B1WWJ3	historical_primary
B8DP14	historical_primary
B8HR51	historical_primary
P84337	historical_primary
Q2JHS2	historical_primary
Q3AM28	historical_primary
Q3AV42	historical_primary
Q4G392	historical_primary
Q5N606	historical_primary
Q72D55	historical_primary
U1PBV3	historical_primary

2.3 Static and MD development set

Member	Split
bmr68	fitting
bmr4082	fitting
bmr4095	fitting
bmr4121	fitting
bmr4237	fitting
bmr4254	fitting
bmr4271	fitting
bmr4296	fitting
bmr4301	fitting
bmr4348	fitting
bmr4421	fitting
bmr4430	fitting
bmr4493	fitting
bmr4510	fitting
bmr4560	selector
bmr4581	fitting
bmr4590	fitting
bmr4603	fitting
bmr4663	fitting
bmr4664	fitting
bmr4678	fitting
bmr4721	fitting
bmr4851	fitting
bmr4863	selector
bmr4869	selector
bmr4935	selector
bmr4976	fitting
bmr4981	fitting
bmr5027	fitting
bmr5031	selector
bmr5038	selector
bmr5053	fitting
bmr5105	fitting
bmr5141	fitting
bmr5174	fitting
bmr5187	fitting
bmr5225	selector
bmr5238	fitting
bmr5276	fitting
bmr5311	fitting
bmr5328	fitting
bmr5329	fitting
bmr5375	selector
bmr5499	fitting
bmr5537	fitting
bmr5563	fitting
bmr5689	fitting
bmr5691	fitting
bmr5763	fitting
bmr5801	fitting
bmr5846	fitting
bmr5847	fitting
bmr5849	selector
bmr5873	fitting
bmr5875	selector
bmr5963	fitting
bmr5977	fitting
bmr6010	fitting
bmr6028	selector
bmr6031	fitting
bmr6122	fitting
bmr6167	fitting
bmr6175	fitting

Member	Split
bmr6188	fitting
bmr6189	fitting
bmr6197	fitting
bmr6208	selector
bmr6209	fitting
bmr6254	fitting
bmr6258	fitting
bmr6269	fitting
bmr6291	fitting
bmr6318	fitting
bmr6355	fitting
bmr6368	fitting
bmr6375	fitting
bmr6376	fitting
bmr6378	fitting
bmr6380	fitting
bmr6383	fitting
bmr6466	fitting
bmr6469	selector
bmr6478	fitting
bmr6505	fitting
bmr6517	fitting
bmr6527	fitting
bmr6552	fitting
bmr6574	fitting
bmr6577	fitting
bmr6611	fitting
bmr6649	fitting
bmr6718	selector
bmr6763	fitting
bmr6781	fitting
bmr6804	fitting
bmr6805	fitting
bmr6816	fitting
bmr6854	fitting
bmr6865	fitting
bmr6912	fitting
bmr6932	fitting
bmr6964	fitting
bmr6976	fitting
bmr6981	fitting
bmr6989	fitting
bmr7000	fitting
bmr7057	fitting
bmr7094	fitting
bmr7104	selector
bmr7109	fitting
bmr7118	fitting
bmr7127	fitting
bmr7175	fitting
bmr7181	fitting
bmr7185	fitting
bmr7223	fitting
bmr7350	fitting
bmr7362	fitting
bmr7366	fitting
bmr7371	fitting
bmr10005	fitting
bmr10028	fitting
bmr10047	fitting
bmr10050	fitting
bmr10056	fitting
bmr10061	fitting
bmr10063	fitting

Member	Split
bmr10069	selector
bmr10077	selector
bmr10087	fitting
bmr10095	fitting
bmr10100	fitting
bmr10111	selector
bmr10112	fitting
bmr10118	fitting
bmr10123	fitting
bmr10138	fitting
bmr10212	fitting
bmr10240	fitting
bmr10241	fitting
bmr10252	selector
bmr10261	selector
bmr10281	fitting
bmr10287	fitting
bmr10289	fitting
bmr10290	fitting
bmr10291	fitting
bmr10292	fitting
bmr10293	fitting
bmr10295	fitting
bmr10296	fitting
bmr10297	fitting
bmr10302	fitting
bmr10329	fitting
bmr11173	selector
bmr11322	selector
bmr15129	fitting
bmr15133	fitting
bmr15142	fitting
bmr15167	fitting
bmr15324	fitting
bmr15332	fitting
bmr15339	fitting
bmr15341	fitting
bmr15355	fitting
bmr15410	fitting
bmr15531	fitting
bmr15591	fitting
bmr15756	fitting
bmr15784	fitting
bmr15884	fitting
bmr16989	fitting
bmr17439	fitting

2.4 Matched tleap and GROMACS preparation comparison set

Member	Split
bmr68	evaluation
bmr4082	fitting
bmr4095	evaluation
bmr4121	evaluation
bmr4237	fitting
bmr4254	fitting
bmr4271	fitting
bmr4296	fitting
bmr4301	evaluation
bmr4348	fitting
bmr4421	fitting
bmr4430	evaluation

Member	Split
bmr4493	evaluation
bmr4510	fitting
bmr4560	selector
bmr4581	evaluation
bmr4590	fitting
bmr4603	fitting
bmr4663	evaluation
bmr4664	fitting
bmr4678	fitting
bmr4721	fitting
bmr4851	fitting
bmr4863	selector
bmr4869	selector
bmr4935	selector
bmr4976	fitting
bmr4981	fitting
bmr5027	fitting
bmr5031	selector
bmr5038	selector
bmr5053	fitting
bmr5105	fitting
bmr5141	fitting
bmr5174	fitting
bmr5187	fitting
bmr5225	selector
bmr5238	fitting
bmr5276	evaluation
bmr5311	fitting
bmr5328	fitting
bmr5329	fitting
bmr5375	selector
bmr5499	fitting
bmr5537	fitting
bmr5563	fitting
bmr5689	fitting
bmr5691	fitting
bmr5763	fitting
bmr5801	fitting
bmr5846	fitting
bmr5847	fitting
bmr5849	selector
bmr5873	fitting
bmr5875	selector
bmr5963	fitting
bmr5977	fitting
bmr6010	fitting
bmr6028	selector
bmr6031	fitting
bmr6122	fitting
bmr6167	fitting
bmr6175	evaluation
bmr6188	fitting
bmr6189	fitting
bmr6197	evaluation
bmr6208	selector
bmr6209	fitting
bmr6254	fitting
bmr6258	fitting
bmr6269	fitting
bmr6291	fitting
bmr6318	fitting
bmr6355	evaluation
bmr6368	fitting
bmr6375	fitting

Member	Split
bmr6376	fitting
bmr6378	fitting
bmr6380	fitting
bmr6383	fitting
bmr6466	evaluation
bmr6469	selector
bmr6478	fitting
bmr6505	fitting
bmr6517	fitting
bmr6527	fitting
bmr6552	fitting
bmr6574	fitting
bmr6577	fitting
bmr6611	fitting
bmr6649	fitting
bmr6718	selector
bmr6763	fitting
bmr6781	fitting
bmr6804	fitting
bmr6805	fitting
bmr6816	fitting
bmr6854	evaluation
bmr6865	fitting
bmr6912	fitting
bmr6932	fitting
bmr6964	fitting
bmr6976	fitting
bmr6981	evaluation
bmr6989	fitting
bmr7000	fitting
bmr7057	fitting
bmr7094	fitting
bmr7104	selector
bmr7109	fitting
bmr7118	fitting
bmr7127	fitting
bmr7175	evaluation
bmr7181	fitting
bmr7185	fitting
bmr7223	evaluation
bmr7350	fitting
bmr7362	fitting
bmr7366	fitting
bmr7371	fitting
bmr10005	fitting
bmr10028	fitting
bmr10047	evaluation
bmr10050	evaluation
bmr10056	fitting
bmr10061	evaluation
bmr10063	evaluation
bmr10069	selector
bmr10077	selector
bmr10087	evaluation
bmr10095	evaluation
bmr10100	evaluation
bmr10111	selector
bmr10112	evaluation
bmr10118	fitting
bmr10123	fitting
bmr10138	fitting
bmr10212	evaluation
bmr10240	evaluation
bmr10241	evaluation

Member	Split
bmr10252	selector
bmr10261	selector
bmr10281	evaluation
bmr10287	evaluation
bmr10289	evaluation
bmr10290	evaluation
bmr10291	evaluation
bmr10292	evaluation
bmr10293	evaluation
bmr10295	evaluation
bmr10296	evaluation
bmr10297	evaluation
bmr10302	evaluation
bmr10329	evaluation
bmr11173	selector
bmr11322	selector
bmr15129	evaluation
bmr15133	fitting
bmr15142	fitting
bmr15167	fitting
bmr15324	evaluation
bmr15332	fitting
bmr15339	fitting
bmr15341	fitting
bmr15355	fitting
bmr15410	evaluation
bmr15531	fitting
bmr15591	fitting
bmr15756	fitting
bmr15784	evaluation
bmr15884	fitting
bmr16989	fitting
bmr17439	evaluation

2.5 Static external set

Member	Like-for-like 386
bmr10004	yes
bmr10008	yes
bmr10011	yes
bmr10013	no
bmr10015	no
bmr10021	no
bmr10023	yes
bmr10024	no
bmr10025	yes
bmr10026	no
bmr10027	yes
bmr10030	yes
bmr10031	yes
bmr10032	yes
bmr10033	yes
bmr10034	yes
bmr10035	yes
bmr10036	yes
bmr10037	yes
bmr10038	no
bmr10039	yes
bmr10040	yes
bmr10041	no
bmr10042	yes
bmr10043	yes

Member	Like-for-like	386
bmr10046	yes	
bmr10049	yes	
bmr10051	no	
bmr10053	no	
bmr10054	yes	
bmr10055	yes	
bmr10057	no	
bmr10058	yes	
bmr10059	yes	
bmr10060	yes	
bmr10065	yes	
bmr10067	yes	
bmr10068	yes	
bmr10071	yes	
bmr10072	yes	
bmr10073	yes	
bmr10074	yes	
bmr10075	yes	
bmr10076	yes	
bmr10079	yes	
bmr10080	yes	
bmr10082	no	
bmr10083	yes	
bmr10084	yes	
bmr10086	yes	
bmr10089	yes	
bmr10090	yes	
bmr10091	yes	
bmr10092	no	
bmr10093	yes	
bmr10094	yes	
bmr10096	no	
bmr10097	yes	
bmr10098	no	
bmr10099	yes	
bmr10101	no	
bmr10102	yes	
bmr10104	yes	
bmr10105	yes	
bmr10106	yes	
bmr10107	yes	
bmr10108	no	
bmr10113	yes	
bmr10114	yes	
bmr10115	yes	
bmr10120	yes	
bmr10121	yes	
bmr10125	yes	
bmr10126	yes	
bmr10127	yes	
bmr10128	yes	
bmr10129	yes	
bmr10130	yes	
bmr10131	yes	
bmr10133	no	
bmr10135	no	
bmr10137	no	
bmr10213	yes	
bmr10235	yes	
bmr10243	no	
bmr10244	no	
bmr10246	no	
bmr10247	no	
bmr10248	yes	

Member	Like-for-like	386
bmr10249	yes	
bmr10250	yes	
bmr10251	yes	
bmr10253	yes	
bmr10254	yes	
bmr10255	yes	
bmr10256	yes	
bmr10257	no	
bmr10258	yes	
bmr10260	yes	
bmr10262	yes	
bmr10263	yes	
bmr10264	yes	
bmr10276	no	
bmr10277	yes	
bmr10278	yes	
bmr10279	yes	
bmr10280	yes	
bmr10288	yes	
bmr10299	yes	
bmr10300	yes	
bmr10311	yes	
bmr10312	yes	
bmr10313	yes	
bmr10314	yes	
bmr10315	yes	
bmr10316	yes	
bmr10317	yes	
bmr10318	yes	
bmr10319	no	
bmr10320	yes	
bmr10321	yes	
bmr10323	yes	
bmr10325	yes	
bmr10326	yes	
bmr10327	yes	
bmr10330	yes	
bmr10332	yes	
bmr10334	yes	
bmr10335	yes	
bmr10336	yes	
bmr1061	yes	
bmr1062	yes	
bmr1093	no	
bmr11011	no	
bmr11062	yes	
bmr11067	no	
bmr11084	yes	
bmr11085	yes	
bmr11091	yes	
bmr11093	no	
bmr11100	yes	
bmr11101	yes	
bmr11102	yes	
bmr11104	yes	
bmr11105	no	
bmr11106	yes	
bmr11108	yes	
bmr11110	yes	
bmr11111	yes	
bmr11112	yes	
bmr11113	no	
bmr11114	yes	
bmr11417	yes	

Member	Like-for-like	386
bmr1162	no	
bmr1209	no	
bmr1333	no	
bmr1336	no	
bmr1344	no	
bmr1381	no	
bmr1396	no	
bmr1397	no	
bmr1398	no	
bmr1399	no	
bmr1414	no	
bmr1442	no	
bmr1444	no	
bmr1451	no	
bmr1452	no	
bmr1453	no	
bmr1454	no	
bmr1479	no	
bmr1480	no	
bmr1494	no	
bmr1495	no	
bmr1496	no	
bmr1497	no	
bmr1500	no	
bmr15001	no	
bmr15002	no	
bmr15007	no	
bmr15013	no	
bmr15014	no	
bmr15020	no	
bmr15036	yes	
bmr15037	no	
bmr15045	no	
bmr15055	yes	
bmr15061	no	
bmr15063	yes	
bmr15064	no	
bmr15065	no	
bmr15066	no	
bmr15067	no	
bmr15070	no	
bmr15071	no	
bmr15073	no	
bmr15074	no	
bmr15076	no	
bmr15077	no	
bmr15078	no	
bmr15079	no	
bmr15082	yes	
bmr15084	no	
bmr15085	no	
bmr15086	yes	
bmr15088	yes	
bmr15089	no	
bmr15090	no	
bmr15094	no	
bmr15100	no	
bmr15101	yes	
bmr15109	no	
bmr15111	no	
bmr15120	no	
bmr15121	no	
bmr15128	no	
bmr15130	no	

Member	Like-for-like	386
bmr15137	no	
bmr15139	no	
bmr15143	no	
bmr15160	no	
bmr15161	no	
bmr15162	no	
bmr1517	no	
bmr15170	no	
bmr15177	no	
bmr15190	yes	
bmr15203	yes	
bmr15204	no	
bmr15210	yes	
bmr15211	yes	
bmr15217	no	
bmr15219	no	
bmr15226	no	
bmr15232	no	
bmr15233	no	
bmr15234	no	
bmr15235	no	
bmr15236	no	
bmr15237	no	
bmr15242	no	
bmr15243	no	
bmr15244	no	
bmr15248	no	
bmr15249	no	
bmr15250	no	
bmr15252	no	
bmr15253	no	
bmr15264	yes	
bmr15265	no	
bmr15270	no	
bmr15281	no	
bmr15288	yes	
bmr15317	yes	
bmr15322	yes	
bmr15325	no	
bmr15327	yes	
bmr15333	no	
bmr15338	no	
bmr15348	yes	
bmr15350	no	
bmr15372	no	
bmr15374	no	
bmr15375	no	
bmr15389	no	
bmr15390	no	
bmr15395	no	
bmr15396	no	
bmr15406	no	
bmr15412	no	
bmr15420	no	
bmr15421	no	
bmr15424	no	
bmr15426	no	
bmr15429	no	
bmr15449	no	
bmr15480	no	
bmr15488	no	
bmr15492	no	
bmr15493	no	
bmr15494	no	

Member	Like-for-like	386
bmr15495	no	
bmr15496	no	
bmr15498	no	
bmr15501	no	
bmr15503	no	
bmr15505	no	
bmr15507	no	
bmr15508	no	
bmr15511	no	
bmr15514	no	
bmr15518	no	
bmr15529	no	
bmr15530	no	
bmr15548	no	
bmr15560	no	
bmr15564	no	
bmr15626	no	
bmr15643	no	
bmr15646	no	
bmr15653	no	
bmr15654	no	
bmr15659	no	
bmr15669	no	
bmr15680	no	
bmr15706	no	
bmr15712	no	
bmr15713	no	
bmr15720	yes	
bmr15722	no	
bmr15726	no	
bmr15727	no	
bmr15729	no	
bmr15730	no	
bmr15741	no	
bmr15757	no	
bmr15783	no	
bmr15785	no	
bmr15792	no	
bmr15800	no	
bmr15826	no	
bmr15854	no	
bmr15882	no	
bmr15885	no	
bmr15892	no	
bmr15928	no	
bmr15932	no	
bmr15945	no	
bmr15949	no	
bmr15952	no	
bmr15953	no	
bmr15956	no	
bmr15959	no	
bmr15965	no	
bmr15969	no	
bmr15971	no	
bmr16006	no	
bmr16007	no	
bmr16010	no	
bmr16011	no	
bmr16024	no	
bmr16042	no	
bmr16046	no	
bmr16047	no	
bmr16048	no	

Member	Like-for-like	386
bmr16049	no	
bmr16059	no	
bmr16074	no	
bmr16090	no	
bmr16113	no	
bmr16142	no	
bmr16143	no	
bmr16146	no	
bmr16169	no	
bmr16170	no	
bmr16171	no	
bmr16172	no	
bmr16173	no	
bmr16204	no	
bmr16205	no	
bmr16235	no	
bmr16254	no	
bmr16265	yes	
bmr16284	no	
bmr16294	no	
bmr16300	no	
bmr16302	no	
bmr16309	no	
bmr16310	no	
bmr1632	no	
bmr1633	no	
bmr16334	no	
bmr16344	no	
bmr16362	no	
bmr16367	no	
bmr16379	no	
bmr16380	no	
bmr16393	no	
bmr16395	no	
bmr1642	yes	
bmr16447	no	
bmr16448	no	
bmr16451	no	
bmr16464	no	
bmr16466	no	
bmr16489	no	
bmr16493	no	
bmr16494	no	
bmr16499	no	
bmr16500	no	
bmr16501	no	
bmr16502	no	
bmr16523	yes	
bmr16524	no	
bmr16525	no	
bmr16526	no	
bmr16527	no	
bmr16528	no	
bmr16537	yes	
bmr16541	no	
bmr16543	no	
bmr16546	no	
bmr16547	no	
bmr16548	no	
bmr16565	no	
bmr16566	no	
bmr1657	yes	
bmr1658	no	
bmr16585	no	

Member	Like-for-like	386
bmr16587	no	
bmr16593	no	
bmr16600	no	
bmr16628	no	
bmr1663	yes	
bmr16630	no	
bmr16634	no	
bmr1666	yes	
bmr16662	no	
bmr16664	no	
bmr16668	no	
bmr16670	no	
bmr16690	yes	
bmr16726	no	
bmr1673	no	
bmr16739	yes	
bmr16740	no	
bmr16741	no	
bmr1675	no	
bmr16838	no	
bmr16847	no	
bmr16849	no	
bmr16866	no	
bmr16869	no	
bmr16878	no	
bmr16891	no	
bmr16898	no	
bmr16904	no	
bmr16925	no	
bmr16931	no	
bmr16933	no	
bmr16981	no	
bmr16982	no	
bmr16984	no	
bmr1700	no	
bmr17008	no	
bmr17010	no	
bmr17013	no	
bmr17136	yes	
bmr17158	no	
bmr17159	no	
bmr17160	no	
bmr17165	no	
bmr17166	no	
bmr17172	no	
bmr17205	yes	
bmr17251	no	
bmr17260	no	
bmr1728	no	
bmr17357	no	
bmr17362	yes	
bmr17393	no	
bmr17415	no	
bmr1743	no	
bmr17476	no	
bmr17507	yes	
bmr1757	no	
bmr1766	yes	
bmr1811	no	
bmr1867	no	
bmr1869	no	
bmr1870	no	
bmr1874	no	
bmr1875	no	

Member	Like-for-like	386
bmr1878	yes	
bmr1911	no	
bmr1918	no	
bmr192	yes	
bmr193	no	
bmr194	no	
bmr196	yes	
bmr199	no	
bmr1991	no	
bmr2039	no	
bmr2061	no	
bmr2065	no	
bmr2151	no	
bmr2196	no	
bmr2204	no	
bmr2205	no	
bmr2206	no	
bmr2208	no	
bmr2217	no	
bmr2219	no	
bmr2227	no	
bmr2262	no	
bmr2281	no	
bmr2410	yes	
bmr2498	no	
bmr2506	no	
bmr2527	no	
bmr2539	no	
bmr2542	no	
bmr2546	no	
bmr260	no	
bmr274	no	
bmr275	no	
bmr280	no	
bmr2856	no	
bmr2868	no	
bmr294	no	
bmr295	no	
bmr2956	no	
bmr298	no	
bmr2999	no	
bmr30	no	
bmr3078	no	
bmr314	no	
bmr32	no	
bmr3427	no	
bmr3440	no	
bmr3441	no	
bmr3442	no	
bmr3449	yes	
bmr3456	no	
bmr349	no	
bmr358	no	
bmr374	no	
bmr375	no	
bmr385	no	
bmr394	no	
bmr395	no	
bmr4012	no	
bmr4022	no	
bmr4031	yes	
bmr4032	no	
bmr4033	no	
bmr4034	no	

Member	Like-for-like 386
bmr4035	no
bmr4036	no
bmr4037	no
bmr4039	no
bmr4041	no
bmr4043	no
bmr4044	no
bmr4045	no
bmr4047	no
bmr4049	no
bmr4052	no
bmr4053	no
bmr4054	no
bmr4055	no
bmr4057	no
bmr4058	no
bmr4059	no
bmr4060	no
bmr4064	no
bmr4069	no
bmr4070	yes
bmr4072	no
bmr4076	no
bmr4078	no
bmr4081	no
bmr4083	no
bmr4085	no
bmr4086	no
bmr4087	no
bmr4090	no
bmr4091	no
bmr4092	no
bmr4093	no
bmr4094	no
bmr4097	no
bmr4098	no
bmr4106	yes
bmr4110	no
bmr4113	no
bmr4114	no
bmr4115	no
bmr4116	no
bmr4117	yes
bmr4127	no
bmr4128	no
bmr4131	no
bmr4133	no
bmr4136	no
bmr4143	no
bmr4149	no
bmr4152	yes
bmr4156	yes
bmr4157	no
bmr4158	no
bmr4159	no
bmr4161	no
bmr4162	no
bmr4163	no
bmr4166	no
bmr4167	no
bmr4168	no
bmr4173	no
bmr4184	no
bmr4186	no

Member	Like-for-like 386
bmr4188	no
bmr4193	no
bmr4195	no
bmr4197	no
bmr4199	no
bmr42	no
bmr4202	no
bmr4204	no
bmr4205	no
bmr4206	yes
bmr4207	no
bmr4209	no
bmr421	no
bmr4210	no
bmr4211	no
bmr4215	no
bmr4216	no
bmr4219	no
bmr4221	no
bmr4222	no
bmr4223	no
bmr4224	yes
bmr4227	no
bmr4228	no
bmr4233	no
bmr4236	no
bmr4241	yes
bmr4242	no
bmr4246	no
bmr4249	no
bmr4255	yes
bmr4257	no
bmr4259	no
bmr4261	no
bmr4267	no
bmr4269	no
bmr4276	no
bmr4278	no
bmr4280	yes
bmr4282	no
bmr4290	no
bmr4292	yes
bmr4295	no
bmr4297	yes
bmr4299	no
bmr4300	no
bmr4302	no
bmr4303	no
bmr4305	yes
bmr4306	no
bmr4308	no
bmr4309	no
bmr4312	no
bmr4316	no
bmr4317	no
bmr4318	yes
bmr4323	no
bmr4326	no
bmr4327	no
bmr4329	no
bmr4331	yes
bmr4332	no
bmr4333	no
bmr4334	no

Member	Like-for-like 386
bmr4335	no
bmr4336	no
bmr4337	no
bmr434	yes
bmr4340	no
bmr4342	no
bmr4343	no
bmr4344	no
bmr435	yes
bmr4350	no
bmr4351	no
bmr4354	no
bmr4357	no
bmr4358	no
bmr4360	no
bmr4364	no
bmr4369	no
bmr4370	no
bmr4376	yes
bmr4378	no
bmr4380	no
bmr4382	no
bmr4383	no
bmr4386	no
bmr4388	no
bmr4393	yes
bmr4394	no
bmr4395	no
bmr4397	no
bmr4404	no
bmr4405	no
bmr4407	no
bmr4410	no
bmr4411	no
bmr4414	no
bmr4417	no
bmr4418	no
bmr4419	no
bmr4422	no
bmr4423	no
bmr4425	no
bmr4426	no
bmr4427	no
bmr4429	no
bmr443	no
bmr4431	no
bmr4437	no
bmr4438	no
bmr4447	yes
bmr4448	no
bmr4449	yes
bmr4452	no
bmr4453	no
bmr4455	no
bmr4458	no
bmr4459	no
bmr4460	no
bmr4465	no
bmr4466	no
bmr4467	no
bmr447	no
bmr4470	yes
bmr4472	no
bmr4473	no

Member	Like-for-like	386
bmr4475	no	
bmr4486	no	
bmr4487	no	
bmr4490	no	
bmr4492	no	
bmr4494	no	
bmr4496	no	
bmr4497	no	
bmr4506	no	
bmr4507	no	
bmr4509	no	
bmr4524	no	
bmr4526	yes	
bmr4540	no	
bmr4553	no	
bmr4554	no	
bmr4557	no	
bmr4558	yes	
bmr4562	no	
bmr4565	no	
bmr4568	no	
bmr4570	no	
bmr4573	yes	
bmr4579	no	
bmr4584	no	
bmr4585	no	
bmr4587	no	
bmr4589	no	
bmr4591	no	
bmr4592	no	
bmr4593	no	
bmr4599	no	
bmr4615	no	
bmr4616	no	
bmr4617	no	
bmr4619	no	
bmr4622	no	
bmr4623	no	
bmr4628	no	
bmr4633	no	
bmr4636	no	
bmr4637	no	
bmr4638	no	
bmr4639	no	
bmr4642	no	
bmr4644	no	
bmr4645	no	
bmr4648	yes	
bmr4649	no	
bmr4656	yes	
bmr4661	no	
bmr4666	no	
bmr4668	yes	
bmr4669	no	
bmr4671	no	
bmr4673	no	
bmr4674	no	
bmr4675	no	
bmr4677	no	
bmr4679	no	
bmr4688	no	
bmr4695	no	
bmr4696	no	
bmr4699	no	

Member	Like-for-like	386
bmr4704	no	
bmr4706	no	
bmr4712	no	
bmr4714	no	
bmr4716	yes	
bmr4717	yes	
bmr4718	no	
bmr4727	no	
bmr4731	no	
bmr4735	no	
bmr4740	no	
bmr4751	no	
bmr4752	no	
bmr4757	yes	
bmr4758	no	
bmr4760	no	
bmr4773	no	
bmr4779	no	
bmr4782	no	
bmr4784	no	
bmr4785	no	
bmr4786	no	
bmr4789	no	
bmr4797	no	
bmr4798	no	
bmr480	no	
bmr4811	no	
bmr4814	yes	
bmr4815	no	
bmr4819	yes	
bmr4820	no	
bmr4825	no	
bmr4827	no	
bmr4828	no	
bmr4829	no	
bmr4831	no	
bmr4834	no	
bmr4836	no	
bmr4839	no	
bmr4840	no	
bmr4843	no	
bmr4848	no	
bmr4850	yes	
bmr4854	no	
bmr4857	no	
bmr4864	no	
bmr4865	no	
bmr4874	no	
bmr4876	no	
bmr4878	no	
bmr4880	yes	
bmr4881	no	
bmr4883	yes	
bmr4884	no	
bmr4886	no	
bmr4887	no	
bmr4891	yes	
bmr4893	no	
bmr4896	no	
bmr4899	no	
bmr4900	no	
bmr4901	yes	
bmr4908	no	
bmr4909	no	

Member	Like-for-like 386
bmr491	no
bmr4911	no
bmr4912	yes
bmr4918	no
bmr4923	no
bmr4924	no
bmr4925	no
bmr4928	no
bmr4929	yes
bmr4930	no
bmr4931	no
bmr4932	no
bmr4933	no
bmr4934	yes
bmr4937	no
bmr4939	no
bmr494	no
bmr4940	no
bmr4944	no
bmr4946	no
bmr4947	no
bmr495	no
bmr4957	no
bmr4959	no
bmr496	no
bmr4961	no
bmr4964	no
bmr4965	no
bmr4969	no
bmr497	no
bmr4971	no
bmr4977	no
bmr4978	no
bmr4980	no
bmr4984	no
bmr4985	no
bmr4986	no
bmr4987	no
bmr4988	no
bmr4989	no
bmr4991	yes
bmr4995	no
bmr5000	yes
bmr5004	yes
bmr5006	no
bmr5008	no
bmr5009	no
bmr5010	yes
bmr5011	no
bmr5018	no
bmr5022	no
bmr5028	yes
bmr5030	no
bmr5039	no
bmr5044	yes
bmr5047	no
bmr5049	yes
bmr5051	yes
bmr5052	no
bmr5056	no
bmr5058	no
bmr5059	no
bmr5060	yes
bmr5065	no

Member	Like-for-like	386
bmr5067	no	
bmr5068	no	
bmr5069	no	
bmr5070	no	
bmr5072	no	
bmr5075	no	
bmr5077	yes	
bmr5081	no	
bmr5082	no	
bmr5083	no	
bmr5092	no	
bmr5093	yes	
bmr5094	no	
bmr5096	no	
bmr5097	no	
bmr5098	no	
bmr5099	no	
bmr5104	no	
bmr5119	no	
bmr5123	no	
bmr5124	no	
bmr5125	no	
bmr5126	yes	
bmr5128	no	
bmr5129	no	
bmr5130	yes	
bmr5131	no	
bmr5132	no	
bmr5133	no	
bmr5142	yes	
bmr5143	no	
bmr5147	no	
bmr5149	no	
bmr5155	no	
bmr5158	no	
bmr5163	no	
bmr5165	yes	
bmr5166	no	
bmr5169	no	
bmr5175	no	
bmr5176	yes	
bmr5177	yes	
bmr5181	no	
bmr5182	no	
bmr5184	no	
bmr5185	no	
bmr5186	no	
bmr5188	no	
bmr5190	no	
bmr5191	yes	
bmr5194	no	
bmr5200	no	
bmr5202	yes	
bmr5204	no	
bmr5208	no	
bmr5209	no	
bmr5210	no	
bmr5213	no	
bmr5217	no	
bmr5218	no	
bmr5219	no	
bmr5220	no	
bmr5226	no	
bmr5232	no	

Member	Like-for-like	386
bmr5235	yes	
bmr5236	no	
bmr5242	no	
bmr5244	no	
bmr5249	no	
bmr5255	no	
bmr5257	no	
bmr5261	no	
bmr5264	no	
bmr5265	no	
bmr5266	yes	
bmr5267	no	
bmr5268	no	
bmr5273	no	
bmr5274	no	
bmr5275	no	
bmr5284	no	
bmr5285	no	
bmr5288	no	
bmr5295	no	
bmr5297	no	
bmr5298	yes	
bmr5309	no	
bmr5312	no	
bmr5313	no	
bmr5317	no	
bmr5318	no	
bmr5320	no	
bmr5322	no	
bmr5323	no	
bmr5324	no	
bmr5326	no	
bmr5334	no	
bmr5338	no	
bmr5340	yes	
bmr5347	no	
bmr5350	yes	
bmr5352	no	
bmr5354	yes	
bmr5355	no	
bmr5356	no	
bmr5364	no	
bmr5368	no	
bmr5369	no	
bmr5392	no	
bmr5393	no	
bmr5398	no	
bmr5399	yes	
bmr5401	yes	
bmr5402	yes	
bmr5408	no	
bmr5410	no	
bmr5454	no	
bmr5455	yes	
bmr5456	no	
bmr5457	no	
bmr5458	yes	
bmr5462	no	
bmr5464	no	
bmr5465	yes	
bmr5469	no	
bmr5471	no	
bmr5472	no	
bmr5473	no	

Member	Like-for-like 386
bmr5474	no
bmr5475	no
bmr5484	no
bmr5485	no
bmr5490	yes
bmr5491	no
bmr5492	no
bmr5495	no
bmr55	no
bmr5501	no
bmr5507	no
bmr5508	no
bmr5509	no
bmr5511	no
bmr5512	yes
bmr5516	no
bmr5518	no
bmr5527	no
bmr5529	no
bmr5535	no
bmr5536	no
bmr5538	yes
bmr554	no
bmr5540	no
bmr555	no
bmr5551	no
bmr5554	no
bmr5555	no
bmr5556	no
bmr5560	no
bmr5561	no
bmr5565	yes
bmr5566	yes
bmr5567	no
bmr5568	yes
bmr5571	no
bmr5572	no
bmr5573	no
bmr5579	no
bmr5583	no
bmr5589	yes
bmr5594	no
bmr5595	no
bmr5596	yes
bmr5598	no
bmr5599	no
bmr5603	yes
bmr5604	no
bmr5606	no
bmr5609	no
bmr5610	no
bmr5615	no
bmr5617	no
bmr5619	no
bmr5621	yes
bmr5629	no
bmr5630	yes
bmr5652	no
bmr5653	yes
bmr5657	no
bmr5662	no
bmr5664	no
bmr5665	no
bmr5666	no

Member	Like-for-like	386
bmr5667	no	
bmr5668	no	
bmr5670	no	
bmr5672	no	
bmr5677	yes	
bmr5678	no	
bmr5682	no	
bmr5687	no	
bmr5690	no	
bmr5692	no	
bmr5698	no	
bmr5701	yes	
bmr5702	no	
bmr5704	no	
bmr5706	no	
bmr5710	no	
bmr5712	no	
bmr5720	yes	
bmr5721	no	
bmr5723	no	
bmr5724	no	
bmr5725	no	
bmr5726	no	
bmr5727	no	
bmr5728	no	
bmr5736	no	
bmr5740	no	
bmr5741	no	
bmr5742	no	
bmr5744	no	
bmr5746	no	
bmr5749	no	
bmr5751	yes	
bmr5753	no	
bmr5754	yes	
bmr5756	no	
bmr5760	no	
bmr5761	no	
bmr5764	no	
bmr5767	no	
bmr5768	no	
bmr5769	yes	
bmr5771	no	
bmr5778	no	
bmr5779	yes	
bmr5783	no	
bmr5784	no	
bmr5786	yes	
bmr5789	yes	
bmr5792	no	
bmr5794	no	
bmr5796	no	
bmr5799	no	
bmr5803	no	
bmr5804	no	
bmr5806	no	
bmr5807	no	
bmr5808	no	
bmr5810	no	
bmr5813	yes	
bmr5817	no	
bmr5820	yes	
bmr5824	no	
bmr5825	no	

Member	Like-for-like	386
bmr5826	no	
bmr5833	yes	
bmr5842	no	
bmr5843	no	
bmr5844	no	
bmr5845	no	
bmr5848	no	
bmr5851	no	
bmr5853	no	
bmr5863	no	
bmr5865	no	
bmr5866	no	
bmr5868	no	
bmr5880	no	
bmr5884	yes	
bmr5886	no	
bmr5888	no	
bmr5891	yes	
bmr5898	yes	
bmr5899	no	
bmr5903	no	
bmr5905	no	
bmr5907	no	
bmr5908	no	
bmr5909	no	
bmr5910	no	
bmr5912	yes	
bmr5917	no	
bmr5918	no	
bmr5920	no	
bmr5921	yes	
bmr5924	yes	
bmr5928	yes	
bmr5929	yes	
bmr5930	yes	
bmr5933	yes	
bmr5935	no	
bmr5936	no	
bmr5937	no	
bmr5940	no	
bmr5942	yes	
bmr5948	no	
bmr5949	no	
bmr5950	no	
bmr5952	no	
bmr5953	no	
bmr5954	no	
bmr5958	no	
bmr5959	no	
bmr5961	no	
bmr5967	no	
bmr5973	no	
bmr5974	yes	
bmr5976	yes	
bmr5995	no	
bmr5996	no	
bmr5997	no	
bmr5998	yes	
bmr5999	no	
bmr60	no	
bmr6000	yes	
bmr6001	yes	
bmr6002	yes	
bmr6004	yes	

Member	Like-for-like	386
bmr6007	no	
bmr6008	no	
bmr6011	no	
bmr6012	no	
bmr6013	yes	
bmr6014	no	
bmr6016	no	
bmr6019	no	
bmr6022	no	
bmr6029	no	
bmr6032	no	
bmr6034	no	
bmr6035	no	
bmr6037	no	
bmr6044	no	
bmr6045	yes	
bmr6046	no	
bmr6047	yes	
bmr6050	no	
bmr6051	no	
bmr6053	no	
bmr6056	no	
bmr6059	no	
bmr6060	no	
bmr6070	no	
bmr6071	yes	
bmr6072	yes	
bmr6074	no	
bmr6075	no	
bmr6080	no	
bmr6082	no	
bmr6084	no	
bmr6089	no	
bmr6090	no	
bmr6091	no	
bmr6092	no	
bmr6093	no	
bmr6096	no	
bmr6097	no	
bmr6102	no	
bmr6103	no	
bmr6108	no	
bmr6110	no	
bmr6113	yes	
bmr6114	no	
bmr6120	yes	
bmr6121	yes	
bmr6123	no	
bmr6127	no	
bmr6129	no	
bmr6130	no	
bmr6132	no	
bmr6133	yes	
bmr6135	no	
bmr6136	no	
bmr6137	no	
bmr6141	no	
bmr6142	no	
bmr6143	no	
bmr6150	no	
bmr6152	yes	
bmr6157	no	
bmr6158	no	
bmr6159	yes	

Member	Like-for-like	386
bmr6160	no	
bmr6165	no	
bmr6166	yes	
bmr6171	yes	
bmr6172	no	
bmr6176	no	
bmr6177	no	
bmr6178	no	
bmr6179	no	
bmr6180	no	
bmr6181	no	
bmr6182	no	
bmr6184	no	
bmr6185	no	
bmr6192	no	
bmr6193	no	
bmr6194	no	
bmr6198	no	
bmr6199	no	
bmr6207	no	
bmr6212	no	
bmr6216	no	
bmr6222	no	
bmr6223	no	
bmr6225	yes	
bmr6231	no	
bmr6233	yes	
bmr6234	no	
bmr6238	no	
bmr6240	yes	
bmr6241	yes	
bmr6247	no	
bmr6248	no	
bmr6250	no	
bmr6252	no	
bmr6255	no	
bmr6256	yes	
bmr6257	no	
bmr6259	yes	
bmr6263	no	
bmr6264	no	
bmr6265	no	
bmr6266	no	
bmr6267	no	
bmr6268	no	
bmr6271	no	
bmr6272	no	
bmr6279	no	
bmr6283	no	
bmr6284	no	
bmr6285	yes	
bmr6287	yes	
bmr6288	no	
bmr6292	no	
bmr6297	no	
bmr6298	yes	
bmr6299	yes	
bmr6300	no	
bmr6302	no	
bmr6303	no	
bmr6304	no	
bmr6305	no	
bmr6306	no	
bmr6309	no	

Member	Like-for-like	386
bmr6310	yes	
bmr6311	no	
bmr6312	yes	
bmr6317	no	
bmr6321	no	
bmr6322	no	
bmr6327	no	
bmr6328	no	
bmr6330	no	
bmr6331	yes	
bmr6334	no	
bmr6336	no	
bmr6338	no	
bmr6339	yes	
bmr6343	yes	
bmr6344	no	
bmr6345	yes	
bmr6349	no	
bmr6351	no	
bmr6354	yes	
bmr6356	yes	
bmr6357	no	
bmr6358	yes	
bmr6359	yes	
bmr6362	no	
bmr6363	no	
bmr6364	yes	
bmr6365	yes	
bmr6366	yes	
bmr6367	yes	
bmr6369	yes	
bmr6370	no	
bmr6371	yes	
bmr6373	no	
bmr6379	no	
bmr6390	no	
bmr6391	no	
bmr6392	no	
bmr6393	no	
bmr6399	no	
bmr6400	no	
bmr6401	no	
bmr6404	no	
bmr6406	yes	
bmr6407	no	
bmr6410	yes	
bmr6415	no	
bmr6416	no	
bmr6419	yes	
bmr6429	no	
bmr6443	no	
bmr6444	no	
bmr6447	yes	
bmr6448	yes	
bmr6449	no	
bmr6451	no	
bmr6455	no	
bmr6460	yes	
bmr6464	yes	
bmr6465	no	
bmr6467	no	
bmr6468	no	
bmr6475	yes	
bmr6476	yes	

Member	Like-for-like	386
bmr6479	no	
bmr6480	no	
bmr6481	no	
bmr6482	no	
bmr6483	no	
bmr6489	no	
bmr6493	no	
bmr6494	no	
bmr6495	no	
bmr6496	no	
bmr6498	no	
bmr6500	no	
bmr6501	no	
bmr6503	no	
bmr6504	no	
bmr6513	no	
bmr6514	no	
bmr6518	yes	
bmr6520	no	
bmr6525	no	
bmr6536	no	
bmr6538	no	
bmr6539	no	
bmr6540	no	
bmr6542	no	
bmr6546	no	
bmr6549	no	
bmr6554	no	
bmr6555	no	
bmr6556	no	
bmr6561	no	
bmr6563	no	
bmr6564	yes	
bmr6568	no	
bmr6569	no	
bmr6570	no	
bmr6571	no	
bmr6575	no	
bmr6578	no	
bmr6579	no	
bmr6580	no	
bmr6585	no	
bmr6587	no	
bmr6588	no	
bmr6589	no	
bmr6590	no	
bmr6592	no	
bmr6594	no	
bmr6596	no	
bmr6597	no	
bmr6598	no	
bmr6600	no	
bmr6603	yes	
bmr6604	no	
bmr6612	no	
bmr6613	no	
bmr6617	no	
bmr6620	yes	
bmr6621	no	
bmr6624	yes	
bmr6625	no	
bmr6626	yes	
bmr6628	no	
bmr6629	yes	

Member	Like-for-like	386
bmr6635	no	
bmr6639	no	
bmr6642	no	
bmr6643	no	
bmr6647	no	
bmr6648	no	
bmr6653	no	
bmr6654	no	
bmr6655	no	
bmr6681	no	
bmr6682	no	
bmr6688	no	
bmr6693	yes	
bmr6695	no	
bmr6705	no	
bmr6712	no	
bmr6717	yes	
bmr6721	yes	
bmr6722	yes	
bmr6724	yes	
bmr6727	no	
bmr6731	yes	
bmr6733	no	
bmr6734	no	
bmr6735	no	
bmr6738	yes	
bmr6739	no	
bmr6740	no	
bmr6742	yes	
bmr6746	yes	
bmr6747	no	
bmr6751	no	
bmr6752	yes	
bmr6753	yes	
bmr6754	no	
bmr6759	yes	
bmr6761	no	
bmr6766	no	
bmr6769	no	
bmr6771	no	
bmr6776	no	
bmr6779	no	
bmr6780	no	
bmr6786	no	
bmr6787	no	
bmr6801	no	
bmr6807	no	
bmr6810	yes	
bmr6811	no	
bmr6820	no	
bmr6821	no	
bmr6822	no	
bmr6824	yes	
bmr6828	no	
bmr6832	no	
bmr6833	no	
bmr6834	yes	
bmr6835	no	
bmr6838	no	
bmr6840	yes	
bmr6843	no	
bmr6847	yes	
bmr6850	no	
bmr6851	yes	

Member	Like-for-like	386
bmr6853	no	
bmr6860	yes	
bmr6866	yes	
bmr6868	no	
bmr6869	yes	
bmr6872	no	
bmr6875	no	
bmr6876	no	
bmr6877	no	
bmr6882	no	
bmr6883	no	
bmr6888	no	
bmr6893	no	
bmr6894	no	
bmr6899	no	
bmr6900	no	
bmr6901	yes	
bmr6902	no	
bmr6905	no	
bmr6907	no	
bmr6908	no	
bmr6910	no	
bmr6914	no	
bmr6916	no	
bmr6917	no	
bmr6918	no	
bmr6919	no	
bmr6920	no	
bmr6922	no	
bmr6925	no	
bmr6926	no	
bmr6928	no	
bmr6929	no	
bmr6939	no	
bmr6940	no	
bmr6944	no	
bmr6945	no	
bmr695	no	
bmr6955	no	
bmr6957	yes	
bmr6960	no	
bmr6969	no	
bmr6970	no	
bmr6973	no	
bmr6980	yes	
bmr6984	no	
bmr6985	yes	
bmr6986	yes	
bmr7003	no	
bmr7004	yes	
bmr7007	yes	
bmr7014	no	
bmr7019	no	
bmr7020	no	
bmr7021	no	
bmr7032	no	
bmr7034	no	
bmr7054	yes	
bmr7055	no	
bmr7058	yes	
bmr7063	yes	
bmr7067	no	
bmr7068	no	
bmr7069	yes	

Member	Like-for-like	386
bmr7070	no	
bmr7072	yes	
bmr7074	no	
bmr7075	no	
bmr7078	no	
bmr7079	yes	
bmr7083	yes	
bmr7084	no	
bmr7086	no	
bmr7087	no	
bmr7089	no	
bmr7091	no	
bmr7092	yes	
bmr7095	yes	
bmr7099	no	
bmr7101	no	
bmr7102	no	
bmr7103	no	
bmr7106	yes	
bmr7107	no	
bmr7110	no	
bmr7112	no	
bmr7114	no	
bmr7115	no	
bmr7117	no	
bmr7119	no	
bmr7121	yes	
bmr7126	no	
bmr7129	no	
bmr7130	no	
bmr7131	no	
bmr7134	no	
bmr7135	no	
bmr7136	no	
bmr7137	yes	
bmr7147	no	
bmr7151	no	
bmr7159	no	
bmr7160	no	
bmr7167	yes	
bmr7171	yes	
bmr7172	yes	
bmr7173	no	
bmr7174	no	
bmr7176	no	
bmr7180	yes	
bmr7182	yes	
bmr7186	no	
bmr7196	no	
bmr7199	no	
bmr7200	no	
bmr7201	yes	
bmr7203	yes	
bmr7207	yes	
bmr7209	yes	
bmr7210	no	
bmr7211	no	
bmr7220	no	
bmr7224	no	
bmr7226	no	
bmr7227	no	
bmr7228	yes	
bmr7229	yes	
bmr7237	no	

Member	Like-for-like	386
bmr7238	no	
bmr7239	no	
bmr7241	no	
bmr7242	yes	
bmr7247	no	
bmr7251	no	
bmr7256	no	
bmr7259	no	
bmr7260	no	
bmr7261	no	
bmr7264	no	
bmr7272	no	
bmr7273	no	
bmr7274	yes	
bmr7281	yes	
bmr7287	yes	
bmr7297	no	
bmr7302	yes	
bmr7307	no	
bmr7308	no	
bmr7310	no	
bmr7314	no	
bmr7315	no	
bmr7319	no	
bmr7320	no	
bmr7321	no	
bmr7322	no	
bmr7324	no	
bmr7330	no	
bmr7339	yes	
bmr7356	no	
bmr7357	no	
bmr7359	no	
bmr7360	no	
bmr7365	no	
bmr7370	no	
bmr7376	no	
bmr7377	no	
bmr7396	yes	
bmr7399	no	
bmr7400	no	
bmr7429	no	
bmr754	no	
bmr79	no	
bmr8	no	
bmr80	no	
bmr915	no	
bmr916	no	
bmr936	no	
bmr937	no	
bmr96	no	
bmr975	no	
bmr979	no	

2.6 Shielding trajectory set

Member	Protein	Frames	Atoms
1L2Y	TC5b Trp-cage	1,501	304
5AWL	CLN025	5,001	166

3 Retained source-deposited assignment

BMRB 4678 deposits the ILE 56 CB and CG1 assignments shown below; the accompanying BMRB validation report flags both values and the CB > CG1 side-chain-order rule break, and the RefDB-referenced labels were retained unchanged in both chemical-shift preparation routes.

3.1 BMR4678 ILE 56 carbon assignments

Atom	BMRB deposit (ppm)	RefDB value used (ppm)	Use
CB	28.27	28.29	Training
CG1	38.39	38.41	Training

4 Shielding designation vocabulary

4.1 Shielding designation support

IUPAC	Element	Region	Headline	$\geq 10/\text{group}$	All 79	Primary 61	Previous 18	<i>n</i>	Proteins
ALA:C	6	backbone	C	yes	scored	scored	scored	268	73
ALA:CA	6	backbone	CA	yes	scored	scored	scored	268	73
ALA:CB	6	sidechain	CB	yes	scored	scored	scored	268	73
ALA:H	1	backbone	H	yes	scored	scored	scored	267	73
ALA:H1	1	sidechain	—	no	preparation excluded	not observed	preparation excluded	0	0
ALA:H2	1	sidechain	—	no	preparation excluded	not observed	preparation excluded	0	0
ALA:H3	1	sidechain	—	no	preparation excluded	not observed	preparation excluded	0	0
ALA:HA	1	backbone	HA	yes	scored	scored	scored	268	73
ALA:HB1	1	sidechain	—	yes	scored	scored	scored	268	73
ALA:HB2	1	sidechain	—	yes	scored	scored	scored	268	73
ALA:HB3	1	sidechain	—	yes	scored	scored	scored	268	73
ALA:N	7	backbone	N	yes	scored	scored	scored	268	73
ALA:O	8	backbone	—	yes	scored	scored	scored	268	73
ALA:OXT	8	sidechain	—	no	preparation excluded	preparation excluded	preparation excluded	0	0
ARG:C	6	backbone	C	yes	scored	scored	scored	338	77
ARG:CA	6	backbone	CA	yes	scored	scored	scored	338	77
ARG:CB	6	sidechain	CB	yes	scored	scored	scored	338	77
ARG:CD	6	sidechain	—	yes	scored	scored	scored	338	77
ARG:CG	6	sidechain	—	yes	scored	scored	scored	338	77
ARG:CZ	6	sidechain	—	yes	scored	scored	scored	338	77
ARG:H	1	backbone	H	yes	scored	scored	scored	336	76
ARG:H1	1	sidechain	—	no	preparation excluded	preparation excluded	not observed	0	0
ARG:H2	1	sidechain	—	no	preparation excluded	preparation excluded	not observed	0	0
ARG:H3	1	sidechain	—	no	preparation excluded	preparation excluded	not observed	0	0
ARG:HA	1	backbone	HA	yes	scored	scored	scored	338	77
ARG:HB2	1	sidechain	—	yes	scored	scored	scored	338	77
ARG:HB3	1	sidechain	—	yes	scored	scored	scored	338	77
ARG:HD2	1	sidechain	—	yes	scored	scored	scored	338	77
ARG:HD3	1	sidechain	—	yes	scored	scored	scored	338	77
ARG:HE	1	sidechain	—	yes	scored	scored	scored	338	77
ARG:HG2	1	sidechain	—	yes	scored	scored	scored	338	77
ARG:HG3	1	sidechain	—	yes	scored	scored	scored	338	77
ARG:HH11	1	sidechain	—	yes	scored	scored	scored	338	77
ARG:HH12	1	sidechain	—	yes	scored	scored	scored	338	77
ARG:HH21	1	sidechain	—	yes	scored	scored	scored	338	77
ARG:HH22	1	sidechain	—	yes	scored	scored	scored	338	77
ARG:N	7	backbone	N	yes	scored	scored	scored	338	77
ARG:NE	7	sidechain	—	yes	scored	scored	scored	338	77

IUPAC	Element	Region	Headline	$\geq 10/\text{group}$	All 79	Primary 61	Previous 18	<i>n</i>	Proteins
ARG:NH1	7	sidechain	—	yes	scored	scored	scored	338	77
ARG:NH2	7	sidechain	—	yes	scored	scored	scored	338	77
ARG:O	8	backbone	—	yes	scored	scored	scored	338	77
ARG:OXT	8	sidechain	—	no	preparation excluded	preparation excluded	preparation excluded	0	0
ASN:C	6	backbone	C	yes	scored	scored	scored	98	54
ASN:CA	6	backbone	CA	yes	scored	scored	scored	98	54
ASN:CB	6	sidechain	CB	yes	scored	scored	scored	98	54
ASN:CG	6	sidechain	—	yes	scored	scored	scored	98	54
ASN:H	1	backbone	H	yes	scored	scored	scored	97	53
ASN:H1	1	sidechain	—	no	preparation excluded	preparation excluded	not observed	0	0
ASN:H2	1	sidechain	—	no	preparation excluded	preparation excluded	not observed	0	0
ASN:H3	1	sidechain	—	no	preparation excluded	preparation excluded	not observed	0	0
ASN:HA	1	backbone	HA	yes	scored	scored	scored	98	54
ASN:HB2	1	sidechain	—	yes	scored	scored	scored	98	54
ASN:HB3	1	sidechain	—	yes	scored	scored	scored	98	54
ASN:HD21	1	sidechain	—	yes	scored	scored	scored	98	54
ASN:HD22	1	sidechain	—	yes	scored	scored	scored	98	54
ASN:N	7	backbone	N	yes	scored	scored	scored	98	54
ASN:ND2	7	sidechain	—	yes	scored	scored	scored	98	54
ASN:O	8	backbone	—	yes	scored	scored	scored	98	54
ASN:OD1	8	sidechain	—	yes	scored	scored	scored	98	54
ASN:OXT	8	sidechain	—	no	preparation excluded	not observed	preparation excluded	0	0
ASP:C	6	backbone	C	yes	scored	scored	scored	154	59
ASP:CA	6	backbone	CA	yes	scored	scored	scored	154	59
ASP:CB	6	sidechain	CB	yes	scored	scored	scored	154	59
ASP:CG	6	sidechain	—	yes	scored	scored	scored	154	59
ASP:H	1	backbone	H	yes	scored	scored	scored	154	59
ASP:H1	1	sidechain	—	no	not observed	not observed	not observed	0	0
ASP:H2	1	sidechain	—	no	not observed	not observed	not observed	0	0
ASP:H3	1	sidechain	—	no	not observed	not observed	not observed	0	0
ASP:HA	1	backbone	HA	yes	scored	scored	scored	154	59
ASP:HB2	1	sidechain	—	yes	scored	scored	scored	154	59
ASP:HB3	1	sidechain	—	yes	scored	scored	scored	154	59
ASP:N	7	backbone	N	yes	scored	scored	scored	154	59
ASP:O	8	backbone	—	yes	scored	scored	scored	154	59
ASP:OD1	8	sidechain	—	yes	scored	scored	scored	154	59
ASP:OD2	8	sidechain	—	yes	scored	scored	scored	154	59
ASP:OXT	8	sidechain	—	no	preparation excluded	preparation excluded	preparation excluded	0	0
CYS:C	6	backbone	C	no	scored	scored	scored	42	29
CYS:CA	6	backbone	CA	no	scored	scored	scored	42	29
CYS:CB	6	sidechain	CB	no	scored	scored	scored	42	29
CYS:H	1	backbone	H	no	scored	scored	scored	42	29

IUPAC	Element	Region	Headline	$\geq 10/\text{group}$	All 79	Primary 61	Previous 18	n	Proteins
CYS:H1	1	sidechain	—	no	not observed	not observed	not observed	0	0
CYS:H2	1	sidechain	—	no	not observed	not observed	not observed	0	0
CYS:H3	1	sidechain	—	no	not observed	not observed	not observed	0	0
CYS:HA	1	backbone	HA	no	scored	scored	scored	42	29
CYS:HB2	1	sidechain	—	no	scored	scored	scored	42	29
CYS:HB3	1	sidechain	—	no	scored	scored	scored	42	29
CYS:HG	1	sidechain	—	no	scored	scored	scored	40	28
CYS:N	7	backbone	N	no	scored	scored	scored	42	29
CYS:O	8	backbone	—	no	scored	scored	scored	42	29
CYS:OXT	8	sidechain	—	no	not observed	not observed	not observed	0	0
CYS:SG	16	sidechain	—	no	preparation excluded	preparation excluded	preparation excluded	0	0
GLN:C	6	backbone	C	yes	scored	scored	scored	71	41
GLN:CA	6	backbone	CA	yes	scored	scored	scored	71	41
GLN:CB	6	sidechain	CB	yes	scored	scored	scored	71	41
GLN:CD	6	sidechain	—	yes	scored	scored	scored	71	41
GLN:CG	6	sidechain	—	yes	scored	scored	scored	71	41
GLN:H	1	backbone	H	yes	scored	scored	scored	71	41
GLN:H1	1	sidechain	—	no	not observed	not observed	not observed	0	0
GLN:H2	1	sidechain	—	no	not observed	not observed	not observed	0	0
GLN:H3	1	sidechain	—	no	not observed	not observed	not observed	0	0
GLN:HA	1	backbone	HA	yes	scored	scored	scored	71	41
GLN:HB2	1	sidechain	—	yes	scored	scored	scored	71	41
GLN:HB3	1	sidechain	—	yes	scored	scored	scored	71	41
GLN:HE21	1	sidechain	—	yes	scored	scored	scored	71	41
GLN:HE22	1	sidechain	—	yes	scored	scored	scored	71	41
GLN:HG2	1	sidechain	—	yes	scored	scored	scored	71	41
GLN:HG3	1	sidechain	—	yes	scored	scored	scored	71	41
GLN:N	7	backbone	N	yes	scored	scored	scored	71	41
GLN:NE2	7	sidechain	—	yes	scored	scored	scored	71	41
GLN:O	8	backbone	—	yes	scored	scored	scored	71	41
GLN:OE1	8	sidechain	—	yes	scored	scored	scored	71	41
GLN:OXT	8	sidechain	—	no	preparation excluded	preparation excluded	not observed	0	0
GLU:C	6	backbone	C	yes	scored	scored	scored	207	64
GLU:CA	6	backbone	CA	yes	scored	scored	scored	207	64
GLU:CB	6	sidechain	CB	yes	scored	scored	scored	207	64
GLU:CD	6	sidechain	—	yes	scored	scored	scored	207	64
GLU:CG	6	sidechain	—	yes	scored	scored	scored	207	64
GLU:H	1	backbone	H	yes	scored	scored	scored	207	64
GLU:H1	1	sidechain	—	no	not observed	not observed	not observed	0	0
GLU:H2	1	sidechain	—	no	not observed	not observed	not observed	0	0
GLU:H3	1	sidechain	—	no	not observed	not observed	not observed	0	0
GLU:HA	1	backbone	HA	yes	scored	scored	scored	207	64

IUPAC	Element	Region	Headline	$\geq 10/\text{group}$	All 79	Primary 61	Previous 18	n	Proteins
GLU:HB2	1	sidechain	—	yes	scored	scored	scored	207	64
GLU:HB3	1	sidechain	—	yes	scored	scored	scored	207	64
GLU:HG2	1	sidechain	—	yes	scored	scored	scored	207	64
GLU:HG3	1	sidechain	—	yes	scored	scored	scored	207	64
GLU:N	7	backbone	N	yes	scored	scored	scored	207	64
GLU:O	8	backbone	—	yes	scored	scored	scored	207	64
GLU:OE1	8	sidechain	—	yes	scored	scored	scored	207	64
GLU:OE2	8	sidechain	—	yes	scored	scored	scored	207	64
GLU:OXT	8	sidechain	—	no	preparation excluded	preparation excluded	preparation excluded	0	0
GLY:C	6	backbone	C	yes	scored	scored	scored	272	74
GLY:CA	6	backbone	CA	yes	scored	scored	scored	272	74
GLY:H	1	backbone	H	yes	scored	scored	scored	272	74
GLY:H1	1	sidechain	—	no	not observed	not observed	not observed	0	0
GLY:H2	1	sidechain	—	no	not observed	not observed	not observed	0	0
GLY:H3	1	sidechain	—	no	not observed	not observed	not observed	0	0
GLY:HA2	1	sidechain	HA	yes	scored	scored	scored	272	74
GLY:HA3	1	sidechain	HA	yes	scored	scored	scored	272	74
GLY:N	7	backbone	N	yes	scored	scored	scored	272	74
GLY:O	8	backbone	—	yes	scored	scored	scored	272	74
GLY:OXT	8	sidechain	—	no	preparation excluded	preparation excluded	not observed	0	0
HIS:C	6	backbone	C	no	scored	scored	scored	65	43
HIS:CA	6	backbone	CA	no	scored	scored	scored	65	43
HIS:CB	6	sidechain	CB	no	scored	scored	scored	65	43
HIS:CD2	6	sidechain	—	no	scored	scored	scored	65	43
HIS:CE1	6	sidechain	—	no	scored	scored	scored	65	43
HIS:CG	6	sidechain	—	no	scored	scored	scored	65	43
HIS:H	1	backbone	H	no	scored	scored	scored	65	43
HIS:H1	1	sidechain	—	no	not observed	not observed	not observed	0	0
HIS:H2	1	sidechain	—	no	not observed	not observed	not observed	0	0
HIS:H3	1	sidechain	—	no	not observed	not observed	not observed	0	0
HIS:HA	1	backbone	HA	no	scored	scored	scored	65	43
HIS:HB2	1	sidechain	—	no	scored	scored	scored	65	43
HIS:HB3	1	sidechain	—	no	scored	scored	scored	65	43
HIS:HD2	1	sidechain	—	no	scored	scored	scored	65	43
HIS:HE1	1	sidechain	—	no	scored	scored	scored	65	43
HIS:HE2	1	sidechain	—	no	scored	scored	scored	65	43
HIS:N	7	backbone	N	no	scored	scored	scored	65	43
HIS:ND1	7	sidechain	—	no	scored	scored	scored	65	43
HIS:NE2	7	sidechain	—	no	scored	scored	scored	65	43
HIS:O	8	backbone	—	no	scored	scored	scored	65	43
HIS:OXT	8	sidechain	—	no	not observed	not observed	not observed	0	0
ILE:C	6	backbone	C	yes	scored	scored	scored	236	77

IUPAC	Element	Region	Headline	$\geq 10/\text{group}$	All 79	Primary 61	Previous 18	<i>n</i>	Proteins
ILE:CA	6	backbone	CA	yes	scored	scored	scored	236	77
ILE:CB	6	sidechain	CB	yes	scored	scored	scored	236	77
ILE:CD1	6	sidechain	—	yes	scored	scored	scored	236	77
ILE:CG1	6	sidechain	—	yes	scored	scored	scored	236	77
ILE:CG2	6	sidechain	—	yes	scored	scored	scored	236	77
ILE:H	1	backbone	H	yes	scored	scored	scored	233	77
ILE:H1	1	sidechain	—	no	preparation excluded	preparation excluded	preparation excluded	0	0
ILE:H2	1	sidechain	—	no	preparation excluded	preparation excluded	preparation excluded	0	0
ILE:H3	1	sidechain	—	no	preparation excluded	preparation excluded	preparation excluded	0	0
ILE:HA	1	backbone	HA	yes	scored	scored	scored	236	77
ILE:HB	1	sidechain	—	yes	scored	scored	scored	236	77
ILE:HD11	1	sidechain	—	yes	scored	scored	scored	236	77
ILE:HD12	1	sidechain	—	yes	scored	scored	scored	236	77
ILE:HD13	1	sidechain	—	yes	scored	scored	scored	236	77
ILE:HG12	1	sidechain	—	yes	scored	scored	scored	236	77
ILE:HG13	1	sidechain	—	yes	scored	scored	scored	236	77
ILE:HG21	1	sidechain	—	yes	scored	scored	scored	236	77
ILE:HG22	1	sidechain	—	yes	scored	scored	scored	236	77
ILE:HG23	1	sidechain	—	yes	scored	scored	scored	236	77
ILE:N	7	backbone	N	yes	scored	scored	scored	236	77
ILE:O	8	backbone	—	yes	scored	scored	scored	236	77
ILE:OXT	8	sidechain	—	no	preparation excluded	preparation excluded	preparation excluded	0	0
LEU:C	6	backbone	C	yes	scored	scored	scored	269	77
LEU:CA	6	backbone	CA	yes	scored	scored	scored	269	77
LEU:CB	6	sidechain	CB	yes	scored	scored	scored	269	77
LEU:CD1	6	sidechain	—	yes	scored	scored	scored	269	77
LEU:CD2	6	sidechain	—	yes	scored	scored	scored	269	77
LEU:CG	6	sidechain	—	yes	scored	scored	scored	269	77
LEU:H	1	backbone	H	yes	scored	scored	scored	267	77
LEU:H1	1	sidechain	—	no	preparation excluded	preparation excluded	not observed	0	0
LEU:H2	1	sidechain	—	no	preparation excluded	preparation excluded	not observed	0	0
LEU:H3	1	sidechain	—	no	preparation excluded	preparation excluded	not observed	0	0
LEU:HA	1	backbone	HA	yes	scored	scored	scored	269	77
LEU:HB2	1	sidechain	—	yes	scored	scored	scored	269	77
LEU:HB3	1	sidechain	—	yes	scored	scored	scored	269	77
LEU:HD11	1	sidechain	—	yes	scored	scored	scored	269	77
LEU:HD12	1	sidechain	—	yes	scored	scored	scored	269	77
LEU:HD13	1	sidechain	—	yes	scored	scored	scored	269	77
LEU:HD21	1	sidechain	—	yes	scored	scored	scored	269	77
LEU:HD22	1	sidechain	—	yes	scored	scored	scored	269	77
LEU:HD23	1	sidechain	—	yes	scored	scored	scored	269	77
LEU:HG	1	sidechain	—	yes	scored	scored	scored	269	77

IUPAC	Element	Region	Headline	$\geq 10/\text{group}$	All 79	Primary 61	Previous 18	<i>n</i>	Proteins
LEU:N	7	backbone	N	yes	scored	scored	scored	269	77
LEU:O	8	backbone	—	yes	scored	scored	scored	269	77
LEU:OXT	8	sidechain	—	no	preparation excluded	preparation excluded	preparation excluded	0	0
LYS:C	6	backbone	C	yes	scored	scored	scored	239	76
LYS:CA	6	backbone	CA	yes	scored	scored	scored	239	76
LYS:CB	6	sidechain	CB	yes	scored	scored	scored	239	76
LYS:CD	6	sidechain	—	yes	scored	scored	scored	239	76
LYS:CE	6	sidechain	—	yes	scored	scored	scored	239	76
LYS:CG	6	sidechain	—	yes	scored	scored	scored	239	76
LYS:H	1	backbone	H	yes	scored	scored	scored	238	75
LYS:H1	1	sidechain	—	no	preparation excluded	preparation excluded	not observed	0	0
LYS:H2	1	sidechain	—	no	preparation excluded	preparation excluded	not observed	0	0
LYS:H3	1	sidechain	—	no	preparation excluded	preparation excluded	not observed	0	0
LYS:HA	1	backbone	HA	yes	scored	scored	scored	239	76
LYS:HB2	1	sidechain	—	yes	scored	scored	scored	239	76
LYS:HB3	1	sidechain	—	yes	scored	scored	scored	239	76
LYS:HD2	1	sidechain	—	yes	scored	scored	scored	239	76
LYS:HD3	1	sidechain	—	yes	scored	scored	scored	239	76
LYS:HE2	1	sidechain	—	yes	scored	scored	scored	239	76
LYS:HE3	1	sidechain	—	yes	scored	scored	scored	239	76
LYS:HG2	1	sidechain	—	yes	scored	scored	scored	239	76
LYS:HG3	1	sidechain	—	yes	scored	scored	scored	239	76
LYS:HZ1	1	sidechain	—	yes	scored	scored	scored	239	76
LYS:HZ2	1	sidechain	—	yes	scored	scored	scored	239	76
LYS:HZ3	1	sidechain	—	yes	scored	scored	scored	239	76
LYS:N	7	backbone	N	yes	scored	scored	scored	239	76
LYS:NZ	7	sidechain	—	yes	scored	scored	scored	239	76
LYS:O	8	backbone	—	yes	scored	scored	scored	239	76
LYS:OXT	8	sidechain	—	no	preparation excluded	preparation excluded	preparation excluded	0	0
MET:C	6	backbone	C	yes	scored	scored	scored	127	75
MET:CA	6	backbone	CA	yes	scored	scored	scored	127	75
MET:CB	6	sidechain	CB	yes	scored	scored	scored	127	75
MET:CE	6	sidechain	—	yes	scored	scored	scored	127	75
MET:CG	6	sidechain	—	yes	scored	scored	scored	127	75
MET:H	1	backbone	H	yes	scored	scored	scored	61	49
MET:H1	1	sidechain	—	no	preparation excluded	preparation excluded	preparation excluded	0	0
MET:H2	1	sidechain	—	no	preparation excluded	preparation excluded	preparation excluded	0	0
MET:H3	1	sidechain	—	no	preparation excluded	preparation excluded	preparation excluded	0	0
MET:HA	1	backbone	HA	yes	scored	scored	scored	127	75
MET:HB2	1	sidechain	—	yes	scored	scored	scored	127	75
MET:HB3	1	sidechain	—	yes	scored	scored	scored	127	75
MET:HE1	1	sidechain	—	yes	scored	scored	scored	127	75

IUPAC	Element	Region	Headline	$\geq 10/\text{group}$	All 79	Primary 61	Previous 18	n	Proteins
MET:HE2	1	sidechain	—	yes	scored	scored	scored	127	75
MET:HE3	1	sidechain	—	yes	scored	scored	scored	127	75
MET:HG2	1	sidechain	—	yes	scored	scored	scored	127	75
MET:HG3	1	sidechain	—	yes	scored	scored	scored	127	75
MET:N	7	backbone	N	yes	scored	scored	scored	127	75
MET:O	8	backbone	—	yes	scored	scored	scored	127	75
MET:OXT	8	sidechain	—	no	preparation excluded	preparation excluded	not observed	0	0
MET:SD	16	sidechain	—	yes	scored	scored	scored	127	75
PHE:C	6	backbone	C	yes	scored	scored	scored	92	61
PHE:CA	6	backbone	CA	yes	scored	scored	scored	92	61
PHE:CB	6	sidechain	CB	yes	scored	scored	scored	92	61
PHE:CD1	6	sidechain	—	yes	scored	scored	scored	92	61
PHE:CD2	6	sidechain	—	yes	scored	scored	scored	92	61
PHE:CE1	6	sidechain	—	yes	scored	scored	scored	92	61
PHE:CE2	6	sidechain	—	yes	scored	scored	scored	92	61
PHE:CG	6	sidechain	—	yes	scored	scored	scored	92	61
PHE:CZ	6	sidechain	—	yes	scored	scored	scored	92	61
PHE:H	1	backbone	H	yes	scored	scored	scored	91	60
PHE:H1	1	sidechain	—	no	preparation excluded	preparation excluded	not observed	0	0
PHE:H2	1	sidechain	—	no	preparation excluded	preparation excluded	not observed	0	0
PHE:H3	1	sidechain	—	no	preparation excluded	preparation excluded	not observed	0	0
PHE:HA	1	backbone	HA	yes	scored	scored	scored	92	61
PHE:HB2	1	sidechain	—	yes	scored	scored	scored	92	61
PHE:HB3	1	sidechain	—	yes	scored	scored	scored	92	61
PHE:HD1	1	sidechain	—	yes	scored	scored	scored	92	61
PHE:HD2	1	sidechain	—	yes	scored	scored	scored	92	61
PHE:HE1	1	sidechain	—	yes	scored	scored	scored	92	61
PHE:HE2	1	sidechain	—	yes	scored	scored	scored	92	61
PHE:HZ	1	sidechain	—	yes	scored	scored	scored	92	61
PHE:N	7	backbone	N	yes	scored	scored	scored	92	61
PHE:O	8	backbone	—	yes	scored	scored	scored	92	61
PHE:OXT	8	sidechain	—	no	not observed	not observed	not observed	0	0
PRO:C	6	backbone	C	yes	scored	scored	scored	97	55
PRO:CA	6	backbone	CA	yes	scored	scored	scored	97	55
PRO:CB	6	sidechain	CB	yes	scored	scored	scored	97	55
PRO:CD	6	sidechain	—	yes	scored	scored	scored	97	55
PRO:CG	6	sidechain	—	yes	scored	scored	scored	97	55
PRO:H2	1	sidechain	—	no	preparation excluded	not observed	preparation excluded	0	0
PRO:H3	1	sidechain	—	no	preparation excluded	not observed	preparation excluded	0	0
PRO:HA	1	backbone	HA	yes	scored	scored	scored	97	55
PRO:HB2	1	sidechain	—	yes	scored	scored	scored	97	55
PRO:HB3	1	sidechain	—	yes	scored	scored	scored	97	55

IUPAC	Element	Region	Headline	$\geq 10/\text{group}$	All 79	Primary 61	Previous 18	<i>n</i>	Proteins
PRO:HD2	1	sidechain	—	yes	scored	scored	scored	97	55
PRO:HD3	1	sidechain	—	yes	scored	scored	scored	97	55
PRO:HG2	1	sidechain	—	yes	scored	scored	scored	97	55
PRO:HG3	1	sidechain	—	yes	scored	scored	scored	97	55
PRO:N	7	backbone	N	yes	scored	scored	scored	97	55
PRO:O	8	backbone	—	yes	scored	scored	scored	97	55
PRO:OXT	8	sidechain	—	no	preparation excluded	preparation excluded	not observed	0	0
SER:C	6	backbone	C	yes	scored	scored	scored	174	69
SER:CA	6	backbone	CA	yes	scored	scored	scored	174	69
SER:CB	6	sidechain	CB	yes	scored	scored	scored	174	69
SER:H	1	backbone	H	yes	scored	scored	scored	174	69
SER:H1	1	sidechain	—	no	not observed	not observed	not observed	0	0
SER:H2	1	sidechain	—	no	not observed	not observed	not observed	0	0
SER:H3	1	sidechain	—	no	not observed	not observed	not observed	0	0
SER:HA	1	backbone	HA	yes	scored	scored	scored	174	69
SER:HB2	1	sidechain	—	yes	scored	scored	scored	174	69
SER:HB3	1	sidechain	—	yes	scored	scored	scored	174	69
SER:HG	1	sidechain	—	yes	scored	scored	scored	174	69
SER:N	7	backbone	N	yes	scored	scored	scored	174	69
SER:O	8	backbone	—	yes	scored	scored	scored	174	69
SER:OG	8	sidechain	—	yes	scored	scored	scored	174	69
SER:OXT	8	sidechain	—	no	preparation excluded	preparation excluded	preparation excluded	0	0
THR:C	6	backbone	C	yes	scored	scored	scored	193	74
THR:CA	6	backbone	CA	yes	scored	scored	scored	193	74
THR:CB	6	sidechain	CB	yes	scored	scored	scored	193	74
THR:CG2	6	sidechain	—	yes	scored	scored	scored	193	74
THR:H	1	backbone	H	yes	scored	scored	scored	193	74
THR:H1	1	sidechain	—	no	not observed	not observed	not observed	0	0
THR:H2	1	sidechain	—	no	not observed	not observed	not observed	0	0
THR:H3	1	sidechain	—	no	not observed	not observed	not observed	0	0
THR:HA	1	backbone	HA	yes	scored	scored	scored	193	74
THR:HB	1	sidechain	—	yes	scored	scored	scored	193	74
THR:HG1	1	sidechain	—	yes	scored	scored	scored	193	74
THR:HG21	1	sidechain	—	yes	scored	scored	scored	193	74
THR:HG22	1	sidechain	—	yes	scored	scored	scored	193	74
THR:HG23	1	sidechain	—	yes	scored	scored	scored	193	74
THR:N	7	backbone	N	yes	scored	scored	scored	193	74
THR:O	8	backbone	—	yes	scored	scored	scored	193	74
THR:OG1	8	sidechain	—	yes	scored	scored	scored	193	74
THR:OXT	8	sidechain	—	no	preparation excluded	preparation excluded	preparation excluded	0	0
TRP:C	6	backbone	C	no	scored	scored	scored	16	15
TRP:CA	6	backbone	CA	no	scored	scored	scored	16	15

IUPAC	Element	Region	Headline	$\geq 10/\text{group}$	All 79	Primary 61	Previous 18	<i>n</i>	Proteins
TRP:CB	6	sidechain	CB	no	scored	scored	scored	16	15
TRP:CD1	6	sidechain	—	no	scored	scored	scored	16	15
TRP:CD2	6	sidechain	—	no	scored	scored	scored	16	15
TRP:CE2	6	sidechain	—	no	scored	scored	scored	16	15
TRP:CE3	6	sidechain	—	no	scored	scored	scored	16	15
TRP:CG	6	sidechain	—	no	scored	scored	scored	16	15
TRP:CH2	6	sidechain	—	no	scored	scored	scored	16	15
TRP:CZ2	6	sidechain	—	no	scored	scored	scored	16	15
TRP:CZ3	6	sidechain	—	no	scored	scored	scored	16	15
TRP:H	1	backbone	H	no	scored	scored	scored	16	15
TRP:HA	1	backbone	HA	no	scored	scored	scored	16	15
TRP:HB2	1	sidechain	—	no	scored	scored	scored	16	15
TRP:HB3	1	sidechain	—	no	scored	scored	scored	16	15
TRP:HD1	1	sidechain	—	no	scored	scored	scored	16	15
TRP:HE1	1	sidechain	—	no	scored	scored	scored	16	15
TRP:HE3	1	sidechain	—	no	scored	scored	scored	16	15
TRP:HH2	1	sidechain	—	no	scored	scored	scored	16	15
TRP:HZ2	1	sidechain	—	no	scored	scored	scored	16	15
TRP:HZ3	1	sidechain	—	no	scored	scored	scored	16	15
TRP:N	7	backbone	N	no	scored	scored	scored	16	15
TRP:NE1	7	sidechain	—	no	scored	scored	scored	16	15
TRP:O	8	backbone	—	no	scored	scored	scored	16	15
TRP:OXT	8	sidechain	—	no	not observed	not observed	not observed	0	0
TYR:C	6	backbone	C	yes	scored	scored	scored	74	51
TYR:CA	6	backbone	CA	yes	scored	scored	scored	74	51
TYR:CB	6	sidechain	CB	yes	scored	scored	scored	74	51
TYR:CD1	6	sidechain	—	yes	scored	scored	scored	74	51
TYR:CD2	6	sidechain	—	yes	scored	scored	scored	74	51
TYR:CE1	6	sidechain	—	yes	scored	scored	scored	74	51
TYR:CE2	6	sidechain	—	yes	scored	scored	scored	74	51
TYR:CG	6	sidechain	—	yes	scored	scored	scored	74	51
TYR:CZ	6	sidechain	—	yes	scored	scored	scored	74	51
TYR:H	1	backbone	H	yes	scored	scored	scored	74	51
TYR:H1	1	sidechain	—	no	not observed	not observed	not observed	0	0
TYR:H2	1	sidechain	—	no	not observed	not observed	not observed	0	0
TYR:H3	1	sidechain	—	no	not observed	not observed	not observed	0	0
TYR:HA	1	backbone	HA	yes	scored	scored	scored	74	51
TYR:HB2	1	sidechain	—	yes	scored	scored	scored	74	51
TYR:HB3	1	sidechain	—	yes	scored	scored	scored	74	51
TYR:HD1	1	sidechain	—	yes	scored	scored	scored	74	51
TYR:HD2	1	sidechain	—	yes	scored	scored	scored	74	51
TYR:HE1	1	sidechain	—	yes	scored	scored	scored	74	51

IUPAC	Element	Region	Headline	$\geq 10/\text{group}$	All 79	Primary 61	Previous 18	<i>n</i>	Proteins
TYR:HE2	1	sidechain	—	yes	scored	scored	scored	74	51
TYR:HH	1	sidechain	—	yes	scored	scored	scored	74	51
TYR:N	7	backbone	N	yes	scored	scored	scored	74	51
TYR:O	8	backbone	—	yes	scored	scored	scored	74	51
TYR:OH	8	sidechain	—	yes	scored	scored	scored	74	51
TYR:OXT	8	sidechain	—	no	preparation excluded	preparation excluded	not observed	0	0
VAL:C	6	backbone	C	yes	scored	scored	scored	282	74
VAL:CA	6	backbone	CA	yes	scored	scored	scored	282	74
VAL:CB	6	sidechain	CB	yes	scored	scored	scored	282	74
VAL:CG1	6	sidechain	—	yes	scored	scored	scored	282	74
VAL:CG2	6	sidechain	—	yes	scored	scored	scored	282	74
VAL:H	1	backbone	H	yes	scored	scored	scored	281	74
VAL:H1	1	sidechain	—	no	preparation excluded	preparation excluded	not observed	0	0
VAL:H2	1	sidechain	—	no	preparation excluded	preparation excluded	not observed	0	0
VAL:H3	1	sidechain	—	no	preparation excluded	preparation excluded	not observed	0	0
VAL:HA	1	backbone	HA	yes	scored	scored	scored	282	74
VAL:HB	1	sidechain	—	yes	scored	scored	scored	282	74
VAL:HG11	1	sidechain	—	yes	scored	scored	scored	282	74
VAL:HG12	1	sidechain	—	yes	scored	scored	scored	282	74
VAL:HG13	1	sidechain	—	yes	scored	scored	scored	282	74
VAL:HG21	1	sidechain	—	yes	scored	scored	scored	282	74
VAL:HG22	1	sidechain	—	yes	scored	scored	scored	282	74
VAL:HG23	1	sidechain	—	yes	scored	scored	scored	282	74
VAL:N	7	backbone	N	yes	scored	scored	scored	282	74
VAL:O	8	backbone	—	yes	scored	scored	scored	282	74
VAL:OXT	8	sidechain	—	no	preparation excluded	preparation excluded	preparation excluded	0	0

5 Shielding IUPAC results

5.1 Historical-primary 61: T0 RMSE (ppm)

IUPAC	<i>n</i>	Proteins	Structure	AIMNet	MOPAC	Joint
ALA:C	186	55	0.495861	0.484571	0.440585	0.417613
ALA:CA	186	55	0.513628	0.509249	0.469568	0.492662
ALA:CB	186	55	0.489618	0.455739	0.425158	0.468573
ALA:H	186	55	0.152856	0.112391	0.135741	0.111549
ALA:HA	186	55	0.098256	0.075562	0.076415	0.075801
ALA:HB1	186	55	0.072565	0.078601	0.067122	0.077253
ALA:HB2	186	55	0.070694	0.065027	0.057966	0.062300
ALA:HB3	186	55	0.068392	0.066682	0.064293	0.059070
ALA:N	186	55	1.140833	1.168437	1.153965	1.173299
ALA:O	186	55	2.741162	2.688395	2.794923	2.807983
ARG:C	297	59	0.537718	0.600442	0.490720	0.493813
ARG:CA	297	59	0.537060	0.518307	0.513330	0.456620
ARG:CB	297	59	0.577400	0.655186	0.520456	0.580449
ARG:CD	297	59	0.532715	0.553543	0.584996	0.522915
ARG:CG	297	59	0.570346	0.513858	0.542232	0.499258
ARG:CZ	297	59	0.450217	0.415180	0.452849	0.366291
ARG:H	295	58	0.131261	0.126579	0.144552	0.096258
ARG:HA	297	59	0.091454	0.078715	0.078954	0.081476
ARG:HB2	297	59	0.076566	0.072740	0.071979	0.076769
ARG:HB3	297	59	0.069282	0.075650	0.062955	0.069056
ARG:HD2	297	59	0.075927	0.060219	0.072020	0.055867
ARG:HD3	297	59	0.074800	0.067226	0.082110	0.055596
ARG:HE	297	59	0.186092	0.151000	0.148047	0.157669
ARG:HG2	297	59	0.084080	0.078249	0.073987	0.068587
ARG:HG3	297	59	0.080200	0.078779	0.069274	0.061835
ARG:HH11	297	59	0.116352	0.085835	0.082458	0.083236
ARG:HH12	297	59	0.108370	0.077137	0.090832	0.080907
ARG:HH21	297	59	0.175202	0.147665	0.157573	0.104460
ARG:HH22	297	59	0.106809	0.097666	0.103054	0.093045
ARG:N	297	59	1.613446	1.397226	0.978799	0.984113
ARG:NE	297	59	1.058982	1.037659	0.925311	0.909582
ARG:NH1	297	59	0.842126	0.887065	0.755660	0.718768
ARG:NH2	297	59	0.953405	0.952931	0.851503	0.875228
ARG:O	297	59	2.833257	2.753622	2.337900	2.931694
ASN:C	61	36	0.457983	0.492976	0.543877	0.655398
ASN:CA	61	36	0.615400	0.575881	0.632241	0.619028
ASN:CB	61	36	0.761775	0.613210	0.673629	0.613106
ASN:CG	61	36	0.614299	0.648266	0.482857	0.446820
ASN:H	60	35	0.133170	0.175277	0.148856	0.156106
ASN:HA	61	36	0.091662	0.090281	0.073646	0.069753
ASN:HB2	61	36	0.092255	0.093336	0.099386	0.088700
ASN:HB3	61	36	0.080523	0.077424	0.075083	0.079880
ASN:HD21	61	36	0.142011	0.103465	0.113610	0.096733
ASN:HD22	61	36	0.137899	0.166334	0.183853	0.172606
ASN:N	61	36	1.314567	1.100554	1.280598	1.240342
ASN:ND2	61	36	1.350799	1.198244	1.408752	1.133691
ASN:O	61	36	3.350217	3.167165	3.734001	3.082468
ASN:OD1	61	36	3.804428	2.928836	3.764421	3.498953
ASP:C	113	43	0.640949	0.544226	0.613764	0.616975
ASP:CA	113	43	0.724804	0.780714	0.755879	0.726067
ASP:CB	113	43	1.176779	0.928217	0.990680	0.922414
ASP:CG	113	43	0.891645	0.668439	0.661231	0.724993
ASP:H	113	43	0.227676	0.183961	0.187726	0.133504
ASP:HA	113	43	0.097269	0.099784	0.082646	0.090740
ASP:HB2	113	43	0.114968	0.096097	0.089738	0.099023
ASP:HB3	113	43	0.095499	0.085535	0.086721	0.083245
ASP:N	113	43	1.370115	1.343831	1.318876	1.177165
ASP:O	113	43	4.113561	3.777935	3.834776	3.739755
ASP:OD1	113	43	4.483756	5.496884	4.112708	5.080270

IUPAC	<i>n</i>	Proteins	Structure	AIMNet	MOPAC	Joint
ASP:OD2	113	43	5.201403	4.339178	3.774939	4.434266
CYS:C	27	21	0.932034	0.887286	0.738389	0.923573
CYS:CA	27	21	0.643192	0.771894	5.135920	1.174751
CYS:CB	27	21	1.367664	1.589688	3.092016	2.906923
CYS:H	27	21	0.191224	0.206516	0.147671	0.157509
CYS:HA	27	21	0.147488	0.109608	0.341125	0.159858
CYS:HB2	27	21	0.133446	0.216740	0.415315	0.252054
CYS:HB3	27	21	0.111692	0.111170	0.104396	0.140879
CYS:HG	25	20	0.199529	0.198762	0.111563	0.193105
CYS:N	27	21	1.614005	1.756189	2.605488	2.550514
CYS:O	27	21	4.585628	4.895727	18.355668	7.922225
GLN:C	52	29	0.569092	0.581190	0.578376	0.516733
GLN:CA	52	29	0.543794	0.485067	0.673065	0.654296
GLN:CB	52	29	0.590975	0.657382	0.576637	0.549856
GLN:CD	52	29	0.618563	0.670866	0.615032	0.521416
GLN:CG	52	29	0.756113	0.804921	0.682267	0.708666
GLN:H	52	29	0.175978	0.190633	0.219073	0.236204
GLN:HA	52	29	0.118452	0.086002	0.072079	0.083450
GLN:HB2	52	29	0.068377	0.086204	0.079517	0.057499
GLN:HB3	52	29	0.086560	0.080622	0.072782	0.089230
GLN:HE21	52	29	0.120288	0.097343	0.083048	0.165827
GLN:HE22	52	29	0.152409	0.125155	0.167472	0.314504
GLN:HG2	52	29	0.078895	0.081592	0.096963	0.078286
GLN:HG3	52	29	0.091261	0.081066	0.147650	0.100814
GLN:N	52	29	1.163792	1.206537	1.087577	1.520765
GLN:NE2	52	29	1.243527	1.249110	2.325148	1.584413
GLN:O	52	29	3.276792	3.068817	2.640710	2.880860
GLN:OE1	52	29	2.921341	3.399983	2.383540	4.025967
GLU:C	129	46	0.511263	0.411835	0.482162	0.402738
GLU:CA	129	46	0.581376	0.478747	0.524302	0.481149
GLU:CB	129	46	0.707345	0.611247	0.692985	0.607848
GLU:CD	129	46	0.691651	0.633126	0.546412	0.622918
GLU:CG	129	46	0.626917	0.595161	0.710292	0.595594
GLU:H	129	46	0.143444	0.147933	0.185685	0.118557
GLU:HA	129	46	0.109267	0.081756	0.073099	0.072842
GLU:HB2	129	46	0.099619	0.087311	0.084714	0.086841
GLU:HB3	129	46	0.089005	0.075281	0.074741	0.074851
GLU:HG2	129	46	0.086793	0.073723	0.074731	0.073267
GLU:HG3	129	46	0.098885	0.081701	0.075774	0.072537
GLU:N	129	46	0.969501	1.050519	1.025192	1.105687
GLU:O	129	46	3.145028	2.490166	2.643103	2.536791
GLU:OE1	129	46	4.643937	3.801053	3.948071	3.759504
GLU:OE2	129	46	3.832783	3.860701	3.639489	3.537084
GLY:C	220	59	0.591071	0.586601	0.602083	0.696929
GLY:CA	220	59	0.592640	0.683428	0.576210	0.572373
GLY:H	220	59	0.163401	0.151204	0.186603	0.149498
GLY:HA2	220	59	0.091172	0.084159	0.085713	0.090803
GLY:HA3	220	59	0.088273	0.084059	0.086386	0.106395
GLY:N	220	59	1.290387	1.411329	1.331372	1.222487
GLY:O	220	59	3.622823	3.767180	4.177722	3.910460
HIS:C	52	35	0.671645	0.671863	0.671557	0.629748
HIS:CA	52	35	0.905800	0.808164	0.807890	0.719143
HIS:CB	52	35	0.543256	0.631750	0.579325	0.579269
HIS:CD2	52	35	0.872516	0.772098	0.690303	0.759396
HIS:CE1	52	35	0.848432	0.633963	0.669671	0.679127
HIS:CG	52	35	1.069549	0.731759	0.711908	0.808866
HIS:H	52	35	0.168346	0.157047	0.175763	0.118244
HIS:HA	52	35	0.111020	0.100612	0.096530	0.085633
HIS:HB2	52	35	0.100306	0.108620	0.094088	0.092418
HIS:HB3	52	35	0.089526	0.098949	0.084886	0.082031
HIS:HD2	52	35	0.142291	0.140125	0.153969	0.134545
HIS:HE1	52	35	0.106518	0.083303	0.063684	0.086649
HIS:HE2	52	35	0.101324	0.082162	0.078698	0.071586
HIS:N	52	35	1.305281	1.527827	1.356357	1.226610

IUPAC	<i>n</i>	Proteins	Structure	AIMNet	MOPAC	Joint
HIS:ND1	52	35	2.037881	2.234313	2.020802	1.955422
HIS:NE2	52	35	1.250581	1.122583	1.236836	1.005255
HIS:O	52	35	4.263476	4.279851	2.630092	3.219697
ILE:C	170	59	0.492777	0.524072	0.471983	0.482274
ILE:CA	170	59	0.591001	0.596718	0.532191	0.581074
ILE:CB	170	59	0.543552	0.617744	0.413057	0.456335
ILE:CD1	170	59	0.497289	0.484142	0.439471	0.551240
ILE:CG1	170	59	0.512348	0.434132	0.482665	0.364930
ILE:CG2	170	59	0.477402	0.493623	0.510529	0.421999
ILE:H	168	59	0.127338	0.128825	0.127912	0.103731
ILE:HA	170	59	0.098224	0.099049	0.090670	0.084080
ILE:HB	170	59	0.079306	0.080951	0.066683	0.070700
ILE:HD11	170	59	0.059545	0.059204	0.064043	0.051970
ILE:HD12	170	59	0.070071	0.060453	0.066070	0.063816
ILE:HD13	170	59	0.076982	0.066278	0.062481	0.065659
ILE:HG12	170	59	0.078817	0.069614	0.081865	0.056538
ILE:HG13	170	59	0.067479	0.068205	0.071585	0.054031
ILE:HG21	170	59	0.073617	0.074402	0.067907	0.066531
ILE:HG22	170	59	0.074791	0.071242	0.068832	0.075075
ILE:HG23	170	59	0.070778	0.075240	0.070018	0.060827
ILE:N	170	59	0.987361	1.140300	1.111665	0.848684
ILE:O	170	59	3.241732	2.732374	2.224538	2.425726
LEU:C	217	61	0.578574	0.604790	0.481180	0.537904
LEU:CA	217	61	0.710060	0.557141	0.493427	0.533803
LEU:CB	217	61	0.583359	0.500359	0.614072	0.540157
LEU:CD1	217	61	0.490739	0.545799	0.516148	0.552412
LEU:CD2	217	61	0.523546	0.484099	0.575182	0.525632
LEU:CG	217	61	0.450861	0.406002	0.437364	0.434552
LEU:H	215	61	0.143860	0.121865	0.139116	0.105402
LEU:HA	217	61	0.079210	0.081584	0.094668	0.083850
LEU:HB2	217	61	0.074540	0.079393	0.071719	0.063263
LEU:HB3	217	61	0.078940	0.079483	0.070089	0.071700
LEU:HD11	217	61	0.060118	0.057501	0.057596	0.053475
LEU:HD12	217	61	0.068230	0.070307	0.057602	0.072119
LEU:HD13	217	61	0.073318	0.064132	0.070638	0.053373
LEU:HD21	217	61	0.060584	0.058050	0.067052	0.055992
LEU:HD22	217	61	0.065552	0.066171	0.065770	0.064942
LEU:HD23	217	61	0.065813	0.066561	0.060846	0.061502
LEU:HG	217	61	0.073819	0.084821	0.076526	0.080587
LEU:N	217	61	1.184655	1.173591	0.931029	1.119730
LEU:O	217	61	3.369088	3.460058	3.179796	3.157147
LYS:C	182	59	0.548223	0.492746	0.426900	0.425005
LYS:CA	182	59	0.552364	0.534112	0.497031	0.472929
LYS:CB	182	59	0.436040	0.533234	0.440566	0.441437
LYS:CD	182	59	0.510438	0.549719	0.509308	0.548718
LYS:CE	182	59	0.538469	0.487019	0.458312	0.466712
LYS:CG	182	59	0.639991	0.598776	0.582623	0.598949
LYS:H	181	58	0.134778	0.123458	0.142790	0.121650
LYS:HA	182	59	0.085256	0.074833	0.069499	0.066428
LYS:HB2	182	59	0.063262	0.066067	0.055227	0.054569
LYS:HB3	182	59	0.080708	0.066805	0.064970	0.055032
LYS:HD2	182	59	0.054422	0.060150	0.051718	0.057509
LYS:HD3	182	59	0.051795	0.049750	0.044424	0.055253
LYS:HE2	182	59	0.075590	0.077839	0.070446	0.069864
LYS:HE3	182	59	0.097423	0.074169	0.063176	0.078065
LYS:HG2	182	59	0.068536	0.077697	0.064841	0.068804
LYS:HG3	182	59	0.068841	0.072143	0.076064	0.069122
LYS:HZ1	182	59	0.098430	0.120375	0.145463	0.140286
LYS:HZ2	182	59	0.082540	0.096905	0.076004	0.086497
LYS:HZ3	182	59	0.136887	0.111016	0.119910	0.092515
LYS:N	182	59	1.000395	1.133444	0.913914	0.848106
LYS:NZ	182	59	0.494790	0.758610	0.537116	0.728994
LYS:O	182	59	2.355268	2.128476	2.105386	2.146298
MET:C	99	57	0.666234	0.657440	0.744666	0.572777

IUPAC	<i>n</i>	Proteins	Structure	AIMNet	MOPAC	Joint
MET:CA	99	57	0.825046	0.873376	0.710525	0.928041
MET:CB	99	57	1.014571	0.949442	0.956600	1.020915
MET:CE	99	57	0.746500	0.761143	0.677940	0.686366
MET:CG	99	57	0.961688	1.066539	0.689204	0.702578
MET:H	48	37	0.140875	0.189418	0.279367	0.381652
MET:HA	99	57	0.142803	0.144512	0.142716	0.164787
MET:HB2	99	57	0.106940	0.094614	0.099620	0.107916
MET:HB3	99	57	0.108111	0.092672	0.121122	0.092392
MET:HE1	99	57	0.078762	0.076815	0.095484	0.071125
MET:HE2	99	57	0.128543	0.101576	0.099942	0.115396
MET:HE3	99	57	0.060936	0.071318	0.076316	0.078992
MET:HG2	99	57	0.108709	0.134550	0.094371	0.095966
MET:HG3	99	57	0.093492	0.090084	0.102452	0.080347
MET:N	99	57	2.170781	3.616600	6.025285	3.326026
MET:O	99	57	3.446476	3.157306	3.179226	3.513843
MET:SD	99	57	4.072769	4.163359	4.778009	4.156511
PHE:C	76	50	0.485623	0.564309	0.605600	0.481237
PHE:CA	76	50	0.930031	0.839991	0.673371	0.643551
PHE:CB	76	50	0.523172	0.622608	0.644303	0.449681
PHE:CD1	76	50	0.723271	0.582287	0.530800	0.539813
PHE:CD2	76	50	0.560748	0.679609	0.506776	0.605847
PHE:CE1	76	50	0.849216	0.652611	0.788066	0.689847
PHE:CE2	76	50	0.519897	0.500907	0.545834	0.403360
PHE:CG	76	50	0.805192	0.554014	0.721795	0.535516
PHE:CZ	76	50	0.481542	0.642412	0.443988	0.343323
PHE:H	75	49	0.146975	0.128454	0.156501	0.132024
PHE:HA	76	50	0.106133	0.102928	0.136642	0.102757
PHE:HB2	76	50	0.109112	0.122515	0.279020	0.262048
PHE:HB3	76	50	0.102320	0.090767	0.096034	0.080845
PHE:HD1	76	50	0.095220	0.078385	0.089961	0.070286
PHE:HD2	76	50	0.080790	0.073682	0.128636	0.074409
PHE:HE1	76	50	0.082560	0.071430	0.063821	0.084212
PHE:HE2	76	50	0.079333	0.080588	0.080132	0.072224
PHE:HZ	76	50	0.061858	0.062536	0.049895	0.065353
PHE:N	76	50	1.187247	1.110583	1.027535	1.197738
PHE:O	76	50	3.647806	3.544103	3.846068	2.791953
PRO:C	76	44	0.665549	0.597986	0.580763	0.647673
PRO:CA	76	44	0.735614	0.651771	0.727611	0.631466
PRO:CB	76	44	0.458336	0.561631	0.441959	0.420757
PRO:CD	76	44	0.901557	0.703234	0.762246	0.805000
PRO:CG	76	44	0.475061	0.437236	0.416553	0.483686
PRO:HA	76	44	0.078787	0.083646	0.071000	0.077481
PRO:HB2	76	44	0.077256	0.074948	0.077480	0.061645
PRO:HB3	76	44	0.074477	0.052032	0.062086	0.065077
PRO:HD2	76	44	0.114081	0.105473	0.103648	0.107519
PRO:HD3	76	44	0.102307	0.089064	0.094464	0.083118
PRO:HG2	76	44	0.087326	0.078089	0.086921	0.071653
PRO:HG3	76	44	0.064670	0.063595	0.061604	0.059936
PRO:N	76	44	1.858022	1.952760	1.906561	1.724866
PRO:O	76	44	3.411153	3.784450	3.112846	3.120091
SER:C	139	54	0.611760	0.569196	0.540516	0.546934
SER:CA	139	54	0.730104	0.730390	0.560950	0.791719
SER:CB	139	54	0.877589	0.975798	1.009378	1.038276
SER:H	139	54	0.170432	0.136806	0.143859	0.128169
SER:HA	139	54	0.111888	0.092254	0.085656	0.100464
SER:HB2	139	54	0.103132	0.090149	0.079567	0.079506
SER:HB3	139	54	0.093832	0.095946	0.083325	0.096307
SER:HG	139	54	0.148292	0.161069	0.155143	0.136145
SER:N	139	54	1.396769	1.291249	1.423432	1.414511
SER:O	139	54	3.328010	3.294908	2.977099	3.564788
SER:OG	139	54	2.355909	2.551737	2.153135	2.368149
THR:C	150	57	0.545259	0.568179	0.495699	0.515233
THR:CA	150	57	0.623160	0.555811	0.537438	0.616069
THR:CB	150	57	0.595640	0.689708	0.609226	0.603796

IUPAC	n	Proteins	Structure	AIMNet	MOPAC	Joint
THR:CG2	150	57	0.539095	0.530563	0.423313	0.437416
THR:H	150	57	0.165150	0.130327	0.174399	0.123431
THR:HA	150	57	0.082503	0.083631	0.063488	0.076396
THR:HB	150	57	0.088152	0.071742	0.091131	0.078639
THR:HG1	150	57	0.170626	0.189649	0.114661	0.151334
THR:HG21	150	57	0.073130	0.067338	0.071501	0.068913
THR:HG22	150	57	0.071951	0.071719	0.071819	0.052281
THR:HG23	150	57	0.060619	0.066138	0.072454	0.076989
THR:N	150	57	1.306463	1.194963	1.363789	1.110011
THR:O	150	57	3.146452	3.091894	2.724285	2.970173
THR:OG1	150	57	2.323947	2.586443	2.230906	2.587968
TRP:C	12	11	0.625406	0.397091	0.867716	0.558963
TRP:CA	12	11	0.811454	0.768922	0.968549	0.782878
TRP:CB	12	11	0.671930	0.954326	0.795371	0.846042
TRP:CD1	12	11	0.521922	1.124972	0.529649	0.663239
TRP:CD2	12	11	0.730926	0.686199	0.451621	0.679768
TRP:CE2	12	11	0.470528	0.576284	0.642614	0.448377
TRP:CE3	12	11	0.694909	0.576825	0.835143	0.529748
TRP:CG	12	11	0.689071	0.569903	0.938217	0.826221
TRP:CH2	12	11	0.477970	0.532661	0.591604	0.454623
TRP:CZ2	12	11	0.419139	0.614149	0.314185	0.355978
TRP:CZ3	12	11	0.489997	0.494821	0.461753	0.686148
TRP:H	12	11	0.177400	0.209585	0.191623	0.149369
TRP:HA	12	11	0.150202	0.071657	0.335407	0.107754
TRP:HB2	12	11	0.094317	0.076152	0.086840	0.110696
TRP:HB3	12	11	0.141071	0.073835	0.067202	0.063926
TRP:HD1	12	11	0.148545	0.157212	0.117822	0.066658
TRP:HE1	12	11	0.273192	0.113182	0.118480	0.123470
TRP:HE3	12	11	0.175434	0.078626	0.072705	0.072111
TRP:HH2	12	11	0.091737	0.086861	0.085469	0.076141
TRP:HZ2	12	11	0.058212	0.081780	0.056105	0.086478
TRP:HZ3	12	11	0.111357	0.061687	0.080557	0.077906
TRP:N	12	11	1.752864	1.662544	1.733476	1.825282
TRP:NE1	12	11	2.387276	1.701584	1.327938	1.186261
TRP:O	12	11	3.531580	4.265339	3.283785	2.983888
TYR:C	52	37	0.519102	0.459507	0.520835	0.445696
TYR:CA	52	37	0.909722	0.639056	0.738831	0.781634
TYR:CB	52	37	0.456304	0.539307	0.366824	0.465937
TYR:CD1	52	37	0.563310	0.535166	0.508062	0.497078
TYR:CD2	52	37	0.506386	0.524320	0.437225	0.425240
TYR:CE1	52	37	0.602501	0.564100	0.475865	0.555442
TYR:CE2	52	37	0.482303	0.497102	0.610707	0.569081
TYR:CG	52	37	0.532020	0.436332	0.596136	0.484191
TYR:CZ	52	37	0.408461	0.754407	0.370044	0.513211
TYR:H	52	37	0.162074	0.118765	0.171759	0.117581
TYR:HA	52	37	0.099393	0.082959	0.079553	0.086425
TYR:HB2	52	37	0.079894	0.064955	0.076991	0.068722
TYR:HB3	52	37	0.056864	0.072669	0.081415	0.066000
TYR:HD1	52	37	0.092000	0.069223	0.086487	0.079623
TYR:HD2	52	37	0.088281	0.065309	0.061010	0.073057
TYR:HE1	52	37	0.091714	0.084471	0.098148	0.101680
TYR:HE2	52	37	0.082691	0.060445	0.067886	0.061170
TYR:HH	52	37	0.089354	0.087768	0.096411	0.137099
TYR:N	52	37	1.025358	1.111573	1.158282	0.948595
TYR:O	52	37	3.391522	2.913272	2.990501	3.278997
TYR:OH	52	37	1.897987	2.128973	1.850802	1.863183
VAL:C	211	57	0.498942	0.544815	0.487542	0.451504
VAL:CA	211	57	0.651799	0.572919	0.490059	0.495156
VAL:CB	211	57	0.562656	0.472506	0.484116	0.580847
VAL:CG1	211	57	0.588121	0.556613	0.668123	0.552247
VAL:CG2	211	57	0.539919	0.476153	0.454524	0.477963
VAL:H	210	57	0.138217	0.120773	0.129008	0.109585
VAL:HA	211	57	0.087818	0.075274	0.083908	0.075750
VAL:HB	211	57	0.070966	0.072227	0.070550	0.066269

IUPAC	<i>n</i>	Proteins	Structure	AIMNet	MOPAC	Joint
VAL:HG11	211	57	0.087429	0.077672	0.079184	0.072123
VAL:HG12	211	57	0.068748	0.069924	0.062278	0.075647
VAL:HG13	211	57	0.076717	0.067619	0.066466	0.064890
VAL:HG21	211	57	0.072729	0.076713	0.065264	0.067917
VAL:HG22	211	57	0.077656	0.071251	0.069275	0.072604
VAL:HG23	211	57	0.078334	0.070136	0.068531	0.072671
VAL:N	211	57	1.124544	1.217793	1.055289	1.023181
VAL:O	211	57	2.644517	3.148151	2.299214	2.416279

5.2 Historical-primary 61: T2 vector RMSE (ppm)

IUPAC	<i>n</i>	Proteins	Structure	AIMNet	MOPAC	Joint
ALA:C	186	55	3.768325	3.343815	3.423274	3.198102
ALA:CA	186	55	2.178508	2.094572	2.114687	2.017073
ALA:CB	186	55	2.011191	2.016653	1.979656	1.897067
ALA:H	186	55	0.613530	0.631196	0.588740	0.575120
ALA:HA	186	55	0.504285	0.506010	0.471390	0.479635
ALA:HB1	186	55	0.529668	0.556862	0.489406	0.478078
ALA:HB2	186	55	0.521799	0.550821	0.561117	0.487296
ALA:HB3	186	55	0.509576	0.503561	0.453664	0.465159
ALA:N	186	55	4.708252	4.546323	4.254755	4.260625
ALA:O	186	55	15.259604	15.208899	12.698901	13.543429
ARG:C	297	59	3.826411	3.593442	3.442043	3.084970
ARG:CA	297	59	2.197366	2.162567	2.157671	1.831352
ARG:CB	297	59	2.466917	2.364841	2.278135	2.177392
ARG:CD	297	59	2.571877	2.571539	2.626896	2.322468
ARG:CG	297	59	2.693258	2.352629	2.410865	2.084629
ARG:CZ	297	59	3.739724	4.155741	3.581919	3.121601
ARG:H	295	58	0.672197	0.683639	0.569149	0.558195
ARG:HA	297	59	0.565020	0.542614	0.500536	0.464159
ARG:HB2	297	59	0.573557	0.564280	0.530656	0.496264
ARG:HB3	297	59	0.576166	0.559884	0.524201	0.503316
ARG:HD2	297	59	0.565069	0.549314	0.521730	0.525117
ARG:HD3	297	59	0.573982	0.539081	0.542323	0.517252
ARG:HE	297	59	0.750589	0.819495	0.746868	0.702882
ARG:HG2	297	59	0.578680	0.578941	0.525162	0.520587
ARG:HG3	297	59	0.587291	0.630971	0.519772	0.511994
ARG:HH11	297	59	0.732829	0.736692	0.646130	0.692443
ARG:HH12	297	59	0.666178	0.607709	0.617740	0.595263
ARG:HH21	297	59	0.751538	0.768175	0.655728	0.727000
ARG:HH22	297	59	0.673856	0.647416	0.593187	0.585072
ARG:N	297	59	5.093793	4.376152	3.813223	4.315594
ARG:NE	297	59	4.591818	4.644030	4.377134	4.476421
ARG:NH1	297	59	3.836668	4.082938	3.605147	3.222984
ARG:NH2	297	59	3.743841	3.810402	3.414662	3.266263
ARG:O	297	59	15.074336	17.940277	14.250920	12.765598
ASN:C	61	36	3.927339	3.737688	3.647538	3.485238
ASN:CA	61	36	2.738917	2.728533	2.635453	2.674228
ASN:CB	61	36	3.518177	3.199196	2.964318	2.981762
ASN:CG	61	36	4.796770	4.456350	4.405359	4.031762
ASN:H	60	35	0.790651	0.774326	0.739567	0.643660
ASN:HA	61	36	0.604338	0.565528	0.558863	0.542173
ASN:HB2	61	36	0.622595	0.565776	0.623548	0.563581
ASN:HB3	61	36	0.645481	0.605631	0.606298	0.557626
ASN:HD21	61	36	0.751757	0.695386	0.623436	0.620947
ASN:HD22	61	36	0.691176	0.721774	0.638116	0.611841
ASN:N	61	36	5.568473	4.875976	5.401652	5.010809
ASN:ND2	61	36	5.575207	6.149236	5.455966	5.223442
ASN:O	61	36	18.542529	17.885618	15.994529	15.720922
ASN:OD1	61	36	19.910429	20.595011	19.076781	16.521958
ASP:C	113	43	4.503450	4.342668	4.223524	4.056846
ASP:CA	113	43	2.987716	3.004973	3.082806	2.748211

IUPAC	<i>n</i>	Proteins	Structure	AIMNet	MOPAC	Joint
ASP:CB	113	43	4.352062	4.068838	3.674684	3.516566
ASP:CG	113	43	5.356539	4.861985	4.511597	4.041043
ASP:H	113	43	0.836578	0.881212	0.772393	0.733339
ASP:HA	113	43	0.639430	0.610246	0.578804	0.543422
ASP:HB2	113	43	0.634632	0.589162	0.558445	0.550837
ASP:HB3	113	43	0.673289	0.629414	0.580011	0.560612
ASP:N	113	43	5.727829	5.799422	5.612302	5.179926
ASP:O	113	43	18.951492	18.073525	16.659597	17.392136
ASP:OD1	113	43	21.204074	20.877237	19.607564	18.924091
ASP:OD2	113	43	19.855995	19.180430	18.378101	18.314610
CYS:C	27	21	5.482808	4.505716	10.444863	11.949161
CYS:CA	27	21	3.264668	3.188327	13.159590	5.448534
CYS:CB	27	21	6.438320	7.648113	13.226213	9.624670
CYS:H	27	21	0.953258	0.779895	1.472711	1.207229
CYS:HA	27	21	0.688021	0.702015	1.148493	1.034254
CYS:HB2	27	21	1.125385	0.937626	1.999429	1.311305
CYS:HB3	27	21	0.968575	0.832489	1.973133	1.660677
CYS:HG	25	20	1.121772	1.357110	0.983564	1.068121
CYS:N	27	21	6.445120	5.879775	12.353271	11.688547
CYS:O	27	21	22.291104	21.799329	60.315436	46.510309
GLN:C	52	29	4.100462	3.860464	4.454970	3.440333
GLN:CA	52	29	2.647974	2.565075	3.170343	2.281435
GLN:CB	52	29	2.715364	2.335727	2.558505	2.322709
GLN:CD	52	29	4.756865	4.247856	6.578923	5.705967
GLN:CG	52	29	3.886407	3.742488	3.932936	3.724217
GLN:H	52	29	0.723548	0.687037	0.927076	0.718292
GLN:HA	52	29	0.604800	0.582373	0.608914	0.494679
GLN:HB2	52	29	0.636045	0.653217	0.596354	0.647552
GLN:HB3	52	29	0.624738	0.571304	0.635746	0.582888
GLN:HE21	52	29	0.708681	0.739999	0.759403	1.252580
GLN:HE22	52	29	0.668081	0.792368	0.798906	0.996555
GLN:HG2	52	29	0.614677	0.591657	0.603209	0.551270
GLN:HG3	52	29	0.665130	0.604049	0.877659	0.778370
GLN:N	52	29	4.652767	4.800805	4.848806	4.434573
GLN:NE2	52	29	5.689078	5.505515	11.882505	11.541365
GLN:O	52	29	15.826354	18.326349	18.592023	14.653151
GLN:OE1	52	29	20.572020	20.277314	19.574710	20.180079
GLU:C	129	46	3.730525	3.643684	3.663032	3.221908
GLU:CA	129	46	2.444217	2.142889	2.372051	2.038385
GLU:CB	129	46	2.842955	2.661082	2.528756	2.419962
GLU:CD	129	46	4.444519	4.733411	4.065983	3.717403
GLU:CG	129	46	3.420682	3.164369	3.078606	2.870845
GLU:H	129	46	0.710073	0.704401	0.613794	0.631927
GLU:HA	129	46	0.562681	0.553425	0.503590	0.459476
GLU:HB2	129	46	0.616407	0.587794	0.524329	0.506047
GLU:HB3	129	46	0.616928	0.584405	0.540281	0.520031
GLU:HG2	129	46	0.640473	0.592019	0.524079	0.508532
GLU:HG3	129	46	0.628797	0.602657	0.533601	0.493449
GLU:N	129	46	4.920079	4.467886	4.431019	4.254966
GLU:O	129	46	15.879952	15.814702	14.431098	13.849107
GLU:OE1	129	46	18.257791	17.858697	15.633165	15.959108
GLU:OE2	129	46	19.802856	18.309482	16.685728	16.415392
GLY:C	220	59	4.337354	4.705342	4.501255	3.852968
GLY:CA	220	59	3.400288	2.845635	2.852698	2.777018
GLY:H	220	59	0.724716	0.708629	0.822299	0.741618
GLY:HA2	220	59	0.618236	0.591975	0.612848	0.556534
GLY:HA3	220	59	0.636682	0.639445	0.651530	0.620885
GLY:N	220	59	5.428769	5.254313	8.623071	5.599261
GLY:O	220	59	18.107239	18.062465	18.569056	17.279381
HIS:C	52	35	4.723094	4.597413	4.001643	3.835771
HIS:CA	52	35	3.326744	3.153320	3.133362	3.057781
HIS:CB	52	35	2.832969	2.826690	2.532590	2.768258
HIS:CD2	52	35	6.170412	5.473363	5.535075	5.853890
HIS:CE1	52	35	4.914181	4.672840	4.663165	4.704394

IUPAC	n	Proteins	Structure	AIMNet	MOPAC	Joint
HIS:CG	52	35	5.037573	4.740061	5.305822	4.997515
HIS:H	52	35	0.803758	0.791563	0.712660	0.699774
HIS:HA	52	35	0.656652	0.560725	0.571151	0.569898
HIS:HB2	52	35	0.658139	0.563475	0.532460	0.572956
HIS:HB3	52	35	0.663770	0.570669	0.533842	0.607231
HIS:HD2	52	35	0.680843	0.600708	0.631365	0.569103
HIS:HE1	52	35	0.554662	0.561749	0.544071	0.525222
HIS:HE2	52	35	0.563615	0.582225	0.480782	0.521944
HIS:N	52	35	5.484138	5.433990	5.044527	4.486523
HIS:ND1	52	35	13.979140	14.347423	11.279868	11.838416
HIS:NE2	52	35	4.504116	5.177165	4.394541	4.409262
HIS:O	52	35	18.675127	18.149240	17.077071	17.274267
ILE:C	170	59	3.448068	3.350285	3.422112	3.060708
ILE:CA	170	59	2.229596	2.113494	2.050882	1.938225
ILE:CB	170	59	2.193547	2.110035	1.925598	1.878928
ILE:CD1	170	59	1.761562	1.632275	1.526804	1.563701
ILE:CG1	170	59	1.939391	1.926053	1.795916	1.697943
ILE:CG2	170	59	1.883529	1.846334	1.667637	1.706898
ILE:H	168	59	0.656969	0.625907	0.566170	0.589002
ILE:HA	170	59	0.575584	0.525668	0.496263	0.502118
ILE:HB	170	59	0.556956	0.519278	0.478181	0.467467
ILE:HD11	170	59	0.536002	0.497015	0.464816	0.462664
ILE:HD12	170	59	0.529102	0.492495	0.497694	0.467722
ILE:HD13	170	59	0.553552	0.491169	0.480242	0.448897
ILE:HG12	170	59	0.567469	0.506601	0.473623	0.469744
ILE:HG13	170	59	0.558063	0.548413	0.491569	0.502780
ILE:HG21	170	59	0.590810	0.505350	0.498384	0.496984
ILE:HG22	170	59	0.542469	0.506161	0.491502	0.457590
ILE:HG23	170	59	0.604384	0.561957	0.529897	0.562684
ILE:N	170	59	4.362920	4.395769	3.962725	4.203838
ILE:O	170	59	17.465351	15.974532	14.343574	13.507811
LEU:C	217	61	3.971040	3.847264	3.608757	3.292551
LEU:CA	217	61	2.517481	2.218928	2.187582	1.919588
LEU:CB	217	61	2.436072	2.393734	2.364052	2.011151
LEU:CD1	217	61	2.068182	1.954713	1.929727	1.926086
LEU:CD2	217	61	2.067416	1.841967	1.876131	1.913053
LEU:CG	217	61	1.702503	1.626940	1.488413	1.678923
LEU:H	215	61	0.700828	0.652404	0.614724	0.587006
LEU:HA	217	61	0.563733	0.602721	0.540340	0.516158
LEU:HB2	217	61	0.579171	0.541815	0.533996	0.508224
LEU:HB3	217	61	0.584975	0.546405	0.530577	0.515405
LEU:HD11	217	61	0.547643	0.529686	0.500270	0.484409
LEU:HD12	217	61	0.588981	0.570476	0.529457	0.500254
LEU:HD13	217	61	0.582916	0.548456	0.585962	0.513990
LEU:HD21	217	61	0.591620	0.531378	0.493929	0.519698
LEU:HD22	217	61	0.580368	0.530812	0.489491	0.483433
LEU:HD23	217	61	0.576302	0.543903	0.512344	0.478821
LEU:HG	217	61	0.569525	0.549350	0.546469	0.507173
LEU:N	217	61	4.474286	3.945857	4.327374	4.222380
LEU:O	217	61	16.457888	16.690435	15.348140	14.936344
LYS:C	182	59	3.763758	3.465619	3.424963	2.945794
LYS:CA	182	59	2.311226	2.245271	2.180772	2.246787
LYS:CB	182	59	2.219333	2.249544	2.021342	1.907428
LYS:CD	182	59	2.333845	2.068628	2.024202	1.754565
LYS:CE	182	59	2.955243	2.744140	2.337178	2.358890
LYS:CG	182	59	2.515635	2.272044	2.137881	2.015177
LYS:H	181	58	0.645305	0.632629	0.608153	0.579764
LYS:HA	182	59	0.563703	0.554503	0.489888	0.487746
LYS:HB2	182	59	0.560799	0.552712	0.495611	0.469791
LYS:HB3	182	59	0.551535	0.545181	0.504113	0.513998
LYS:HD2	182	59	0.543429	0.538616	0.487676	0.463090
LYS:HD3	182	59	0.561723	0.502851	0.454311	0.440610
LYS:HE2	182	59	0.567229	0.543184	0.474309	0.494876
LYS:HE3	182	59	0.591871	0.594161	0.518242	0.494956

IUPAC	n	Proteins	Structure	AIMNet	MOPAC	Joint
LYS:HG2	182	59	0.528153	0.529977	0.455838	0.489361
LYS:HG3	182	59	0.549356	0.550102	0.497775	0.489866
LYS:HZ1	182	59	0.636847	0.706791	0.579810	0.602684
LYS:HZ2	182	59	0.677082	0.698087	0.597129	0.631768
LYS:HZ3	182	59	0.632830	0.699472	0.610272	0.537903
LYS:N	182	59	4.855051	4.574720	4.353533	4.592443
LYS:NZ	182	59	2.121634	2.478435	2.309299	2.067388
LYS:O	182	59	17.296921	14.592457	14.360712	14.275362
MET:C	99	57	4.507095	4.060561	5.491967	4.593926
MET:CA	99	57	3.127176	3.185367	3.410339	2.990677
MET:CB	99	57	4.416117	4.053933	4.233092	4.194318
MET:CE	99	57	3.211386	3.269135	2.951884	2.905366
MET:CG	99	57	3.920441	3.651577	3.547751	3.372883
MET:H	48	37	0.701169	0.680331	1.162449	0.954361
MET:HA	99	57	0.778508	0.751586	0.763134	0.746615
MET:HB2	99	57	0.722635	0.711013	0.655132	0.647333
MET:HB3	99	57	0.765214	0.711969	0.778841	0.727713
MET:HE1	99	57	0.748403	0.733702	0.757081	0.719763
MET:HE2	99	57	0.735462	0.784214	0.709932	0.741171
MET:HE3	99	57	0.723708	0.773628	0.712681	0.692306
MET:HG2	99	57	0.839998	0.919016	0.822546	0.687956
MET:HG3	99	57	0.842852	0.829890	0.828525	0.722060
MET:N	99	57	4.890870	4.854750	5.157033	5.213377
MET:O	99	57	19.773152	19.796887	24.604307	16.132822
MET:SD	99	57	19.630894	18.745859	17.702850	18.105656
PHE:C	76	50	3.812491	3.863442	3.869788	3.125293
PHE:CA	76	50	3.090417	3.044088	4.000377	2.819432
PHE:CB	76	50	2.863530	2.957027	3.718316	3.741543
PHE:CD1	76	50	5.895427	5.573195	5.727310	5.678852
PHE:CD2	76	50	5.006838	5.779156	6.941231	5.762357
PHE:CE1	76	50	5.230311	5.203098	6.055975	4.455780
PHE:CE2	76	50	3.805647	6.323791	5.718803	3.602611
PHE:CG	76	50	6.170180	5.217847	6.714539	5.531177
PHE:CZ	76	50	5.473000	7.081192	5.827091	5.550859
PHE:H	75	49	0.684063	0.647157	0.850221	0.702060
PHE:HA	76	50	0.572568	0.554219	0.633367	0.621863
PHE:HB2	76	50	0.575897	0.603194	1.159580	1.324919
PHE:HB3	76	50	0.609539	0.640805	0.692026	0.754892
PHE:HD1	76	50	0.612686	0.638562	0.659661	0.589321
PHE:HD2	76	50	0.607505	0.638851	0.703832	0.596727
PHE:HE1	76	50	0.531958	0.566871	0.481179	0.506937
PHE:HE2	76	50	0.664674	0.566458	0.542881	0.576855
PHE:HZ	76	50	0.592581	0.696171	0.606661	0.543077
PHE:N	76	50	5.317988	5.055035	5.714205	4.615631
PHE:O	76	50	17.388685	16.626769	17.731363	14.117408
PRO:C	76	44	4.375463	3.901653	3.946224	3.865270
PRO:CA	76	44	2.870710	2.823873	3.103368	2.765952
PRO:CB	76	44	2.793080	2.520275	2.643951	2.396738
PRO:CD	76	44	3.365705	3.201697	3.206055	2.984494
PRO:CG	76	44	2.472610	2.533191	2.388060	2.310205
PRO:HA	76	44	0.609549	0.601525	0.565691	0.560673
PRO:HB2	76	44	0.618309	0.528387	0.527909	0.545562
PRO:HB3	76	44	0.595856	0.564151	0.515054	0.511472
PRO:HD2	76	44	0.715517	0.687014	0.669734	0.640688
PRO:HD3	76	44	0.700273	0.651895	0.612818	0.589426
PRO:HG2	76	44	0.543137	0.539449	0.511666	0.496327
PRO:HG3	76	44	0.532958	0.548011	0.509810	0.528660
PRO:N	76	44	7.290011	6.869577	6.544425	6.300061
PRO:O	76	44	18.067170	20.407801	17.373892	17.530033
SER:C	139	54	4.856061	4.109366	4.533160	4.029575
SER:CA	139	54	2.802936	2.670673	2.718336	2.596931
SER:CB	139	54	3.734205	3.428104	3.666386	3.420317
SER:H	139	54	0.708311	0.687737	0.768431	0.675681
SER:HA	139	54	0.590748	0.552654	0.616498	0.620718

IUPAC	<i>n</i>	Proteins	Structure	AIMNet	MOPAC	Joint
SER:HB2	139	54	0.626962	0.594815	0.594205	0.602655
SER:HB3	139	54	0.633979	0.602190	0.579824	0.647078
SER:HG	139	54	0.977828	0.974323	0.972036	0.928243
SER:N	139	54	5.413732	5.498328	5.585890	5.468827
SER:O	139	54	20.946697	20.838936	16.818765	17.153063
SER:OG	139	54	9.046183	8.743360	9.122449	8.420537
THR:C	150	57	4.139522	3.923103	3.930753	3.682136
THR:CA	150	57	2.459334	2.398292	2.492146	2.154608
THR:CB	150	57	4.146307	2.900483	3.151417	2.709294
THR:CG2	150	57	2.115731	1.972154	2.174469	1.825261
THR:H	150	57	0.767284	0.690947	0.702670	0.692219
THR:HA	150	57	0.565416	0.562547	0.526393	0.470690
THR:HB	150	57	0.611818	0.580626	0.616394	0.651604
THR:HG1	150	57	0.852905	0.988104	0.931520	0.775226
THR:HG21	150	57	0.568064	0.530130	0.534324	0.511007
THR:HG22	150	57	0.556469	0.510353	0.545401	0.509010
THR:HG23	150	57	0.542855	0.492730	0.509467	0.554422
THR:N	150	57	5.157109	5.147872	5.080107	5.105037
THR:O	150	57	18.047636	16.738302	16.488165	15.262540
THR:OG1	150	57	7.174995	7.390503	7.508895	7.079127
TRP:C	12	11	4.339845	3.921255	5.865279	4.339519
TRP:CA	12	11	3.817506	3.723647	4.180739	2.781945
TRP:CB	12	11	3.412338	3.290989	3.019504	3.146633
TRP:CD1	12	11	6.157451	6.944562	7.469304	7.814677
TRP:CD2	12	11	5.780417	6.208172	5.654593	7.042109
TRP:CE2	12	11	7.954969	5.324174	7.434275	8.226933
TRP:CE3	12	11	6.841344	6.588879	7.256589	7.118074
TRP:CG	12	11	7.471184	5.361041	6.840440	6.501916
TRP:CH2	12	11	7.873052	11.000693	6.974462	6.225274
TRP:CZ2	12	11	4.887001	6.666458	11.162404	5.831433
TRP:CZ3	12	11	5.764401	5.811940	8.364710	4.966530
TRP:H	12	11	0.809766	0.667231	0.830813	0.817441
TRP:HA	12	11	0.661492	0.711793	0.884209	0.758848
TRP:HB2	12	11	0.810095	0.708461	0.665331	0.729272
TRP:HB3	12	11	0.821431	0.763667	0.757474	0.756906
TRP:HD1	12	11	0.960995	0.854569	0.865959	0.805728
TRP:HE1	12	11	0.832498	0.892862	0.857826	0.760655
TRP:HE3	12	11	0.703562	0.714948	0.698486	0.674469
TRP:HH2	12	11	0.821750	0.635216	0.604581	0.652242
TRP:HZ2	12	11	0.902989	1.283397	0.668548	0.631676
TRP:HZ3	12	11	0.707614	0.579947	0.580875	0.564103
TRP:N	12	11	6.161767	5.133677	6.239351	6.199048
TRP:NE1	12	11	6.439525	7.491125	5.612421	6.012795
TRP:O	12	11	19.757143	20.467941	21.379573	16.047217
TYR:C	52	37	3.608673	3.752852	3.352642	3.535099
TYR:CA	52	37	3.089424	2.728874	2.682745	2.352537
TYR:CB	52	37	2.403316	2.089465	2.207465	2.045340
TYR:CD1	52	37	4.963503	5.450515	5.398656	4.900367
TYR:CD2	52	37	4.523235	5.513691	5.088782	4.646503
TYR:CE1	52	37	3.423150	4.210279	4.698403	3.576660
TYR:CE2	52	37	3.410286	4.219719	3.850242	3.965365
TYR:CG	52	37	4.333345	5.132404	4.179858	3.782426
TYR:CZ	52	37	4.456622	4.811843	4.538382	4.825718
TYR:H	52	37	0.847964	0.774088	0.765621	0.751342
TYR:HA	52	37	0.659687	0.626638	0.582190	0.648515
TYR:HB2	52	37	0.675565	0.610742	0.629266	0.636483
TYR:HB3	52	37	0.664387	0.588013	0.580877	0.559334
TYR:HD1	52	37	0.581557	0.521374	0.529568	0.543757
TYR:HD2	52	37	0.643493	0.687014	0.666224	0.657688
TYR:HE1	52	37	0.580186	0.534045	0.571976	0.552658
TYR:HE2	52	37	0.559646	0.572011	0.562561	0.599470
TYR:HH	52	37	0.795344	0.841509	0.794065	0.681540
TYR:N	52	37	4.860455	4.891687	5.046536	5.332877
TYR:O	52	37	17.560992	17.087775	17.528222	15.599264

IUPAC	<i>n</i>	Proteins	Structure	AIMNet	MOPAC	Joint
TYR:OH	52	37	6.591689	7.498022	6.855650	7.094714
VAL:C	211	57	3.704158	3.686725	3.062466	3.070978
VAL:CA	211	57	2.323099	2.104979	2.139901	1.954052
VAL:CB	211	57	2.102819	2.006897	1.927719	2.058817
VAL:CG1	211	57	2.046530	2.029467	2.042521	1.878613
VAL:CG2	211	57	2.113071	2.086797	2.095176	2.049010
VAL:H	210	57	0.690267	0.642737	0.657149	0.604980
VAL:HA	211	57	0.533442	0.552462	0.453728	0.445987
VAL:HB	211	57	0.579276	0.547299	0.520248	0.494821
VAL:HG11	211	57	0.524456	0.512910	0.475423	0.459548
VAL:HG12	211	57	0.536172	0.496556	0.466521	0.472339
VAL:HG13	211	57	0.559149	0.554290	0.527178	0.510617
VAL:HG21	211	57	0.568928	0.531615	0.535699	0.499266
VAL:HG22	211	57	0.577760	0.506005	0.506403	0.480662
VAL:HG23	211	57	0.554850	0.529840	0.484546	0.471486
VAL:N	211	57	4.663881	4.572014	4.308909	4.443537
VAL:O	211	57	16.850580	16.075980	13.943659	13.180819

5.3 Previous-paper 18: T0 RMSE (ppm)

IUPAC	<i>n</i>	Proteins	Structure	AIMNet	MOPAC	Joint
ALA:C	82	18	0.489116	0.491729	0.427815	0.468877
ALA:CA	82	18	0.450440	0.494654	0.394981	0.379164
ALA:CB	82	18	0.462332	0.436783	0.416957	0.437532
ALA:H	81	18	0.171499	0.117845	0.134456	0.101624
ALA:HA	82	18	0.092821	0.083886	0.070792	0.063980
ALA:HB1	82	18	0.078465	0.072027	0.067102	0.056617
ALA:HB2	82	18	0.077362	0.069097	0.060884	0.068557
ALA:HB3	82	18	0.060139	0.063071	0.045801	0.056742
ALA:N	82	18	1.092426	1.009982	0.991302	0.937197
ALA:O	82	18	2.517077	2.743281	2.470046	2.499828
ARG:C	41	18	0.544817	0.421896	0.391997	0.374530
ARG:CA	41	18	0.670744	0.475331	0.610977	0.531259
ARG:CB	41	18	0.631371	0.616648	0.539481	0.529167
ARG:CD	41	18	0.596880	0.606968	0.547474	0.506202
ARG:CG	41	18	0.586329	0.643321	0.444973	0.639159
ARG:CZ	41	18	0.619898	0.527706	0.498419	0.462289
ARG:H	41	18	0.123420	0.097415	0.139725	0.110206
ARG:HA	41	18	0.094942	0.084479	0.078439	0.084884
ARG:HB2	41	18	0.067232	0.064501	0.067507	0.068991
ARG:HB3	41	18	0.060555	0.068740	0.067842	0.065747
ARG:HD2	41	18	0.089089	0.081844	0.067024	0.064846
ARG:HD3	41	18	0.144755	0.074373	0.060220	0.073779
ARG:HE	41	18	0.142853	0.154592	0.140316	0.181481
ARG:HG2	41	18	0.094027	0.056636	0.066241	0.063230
ARG:HG3	41	18	0.066624	0.065452	0.060479	0.066282
ARG:HH11	41	18	0.132369	0.106358	0.127975	0.089653
ARG:HH12	41	18	0.130675	0.104448	0.081491	0.082671
ARG:HH21	41	18	0.191296	0.144794	0.189862	0.157027
ARG:HH22	41	18	0.101603	0.090462	0.083913	0.079505
ARG:N	41	18	0.920886	1.011898	0.872542	0.556216
ARG:NE	41	18	1.100300	1.256121	1.107526	1.156485
ARG:NH1	41	18	0.850212	0.958675	0.835153	0.863541
ARG:NH2	41	18	0.948686	1.093168	0.850671	0.943433
ARG:O	41	18	1.869480	2.081320	2.731273	2.274891
ASN:C	37	18	0.662854	0.522255	0.615893	0.514728
ASN:CA	37	18	0.602807	0.455462	0.431124	0.494118
ASN:CB	37	18	0.600931	0.523731	0.415012	0.402733
ASN:CG	37	18	0.441397	0.339483	0.499112	0.340591
ASN:H	37	18	0.135339	0.125235	0.122248	0.098985
ASN:HA	37	18	0.090481	0.088745	0.057852	0.088312
ASN:HB2	37	18	0.082323	0.081395	0.088596	0.065813

IUPAC	<i>n</i>	Proteins	Structure	AIMNet	MOPAC	Joint
ASN:HB3	37	18	0.087627	0.066014	0.072721	0.064136
ASN:HD21	37	18	0.114050	0.088222	0.115485	0.084975
ASN:HD22	37	18	0.136141	0.089017	0.104650	0.102304
ASN:N	37	18	1.253449	1.209021	1.279745	1.410702
ASN:ND2	37	18	1.493960	1.101572	1.163587	1.095338
ASN:O	37	18	3.777460	3.205303	2.865314	2.796140
ASN:OD1	37	18	3.327500	2.092068	2.750873	3.357470
ASP:C	41	16	0.607369	0.551999	0.541549	0.564539
ASP:CA	41	16	0.858375	0.653265	0.656881	0.602122
ASP:CB	41	16	0.803144	0.657727	0.663939	0.654530
ASP:CG	41	16	0.754020	0.677998	0.599710	0.625091
ASP:H	41	16	0.170326	0.183522	0.163813	0.149422
ASP:HA	41	16	0.111469	0.094320	0.080081	0.080173
ASP:HB2	41	16	0.117533	0.092967	0.086967	0.087095
ASP:HB3	41	16	0.087018	0.070707	0.085678	0.083501
ASP:N	41	16	1.094542	1.149580	1.281218	1.058228
ASP:O	41	16	3.564708	3.592715	3.504490	2.706249
ASP:OD1	41	16	4.392673	3.097360	4.087546	3.376106
ASP:OD2	41	16	4.222609	4.121239	3.259411	3.787234
CYS:C	15	8	0.865904	0.867581	0.784607	0.967330
CYS:CA	15	8	0.704998	0.592156	0.575405	0.482688
CYS:CB	15	8	0.933438	0.811442	0.828717	0.870326
CYS:H	15	8	0.221606	0.263969	0.212706	0.267698
CYS:HA	15	8	0.105666	0.111301	0.114384	0.071033
CYS:HB2	15	8	0.107170	0.114842	0.112054	0.104514
CYS:HB3	15	8	0.113099	0.104726	0.121843	0.109381
CYS:HG	15	8	0.233217	0.358873	0.248689	0.373052
CYS:N	15	8	2.267502	2.212376	1.881751	2.001420
CYS:O	15	8	4.005185	2.812131	3.197420	3.677926
GLN:C	19	12	0.610418	0.537033	0.524314	0.548625
GLN:CA	19	12	0.454532	0.544040	0.581266	0.463351
GLN:CB	19	12	0.569611	0.498723	0.415614	0.438117
GLN:CD	19	12	0.590522	0.495819	0.412093	0.525080
GLN:CG	19	12	0.722766	0.729726	0.654403	0.719995
GLN:H	19	12	0.195250	0.148263	0.140749	0.160506
GLN:HA	19	12	0.099086	0.079796	0.081689	0.074007
GLN:HB2	19	12	0.073679	0.080852	0.074616	0.068106
GLN:HB3	19	12	0.076662	0.080553	0.076089	0.076376
GLN:HE21	19	12	0.144286	0.095202	0.094285	0.108939
GLN:HE22	19	12	0.214728	0.142971	0.153905	0.106356
GLN:HG2	19	12	0.079552	0.095437	0.080962	0.082149
GLN:HG3	19	12	0.145947	0.144938	0.126383	0.099938
GLN:N	19	12	0.873726	0.930590	1.134222	1.135909
GLN:NE2	19	12	1.461782	1.320023	1.388165	1.114925
GLN:O	19	12	2.433920	3.100905	2.986972	2.388558
GLN:OE1	19	12	2.807596	3.003102	2.338889	2.053994
GLU:C	78	18	0.461122	0.510880	0.478948	0.488491
GLU:CA	78	18	0.553125	0.424464	0.445729	0.423403
GLU:CB	78	18	0.486791	0.457977	0.497919	0.424885
GLU:CD	78	18	0.769450	0.487987	0.542958	0.441202
GLU:CG	78	18	0.504764	0.552257	0.573899	0.565164
GLU:H	78	18	0.159410	0.153969	0.137859	0.140299
GLU:HA	78	18	0.085592	0.073887	0.067231	0.078710
GLU:HB2	78	18	0.103351	0.088305	0.074244	0.075104
GLU:HB3	78	18	0.067850	0.067651	0.062752	0.062521
GLU:HG2	78	18	0.082684	0.058722	0.075853	0.072625
GLU:HG3	78	18	0.088834	0.074806	0.073575	0.062826
GLU:N	78	18	1.105980	0.984204	1.113818	1.077031
GLU:O	78	18	2.376080	2.301290	2.368032	2.612955
GLU:OE1	78	18	4.032623	3.777387	3.573991	2.873637
GLU:OE2	78	18	3.004717	2.950272	2.717631	3.028577
GLY:C	52	15	0.636612	0.704126	0.647427	0.677461
GLY:CA	52	15	0.583144	0.513912	0.536430	0.481463
GLY:H	52	15	0.160135	0.121136	0.130777	0.126332

IUPAC	<i>n</i>	Proteins	Structure	AIMNet	MOPAC	Joint
GLY:HA2	52	15	0.086382	0.074071	0.061705	0.062361
GLY:HA3	52	15	0.100877	0.089635	0.091561	0.086752
GLY:N	52	15	1.274009	1.371868	1.189315	1.344261
GLY:O	52	15	3.350149	3.794871	3.568546	3.050810
HIS:C	13	8	0.644440	0.599015	0.570150	0.749041
HIS:CA	13	8	1.286547	0.896358	0.910871	0.580404
HIS:CB	13	8	0.684982	0.735538	0.692932	0.625060
HIS:CD2	13	8	0.712659	1.163630	0.927434	0.971257
HIS:CE1	13	8	1.163119	0.922722	1.125199	1.144924
HIS:CG	13	8	0.998933	0.612035	0.860586	1.019989
HIS:H	13	8	0.097539	0.229110	0.253474	0.144938
HIS:HA	13	8	0.099178	0.097660	0.099566	0.124040
HIS:HB2	13	8	0.120601	0.132435	0.117922	0.128757
HIS:HB3	13	8	0.142882	0.090375	0.132754	0.116474
HIS:HD2	13	8	0.096229	0.059833	0.084179	0.074818
HIS:HE1	13	8	0.117557	0.120230	0.090472	0.093872
HIS:HE2	13	8	0.105894	0.170985	0.123465	0.101397
HIS:N	13	8	1.680451	1.070724	1.469619	1.437480
HIS:ND1	13	8	2.836918	2.358021	3.015552	3.090023
HIS:NE2	13	8	1.649987	1.860378	1.135048	1.325546
HIS:O	13	8	4.416665	3.741305	3.589269	2.728029
ILE:C	66	18	0.522202	0.449945	0.371128	0.446397
ILE:CA	66	18	0.590454	0.515750	0.451960	0.441340
ILE:CB	66	18	0.451763	0.349475	0.367362	0.400198
ILE:CD1	66	18	0.403240	0.494153	0.440957	0.338282
ILE:CG1	66	18	0.503567	0.447683	0.386534	0.354109
ILE:CG2	66	18	0.502225	0.460990	0.405770	0.483096
ILE:H	65	18	0.121210	0.119811	0.113098	0.090958
ILE:HA	66	18	0.084956	0.081467	0.073943	0.076766
ILE:HB	66	18	0.075402	0.084424	0.074530	0.075592
ILE:HD11	66	18	0.055653	0.059584	0.054332	0.047401
ILE:HD12	66	18	0.057832	0.063383	0.059745	0.056400
ILE:HD13	66	18	0.079548	0.058478	0.051894	0.063779
ILE:HG12	66	18	0.058844	0.070244	0.055217	0.047172
ILE:HG13	66	18	0.066198	0.069904	0.059547	0.059721
ILE:HG21	66	18	0.055187	0.051487	0.049737	0.044957
ILE:HG22	66	18	0.052160	0.053948	0.042001	0.059084
ILE:HG23	66	18	0.053411	0.059670	0.055277	0.061378
ILE:N	66	18	0.845308	1.364075	1.008406	0.818155
ILE:O	66	18	2.439984	2.293536	2.476496	2.181127
LEU:C	52	16	0.436504	0.404685	0.505291	0.440639
LEU:CA	52	16	0.510381	0.560863	0.464638	0.517316
LEU:CB	52	16	0.467583	0.408986	0.371843	0.490575
LEU:CD1	52	16	0.331019	0.319841	0.440030	0.343681
LEU:CD2	52	16	0.410714	0.561223	0.535958	0.448482
LEU:CG	52	16	0.356379	0.351753	0.346477	0.437731
LEU:H	52	16	0.154288	0.121284	0.126410	0.098083
LEU:HA	52	16	0.100976	0.110174	0.083456	0.085959
LEU:HB2	52	16	0.064496	0.047359	0.048951	0.064104
LEU:HB3	52	16	0.069784	0.070716	0.064759	0.065624
LEU:HD11	52	16	0.036283	0.053080	0.045521	0.038324
LEU:HD12	52	16	0.047406	0.044398	0.050372	0.047538
LEU:HD13	52	16	0.063687	0.043039	0.065790	0.046074
LEU:HD21	52	16	0.060121	0.060345	0.057356	0.045809
LEU:HD22	52	16	0.056182	0.054295	0.062456	0.051259
LEU:HD23	52	16	0.052662	0.070545	0.049987	0.042536
LEU:HG	52	16	0.063377	0.054959	0.056244	0.050924
LEU:N	52	16	0.875419	1.007085	0.895522	0.940413
LEU:O	52	16	2.607759	2.620877	2.322559	2.290063
LYS:C	57	17	0.658990	0.487144	0.407168	0.468082
LYS:CA	57	17	0.472017	0.565120	0.501933	0.501219
LYS:CB	57	17	0.479387	0.422530	0.478796	0.494369
LYS:CD	57	17	0.344311	0.324307	0.332012	0.356756
LYS:CE	57	17	0.495880	0.574104	0.478854	0.481124

IUPAC	<i>n</i>	Proteins	Structure	AIMNet	MOPAC	Joint
LYS:CG	57	17	0.528850	0.444751	0.495897	0.524914
LYS:H	57	17	0.129075	0.117344	0.123443	0.110492
LYS:HA	57	17	0.096203	0.079313	0.061984	0.071173
LYS:HB2	57	17	0.075574	0.059097	0.059863	0.069786
LYS:HB3	57	17	0.074585	0.076523	0.060295	0.051465
LYS:HD2	57	17	0.065625	0.073216	0.074194	0.069142
LYS:HD3	57	17	0.083389	0.078994	0.091707	0.066745
LYS:HE2	57	17	0.095658	0.088796	0.101889	0.079115
LYS:HE3	57	17	0.083231	0.091618	0.083551	0.072759
LYS:HG2	57	17	0.078746	0.066556	0.061434	0.062410
LYS:HG3	57	17	0.090247	0.076285	0.062778	0.069430
LYS:HZ1	57	17	0.105255	0.121972	0.074728	0.152146
LYS:HZ2	57	17	0.115231	0.073792	0.065831	0.066411
LYS:HZ3	57	17	0.115012	0.114728	0.105674	0.100236
LYS:N	57	17	0.955247	0.960797	0.801603	0.802105
LYS:NZ	57	17	0.789106	0.764164	0.732299	0.590095
LYS:O	57	17	2.391380	2.427357	2.224965	2.079000
MET:C	28	18	0.617610	0.557594	0.454408	0.502393
MET:CA	28	18	0.920033	0.674145	0.736295	0.547060
MET:CB	28	18	1.360771	1.124845	1.004116	1.068597
MET:CE	28	18	0.889157	0.861373	0.975849	0.732125
MET:CG	28	18	0.828342	0.695531	0.688938	0.728895
MET:H	13	12	0.132767	0.118964	0.102191	0.115676
MET:HA	28	18	0.112825	0.103468	0.083072	0.073131
MET:HB2	28	18	0.073475	0.098262	0.093879	0.090907
MET:HB3	28	18	0.114826	0.114805	0.102714	0.104397
MET:HE1	28	18	0.089269	0.110430	0.092417	0.080883
MET:HE2	28	18	0.053676	0.080898	0.071582	0.067861
MET:HE3	28	18	0.118730	0.179548	0.118616	0.182299
MET:HG2	28	18	0.124606	0.102658	0.064925	0.089941
MET:HG3	28	18	0.102539	0.101822	0.072979	0.068296
MET:N	28	18	2.125766	3.748555	2.576679	2.343747
MET:O	28	18	2.745957	3.760935	2.575838	3.149237
MET:SD	28	18	3.420395	3.876734	3.380245	4.063864
PHE:C	16	11	0.520739	0.478408	0.500814	0.252671
PHE:CA	16	11	0.614379	0.613531	0.537846	0.661307
PHE:CB	16	11	0.482494	0.653631	0.416657	0.479758
PHE:CD1	16	11	0.466712	0.560442	0.347673	0.440650
PHE:CD2	16	11	0.447046	0.727024	0.614061	0.601982
PHE:CE1	16	11	0.614917	0.460416	0.424166	0.494254
PHE:CE2	16	11	0.565489	0.370386	0.540328	0.302204
PHE:CG	16	11	0.724102	0.471927	0.752504	0.616036
PHE:CZ	16	11	0.416351	0.581277	0.463588	0.430756
PHE:H	16	11	0.193853	0.127604	0.149391	0.158662
PHE:HA	16	11	0.104410	0.104237	0.074952	0.089554
PHE:HB2	16	11	0.107846	0.058245	0.091505	0.086823
PHE:HB3	16	11	0.091683	0.099454	0.110535	0.104164
PHE:HD1	16	11	0.101617	0.057525	0.107529	0.069888
PHE:HD2	16	11	0.096903	0.073642	0.083407	0.055523
PHE:HE1	16	11	0.070724	0.072377	0.057023	0.073890
PHE:HE2	16	11	0.068863	0.082738	0.048008	0.074409
PHE:HZ	16	11	0.110350	0.059213	0.052173	0.068539
PHE:N	16	11	0.880398	0.799359	0.925197	0.899968
PHE:O	16	11	3.639961	3.634601	3.386579	2.659162
PRO:C	21	11	0.662056	0.504656	0.556805	0.542767
PRO:CA	21	11	0.673083	0.744445	0.821544	0.738859
PRO:CB	21	11	0.474115	0.524145	0.606694	0.666679
PRO:CD	21	11	0.689831	0.525269	0.548322	0.611147
PRO:CG	21	11	0.495561	0.540893	0.444521	0.435694
PRO:HA	21	11	0.084980	0.069434	0.104097	0.092390
PRO:HB2	21	11	0.087891	0.060043	0.063197	0.049321
PRO:HB3	21	11	0.086494	0.060926	0.055208	0.063451
PRO:HD2	21	11	0.078430	0.125974	0.099070	0.082628
PRO:HD3	21	11	0.085204	0.070207	0.078262	0.057361

IUPAC	<i>n</i>	Proteins	Structure	AIMNet	MOPAC	Joint
PRO:HG2	21	11	0.113944	0.086737	0.116643	0.095073
PRO:HG3	21	11	0.049936	0.051233	0.092324	0.067164
PRO:N	21	11	1.913868	2.561568	2.082384	2.072088
PRO:O	21	11	2.969043	4.041369	4.583111	3.798197
SER:C	35	15	0.753310	0.611530	0.583153	0.613344
SER:CA	35	15	0.650529	0.727185	0.558945	0.548007
SER:CB	35	15	0.955099	0.681350	0.644459	0.693289
SER:H	35	15	0.163618	0.170399	0.133650	0.144078
SER:HA	35	15	0.092315	0.093616	0.067514	0.099386
SER:HB2	35	15	0.086152	0.068291	0.079530	0.069522
SER:HB3	35	15	0.115696	0.104915	0.086493	0.089342
SER:HG	35	15	0.180465	0.133564	0.150043	0.148392
SER:N	35	15	1.238250	1.206334	1.117874	1.271041
SER:O	35	15	3.607085	3.891859	3.093181	3.317974
SER:OG	35	15	1.841007	2.032675	2.202352	2.159906
THR:C	43	17	0.585197	0.410386	0.525933	0.451043
THR:CA	43	17	0.540772	0.563357	0.632383	0.608625
THR:CB	43	17	0.779753	0.707119	0.613180	0.737046
THR:CG2	43	17	0.473236	0.515406	0.495257	0.404895
THR:H	43	17	0.138169	0.117443	0.124832	0.109477
THR:HA	43	17	0.096307	0.082371	0.081428	0.065622
THR:HB	43	17	0.071373	0.077613	0.069424	0.074800
THR:HG1	43	17	0.134710	0.114521	0.106689	0.128474
THR:HG21	43	17	0.075313	0.068446	0.072274	0.067281
THR:HG22	43	17	0.056871	0.062545	0.059034	0.045735
THR:HG23	43	17	0.052550	0.072274	0.048474	0.060855
THR:N	43	17	0.941329	1.043024	0.842683	0.851960
THR:O	43	17	3.415222	2.711966	2.971628	2.949058
THR:OG1	43	17	2.218086	2.477043	2.232039	1.960074
TRP:C	4	4	0.372286	0.495312	0.336523	0.274141
TRP:CA	4	4	1.075433	1.214563	0.737172	0.861932
TRP:CB	4	4	0.404480	0.547823	0.363791	0.396899
TRP:CD1	4	4	0.584252	1.081054	0.820329	0.746206
TRP:CD2	4	4	0.385442	0.152999	0.696363	0.328112
TRP:CE2	4	4	0.184006	0.636465	0.233232	0.738655
TRP:CE3	4	4	0.552287	0.687839	0.344941	0.563859
TRP:CG	4	4	1.222650	0.651493	0.363980	0.851262
TRP:CH2	4	4	0.589444	0.492690	0.693768	0.759737
TRP:CZ2	4	4	0.537495	0.359338	0.403670	0.213516
TRP:CZ3	4	4	0.267771	0.563423	0.771557	0.649016
TRP:H	4	4	0.170017	0.194775	0.078155	0.206576
TRP:HA	4	4	0.135652	0.075549	0.063967	0.092376
TRP:HB2	4	4	0.116997	0.057747	0.065495	0.042401
TRP:HB3	4	4	0.142629	0.072919	0.145533	0.078197
TRP:HD1	4	4	0.260658	0.121365	0.136367	0.076868
TRP:HE1	4	4	0.174488	0.054220	0.098418	0.121254
TRP:HE3	4	4	0.137398	0.045804	0.046431	0.074941
TRP:HH2	4	4	0.156212	0.104553	0.147440	0.117936
TRP:HZ2	4	4	0.078997	0.072022	0.071226	0.076391
TRP:HZ3	4	4	0.122333	0.044525	0.057037	0.074808
TRP:N	4	4	1.062635	1.402701	1.115683	1.213394
TRP:NE1	4	4	0.224640	0.679094	0.631982	0.854697
TRP:O	4	4	1.922536	2.185245	0.740195	1.527640
TYR:C	22	14	0.597977	0.347292	0.402922	0.467467
TYR:CA	22	14	0.587401	0.474769	0.572923	0.478096
TYR:CB	22	14	0.693370	0.531059	0.563157	0.649684
TYR:CD1	22	14	0.488562	0.440619	0.604748	0.539453
TYR:CD2	22	14	0.712126	0.700621	0.576817	0.462701
TYR:CE1	22	14	0.763490	0.840176	0.653845	0.634609
TYR:CE2	22	14	0.587576	0.824369	0.764290	0.584483
TYR:CG	22	14	0.403764	0.456030	0.606681	0.564737
TYR:CZ	22	14	0.685683	0.551322	0.613137	0.562077
TYR:H	22	14	0.159033	0.131527	0.175118	0.116453
TYR:HA	22	14	0.072350	0.063518	0.065772	0.069133

IUPAC	<i>n</i>	Proteins	Structure	AIMNet	MOPAC	Joint
TYR:HB2	22	14	0.078137	0.068517	0.079848	0.075991
TYR:HB3	22	14	0.061526	0.067536	0.068058	0.070665
TYR:HD1	22	14	0.079585	0.077330	0.059347	0.058845
TYR:HD2	22	14	0.075932	0.072248	0.062747	0.073331
TYR:HE1	22	14	0.072736	0.055145	0.091362	0.052847
TYR:HE2	22	14	0.072264	0.087533	0.073315	0.080069
TYR:HH	22	14	0.133855	0.247756	0.205683	0.192408
TYR:N	22	14	1.013776	1.045906	0.908274	0.764576
TYR:O	22	14	2.217741	2.804410	2.016358	1.994152
TYR:OH	22	14	1.810259	1.536943	1.642389	1.418038
VAL:C	71	17	0.436208	0.477822	0.385539	0.423416
VAL:CA	71	17	0.569101	0.556536	0.490527	0.433861
VAL:CB	71	17	0.569536	0.506909	0.529782	0.576934
VAL:CG1	71	17	0.406072	0.508729	0.482444	0.414713
VAL:CG2	71	17	0.477340	0.446420	0.544878	0.524330
VAL:H	71	17	0.130359	0.097212	0.110301	0.100418
VAL:HA	71	17	0.078833	0.071081	0.064731	0.066851
VAL:HB	71	17	0.075038	0.058122	0.073324	0.058526
VAL:HG11	71	17	0.070359	0.058916	0.055384	0.059228
VAL:HG12	71	17	0.065137	0.066025	0.065434	0.065288
VAL:HG13	71	17	0.064404	0.052280	0.062739	0.065137
VAL:HG21	71	17	0.067998	0.068765	0.069342	0.056700
VAL:HG22	71	17	0.088707	0.080431	0.076909	0.095183
VAL:HG23	71	17	0.081136	0.093057	0.067291	0.082908
VAL:N	71	17	0.958876	0.979079	1.014105	1.051703
VAL:O	71	17	2.933990	2.594393	2.166332	2.108111

5.4 Previous-paper 18: T2 vector RMSE (ppm)

IUPAC	<i>n</i>	Proteins	Structure	AIMNet	MOPAC	Joint
ALA:C	82	18	3.367451	3.360892	3.025154	2.667537
ALA:CA	82	18	2.155966	1.998675	2.052131	1.829459
ALA:CB	82	18	1.845587	1.808810	1.904643	1.779589
ALA:H	81	18	0.639838	0.598634	0.598793	0.584742
ALA:HA	82	18	0.561658	0.535838	0.470004	0.457670
ALA:HB1	82	18	0.561015	0.516700	0.502780	0.468767
ALA:HB2	82	18	0.548006	0.526348	0.506128	0.503589
ALA:HB3	82	18	0.537485	0.518271	0.493277	0.484951
ALA:N	82	18	4.352518	4.336591	4.043884	3.787941
ALA:O	82	18	15.670861	15.417955	13.916988	13.108161
ARG:C	41	18	3.510529	3.519308	3.091357	3.023228
ARG:CA	41	18	2.433588	1.985782	1.976812	1.921173
ARG:CB	41	18	2.342218	2.279003	2.205187	2.028587
ARG:CD	41	18	3.401415	2.674296	2.668550	2.489269
ARG:CG	41	18	2.621353	2.421482	2.158597	2.154367
ARG:CZ	41	18	4.027106	4.449526	3.833125	3.113538
ARG:H	41	18	0.668177	0.603144	0.556776	0.581650
ARG:HA	41	18	0.563414	0.507319	0.495378	0.495294
ARG:HB2	41	18	0.544000	0.516748	0.477501	0.424158
ARG:HB3	41	18	0.579048	0.490605	0.461801	0.441488
ARG:HD2	41	18	0.583333	0.484727	0.510216	0.465222
ARG:HD3	41	18	0.605059	0.523928	0.559423	0.467701
ARG:HE	41	18	0.635033	0.603738	0.667941	0.676509
ARG:HG2	41	18	0.541071	0.549676	0.489321	0.479988
ARG:HG3	41	18	0.560423	0.571214	0.487839	0.496405
ARG:HH11	41	18	0.813114	0.746119	0.788717	0.678722
ARG:HH12	41	18	0.659727	0.586137	0.571709	0.613686
ARG:HH21	41	18	0.765939	0.792325	0.693858	0.665569
ARG:HH22	41	18	0.616765	0.588628	0.519780	0.555082
ARG:N	41	18	4.226379	4.094902	3.627807	3.752304
ARG:NE	41	18	5.144878	4.748661	4.892125	4.553418
ARG:NH1	41	18	4.286157	4.284671	4.076734	4.239170

IUPAC	<i>n</i>	Proteins	Structure	AIMNet	MOPAC	Joint
ARG:NH2	41	18	4.050684	4.209902	3.858071	3.988152
ARG:O	41	18	14.171794	15.521445	13.576591	12.897334
ASN:C	37	18	4.041890	3.561375	3.470880	3.447918
ASN:CA	37	18	2.691860	2.442127	2.333408	1.984306
ASN:CB	37	18	2.595160	2.826694	2.416461	2.601646
ASN:CG	37	18	4.469908	3.943132	4.054174	3.420211
ASN:H	37	18	0.693876	0.610723	0.544365	0.570629
ASN:HA	37	18	0.472767	0.447232	0.416817	0.398243
ASN:HB2	37	18	0.595072	0.546184	0.461956	0.538325
ASN:HB3	37	18	0.558844	0.502329	0.477730	0.429878
ASN:HD21	37	18	0.680819	0.662412	0.511695	0.546178
ASN:HD22	37	18	0.640168	0.678806	0.505162	0.520666
ASN:N	37	18	5.132278	4.413713	4.553644	4.531691
ASN:ND2	37	18	4.977680	4.522771	4.728079	4.202188
ASN:O	37	18	16.224363	15.698920	14.304003	12.996102
ASN:OD1	37	18	15.789411	19.176355	17.865684	15.033697
ASP:C	41	16	4.267819	5.201768	4.130316	3.887111
ASP:CA	41	16	3.036290	2.699580	2.460576	2.445473
ASP:CB	41	16	3.742323	3.557428	3.148706	2.858858
ASP:CG	41	16	4.682001	4.241894	4.325689	3.879867
ASP:H	41	16	0.703221	0.692183	0.618819	0.575415
ASP:HA	41	16	0.599276	0.538128	0.542681	0.471292
ASP:HB2	41	16	0.625968	0.567589	0.582455	0.514750
ASP:HB3	41	16	0.655705	0.603902	0.616663	0.508559
ASP:N	41	16	5.345403	5.369990	4.973507	4.408058
ASP:O	41	16	18.239144	19.821984	17.394560	18.382280
ASP:OD1	41	16	18.961822	17.459479	17.157612	17.551571
ASP:OD2	41	16	18.175836	18.393487	16.915266	16.390895
CYS:C	15	8	4.484620	4.123470	3.709426	4.349372
CYS:CA	15	8	2.971402	3.235342	2.878975	2.668799
CYS:CB	15	8	3.627139	4.071341	3.914400	3.868969
CYS:H	15	8	0.883415	0.849030	0.881050	0.975764
CYS:HA	15	8	0.611961	0.657664	0.696410	0.607193
CYS:HB2	15	8	0.780589	0.802791	0.724755	0.894388
CYS:HB3	15	8	0.828839	0.883127	0.803835	0.843622
CYS:HG	15	8	1.072112	1.552734	1.145869	1.449485
CYS:N	15	8	5.664747	6.015492	5.643067	5.904498
CYS:O	15	8	17.739452	22.664623	17.774160	16.967185
GLN:C	19	12	3.862776	3.549449	3.637143	3.435553
GLN:CA	19	12	2.462977	2.003464	2.280587	2.386322
GLN:CB	19	12	2.400748	2.250637	2.021203	2.135439
GLN:CD	19	12	4.861613	3.985266	4.853902	3.855425
GLN:CG	19	12	3.365087	3.503015	3.231631	2.843249
GLN:H	19	12	0.643517	0.601962	0.717465	0.601075
GLN:HA	19	12	0.511074	0.533574	0.423251	0.449560
GLN:HB2	19	12	0.566348	0.514145	0.479798	0.466358
GLN:HB3	19	12	0.524968	0.539976	0.508570	0.495750
GLN:HE21	19	12	0.780813	0.627035	0.704433	0.687928
GLN:HE22	19	12	0.692500	0.677769	0.632710	0.603195
GLN:HG2	19	12	0.563055	0.521500	0.522314	0.502561
GLN:HG3	19	12	0.662180	0.557722	0.517035	0.509741
GLN:N	19	12	5.085711	5.370537	5.109706	4.830784
GLN:NE2	19	12	5.004314	4.592013	6.196438	4.550399
GLN:O	19	12	13.734481	16.886031	16.905591	12.837744
GLN:OE1	19	12	18.590430	21.199843	17.431563	14.098657
GLU:C	78	18	3.333262	3.318252	3.189646	2.910631
GLU:CA	78	18	2.112076	1.919240	1.975105	1.814463
GLU:CB	78	18	2.693866	2.529662	2.367585	2.215453
GLU:CD	78	18	4.497462	4.248357	3.455109	3.294488
GLU:CG	78	18	3.448631	2.925310	2.749295	2.770478
GLU:H	78	18	0.673521	0.646232	0.585889	0.667372
GLU:HA	78	18	0.496132	0.485223	0.453210	0.439081
GLU:HB2	78	18	0.592810	0.544787	0.484269	0.489331
GLU:HB3	78	18	0.567028	0.536067	0.474023	0.497847

IUPAC	<i>n</i>	Proteins	Structure	AIMNet	MOPAC	Joint
GLU:HG2	78	18	0.600556	0.578098	0.521486	0.504686
GLU:HG3	78	18	0.542506	0.557470	0.476596	0.456840
GLU:N	78	18	3.974658	3.783763	3.812577	3.661127
GLU:O	78	18	14.638606	13.628356	13.751628	12.316260
GLU:OE1	78	18	16.434534	14.999132	15.242841	14.032754
GLU:OE2	78	18	16.608170	16.936551	15.268560	13.632489
GLY:C	52	15	4.242020	3.925127	4.258298	3.668302
GLY:CA	52	15	2.712285	2.426218	2.212475	2.071748
GLY:H	52	15	0.705438	0.628750	0.657346	0.600518
GLY:HA2	52	15	0.590909	0.551860	0.515852	0.460728
GLY:HA3	52	15	0.607938	0.560845	0.503225	0.480272
GLY:N	52	15	4.922391	4.726613	4.676058	4.918717
GLY:O	52	15	17.884137	18.805060	14.465584	15.274780
HIS:C	13	8	5.717699	4.947465	4.801551	4.573142
HIS:CA	13	8	3.331781	2.936016	2.719466	2.748870
HIS:CB	13	8	3.311399	3.079406	3.162381	3.147525
HIS:CD2	13	8	7.666704	5.784884	5.226538	6.646887
HIS:CE1	13	8	5.958953	7.198994	6.646233	6.032666
HIS:CG	13	8	5.714184	5.019506	5.865608	5.546820
HIS:H	13	8	0.659037	0.688014	0.714829	0.676882
HIS:HA	13	8	0.666158	0.569118	0.527568	0.443149
HIS:HB2	13	8	0.669370	0.606708	0.696580	0.608841
HIS:HB3	13	8	0.728138	0.614704	0.620900	0.700770
HIS:HD2	13	8	0.547043	0.566971	0.592273	0.486283
HIS:HE1	13	8	0.791571	0.699796	0.830376	0.693723
HIS:HE2	13	8	0.755225	0.706141	0.902351	0.991072
HIS:N	13	8	5.136543	5.694225	5.317320	5.730396
HIS:ND1	13	8	17.616786	17.738604	13.943489	13.636090
HIS:NE2	13	8	6.180942	5.746873	6.075335	5.350801
HIS:O	13	8	21.153421	16.156336	19.359787	19.826622
ILE:C	66	18	3.507867	3.530254	3.388399	2.978880
ILE:CA	66	18	1.880137	2.094246	1.939156	1.829842
ILE:CB	66	18	1.785955	1.719727	1.719295	1.650064
ILE:CD1	66	18	1.631656	1.492914	1.438909	1.563333
ILE:CG1	66	18	1.930657	1.704334	1.647962	1.729153
ILE:CG2	66	18	1.500685	1.660489	1.480730	1.377050
ILE:H	65	18	0.662115	0.598230	0.575584	0.566000
ILE:HA	66	18	0.529307	0.495758	0.472683	0.463735
ILE:HB	66	18	0.548385	0.515617	0.500635	0.480087
ILE:HD11	66	18	0.527009	0.513001	0.482103	0.478163
ILE:HD12	66	18	0.538656	0.509108	0.455150	0.471635
ILE:HD13	66	18	0.572757	0.523053	0.469105	0.454179
ILE:HG12	66	18	0.551484	0.518265	0.482851	0.479981
ILE:HG13	66	18	0.563886	0.529251	0.473437	0.470137
ILE:HG21	66	18	0.537334	0.480771	0.458349	0.464634
ILE:HG22	66	18	0.525475	0.509903	0.495444	0.464766
ILE:HG23	66	18	0.559906	0.487792	0.489242	0.472215
ILE:N	66	18	4.057046	4.209997	3.862390	3.750639
ILE:O	66	18	13.234491	13.991125	12.733384	10.321425
LEU:C	52	16	3.414714	3.163191	2.947341	2.698429
LEU:CA	52	16	2.076512	2.068888	1.836739	1.900907
LEU:CB	52	16	2.253154	2.158713	1.983130	1.857524
LEU:CD1	52	16	1.680698	1.490668	1.550859	1.493515
LEU:CD2	52	16	1.728724	1.620890	1.855967	1.513571
LEU:CG	52	16	1.468870	1.349547	1.188967	1.312171
LEU:H	52	16	0.630808	0.595946	0.578061	0.539356
LEU:HA	52	16	0.536753	0.484276	0.453791	0.453235
LEU:HB2	52	16	0.548397	0.507137	0.462473	0.458371
LEU:HB3	52	16	0.580168	0.501551	0.455584	0.477995
LEU:HD11	52	16	0.498592	0.495703	0.454049	0.426873
LEU:HD12	52	16	0.537536	0.501677	0.443273	0.448660
LEU:HD13	52	16	0.522975	0.501224	0.442764	0.450681
LEU:HD21	52	16	0.539986	0.517648	0.477167	0.468132
LEU:HD22	52	16	0.539742	0.499828	0.458923	0.444324

IUPAC	<i>n</i>	Proteins	Structure	AIMNet	MOPAC	Joint
LEU:HD23	52	16	0.512588	0.507896	0.426527	0.459492
LEU:HG	52	16	0.539490	0.484144	0.439723	0.491241
LEU:N	52	16	4.144926	4.070763	3.783267	4.069268
LEU:O	52	16	14.375042	13.938175	12.645562	12.541985
LYS:C	57	17	3.275376	2.984658	2.909724	2.917806
LYS:CA	57	17	1.989529	1.914221	1.985127	1.672974
LYS:CB	57	17	1.980857	2.043586	1.754754	1.789544
LYS:CD	57	17	2.180141	2.043728	2.032094	1.800024
LYS:CE	57	17	2.825839	2.643263	2.581658	2.369749
LYS:CG	57	17	2.423649	2.232325	1.928262	1.934632
LYS:H	57	17	0.646814	0.569473	0.512516	0.533259
LYS:HA	57	17	0.511550	0.508179	0.436631	0.430928
LYS:HB2	57	17	0.548334	0.474262	0.476026	0.486374
LYS:HB3	57	17	0.504783	0.529991	0.428180	0.502760
LYS:HD2	57	17	0.503328	0.477949	0.427522	0.432453
LYS:HD3	57	17	0.523731	0.528051	0.427421	0.437822
LYS:HE2	57	17	0.661020	0.604186	0.548552	0.573171
LYS:HE3	57	17	0.542508	0.505452	0.452740	0.468412
LYS:HG2	57	17	0.532718	0.492261	0.444204	0.429431
LYS:HG3	57	17	0.534981	0.484660	0.430601	0.462008
LYS:HZ1	57	17	0.626829	0.671348	0.513973	0.524157
LYS:HZ2	57	17	0.642023	0.719960	0.569007	0.594817
LYS:HZ3	57	17	0.682683	0.761058	0.603364	0.612087
LYS:N	57	17	4.211783	3.941021	3.788425	4.021895
LYS:NZ	57	17	2.438003	2.336005	2.119303	2.433378
LYS:O	57	17	14.275068	13.362887	11.743030	11.491354
MET:C	28	18	3.980739	4.187327	3.949285	4.118036
MET:CA	28	18	2.700027	2.658734	2.528322	2.408320
MET:CB	28	18	5.304706	4.439313	3.915196	3.475128
MET:CE	28	18	3.448077	2.811394	2.644518	2.909107
MET:CG	28	18	3.624757	3.494457	3.077125	3.139589
MET:H	13	12	0.666210	0.677426	0.526548	0.560146
MET:HA	28	18	0.554535	0.561989	0.503104	0.513553
MET:HB2	28	18	0.727998	0.622831	0.632883	0.573059
MET:HB3	28	18	0.761853	0.729080	0.573654	0.546466
MET:HE1	28	18	0.666043	0.719921	0.640421	0.607962
MET:HE2	28	18	0.688999	0.626384	0.582675	0.558856
MET:HE3	28	18	1.033657	1.055030	1.045866	0.854004
MET:HG2	28	18	0.724484	0.712662	0.607842	0.662119
MET:HG3	28	18	0.817656	0.720294	0.679373	0.631748
MET:N	28	18	4.816680	4.869926	4.810255	4.373065
MET:O	28	18	17.692008	18.197465	14.410851	13.362655
MET:SD	28	18	19.056252	16.780814	17.792082	17.046448
PHE:C	16	11	3.600460	4.086915	3.615221	3.852388
PHE:CA	16	11	2.932099	2.889713	2.611842	2.722835
PHE:CB	16	11	2.522926	2.125551	2.390532	2.173080
PHE:CD1	16	11	6.604315	6.093570	6.107356	6.169455
PHE:CD2	16	11	5.662236	5.451442	5.152778	4.326532
PHE:CE1	16	11	5.439736	5.378085	4.758255	4.525593
PHE:CE2	16	11	3.997286	6.581096	4.327392	3.734786
PHE:CG	16	11	6.784828	5.134368	5.493567	5.005445
PHE:CZ	16	11	6.476634	6.837091	5.409592	5.220639
PHE:H	16	11	0.715964	0.743519	0.646822	0.557141
PHE:HA	16	11	0.558585	0.590510	0.559183	0.542804
PHE:HB2	16	11	0.563155	0.609774	0.508579	0.508883
PHE:HB3	16	11	0.768398	0.653578	0.631355	0.578333
PHE:HD1	16	11	0.702901	0.601513	0.631787	0.638285
PHE:HD2	16	11	0.611374	0.557653	0.598167	0.622637
PHE:HE1	16	11	0.724765	0.518962	0.576439	0.554677
PHE:HE2	16	11	0.601359	0.575891	0.543944	0.579993
PHE:HZ	16	11	0.680229	0.577032	0.680670	0.597335
PHE:N	16	11	5.749710	4.997496	5.584306	4.429271
PHE:O	16	11	20.629538	17.922226	19.052321	16.465707
PRO:C	21	11	4.463115	3.907332	3.508280	4.038176

IUPAC	<i>n</i>	Proteins	Structure	AIMNet	MOPAC	Joint
PRO:CA	21	11	4.106725	3.420355	3.383417	3.129069
PRO:CB	21	11	2.504306	2.770641	2.423798	2.291381
PRO:CD	21	11	4.082771	3.195414	3.394571	3.134727
PRO:CG	21	11	2.855726	2.899835	2.880044	2.388840
PRO:HA	21	11	0.652011	0.635791	0.520580	0.606910
PRO:HB2	21	11	0.536982	0.556081	0.531709	0.627230
PRO:HB3	21	11	0.459107	0.567311	0.486780	0.521191
PRO:HD2	21	11	0.688627	0.639981	0.587114	0.678254
PRO:HD3	21	11	0.736650	0.649012	0.579794	0.577286
PRO:HG2	21	11	0.576633	0.665649	0.544020	0.575248
PRO:HG3	21	11	0.588757	0.637713	0.553493	0.464069
PRO:N	21	11	6.985357	6.390948	6.525323	6.448427
PRO:O	21	11	21.249267	21.165674	19.705883	19.117760
SER:C	35	15	4.663545	4.371003	3.976834	3.751600
SER:CA	35	15	2.479580	2.747502	2.610740	2.407045
SER:CB	35	15	3.593042	3.425772	3.658221	2.936980
SER:H	35	15	0.687857	0.653483	0.610991	0.591214
SER:HA	35	15	0.549380	0.546093	0.473249	0.441032
SER:HB2	35	15	0.625883	0.551608	0.545037	0.525850
SER:HB3	35	15	0.594171	0.561271	0.572424	0.582695
SER:HG	35	15	0.890401	1.012714	0.756623	0.803523
SER:N	35	15	5.321455	5.302384	5.157561	5.127703
SER:O	35	15	18.670065	19.231609	16.119119	13.542103
SER:OG	35	15	8.417044	7.579199	7.347907	7.748392
THR:C	43	17	3.670997	3.360656	3.280741	2.769052
THR:CA	43	17	2.441810	2.062642	2.208548	1.912958
THR:CB	43	17	3.795906	2.715928	2.856198	2.363355
THR:CG2	43	17	2.062323	1.946611	1.710838	1.763814
THR:H	43	17	0.675423	0.631911	0.605034	0.574621
THR:HA	43	17	0.540751	0.561737	0.469102	0.433446
THR:HB	43	17	0.568251	0.527330	0.508996	0.501346
THR:HG1	43	17	0.728812	0.784231	0.674651	0.725037
THR:HG21	43	17	0.533356	0.577082	0.480646	0.495257
THR:HG22	43	17	0.503944	0.505161	0.456478	0.474393
THR:HG23	43	17	0.526603	0.515861	0.452393	0.514562
THR:N	43	17	4.357997	3.908070	4.472547	4.334017
THR:O	43	17	15.035822	14.534242	11.282602	13.543181
THR:OG1	43	17	7.314299	7.528964	6.791523	6.406940
TRP:C	4	4	3.887040	4.172436	3.027799	3.698201
TRP:CA	4	4	3.264378	4.105702	3.711739	2.648338
TRP:CB	4	4	3.129485	2.734396	2.631940	1.871987
TRP:CD1	4	4	8.151421	6.470753	8.804482	5.826465
TRP:CD2	4	4	5.030897	4.662104	3.666722	7.107002
TRP:CE2	4	4	4.297173	4.693458	6.702613	4.834979
TRP:CE3	4	4	5.298427	4.628891	4.812209	4.185140
TRP:CG	4	4	8.250471	4.752543	7.742379	4.663806
TRP:CH2	4	4	3.403296	4.997374	3.398999	3.473212
TRP:CZ2	4	4	3.938313	4.240329	4.025827	4.495223
TRP:CZ3	4	4	3.988232	4.329168	4.395275	4.602316
TRP:H	4	4	0.929187	0.505648	0.896725	0.713714
TRP:HA	4	4	0.751953	0.739270	0.814946	0.673792
TRP:HB2	4	4	0.644493	0.698775	0.579256	0.540992
TRP:HB3	4	4	0.837197	0.655417	0.629431	0.622378
TRP:HD1	4	4	0.728479	0.941984	0.759546	0.941493
TRP:HE1	4	4	1.092013	0.801763	0.441158	0.590232
TRP:HE3	4	4	0.535629	0.722113	0.648889	0.593003
TRP:HH2	4	4	0.553141	0.655403	0.477209	0.582847
TRP:HZ2	4	4	0.793101	0.474939	0.377177	0.345284
TRP:HZ3	4	4	0.504141	0.566536	0.575170	0.521360
TRP:N	4	4	6.841687	5.478764	5.111943	5.091567
TRP:NE1	4	4	3.812288	7.133397	4.534411	5.864813
TRP:O	4	4	14.374227	15.617952	9.082824	10.663508
TYR:C	22	14	3.391670	4.061397	3.068287	3.098576
TYR:CA	22	14	2.688305	2.451291	2.119961	2.027679

IUPAC	n	Proteins	Structure	AIMNet	MOPAC	Joint
TYR:CB	22	14	2.124096	2.349200	2.067948	2.097187
TYR:CD1	22	14	5.047085	4.478803	5.186010	4.253518
TYR:CD2	22	14	4.685393	5.155529	5.048250	4.734277
TYR:CE1	22	14	4.290188	4.251298	4.489285	3.679109
TYR:CE2	22	14	4.187906	4.932020	3.666388	3.619245
TYR:CG	22	14	4.006117	5.199219	4.343097	4.059159
TYR:CZ	22	14	5.778050	5.642994	5.878439	4.743176
TYR:H	22	14	0.645102	0.671934	0.675314	0.559187
TYR:HA	22	14	0.575049	0.534814	0.501972	0.465919
TYR:HB2	22	14	0.628687	0.615201	0.543232	0.537848
TYR:HB3	22	14	0.622588	0.572286	0.545738	0.495167
TYR:HD1	22	14	0.577942	0.478498	0.475880	0.444168
TYR:HD2	22	14	0.635287	0.612465	0.535278	0.575759
TYR:HE1	22	14	0.539042	0.506613	0.519802	0.484485
TYR:HE2	22	14	0.522884	0.502297	0.502043	0.491192
TYR:HH	22	14	0.742041	0.802294	0.813483	0.855731
TYR:N	22	14	4.630731	4.197282	3.918278	4.268859
TYR:O	22	14	14.295029	13.970771	12.772600	12.328794
TYR:OH	22	14	9.875293	9.776788	8.577435	8.050730
VAL:C	71	17	3.665088	3.457923	3.127316	2.639351
VAL:CA	71	17	2.173462	2.123858	1.844266	1.853677
VAL:CB	71	17	1.992692	1.870468	1.770204	1.666742
VAL:CG1	71	17	1.807409	1.818331	1.698903	1.667864
VAL:CG2	71	17	1.986415	1.990047	1.937764	1.936752
VAL:H	71	17	0.630131	0.598504	0.581125	0.574716
VAL:HA	71	17	0.577479	0.532190	0.476944	0.454059
VAL:HB	71	17	0.613101	0.541403	0.515731	0.473919
VAL:HG11	71	17	0.554316	0.499911	0.454433	0.443128
VAL:HG12	71	17	0.562463	0.523113	0.460972	0.475141
VAL:HG13	71	17	0.565789	0.500468	0.488330	0.481728
VAL:HG21	71	17	0.595522	0.522570	0.504072	0.480892
VAL:HG22	71	17	0.605905	0.549925	0.501316	0.477549
VAL:HG23	71	17	0.577021	0.541417	0.495330	0.487577
VAL:N	71	17	4.547621	4.490088	4.132361	4.427438
VAL:O	71	17	16.108597	17.118595	14.316060	13.767291

5.5 All 79: T0 RMSE (ppm)

IUPAC	n	Proteins	Structure	AIMNet	MOPAC	Joint
ALA:C	268	73	0.493807	0.486772	0.436717	0.433942
ALA:CA	268	73	0.495151	0.504828	0.448067	0.460912
ALA:CB	268	73	0.481434	0.450024	0.422665	0.459298
ALA:H	267	73	0.158743	0.114073	0.135352	0.108634
ALA:HA	268	73	0.096626	0.078203	0.074739	0.072389
ALA:HB1	268	73	0.074420	0.076650	0.067116	0.071573
ALA:HB2	268	73	0.072799	0.066299	0.058874	0.064279
ALA:HB3	268	73	0.065977	0.065599	0.059251	0.058368
ALA:N	268	73	1.126243	1.122332	1.106736	1.106421
ALA:O	268	73	2.674593	2.705307	2.699675	2.717409
ARG:C	338	77	0.538584	0.581712	0.479828	0.480923
ARG:CA	338	77	0.554995	0.513286	0.526142	0.466311
ARG:CB	338	77	0.584212	0.650633	0.522801	0.574473
ARG:CD	338	77	0.540904	0.560295	0.580574	0.520916
ARG:CG	338	77	0.572308	0.531246	0.531384	0.518245
ARG:CZ	338	77	0.474047	0.430400	0.458618	0.379233
ARG:H	336	76	0.130329	0.123390	0.143972	0.098066
ARG:HA	338	77	0.091885	0.079436	0.078891	0.081897
ARG:HB2	338	77	0.075495	0.071791	0.071451	0.075868
ARG:HB3	338	77	0.068283	0.074846	0.063568	0.068663
ARG:HD2	338	77	0.077643	0.063238	0.071433	0.057031
ARG:HD3	338	77	0.086360	0.068133	0.079776	0.058105
ARG:HE	338	77	0.181397	0.151441	0.147131	0.160746

IUPAC	<i>n</i>	Proteins	Structure	AIMNet	MOPAC	Joint
ARG:HG2	338	77	0.085348	0.075956	0.073091	0.067959
ARG:HG3	338	77	0.078678	0.077285	0.068268	0.062391
ARG:HH11	338	77	0.118410	0.088578	0.089225	0.084041
ARG:HH12	338	77	0.111314	0.080943	0.089751	0.081123
ARG:HH21	338	77	0.177232	0.147319	0.161834	0.112157
ARG:HH22	338	77	0.106191	0.096820	0.100925	0.091509
ARG:N	338	77	1.546060	1.356332	0.966533	0.942618
ARG:NE	338	77	1.064079	1.066547	0.949280	0.942983
ARG:NH1	338	77	0.843111	0.896056	0.765743	0.737845
ARG:NH2	338	77	0.952834	0.971022	0.851402	0.883782
ARG:O	338	77	2.734512	2.681070	2.389071	2.860073
ASN:C	98	54	0.544467	0.504230	0.572133	0.606136
ASN:CA	98	54	0.610676	0.533620	0.564788	0.575065
ASN:CB	98	54	0.705371	0.581048	0.589475	0.543337
ASN:CG	98	54	0.555381	0.552355	0.489058	0.409960
ASN:H	97	53	0.134001	0.158069	0.139308	0.137153
ASN:HA	98	54	0.091218	0.089704	0.068115	0.077286
ASN:HB2	98	54	0.088636	0.089016	0.095456	0.080824
ASN:HB3	98	54	0.083277	0.073325	0.074200	0.074328
ASN:HD21	98	54	0.132151	0.097989	0.114321	0.092470
ASN:HD22	98	54	0.137238	0.142173	0.158666	0.149987
ASN:N	98	54	1.291832	1.142716	1.280276	1.307273
ASN:ND2	98	54	1.406563	1.162690	1.321545	1.119365
ASN:O	98	54	3.517626	3.181618	3.431962	2.977601
ASN:OD1	98	54	3.631730	2.644213	3.417263	3.446218
ASP:C	154	59	0.632183	0.546306	0.595394	0.603460
ASP:CA	154	59	0.762654	0.748905	0.730833	0.695231
ASP:CB	154	59	1.089889	0.864510	0.915157	0.859291
ASP:CG	154	59	0.857165	0.670997	0.645425	0.699790
ASP:H	154	59	0.213915	0.183844	0.181667	0.137922
ASP:HA	154	59	0.101245	0.098359	0.081971	0.088051
ASP:HB2	154	59	0.115656	0.095274	0.089009	0.095992
ASP:HB3	154	59	0.093317	0.081850	0.086445	0.083313
ASP:N	154	59	1.302456	1.294964	1.308956	1.146706
ASP:O	154	59	3.974848	3.729521	3.749685	3.494585
ASP:OD1	154	59	4.459688	4.972468	4.106024	4.687476
ASP:OD2	154	59	4.959719	4.282239	3.644817	4.271588
CYS:C	42	29	0.908969	0.880299	0.755220	0.939435
CYS:CA	42	29	0.665924	0.712923	4.132230	0.985078
CYS:CB	42	29	1.230304	1.363718	2.528112	2.388053
CYS:H	42	29	0.202598	0.228698	0.173715	0.203820
CYS:HA	42	29	0.134058	0.110216	0.281922	0.135018
CYS:HB2	42	29	0.124699	0.186840	0.339659	0.211525
CYS:HB3	42	29	0.112196	0.108912	0.110942	0.130506
CYS:HG	40	28	0.212788	0.270162	0.175987	0.274762
CYS:N	42	29	1.873746	1.931522	2.372492	2.369064
CYS:O	42	29	4.387152	4.269942	14.840803	6.721451
GLN:C	71	41	0.580439	0.569709	0.564416	0.525457
GLN:CA	71	41	0.521407	0.501529	0.649771	0.609093
GLN:CB	71	41	0.585334	0.618922	0.538287	0.522302
GLN:CD	71	41	0.611185	0.628816	0.567876	0.522399
GLN:CG	71	41	0.747335	0.785504	0.674923	0.711716
GLN:H	71	41	0.181336	0.180273	0.201124	0.218531
GLN:HA	71	41	0.113593	0.084386	0.074772	0.081031
GLN:HB2	71	41	0.069835	0.084805	0.078236	0.060520
GLN:HB3	71	41	0.084025	0.080604	0.073682	0.085978
GLN:HE21	71	41	0.127155	0.096775	0.086199	0.152694
GLN:HE22	71	41	0.171322	0.130162	0.163951	0.274718
GLN:HG2	71	41	0.079072	0.085517	0.092951	0.079338
GLN:HG3	71	41	0.108628	0.102151	0.142271	0.100580
GLN:N	71	41	1.093733	1.139261	1.100253	1.427977
GLN:NE2	71	41	1.305514	1.268475	2.115474	1.473508
GLN:O	71	41	3.073967	3.077437	2.737667	2.757743
GLN:OE1	71	41	2.891341	3.298458	2.371673	3.605544

IUPAC	<i>n</i>	Proteins	Structure	AIMNet	MOPAC	Joint
GLU:C	207	64	0.492969	0.451713	0.480953	0.437031
GLU:CA	207	64	0.570895	0.459047	0.496158	0.460241
GLU:CB	207	64	0.633321	0.558454	0.626652	0.546150
GLU:CD	207	64	0.721952	0.582696	0.545113	0.561394
GLU:CG	207	64	0.583897	0.579368	0.662204	0.584314
GLU:H	207	64	0.149661	0.150236	0.169258	0.127187
GLU:HA	207	64	0.100999	0.078883	0.070945	0.075107
GLU:HB2	207	64	0.101041	0.087687	0.080928	0.082614
GLU:HB3	207	64	0.081679	0.072501	0.070463	0.070459
GLU:HG2	207	64	0.085268	0.068457	0.075156	0.073026
GLU:HG3	207	64	0.095222	0.079173	0.074953	0.069039
GLU:N	207	64	1.023068	1.026034	1.059458	1.094977
GLU:O	207	64	2.879491	2.420726	2.542949	2.565756
GLU:OE1	207	64	4.423517	3.792153	3.811427	3.452491
GLU:OE2	207	64	3.543551	3.545200	3.322292	3.354536
GLY:C	272	74	0.600044	0.610819	0.611012	0.693249
GLY:CA	272	74	0.590836	0.654424	0.568820	0.556143
GLY:H	272	74	0.162782	0.145935	0.177294	0.145354
GLY:HA2	272	74	0.090276	0.082326	0.081671	0.086095
GLY:HA3	272	74	0.090818	0.085153	0.087399	0.102930
GLY:N	272	74	1.287272	1.403871	1.305410	1.246688
GLY:O	272	74	3.572304	3.772490	4.068320	3.761336
HIS:C	65	43	0.666293	0.657939	0.652538	0.655347
HIS:CA	65	43	0.993690	0.826556	0.829510	0.693619
HIS:CB	65	43	0.574406	0.653827	0.603759	0.588713
HIS:CD2	65	43	0.842973	0.864705	0.743802	0.806234
HIS:CE1	65	43	0.920021	0.701292	0.782293	0.794444
HIS:CG	65	43	1.055803	0.709433	0.744024	0.855270
HIS:H	65	43	0.156765	0.173866	0.193814	0.124043
HIS:HA	65	43	0.108754	0.100029	0.097145	0.094571
HIS:HB2	65	43	0.104680	0.113783	0.099313	0.100740
HIS:HB3	65	43	0.102445	0.097294	0.096380	0.089981
HIS:HD2	65	43	0.134348	0.128156	0.142767	0.124906
HIS:HE1	65	43	0.108815	0.091883	0.069868	0.088141
HIS:HE2	65	43	0.102254	0.106055	0.089462	0.078460
HIS:N	65	43	1.388449	1.447996	1.379753	1.271585
HIS:ND1	65	43	2.220808	2.259597	2.255133	2.229033
HIS:NE2	65	43	1.340020	1.303977	1.217159	1.076961
HIS:O	65	43	4.294551	4.177699	2.847890	3.127553
ILE:C	236	77	0.501180	0.504440	0.446081	0.472515
ILE:CA	236	77	0.590849	0.575223	0.511024	0.545613
ILE:CB	236	77	0.519518	0.555916	0.400803	0.441356
ILE:CD1	236	77	0.472875	0.486962	0.439887	0.500888
ILE:CG1	236	77	0.509908	0.437964	0.457819	0.361936
ILE:CG2	236	77	0.484472	0.484718	0.483523	0.439941
ILE:H	233	77	0.125659	0.126375	0.123957	0.100331
ILE:HA	236	77	0.094701	0.094462	0.086319	0.082100
ILE:HB	236	77	0.078233	0.081937	0.068967	0.072101
ILE:HD11	236	77	0.058482	0.059310	0.061482	0.050734
ILE:HD12	236	77	0.066874	0.061286	0.064364	0.061831
ILE:HD13	236	77	0.077708	0.064192	0.059710	0.065139
ILE:HG12	236	77	0.073778	0.069791	0.075368	0.054083
ILE:HG13	236	77	0.067124	0.068684	0.068432	0.055681
ILE:HG21	236	77	0.068961	0.068767	0.063353	0.061267
ILE:HG22	236	77	0.069211	0.066858	0.062499	0.070967
ILE:HG23	236	77	0.066380	0.071229	0.066227	0.060982
ILE:N	236	77	0.949777	1.207067	1.083779	0.840258
ILE:O	236	77	3.038895	2.617071	2.297785	2.359877
LEU:C	269	77	0.553959	0.571596	0.485934	0.520521
LEU:CA	269	77	0.676074	0.557863	0.487994	0.530656
LEU:CB	269	77	0.562839	0.484042	0.575256	0.530933
LEU:CD1	269	77	0.464169	0.509986	0.502334	0.518654
LEU:CD2	269	77	0.503709	0.499937	0.567811	0.511626
LEU:CG	269	77	0.434203	0.396095	0.421327	0.435168

IUPAC	<i>n</i>	Proteins	Structure	AIMNet	MOPAC	Joint
LEU:H	267	77	0.145949	0.121752	0.136734	0.104017
LEU:HA	269	77	0.083859	0.087839	0.092606	0.084262
LEU:HB2	269	77	0.072707	0.074286	0.067916	0.063427
LEU:HB3	269	77	0.077255	0.077865	0.069091	0.070566
LEU:HD11	269	77	0.056303	0.056673	0.055467	0.050899
LEU:HD12	269	77	0.064729	0.066095	0.056277	0.068063
LEU:HD13	269	77	0.071558	0.060629	0.069727	0.052042
LEU:HD21	269	77	0.060495	0.058501	0.065290	0.054173
LEU:HD22	269	77	0.063848	0.064047	0.065142	0.062531
LEU:HD23	269	77	0.063483	0.067349	0.058903	0.058318
LEU:HG	269	77	0.071918	0.079923	0.073046	0.075764
LEU:N	269	77	1.131486	1.143297	0.924271	1.087374
LEU:O	269	77	3.235913	3.314445	3.033035	3.009077
LYS:C	239	76	0.576576	0.491416	0.422277	0.435666
LYS:CA	239	76	0.534300	0.541668	0.498204	0.479827
LYS:CB	239	76	0.446760	0.509023	0.449979	0.454621
LYS:CD	239	76	0.476111	0.505177	0.473096	0.509546
LYS:CE	239	76	0.528624	0.509143	0.463294	0.470189
LYS:CG	239	76	0.615310	0.565862	0.563154	0.582148
LYS:H	238	75	0.133434	0.122022	0.138403	0.119073
LYS:HA	239	76	0.087991	0.075926	0.067783	0.067590
LYS:HB2	239	76	0.066406	0.064473	0.056367	0.058558
LYS:HB3	239	76	0.079291	0.069247	0.063886	0.054203
LYS:HD2	239	76	0.057293	0.063511	0.057877	0.060487
LYS:HD3	239	76	0.060839	0.058077	0.059233	0.058200
LYS:HE2	239	76	0.080829	0.080587	0.079088	0.072178
LYS:HE3	239	76	0.094233	0.078683	0.068587	0.076833
LYS:HG2	239	76	0.071104	0.075190	0.064045	0.067334
LYS:HG3	239	76	0.074507	0.073152	0.073115	0.069195
LYS:HZ1	239	76	0.100100	0.120757	0.132079	0.143204
LYS:HZ2	239	76	0.091405	0.091922	0.073706	0.082154
LYS:HZ3	239	76	0.132000	0.111913	0.116672	0.094414
LYS:N	239	76	0.989814	1.094744	0.888419	0.837364
LYS:NZ	239	76	0.578737	0.759939	0.589564	0.698380
LYS:O	239	76	2.363931	2.203442	2.134513	2.130441
MET:C	127	75	0.655824	0.636773	0.691226	0.558023
MET:CA	127	75	0.846904	0.833554	0.716286	0.858695
MET:CB	127	75	1.100300	0.990786	0.967276	1.031617
MET:CE	127	75	0.780197	0.784343	0.753807	0.696713
MET:CG	127	75	0.933927	0.996681	0.689146	0.708464
MET:H	61	49	0.139187	0.176774	0.252267	0.342736
MET:HA	127	75	0.136759	0.136527	0.131904	0.149489
MET:HB2	127	75	0.100524	0.095430	0.098383	0.104404
MET:HB3	127	75	0.109627	0.097982	0.117312	0.095169
MET:HE1	127	75	0.081195	0.085372	0.094816	0.073388
MET:HE2	127	75	0.116256	0.097395	0.094425	0.106750
MET:HE3	127	75	0.077476	0.105225	0.087419	0.110412
MET:HG2	127	75	0.112407	0.128202	0.088723	0.094671
MET:HG3	127	75	0.095560	0.092799	0.096728	0.077851
MET:N	127	75	2.160937	3.646103	5.455620	3.136012
MET:O	127	75	3.304815	3.299892	3.056449	3.436783
MET:SD	127	75	3.938236	4.101887	4.507245	4.136263
PHE:C	92	61	0.491910	0.550334	0.588717	0.449906
PHE:CA	92	61	0.883275	0.805195	0.651829	0.646674
PHE:CB	92	61	0.516328	0.628113	0.610838	0.455054
PHE:CD1	92	61	0.685584	0.578547	0.503757	0.523917
PHE:CD2	92	61	0.542688	0.688090	0.527006	0.605177
PHE:CE1	92	61	0.813332	0.623456	0.737787	0.660008
PHE:CE2	92	61	0.528109	0.480760	0.544881	0.387668
PHE:CG	92	61	0.791687	0.540634	0.727229	0.550367
PHE:CZ	92	61	0.470853	0.632204	0.447458	0.360057
PHE:H	91	60	0.156240	0.128305	0.155275	0.137083
PHE:HA	92	61	0.105835	0.103157	0.128066	0.100585
PHE:HB2	92	61	0.108892	0.113972	0.256454	0.240910

IUPAC	<i>n</i>	Proteins	Structure	AIMNet	MOPAC	Joint
PHE:HB3	92	61	0.100551	0.092336	0.098709	0.085360
PHE:HD1	92	61	0.096363	0.075174	0.093254	0.070217
PHE:HD2	92	61	0.083815	0.073675	0.121981	0.071484
PHE:HE1	92	61	0.080627	0.071595	0.062691	0.082509
PHE:HE2	92	61	0.077614	0.080966	0.075533	0.072609
PHE:HZ	92	61	0.072655	0.061971	0.050298	0.065918
PHE:N	92	61	1.139831	1.063023	1.010482	1.151496
PHE:O	92	61	3.646443	3.560007	3.770182	2.769316
PRO:C	97	55	0.664795	0.579058	0.575661	0.626453
PRO:CA	97	55	0.722535	0.672918	0.748947	0.656208
PRO:CB	97	55	0.461798	0.553731	0.482418	0.484698
PRO:CD	97	55	0.860151	0.668734	0.721333	0.767197
PRO:CG	97	55	0.479573	0.461656	0.422765	0.473708
PRO:HA	97	55	0.080168	0.080781	0.079345	0.080942
PRO:HB2	97	55	0.079679	0.071983	0.074620	0.059195
PRO:HB3	97	55	0.077238	0.054082	0.060663	0.064728
PRO:HD2	97	55	0.107371	0.110235	0.102674	0.102643
PRO:HD3	97	55	0.098856	0.085336	0.091201	0.078264
PRO:HG2	97	55	0.093732	0.080040	0.094155	0.077328
PRO:HG3	97	55	0.061779	0.061132	0.069418	0.061573
PRO:N	97	55	1.870254	2.099590	1.945973	1.805709
PRO:O	97	55	3.320435	3.841529	3.484175	3.278814
SER:C	174	69	0.642742	0.577961	0.549358	0.560924
SER:CA	174	69	0.714809	0.729746	0.560547	0.749094
SER:CB	174	69	0.893720	0.924138	0.947337	0.978702
SER:H	174	69	0.169083	0.144194	0.141865	0.131524
SER:HA	174	69	0.108235	0.092530	0.082329	0.100248
SER:HB2	174	69	0.099949	0.086199	0.079559	0.077601
SER:HB3	174	69	0.098620	0.097816	0.083972	0.094947
SER:HG	174	69	0.155300	0.155927	0.154130	0.138695
SER:N	174	69	1.366361	1.274623	1.367466	1.386845
SER:O	174	69	3.385994	3.423358	3.000809	3.516533
SER:OG	174	69	2.261774	2.456157	2.163125	2.327758
THR:C	193	74	0.554406	0.537052	0.502592	0.501644
THR:CA	193	74	0.605775	0.557501	0.559987	0.614419
THR:CB	193	74	0.641253	0.693625	0.610109	0.635906
THR:CG2	193	74	0.525138	0.527224	0.440361	0.430383
THR:H	193	74	0.159534	0.127569	0.164652	0.120462
THR:HA	193	74	0.085771	0.083352	0.067897	0.074131
THR:HB	193	74	0.084702	0.073091	0.086767	0.077800
THR:HG1	193	74	0.163309	0.175714	0.112933	0.146550
THR:HG21	193	74	0.073622	0.067587	0.071674	0.068553
THR:HG22	193	74	0.068878	0.069779	0.069175	0.050896
THR:HG23	193	74	0.058917	0.067553	0.067849	0.073701
THR:N	193	74	1.234498	1.162832	1.266391	1.057981
THR:O	193	74	3.208283	3.011399	2.781298	2.965481
THR:OG1	193	74	2.300783	2.562473	2.231159	2.461979
TRP:C	16	15	0.572712	0.423786	0.770071	0.503108
TRP:CA	16	15	0.884863	0.901233	0.916199	0.803371
TRP:CB	16	15	0.616051	0.870678	0.712423	0.759093
TRP:CD1	16	15	0.538182	1.114155	0.615330	0.684923
TRP:CD2	16	15	0.661688	0.599170	0.523642	0.611128
TRP:CE2	16	15	0.417747	0.591903	0.568607	0.535896
TRP:CE3	16	15	0.662140	0.606486	0.743535	0.538478
TRP:CG	16	15	0.854302	0.591357	0.832651	0.832552
TRP:CH2	16	15	0.508136	0.522954	0.618728	0.547094
TRP:CZ2	16	15	0.451645	0.561396	0.338780	0.326247
TRP:CZ3	16	15	0.444970	0.512833	0.555641	0.677056
TRP:H	16	15	0.175583	0.205982	0.170489	0.165535
TRP:HA	16	15	0.146700	0.072650	0.292227	0.104123
TRP:HB2	16	15	0.100468	0.071993	0.082027	0.098182
TRP:HB3	16	15	0.141462	0.073607	0.093178	0.067776
TRP:HD1	16	15	0.183125	0.149061	0.122721	0.069352
TRP:HE1	16	15	0.252165	0.101698	0.113797	0.122920

IUPAC	n	Proteins	Structure	AIMNet	MOPAC	Joint
TRP:HE3	16	15	0.166740	0.071841	0.067108	0.072828
TRP:HH2	16	15	0.111411	0.091605	0.104467	0.088461
TRP:HZ2	16	15	0.064044	0.079453	0.060242	0.084070
TRP:HZ3	16	15	0.114200	0.057876	0.075368	0.077143
TRP:N	16	15	1.608322	1.601540	1.601528	1.693169
TRP:NE1	16	15	2.070491	1.512228	1.192651	1.112672
TRP:O	16	15	3.205944	3.852098	2.867822	2.694645
TYR:C	74	51	0.543747	0.429222	0.488760	0.452278
TYR:CA	74	51	0.827123	0.594971	0.693665	0.705176
TYR:CB	74	51	0.537811	0.536868	0.434560	0.527296
TYR:CD1	74	51	0.542165	0.508895	0.538622	0.510044
TYR:CD2	74	51	0.575290	0.582336	0.482958	0.436713
TYR:CE1	74	51	0.654512	0.658382	0.534999	0.580108
TYR:CE2	74	51	0.515849	0.612930	0.660110	0.573703
TYR:CG	74	51	0.497357	0.442280	0.599290	0.509469
TYR:CZ	74	51	0.506968	0.700211	0.456057	0.528211
TYR:H	74	51	0.161176	0.122698	0.172765	0.117247
TYR:HA	74	51	0.092186	0.077689	0.075718	0.081667
TYR:HB2	74	51	0.079376	0.066034	0.077851	0.070961
TYR:HB3	74	51	0.058289	0.071181	0.077684	0.067421
TYR:HD1	74	51	0.088491	0.071729	0.079394	0.074057
TYR:HD2	74	51	0.084798	0.067447	0.061531	0.073138
TYR:HE1	74	51	0.086508	0.076929	0.096180	0.089975
TYR:HE2	74	51	0.079733	0.069608	0.069544	0.067345
TYR:HH	74	51	0.104581	0.153825	0.138235	0.155610
TYR:N	74	51	1.021928	1.092462	1.089962	0.897835
TYR:O	74	51	3.089499	2.881337	2.737345	2.955940
TYR:OH	74	51	1.872335	1.971622	1.791376	1.742760
VAL:C	282	74	0.483914	0.528748	0.463977	0.444599
VAL:CA	282	74	0.631998	0.568839	0.490177	0.480461
VAL:CB	282	74	0.564396	0.481399	0.496010	0.579864
VAL:CG1	282	74	0.548012	0.544954	0.626578	0.521050
VAL:CG2	282	74	0.524866	0.468845	0.478881	0.490050
VAL:H	281	74	0.136274	0.115276	0.124547	0.107342
VAL:HA	282	74	0.085645	0.074241	0.079516	0.073611
VAL:HB	282	74	0.072013	0.068948	0.071259	0.064408
VAL:HG11	282	74	0.083461	0.073403	0.073917	0.069103
VAL:HG12	282	74	0.067857	0.068963	0.063088	0.073178
VAL:HG13	282	74	0.073811	0.064103	0.065547	0.064953
VAL:HG21	282	74	0.071568	0.074792	0.066314	0.065275
VAL:HG22	282	74	0.080582	0.073670	0.071274	0.078900
VAL:HG23	282	74	0.079049	0.076556	0.068221	0.075380
VAL:N	282	74	1.085218	1.162319	1.045073	1.030436
VAL:O	282	74	2.720302	3.018315	2.266492	2.342513

5.6 All 79: T2 vector RMSE (ppm)

IUPAC	n	Proteins	Structure	AIMNet	MOPAC	Joint
ALA:C	268	73	3.650347	3.349050	3.306555	3.045595
ALA:CA	268	73	2.171636	2.065703	2.095745	1.961575
ALA:CB	268	73	1.962006	1.955406	1.957010	1.861909
ALA:H	267	73	0.621629	0.621498	0.591808	0.578056
ALA:HA	268	73	0.522509	0.515320	0.470966	0.473022
ALA:HB1	268	73	0.539453	0.544888	0.493537	0.475249
ALA:HB2	268	73	0.529955	0.543450	0.544882	0.492338
ALA:HB3	268	73	0.518275	0.508107	0.466142	0.471303
ALA:N	268	73	4.602328	4.483193	4.191361	4.121758
ALA:O	268	73	15.386604	15.273168	13.083646	13.411750
ARG:C	338	77	3.789497	3.584531	3.401432	3.077547
ARG:CA	338	77	2.227356	2.141900	2.136549	1.842481
ARG:CB	338	77	2.452129	2.354595	2.269412	2.159888
ARG:CD	338	77	2.686189	2.584222	2.631984	2.343334

IUPAC	<i>n</i>	Proteins	Structure	AIMNet	MOPAC	Joint
ARG:CG	338	77	2.684639	2.361088	2.381689	2.093212
ARG:CZ	338	77	3.775750	4.192475	3.613322	3.120624
ARG:H	336	76	0.671707	0.674332	0.567654	0.561109
ARG:HA	338	77	0.564825	0.538456	0.499913	0.468046
ARG:HB2	338	77	0.570054	0.558730	0.524495	0.488085
ARG:HB3	338	77	0.576516	0.551944	0.517033	0.496227
ARG:HD2	338	77	0.567316	0.541890	0.520347	0.518221
ARG:HD3	338	77	0.577841	0.537266	0.544426	0.511498
ARG:HE	338	77	0.737538	0.796445	0.737744	0.699736
ARG:HG2	338	77	0.574249	0.575470	0.520946	0.515832
ARG:HG3	338	77	0.584098	0.624027	0.516004	0.510128
ARG:HH11	338	77	0.743030	0.737842	0.665057	0.690793
ARG:HH12	338	77	0.665399	0.605133	0.612341	0.597528
ARG:HH21	338	77	0.753300	0.771145	0.660471	0.719828
ARG:HH22	338	77	0.667191	0.640573	0.584773	0.581517
ARG:N	338	77	4.996606	4.343007	3.791215	4.251245
ARG:NE	338	77	4.662403	4.656847	4.442786	4.485831
ARG:NH1	338	77	3.893958	4.107937	3.665586	3.362655
ARG:NH2	338	77	3.782388	3.861066	3.471468	3.362100
ARG:O	338	77	14.967757	17.664529	14.170833	12.781650
ASN:C	98	54	3.970976	3.672116	3.581865	3.471195
ASN:CA	98	54	2.721246	2.624076	2.525664	2.436809
ASN:CB	98	54	3.201118	3.063884	2.770235	2.844224
ASN:CG	98	54	4.676049	4.269839	4.276159	3.812415
ASN:H	97	53	0.755201	0.716342	0.671833	0.616824
ASN:HA	98	54	0.558318	0.524013	0.509905	0.492796
ASN:HB2	98	54	0.612349	0.558460	0.567967	0.554181
ASN:HB3	98	54	0.614209	0.568838	0.561228	0.513145
ASN:HD21	98	54	0.725789	0.683124	0.583767	0.593825
ASN:HD22	98	54	0.672373	0.705859	0.591441	0.579107
ASN:N	98	54	5.407923	4.706786	5.098087	4.835499
ASN:ND2	98	54	5.357447	5.591038	5.193154	4.863132
ASN:O	98	54	17.703008	17.092930	15.378122	14.751424
ASN:OD1	98	54	18.462936	20.071182	18.628784	15.976362
ASP:C	154	59	4.441939	4.587132	4.198911	4.012358
ASP:CA	154	59	3.000725	2.926781	2.930082	2.670966
ASP:CB	154	59	4.198388	3.939174	3.542288	3.354083
ASP:CG	154	59	5.185532	4.704886	4.462859	3.998767
ASP:H	154	59	0.803239	0.835076	0.734649	0.694809
ASP:HA	154	59	0.628990	0.591905	0.569411	0.525187
ASP:HB2	154	59	0.632337	0.583497	0.564937	0.541464
ASP:HB3	154	59	0.668653	0.622724	0.589992	0.547238
ASP:N	154	59	5.628553	5.688260	5.449552	4.986114
ASP:O	154	59	18.764483	18.555124	16.858399	17.661168
ASP:OD1	154	59	20.630928	20.024375	18.986209	18.568592
ASP:OD2	154	59	19.422881	18.974108	18.000260	17.822745
CYS:C	42	29	5.148576	4.373037	8.662958	9.926974
CYS:CA	42	29	3.163053	3.205197	10.690503	4.650585
CYS:CB	42	29	5.598778	6.597195	10.859516	8.055845
CYS:H	42	29	0.928917	0.805268	1.292869	1.130019
CYS:HA	42	29	0.661861	0.686504	1.010525	0.905165
CYS:HB2	42	29	1.015769	0.891814	1.660589	1.179447
CYS:HB3	42	29	0.921106	0.850920	1.653352	1.423756
CYS:HG	40	28	1.103411	1.433601	1.047380	1.225124
CYS:N	42	29	6.177742	5.928602	10.463033	10.013971
CYS:O	42	29	20.780280	22.112250	49.512747	38.645187
GLN:C	71	41	4.038228	3.779743	4.251559	3.439054
GLN:CA	71	41	2.599758	2.427551	2.958579	2.309970
GLN:CB	71	41	2.634855	2.313263	2.426408	2.274106
GLN:CD	71	41	4.785121	4.179202	6.164783	5.274762
GLN:CG	71	41	3.754000	3.679931	3.758110	3.510199
GLN:H	71	41	0.703024	0.665337	0.875912	0.688881
GLN:HA	71	41	0.581202	0.569724	0.565237	0.483018
GLN:HB2	71	41	0.618164	0.619070	0.567514	0.604410

IUPAC	<i>n</i>	Proteins	Structure	AIMNet	MOPAC	Joint
GLN:HB3	71	41	0.599668	0.563091	0.604342	0.560898
GLN:HE21	71	41	0.728684	0.711529	0.745090	1.129486
GLN:HE22	71	41	0.674702	0.763388	0.758010	0.908143
GLN:HG2	71	41	0.601297	0.573724	0.582663	0.538667
GLN:HG3	71	41	0.664342	0.592007	0.797302	0.716423
GLN:N	71	41	4.772476	4.959686	4.919981	4.543988
GLN:NE2	71	41	5.514171	5.276578	10.662297	10.153733
GLN:O	71	41	15.294620	17.952240	18.156081	14.190116
GLN:OE1	71	41	20.060926	20.528251	19.024865	18.746989
GLU:C	207	64	3.586003	3.524587	3.492197	3.108277
GLU:CA	207	64	2.324641	2.061466	2.230786	1.957019
GLU:CB	207	64	2.787713	2.612338	2.469260	2.344995
GLU:CD	207	64	4.464542	4.556704	3.847205	3.563941
GLU:CG	207	64	3.431240	3.076470	2.958824	2.833443
GLU:H	207	64	0.696525	0.683064	0.603430	0.645511
GLU:HA	207	64	0.538571	0.528760	0.485220	0.451899
GLU:HB2	207	64	0.607623	0.571968	0.509604	0.499814
GLU:HB3	207	64	0.598614	0.566675	0.516313	0.511784
GLU:HG2	207	64	0.625731	0.586812	0.523104	0.507086
GLU:HG3	207	64	0.597746	0.586039	0.512865	0.479982
GLU:N	207	64	4.586771	4.223133	4.208666	4.041459
GLU:O	207	64	15.423933	15.028254	14.178889	13.292283
GLU:OE1	207	64	17.592966	16.838295	15.487241	15.261810
GLU:OE2	207	64	18.663378	17.804580	16.166316	15.425822
GLY:C	272	74	4.319291	4.566502	4.455832	3.818355
GLY:CA	272	74	3.279934	2.770366	2.741884	2.656702
GLY:H	272	74	0.721070	0.694069	0.793420	0.716793
GLY:HA2	272	74	0.613106	0.584519	0.595527	0.539535
GLY:HA3	272	74	0.631288	0.625183	0.625900	0.596571
GLY:N	272	74	5.335678	5.157605	8.020109	5.475700
GLY:O	272	74	18.064800	18.206774	17.857620	16.914526
HIS:C	65	43	4.938067	4.669524	4.173907	3.994150
HIS:CA	65	43	3.327752	3.111074	3.055072	2.998546
HIS:CB	65	43	2.934901	2.879009	2.670457	2.848154
HIS:CD2	65	43	6.497296	5.537069	5.474759	6.020851
HIS:CE1	65	43	5.140152	5.275742	5.121579	4.998366
HIS:CG	65	43	5.179971	4.797252	5.422404	5.112100
HIS:H	65	43	0.776974	0.771965	0.713095	0.695256
HIS:HA	65	43	0.658565	0.562414	0.562704	0.546903
HIS:HB2	65	43	0.660401	0.572383	0.569083	0.580311
HIS:HB3	65	43	0.677133	0.579744	0.552352	0.627056
HIS:HD2	65	43	0.656269	0.594114	0.623743	0.553532
HIS:HE1	65	43	0.609456	0.591939	0.612140	0.562972
HIS:HE2	65	43	0.606797	0.609028	0.589719	0.643728
HIS:N	65	43	5.416404	5.487025	5.100253	4.761366
HIS:ND1	65	43	14.778475	15.086764	11.860545	12.219127
HIS:NE2	65	43	4.885740	5.296012	4.778235	4.612969
HIS:O	65	43	19.196399	17.768550	17.557373	17.814018
ILE:C	236	77	3.464895	3.401574	3.412718	3.038046
ILE:CA	236	77	2.137628	2.108129	2.020259	1.908534
ILE:CB	236	77	2.087590	2.008536	1.870197	1.817828
ILE:CD1	236	77	1.726218	1.594529	1.502741	1.563598
ILE:CG1	236	77	1.936952	1.866701	1.755795	1.706729
ILE:CG2	236	77	1.784754	1.796298	1.617543	1.621425
ILE:H	233	77	0.658409	0.618311	0.568811	0.582676
ILE:HA	236	77	0.563025	0.517478	0.489783	0.491686
ILE:HB	236	77	0.554572	0.518257	0.484565	0.471031
ILE:HD11	236	77	0.533503	0.501537	0.469714	0.467050
ILE:HD12	236	77	0.531791	0.497197	0.486171	0.468819
ILE:HD13	236	77	0.558989	0.500291	0.477154	0.450381
ILE:HG12	236	77	0.563044	0.509890	0.476222	0.472629
ILE:HG13	236	77	0.559698	0.543122	0.486566	0.493869
ILE:HG21	236	77	0.576355	0.498599	0.487519	0.488153
ILE:HG22	236	77	0.537771	0.507210	0.492608	0.459609

IUPAC	<i>n</i>	Proteins	Structure	AIMNet	MOPAC	Joint
ILE:HG23	236	77	0.592282	0.542239	0.518849	0.538915
ILE:N	236	77	4.279582	4.344616	3.934923	4.082167
ILE:O	236	77	16.392505	15.445526	13.912051	12.697501
LEU:C	269	77	3.869738	3.724836	3.490685	3.186352
LEU:CA	269	77	2.438463	2.190725	2.124283	1.915991
LEU:CB	269	77	2.401799	2.350135	2.295350	1.982382
LEU:CD1	269	77	1.999142	1.873990	1.862507	1.850368
LEU:CD2	269	77	2.006407	1.801348	1.872250	1.842595
LEU:CG	269	77	1.659905	1.577126	1.435406	1.614535
LEU:H	267	77	0.687750	0.641798	0.607757	0.578034
LEU:HA	269	77	0.558619	0.581708	0.524723	0.504606
LEU:HB2	269	77	0.573351	0.535287	0.520936	0.498975
LEU:HB3	269	77	0.584049	0.538026	0.516929	0.508388
LEU:HD11	269	77	0.538509	0.523289	0.491674	0.473832
LEU:HD12	269	77	0.579392	0.557839	0.513925	0.490704
LEU:HD13	269	77	0.571819	0.539648	0.561137	0.502375
LEU:HD21	269	77	0.581996	0.528751	0.490733	0.510136
LEU:HD22	269	77	0.572739	0.524965	0.483732	0.476123
LEU:HD23	269	77	0.564547	0.537131	0.496911	0.475146
LEU:HG	269	77	0.563844	0.537362	0.527521	0.504133
LEU:N	269	77	4.412535	3.970308	4.227657	4.193218
LEU:O	269	77	16.076311	16.194910	14.864071	14.504345
LYS:C	239	76	3.653215	3.357176	3.309374	2.939143
LYS:CA	239	76	2.238705	2.170907	2.135740	2.124060
LYS:CB	239	76	2.164845	2.202174	1.961056	1.879985
LYS:CD	239	76	2.298121	2.062717	2.026087	1.765513
LYS:CE	239	76	2.924901	2.720421	2.397750	2.361484
LYS:CG	239	76	2.494005	2.262635	2.089799	1.996263
LYS:H	238	75	0.645667	0.618092	0.586670	0.568973
LYS:HA	239	76	0.551713	0.543813	0.477726	0.474813
LYS:HB2	239	76	0.557852	0.535047	0.491011	0.473798
LYS:HB3	239	76	0.540752	0.541597	0.487080	0.511340
LYS:HD2	239	76	0.534138	0.524785	0.474023	0.455971
LYS:HD3	239	76	0.552899	0.508974	0.448045	0.439947
LYS:HE2	239	76	0.590951	0.558338	0.493032	0.514632
LYS:HE3	239	76	0.580479	0.574250	0.503395	0.488757
LYS:HG2	239	76	0.529245	0.521230	0.453091	0.475754
LYS:HG3	239	76	0.545962	0.535222	0.482604	0.483368
LYS:HZ1	239	76	0.634472	0.698501	0.564805	0.584914
LYS:HZ2	239	76	0.668888	0.703366	0.590544	0.623155
LYS:HZ3	239	76	0.645069	0.714642	0.608631	0.556494
LYS:N	239	76	4.709621	4.431823	4.225627	4.462999
LYS:NZ	239	76	2.201219	2.445220	2.265433	2.160313
LYS:O	239	76	16.626177	14.308810	13.781635	13.663004
MET:C	127	75	4.396467	4.088847	5.191392	4.493339
MET:CA	127	75	3.038166	3.077014	3.236600	2.872447
MET:CB	127	75	4.626715	4.141981	4.165090	4.046755
MET:CE	127	75	3.265044	3.173894	2.886932	2.906191
MET:CG	127	75	3.857199	3.617523	3.449513	3.322856
MET:H	61	49	0.693867	0.679713	1.059432	0.885193
MET:HA	127	75	0.735016	0.714124	0.713990	0.701913
MET:HB2	127	75	0.723820	0.692537	0.650292	0.631709
MET:HB3	127	75	0.764474	0.715777	0.738518	0.691845
MET:HE1	127	75	0.731043	0.730686	0.732958	0.696657
MET:HE2	127	75	0.725474	0.752268	0.683913	0.705038
MET:HE3	127	75	0.802399	0.843773	0.798181	0.731036
MET:HG2	127	75	0.815937	0.877700	0.780303	0.682344
MET:HG3	127	75	0.837362	0.807008	0.798040	0.703146
MET:N	127	75	4.874610	4.858100	5.082612	5.040165
MET:O	127	75	19.333578	19.455561	22.752806	15.564502
MET:SD	127	75	19.505656	18.330731	17.722562	17.877523
PHE:C	92	61	3.776471	3.903226	3.826732	3.263402
PHE:CA	92	61	3.063471	3.017808	3.795559	2.802872
PHE:CB	92	61	2.807265	2.830025	3.523525	3.519343

IUPAC	<i>n</i>	Proteins	Structure	AIMNet	MOPAC	Joint
PHE:CD1	92	61	6.024707	5.667128	5.795196	5.767173
PHE:CD2	92	61	5.126842	5.723510	6.664760	5.539448
PHE:CE1	92	61	5.267331	5.233950	5.850997	4.468000
PHE:CE2	92	61	3.839662	6.369287	5.502153	3.625944
PHE:CG	92	61	6.281397	5.203425	6.518645	5.443394
PHE:CZ	92	61	5.660342	7.039348	5.756658	5.494855
PHE:H	91	60	0.689779	0.665112	0.818130	0.678825
PHE:HA	92	61	0.570161	0.560699	0.621102	0.608851
PHE:HB2	92	61	0.573701	0.604343	1.075063	1.222767
PHE:HB3	92	61	0.640005	0.643045	0.681862	0.727272
PHE:HD1	92	61	0.629305	0.632275	0.654899	0.598124
PHE:HD2	92	61	0.608180	0.625487	0.686624	0.601313
PHE:HE1	92	61	0.570192	0.558834	0.499054	0.515557
PHE:HE2	92	61	0.654103	0.568110	0.543066	0.577402
PHE:HZ	92	61	0.608732	0.676959	0.620167	0.552896
PHE:N	92	61	5.395552	5.045076	5.691826	4.583765
PHE:O	92	61	17.994289	16.859218	17.968072	14.553053
PRO:C	97	55	4.394588	3.902883	3.855633	3.903353
PRO:CA	97	55	3.179319	2.963209	3.166099	2.848494
PRO:CB	97	55	2.733151	2.576542	2.597872	2.374325
PRO:CD	97	55	3.533310	3.200338	3.247796	3.017653
PRO:CG	97	55	2.560419	2.616928	2.502788	2.327455
PRO:HA	97	55	0.618989	0.609107	0.556235	0.571001
PRO:HB2	97	55	0.601635	0.534504	0.528734	0.564246
PRO:HB3	97	55	0.569045	0.564837	0.509066	0.513591
PRO:HD2	97	55	0.709782	0.677109	0.652735	0.649005
PRO:HD3	97	55	0.708307	0.651272	0.605821	0.586819
PRO:HG2	97	55	0.550562	0.569149	0.518841	0.514441
PRO:HG3	97	55	0.545522	0.568632	0.519579	0.515363
PRO:N	97	55	7.225144	6.768828	6.540295	6.332476
PRO:O	97	55	18.801809	20.574245	17.904535	17.885726
SER:C	174	69	4.817955	4.163315	4.426876	3.975223
SER:CA	174	69	2.740959	2.686304	2.697038	2.559867
SER:CB	174	69	3.706242	3.427635	3.664745	3.328737
SER:H	174	69	0.704245	0.680985	0.739460	0.659560
SER:HA	174	69	0.582663	0.551341	0.590482	0.588995
SER:HB2	174	69	0.626745	0.586379	0.584647	0.588013
SER:HB3	174	69	0.626175	0.594186	0.578343	0.634653
SER:HG	174	69	0.960881	0.982166	0.932711	0.904538
SER:N	174	69	5.395297	5.459479	5.502411	5.401941
SER:O	174	69	20.509068	20.525739	16.680389	16.490372
SER:OG	174	69	8.923197	8.521977	8.794317	8.289715
THR:C	193	74	4.039843	3.804996	3.795581	3.499391
THR:CA	193	74	2.455440	2.327704	2.431826	2.103174
THR:CB	193	74	4.070851	2.860395	3.088087	2.636153
THR:CG2	193	74	2.103950	1.966492	2.080139	1.811751
THR:H	193	74	0.747795	0.678239	0.682128	0.667814
THR:HA	193	74	0.560014	0.562366	0.514182	0.462652
THR:HB	193	74	0.602384	0.569184	0.594149	0.621281
THR:HG1	193	74	0.826872	0.946491	0.880800	0.764329
THR:HG21	193	74	0.560517	0.540944	0.522842	0.507540
THR:HG22	193	74	0.545205	0.509201	0.526890	0.501504
THR:HG23	193	74	0.539276	0.497977	0.497319	0.545794
THR:N	193	74	4.990161	4.898888	4.951203	4.943677
THR:O	193	74	17.421748	16.273109	15.480677	14.896662
THR:OG1	193	74	7.206265	7.421575	7.355126	6.935008
TRP:C	16	15	4.231189	3.985535	5.300284	4.188405
TRP:CA	16	15	3.687011	3.822742	4.068561	2.749152
TRP:CB	16	15	3.343869	3.161042	2.927427	2.881329
TRP:CD1	16	15	6.711711	6.829192	7.824488	7.368094
TRP:CD2	16	15	5.602446	5.860022	5.228961	7.058388
TRP:CE2	16	15	7.216479	5.173709	7.258277	7.523701
TRP:CE3	16	15	6.490095	6.157650	6.729257	6.509922
TRP:CG	16	15	7.673429	5.215577	7.076710	6.094584

IUPAC	<i>n</i>	Proteins	Structure	AIMNet	MOPAC	Joint
TRP:CH2	16	15	7.027398	9.849105	6.274603	5.664038
TRP:CZ2	16	15	4.667940	6.150313	9.874273	5.527745
TRP:CZ3	16	15	5.375661	5.478997	7.570066	4.878026
TRP:H	16	15	0.841212	0.630728	0.847772	0.792783
TRP:HA	16	15	0.685228	0.718761	0.867412	0.738503
TRP:HB2	16	15	0.772032	0.706052	0.644890	0.687056
TRP:HB3	16	15	0.825400	0.738094	0.727579	0.725616
TRP:HD1	16	15	0.908463	0.877240	0.840620	0.841725
TRP:HE1	16	15	0.904385	0.870981	0.774955	0.721831
TRP:HE3	16	15	0.665563	0.716746	0.686422	0.655053
TRP:HH2	16	15	0.763509	0.640323	0.575387	0.635604
TRP:HZ2	16	15	0.876809	1.136539	0.608920	0.573643
TRP:HZ3	16	15	0.662629	0.576623	0.579454	0.553726
TRP:N	16	15	6.338588	5.222087	5.977467	5.941562
TRP:NE1	16	15	5.893556	7.403313	5.363270	5.976143
TRP:O	16	15	18.558371	19.369630	19.064077	14.884975
TYR:C	74	51	3.545546	3.847167	3.270688	3.411162
TYR:CA	74	51	2.975826	2.649389	2.528549	2.260839
TYR:CB	74	51	2.323812	2.169934	2.166926	2.060890
TYR:CD1	74	51	4.988498	5.180701	5.336322	4.717334
TYR:CD2	74	51	4.572045	5.409688	5.076766	4.672771
TYR:CE1	74	51	3.702190	4.222515	4.637218	3.607421
TYR:CE2	74	51	3.658775	4.443428	3.796513	3.865703
TYR:CG	74	51	4.238700	5.152358	4.229047	3.866767
TYR:CZ	74	51	4.886946	5.073185	4.974628	4.801327
TYR:H	74	51	0.793093	0.745183	0.739925	0.699749
TYR:HA	74	51	0.635703	0.600807	0.559544	0.600062
TYR:HB2	74	51	0.661975	0.612071	0.604967	0.608830
TYR:HB3	74	51	0.652240	0.583382	0.570656	0.541053
TYR:HD1	74	51	0.580484	0.509005	0.514193	0.516160
TYR:HD2	74	51	0.641064	0.665724	0.630143	0.634437
TYR:HE1	74	51	0.568265	0.526039	0.556976	0.533302
TYR:HE2	74	51	0.548974	0.552205	0.545271	0.569434
TYR:HH	74	51	0.779878	0.830044	0.799887	0.737636
TYR:N	74	51	4.793309	4.695981	4.739249	5.040066
TYR:O	74	51	16.657054	16.223773	16.260327	14.703147
TYR:OH	74	51	7.715279	8.241572	7.409444	7.391862
VAL:C	282	74	3.694360	3.630477	3.078922	2.968224
VAL:CA	282	74	2.286347	2.109748	2.069450	1.929272
VAL:CB	282	74	2.075643	1.973437	1.889298	1.967476
VAL:CG1	282	74	1.989035	1.978432	1.961684	1.827842
VAL:CG2	282	74	2.081909	2.062866	2.056679	2.021334
VAL:H	281	74	0.675578	0.631853	0.638794	0.597478
VAL:HA	282	74	0.544864	0.547429	0.459684	0.448033
VAL:HB	282	74	0.587975	0.545820	0.519114	0.489643
VAL:HG11	282	74	0.532132	0.509669	0.470227	0.455470
VAL:HG12	282	74	0.542911	0.503374	0.465130	0.473046
VAL:HG13	282	74	0.560828	0.541243	0.517672	0.503500
VAL:HG21	282	74	0.575740	0.529353	0.527915	0.494704
VAL:HG22	282	74	0.584974	0.517414	0.505127	0.479880
VAL:HG23	282	74	0.560515	0.532779	0.487283	0.475588
VAL:N	282	74	4.634884	4.551526	4.265148	4.439489
VAL:O	282	74	16.666881	16.344748	14.038350	13.330907

6 Shielding trajectory IUPAC results

6.1 TC5b Trp-cage (1L2Y): IUPAC support

IUPAC	Element	Atoms	Status	<i>n</i>
ARG:C	6	1	scored	1501
ARG:CA	6	1	scored	1501

IUPAC	Element	Atoms	Status	<i>n</i>
ARG:CB	6	1	scored	1501
ARG:CD	6	1	scored	1501
ARG:CG	6	1	scored	1501
ARG:CZ	6	1	scored	1501
ARG:H	1	1	scored	1501
ARG:HA	1	1	scored	1501
ARG:HB2	1	1	scored	1501
ARG:HB3	1	1	scored	1501
ARG:HD2	1	1	scored	1501
ARG:HD3	1	1	scored	1501
ARG:HE	1	1	scored	1501
ARG:HG2	1	1	scored	1501
ARG:HG3	1	1	scored	1501
ARG:HH11	1	1	scored	1501
ARG:HH12	1	1	scored	1501
ARG:HH21	1	1	scored	1501
ARG:HH22	1	1	scored	1501
ARG:N	7	1	scored	1501
ARG:NE	7	1	scored	1501
ARG:NH1	7	1	scored	1501
ARG:NH2	7	1	scored	1501
ARG:O	8	1	scored	1501
ASN:C	6	1	scored	1501
ASN:CA	6	1	scored	1501
ASN:CB	6	1	scored	1501
ASN:CG	6	1	scored	1501
ASN:H1	1	1	preparation excluded	0
ASN:H2	1	1	preparation excluded	0
ASN:H3	1	1	preparation excluded	0
ASN:HA	1	1	scored	1501
ASN:HB2	1	1	scored	1501
ASN:HB3	1	1	scored	1501
ASN:HD21	1	1	scored	1501
ASN:HD22	1	1	scored	1501
ASN:N	7	1	scored	1501
ASN:ND2	7	1	scored	1501
ASN:O	8	1	scored	1501
ASN:OD1	8	1	scored	1501
ASP:C	6	1	scored	1501
ASP:CA	6	1	scored	1501
ASP:CB	6	1	scored	1501
ASP:CG	6	1	scored	1501
ASP:H	1	1	scored	1501
ASP:HA	1	1	scored	1501
ASP:HB2	1	1	scored	1501
ASP:HB3	1	1	scored	1501
ASP:N	7	1	scored	1501
ASP:O	8	1	scored	1501
ASP:OD1	8	1	scored	1501
ASP:OD2	8	1	scored	1501
GLN:C	6	1	scored	1501
GLN:CA	6	1	scored	1501
GLN:CB	6	1	scored	1501
GLN:CD	6	1	scored	1501
GLN:CG	6	1	scored	1501
GLN:H	1	1	scored	1501
GLN:HA	1	1	scored	1501
GLN:HB2	1	1	scored	1501
GLN:HB3	1	1	scored	1501
GLN:HE21	1	1	scored	1501
GLN:HE22	1	1	scored	1501
GLN:HG2	1	1	scored	1501
GLN:HG3	1	1	scored	1501
GLN:N	7	1	scored	1501

IUPAC	Element	Atoms	Status	<i>n</i>
GLN:NE2	7	1	scored	1501
GLN:O	8	1	scored	1501
GLN:OE1	8	1	scored	1501
GLY:C	6	3	scored	4503
GLY:CA	6	3	scored	4503
GLY:H	1	3	scored	4503
GLY:HA2	1	3	scored	4503
GLY:HA3	1	3	scored	4503
GLY:N	7	3	scored	4503
GLY:O	8	3	scored	4503
ILE:C	6	1	scored	1501
ILE:CA	6	1	scored	1501
ILE:CB	6	1	scored	1501
ILE:CD1	6	1	scored	1501
ILE:CG1	6	1	scored	1501
ILE:CG2	6	1	scored	1501
ILE:H	1	1	scored	1501
ILE:HA	1	1	scored	1501
ILE:HB	1	1	scored	1501
ILE:HD11	1	1	scored	1501
ILE:HD12	1	1	scored	1501
ILE:HD13	1	1	scored	1501
ILE:HG12	1	1	scored	1501
ILE:HG13	1	1	scored	1501
ILE:HG21	1	1	scored	1501
ILE:HG22	1	1	scored	1501
ILE:HG23	1	1	scored	1501
ILE:N	7	1	scored	1501
ILE:O	8	1	scored	1501
LEU:C	6	2	scored	3002
LEU:CA	6	2	scored	3002
LEU:CB	6	2	scored	3002
LEU:CD1	6	2	scored	3002
LEU:CD2	6	2	scored	3002
LEU:CG	6	2	scored	3002
LEU:H	1	2	scored	3002
LEU:HA	1	2	scored	3002
LEU:HB2	1	2	scored	3002
LEU:HB3	1	2	scored	3002
LEU:HD11	1	2	scored	3002
LEU:HD12	1	2	scored	3002
LEU:HD13	1	2	scored	3002
LEU:HD21	1	2	scored	3002
LEU:HD22	1	2	scored	3002
LEU:HD23	1	2	scored	3002
LEU:HG	1	2	scored	3002
LEU:N	7	2	scored	3002
LEU:O	8	2	scored	3002
LYS:C	6	1	scored	1501
LYS:CA	6	1	scored	1501
LYS:CB	6	1	scored	1501
LYS:CD	6	1	scored	1501
LYS:CE	6	1	scored	1501
LYS:CG	6	1	scored	1501
LYS:H	1	1	scored	1501
LYS:HA	1	1	scored	1501
LYS:HB2	1	1	scored	1501
LYS:HB3	1	1	scored	1501
LYS:HD2	1	1	scored	1501
LYS:HD3	1	1	scored	1501
LYS:HE2	1	1	scored	1501
LYS:HE3	1	1	scored	1501
LYS:HG2	1	1	scored	1501
LYS:HG3	1	1	scored	1501

IUPAC	Element	Atoms	Status	<i>n</i>
LYS:HZ1	1	1	scored	1501
LYS:HZ2	1	1	scored	1501
LYS:HZ3	1	1	scored	1501
LYS:N	7	1	scored	1501
LYS:NZ	7	1	scored	1501
LYS:O	8	1	scored	1501
PRO:C	6	4	scored	6004
PRO:CA	6	4	scored	6004
PRO:CB	6	4	scored	6004
PRO:CD	6	4	scored	6004
PRO:CG	6	4	scored	6004
PRO:HA	1	4	scored	6004
PRO:HB2	1	4	scored	6004
PRO:HB3	1	4	scored	6004
PRO:HD2	1	4	scored	6004
PRO:HD3	1	4	scored	6004
PRO:HG2	1	4	scored	6004
PRO:HG3	1	4	scored	6004
PRO:N	7	4	scored	6004
PRO:O	8	4	scored	6004
SER:C	6	3	scored	4503
SER:CA	6	3	scored	4503
SER:CB	6	3	scored	4503
SER:H	1	3	scored	4503
SER:HA	1	3	scored	4503
SER:HB2	1	3	scored	4503
SER:HB3	1	3	scored	4503
SER:HG	1	3	scored	4503
SER:N	7	3	scored	4503
SER:O	8	3	scored	4503
SER:OG	8	3	scored	4503
SER:OXT	8	1	preparation excluded	0
TRP:C	6	1	scored	1501
TRP:CA	6	1	scored	1501
TRP:CB	6	1	scored	1501
TRP:CD1	6	1	scored	1501
TRP:CD2	6	1	scored	1501
TRP:CE2	6	1	scored	1501
TRP:CE3	6	1	scored	1501
TRP:CG	6	1	scored	1501
TRP:CH2	6	1	scored	1501
TRP:CZ2	6	1	scored	1501
TRP:CZ3	6	1	scored	1501
TRP:H	1	1	scored	1501
TRP:HA	1	1	scored	1501
TRP:HB2	1	1	scored	1501
TRP:HB3	1	1	scored	1501
TRP:HD1	1	1	scored	1501
TRP:HE1	1	1	scored	1501
TRP:HE3	1	1	scored	1501
TRP:HH2	1	1	scored	1501
TRP:HZ2	1	1	scored	1501
TRP:HZ3	1	1	scored	1501
TRP:N	7	1	scored	1501
TRP:NE1	7	1	scored	1501
TRP:O	8	1	scored	1501
TYR:C	6	1	scored	1501
TYR:CA	6	1	scored	1501
TYR:CB	6	1	scored	1501
TYR:CD1	6	1	scored	1501
TYR:CD2	6	1	scored	1501
TYR:CE1	6	1	scored	1501
TYR:CE2	6	1	scored	1501
TYR:CG	6	1	scored	1501

IUPAC	Element	Atoms	Status	n
TYR:CZ	6	1	scored	1501
TYR:H	1	1	scored	1501
TYR:HA	1	1	scored	1501
TYR:HB2	1	1	scored	1501
TYR:HB3	1	1	scored	1501
TYR:HD1	1	1	scored	1501
TYR:HD2	1	1	scored	1501
TYR:HE1	1	1	scored	1501
TYR:HE2	1	1	scored	1501
TYR:HH	1	1	scored	1501
TYR:N	7	1	scored	1501
TYR:O	8	1	scored	1501
TYR:OH	8	1	scored	1501

6.2 TC5b Trp-cage (1L2Y): T0 RMSE (ppm)

IUPAC	n	Structure	AIMNet	MOPAC	Joint
ARG:C	1501	4.577624	4.203927	4.433691	4.357835
ARG:CA	1501	3.874490	3.622985	3.776766	3.562267
ARG:CB	1501	2.752772	3.010387	2.667314	2.472139
ARG:CD	1501	3.469991	3.427391	3.268040	3.254588
ARG:CG	1501	2.480334	2.453331	2.340403	2.203040
ARG:CZ	1501	3.974044	4.248323	3.966491	3.764187
ARG:H	1501	0.482396	0.480754	0.437021	0.461361
ARG:HA	1501	0.282442	0.236809	0.252258	0.268729
ARG:HB2	1501	0.237585	0.254246	0.262773	0.225237
ARG:HB3	1501	0.226614	0.240235	0.213439	0.207603
ARG:HD2	1501	0.257164	0.245245	0.259741	0.234536
ARG:HD3	1501	0.230109	0.235408	0.236851	0.230136
ARG:HE	1501	0.367520	0.324574	0.396061	0.320793
ARG:HG2	1501	0.239620	0.217704	0.213995	0.219474
ARG:HG3	1501	0.227535	0.235082	0.226098	0.222821
ARG:HH11	1501	0.379078	0.400784	0.416634	0.352324
ARG:HH12	1501	0.411784	0.384351	0.361332	0.386993
ARG:HH21	1501	0.460010	0.365874	0.434622	0.385456
ARG:HH22	1501	0.408033	0.366851	0.380745	0.386641
ARG:N	1501	6.220201	5.719381	6.387170	5.648934
ARG:NE	1501	8.478155	6.348631	6.242700	5.642648
ARG:NH1	1501	8.016425	4.298967	4.159931	3.811033
ARG:NH2	1501	7.308476	4.257410	4.599707	3.716690
ARG:O	1501	12.972678	13.332371	13.422917	13.572115
ASN:C	1501	4.506836	4.390867	4.268757	4.390306
ASN:CA	1501	4.047782	4.075090	3.920432	3.725744
ASN:CB	1501	3.168263	4.035074	3.347195	3.518884
ASN:CG	1501	5.025279	4.751995	4.980727	4.767532
ASN:HA	1501	0.236286	0.262108	0.224581	0.255952
ASN:HB2	1501	0.312005	0.225711	0.268044	0.237249
ASN:HB3	1501	0.242616	0.291373	0.245049	0.248866
ASN:HD21	1501	0.583541	0.685086	0.403135	0.395349
ASN:HD22	1501	0.527831	0.574656	0.363251	0.302759
ASN:N	1501	4.796419	10.906124	3.445858	8.216338
ASN:ND2	1501	8.629310	6.919529	5.450859	5.068956
ASN:O	1501	14.628470	14.817702	15.595213	16.162345
ASN:OD1	1501	15.202447	16.144384	14.480456	14.994682
ASP:C	1501	4.382650	4.166198	4.420025	4.300443
ASP:CA	1501	3.367565	3.136368	3.383493	3.110592
ASP:CB	1501	2.901690	2.666546	2.672800	2.512501
ASP:CG	1501	4.336494	4.349203	4.271211	4.308181
ASP:H	1501	0.365469	0.411529	0.384137	0.450927
ASP:HA	1501	0.261723	0.242554	0.229521	0.251504
ASP:HB2	1501	0.241562	0.242462	0.214839	0.223976
ASP:HB3	1501	0.274092	0.248695	0.223732	0.218156

IUPAC	<i>n</i>	Structure	AIMNet	MOPAC	Joint
ASP:N	1501	5.822400	5.607744	5.679165	5.432113
ASP:O	1501	13.646608	13.816231	14.772167	14.421444
ASP:OD1	1501	13.422263	14.919476	13.593272	13.541877
ASP:OD2	1501	13.870815	14.008315	14.877926	14.427989
GLN:C	1501	4.568002	4.272940	4.456655	4.497752
GLN:CA	1501	3.498212	3.299248	3.399527	3.203633
GLN:CB	1501	2.873381	3.058280	2.807228	2.704546
GLN:CD	1501	4.627789	4.402334	4.741652	4.593343
GLN:CG	1501	3.178931	3.117077	2.921769	2.792405
GLN:H	1501	0.374281	0.381616	0.372490	0.346074
GLN:HA	1501	0.249827	0.234052	0.214786	0.265166
GLN:HB2	1501	0.242542	0.213822	0.224207	0.219447
GLN:HB3	1501	0.316185	0.247513	0.232211	0.208778
GLN:HE21	1501	0.405319	0.550702	0.413082	0.360736
GLN:HE22	1501	0.378745	0.477861	0.346286	0.321506
GLN:HG2	1501	0.233448	0.247669	0.243208	0.246479
GLN:HG3	1501	0.229531	0.235114	0.237630	0.231934
GLN:N	1501	6.421933	5.728525	5.872894	5.688590
GLN:NE2	1501	7.697791	5.628514	5.001871	4.669202
GLN:O	1501	13.103894	13.511676	13.670007	14.047049
GLN:OE1	1501	14.431926	13.679853	14.440191	14.340477
GLY:C	4503	4.445755	4.264076	4.350636	4.349534
GLY:CA	4503	3.188126	3.071024	3.244227	3.225111
GLY:H	4503	0.404712	0.373853	0.422949	0.414423
GLY:HA2	4503	0.304303	0.267344	0.273504	0.225495
GLY:HA3	4503	0.250439	0.255268	0.261319	0.218606
GLY:N	4503	6.358542	5.974792	7.220541	5.868370
GLY:O	4503	14.059681	13.887039	13.824142	14.674586
ILE:C	1501	4.599043	4.269908	4.410432	4.471588
ILE:CA	1501	3.716340	3.421618	3.606158	3.251920
ILE:CB	1501	3.101538	3.060547	3.024205	2.923641
ILE:CD1	1501	1.902733	1.633456	1.670946	1.500694
ILE:CG1	1501	2.926647	2.981562	2.715114	2.832003
ILE:CG2	1501	2.053955	1.908061	1.910538	1.782562
ILE:H	1501	0.364311	0.406774	0.370019	0.362417
ILE:HA	1501	0.253937	0.220293	0.207457	0.245076
ILE:HB	1501	0.213241	0.246220	0.242642	0.232216
ILE:HD11	1501	0.217354	0.198645	0.207433	0.189618
ILE:HD12	1501	0.220300	0.199504	0.237675	0.200813
ILE:HD13	1501	0.235976	0.225516	0.248584	0.228108
ILE:HG12	1501	0.257515	0.277725	0.298962	0.258871
ILE:HG13	1501	0.243743	0.247361	0.225348	0.210872
ILE:HG21	1501	0.267738	0.249408	0.337253	0.215027
ILE:HG22	1501	0.274768	0.230053	0.245889	0.200565
ILE:HG23	1501	0.236841	0.217620	0.230608	0.207075
ILE:N	1501	6.061530	5.687035	5.404862	5.437109
ILE:O	1501	12.264624	12.977962	13.312655	13.803463
LEU:C	3002	4.408149	4.222573	4.252748	4.275906
LEU:CA	3002	3.496844	3.283614	3.251087	3.146210
LEU:CB	3002	2.826787	2.882182	2.697914	2.701988
LEU:CD1	3002	2.318897	2.235136	2.153039	1.905878
LEU:CD2	3002	2.203419	2.042469	2.029809	1.914783
LEU:CG	3002	2.903549	2.947441	2.782068	2.810069
LEU:H	3002	0.362859	0.379216	0.360206	0.361344
LEU:HA	3002	0.225385	0.220594	0.233169	0.243855
LEU:HB2	3002	0.278074	0.245024	0.298346	0.218723
LEU:HB3	3002	0.237616	0.258980	0.241199	0.218394
LEU:HD11	3002	0.247642	0.252710	0.268312	0.235916
LEU:HD12	3002	0.228895	0.235681	0.224448	0.204938
LEU:HD13	3002	0.218288	0.227918	0.255509	0.215050
LEU:HD21	3002	0.257088	0.263668	0.251380	0.230597
LEU:HD22	3002	0.265041	0.233513	0.228945	0.222548
LEU:HD23	3002	0.270189	0.264147	0.275486	0.243313
LEU:HG	3002	0.213788	0.237210	0.215666	0.200524

IUPAC	<i>n</i>	Structure	AIMNet	MOPAC	Joint
LEU:N	3002	6.926823	6.167594	6.037990	6.124078
LEU:O	3002	13.356840	12.929028	13.527469	12.884419
LYS:C	1501	4.471900	4.216957	4.275937	4.367727
LYS:CA	1501	3.439418	3.274953	3.247310	3.061726
LYS:CB	1501	2.611372	2.707988	2.637174	2.441196
LYS:CD	1501	2.540731	2.480496	2.397937	2.364770
LYS:CE	1501	3.863660	3.783548	3.712117	3.570778
LYS:CG	1501	2.599319	2.571495	2.327456	2.341715
LYS:H	1501	0.357987	0.426688	0.360575	0.404493
LYS:HA	1501	0.232440	0.209000	0.204678	0.219757
LYS:HB2	1501	0.209890	0.211697	0.213787	0.200067
LYS:HB3	1501	0.243951	0.220156	0.210267	0.216662
LYS:HD2	1501	0.193952	0.203543	0.205822	0.200101
LYS:HD3	1501	0.196279	0.205571	0.208662	0.198869
LYS:HE2	1501	0.235797	0.230431	0.234312	0.218943
LYS:HE3	1501	0.236315	0.239657	0.238128	0.216101
LYS:HG2	1501	0.205201	0.203729	0.199187	0.192398
LYS:HG3	1501	0.193050	0.219681	0.206255	0.202290
LYS:HZ1	1501	0.239081	0.272497	0.258960	0.276933
LYS:HZ2	1501	0.261730	0.371369	0.283950	0.339266
LYS:HZ3	1501	0.242957	0.281661	0.256198	0.285117
LYS:N	1501	5.906211	5.565928	5.726523	5.382838
LYS:NZ	1501	3.087841	3.343055	2.851535	3.043343
LYS:O	1501	13.974977	14.194805	14.402378	15.603576
PRO:C	6004	4.629996	4.420785	4.544290	4.489326
PRO:CA	6004	4.166620	4.098500	4.059376	3.815262
PRO:CB	6004	2.962801	2.971849	2.974633	2.899022
PRO:CD	6004	3.674348	3.569188	3.551261	3.383327
PRO:CG	6004	3.073951	2.971474	2.916628	2.805198
PRO:HA	6004	0.298644	0.302594	0.305034	0.278393
PRO:HB2	6004	0.230774	0.224798	0.260873	0.222311
PRO:HB3	6004	0.242150	0.262986	0.263604	0.250390
PRO:HD2	6004	0.291684	0.288549	0.285505	0.264128
PRO:HD3	6004	0.278970	0.265954	0.276089	0.264584
PRO:HG2	6004	0.223134	0.216818	0.227645	0.225401
PRO:HG3	6004	0.219879	0.215231	0.227693	0.239394
PRO:N	6004	8.295898	7.631761	7.439598	7.491900
PRO:O	6004	13.186705	13.198493	14.490209	14.817280
SER:C	4503	4.331738	4.402339	4.305670	4.355396
SER:CA	4503	5.123511	5.037951	4.100812	3.908661
SER:CB	4503	2.972329	3.338911	3.001609	2.926839
SER:H	4503	0.411398	0.429938	0.396702	0.408770
SER:HA	4503	0.272606	0.281859	0.254580	0.262008
SER:HB2	4503	0.286250	0.325971	0.306941	0.284807
SER:HB3	4503	0.272025	0.373383	0.274966	0.289330
SER:HG	4503	0.613802	0.634588	0.631199	0.568904
SER:N	4503	7.275777	6.127743	6.388121	5.978614
SER:O	4503	15.693592	14.044486	14.357448	14.832417
SER:OG	4503	9.801271	8.781536	7.704367	9.471426
TRP:C	1501	4.561770	4.518316	4.756950	4.719941
TRP:CA	1501	3.542450	3.248182	3.602561	3.374708
TRP:CB	1501	2.325673	2.399643	2.705085	2.602881
TRP:CD1	1501	4.737452	3.968942	3.656925	3.702384
TRP:CD2	1501	4.311022	4.239577	4.930347	3.914364
TRP:CE2	1501	3.801116	3.781880	4.383469	3.679573
TRP:CE3	1501	3.851506	3.969243	3.585493	3.452605
TRP:CG	1501	4.996524	5.258174	4.966911	4.848765
TRP:CH2	1501	3.480126	3.755875	3.233784	3.650969
TRP:CZ2	1501	3.768921	3.810134	3.870524	3.444640
TRP:CZ3	1501	3.943909	4.238785	3.921304	3.716953
TRP:H	1501	0.421551	0.411949	0.384696	0.424240
TRP:HA	1501	0.251417	0.226943	0.236026	0.236791
TRP:HB2	1501	0.295024	0.282872	0.342401	0.260912
TRP:HB3	1501	0.306114	0.335044	0.265839	0.283458

IUPAC	<i>n</i>	Structure	AIMNet	MOPAC	Joint
TRP:HD1	1501	0.597925	0.487339	0.644809	0.426332
TRP:HE1	1501	0.546303	0.426855	0.500708	0.346820
TRP:HE3	1501	0.324662	0.342340	0.332691	0.398881
TRP:HH2	1501	0.335955	0.355014	0.274616	0.298204
TRP:HZ2	1501	0.418955	0.429197	0.349604	0.362148
TRP:HZ3	1501	0.404317	0.527332	0.496953	0.424186
TRP:N	1501	6.104393	5.834737	5.823654	5.911735
TRP:NE1	1501	6.062557	7.471299	6.624812	7.883030
TRP:O	1501	11.856910	11.965985	12.272109	12.409699
TYR:C	1501	4.270102	4.073932	4.238524	4.214712
TYR:CA	1501	3.497920	3.597184	3.492650	3.477063
TYR:CB	1501	2.548999	2.571725	2.530013	2.441379
TYR:CD1	1501	3.913385	3.949634	3.695690	3.757259
TYR:CD2	1501	3.744631	3.753619	3.630633	3.567868
TYR:CE1	1501	4.319660	4.197980	3.922430	4.010281
TYR:CE2	1501	5.333793	5.704081	5.103452	4.470457
TYR:CG	1501	4.042237	4.061228	4.144413	4.064928
TYR:CZ	1501	4.111352	3.984305	4.138746	3.994816
TYR:H	1501	0.385585	0.435179	0.355520	0.405273
TYR:HA	1501	0.237920	0.221710	0.205992	0.223585
TYR:HB2	1501	0.250286	0.252425	0.310489	0.292892
TYR:HB3	1501	0.415508	0.327673	0.282528	0.281681
TYR:HD1	1501	0.282200	0.277183	0.299331	0.283733
TYR:HD2	1501	0.370095	0.296917	0.382300	0.295884
TYR:HE1	1501	0.347000	0.374898	0.327395	0.366951
TYR:HE2	1501	0.303151	0.296999	0.331883	0.301670
TYR:HH	1501	0.360524	0.348307	0.391287	0.374573
TYR:N	1501	5.986468	5.773500	5.924470	5.661254
TYR:O	1501	13.258272	11.452354	12.810909	12.634422
TYR:OH	1501	6.914095	7.923382	6.759693	6.809075

6.3 TC5b Trp-cage (1L2Y): T2 vector RMSE (ppm)

IUPAC	<i>n</i>	Structure	AIMNet	MOPAC	Joint
ARG:C	1501	9.518959	9.538556	7.908289	8.151677
ARG:CA	1501	9.459962	9.192070	8.467499	8.320668
ARG:CB	1501	10.839898	10.042363	9.080324	9.666917
ARG:CD	1501	10.684258	10.144152	9.590152	8.901263
ARG:CG	1501	10.092656	9.299260	8.700414	8.723276
ARG:CZ	1501	12.811040	10.794747	9.497488	9.234329
ARG:H	1501	1.355294	1.388604	1.339438	1.141078
ARG:HA	1501	1.223530	1.052369	0.951837	0.951296
ARG:HB2	1501	1.069845	1.047634	0.986508	0.905585
ARG:HB3	1501	1.059262	0.985020	1.025543	0.870742
ARG:HD2	1501	1.141011	1.136533	0.988030	0.915224
ARG:HD3	1501	1.132529	1.143891	1.093722	0.959640
ARG:HE	1501	1.320655	1.382453	1.167162	1.275231
ARG:HG2	1501	1.267129	1.036259	0.989800	0.983424
ARG:HG3	1501	1.195766	1.067456	0.964991	1.038734
ARG:HH11	1501	1.569232	1.319630	1.074428	1.503625
ARG:HH12	1501	1.594595	1.351408	1.118308	1.071219
ARG:HH21	1501	1.592605	1.452155	1.054685	1.523142
ARG:HH22	1501	1.318981	1.235472	1.044747	1.091155
ARG:N	1501	15.638909	16.191380	15.992046	15.008973
ARG:NE	1501	17.631691	17.196885	16.445738	14.823390
ARG:NH1	1501	17.173321	16.379561	13.152520	13.848762
ARG:NH2	1501	15.564055	13.896769	12.486864	13.219825
ARG:O	1501	33.366717	37.042164	33.216121	34.396828
ASN:C	1501	13.242369	12.064327	10.925944	14.083171
ASN:CA	1501	10.637098	10.110055	8.870112	9.384824
ASN:CB	1501	11.439709	10.372068	8.510614	10.515301
ASN:CG	1501	14.036326	12.798766	12.335151	12.611538

IUPAC	<i>n</i>	Structure	AIMNet	MOPAC	Joint
ASN:HA	1501	0.965316	1.230083	1.096337	1.023622
ASN:HB2	1501	1.010120	0.944361	1.044597	0.990305
ASN:HB3	1501	1.138090	1.097747	1.079034	1.006164
ASN:HD21	1501	3.218777	1.972741	1.936561	2.148155
ASN:HD22	1501	2.455606	2.175454	2.135706	2.417357
ASN:N	1501	15.605299	15.000286	17.018516	18.369223
ASN:ND2	1501	49.217297	34.356930	26.779497	22.024907
ASN:O	1501	39.150230	40.002011	35.762028	41.312629
ASN:OD1	1501	46.031195	46.525965	39.296747	42.758387
ASP:C	1501	10.044175	9.785649	8.647395	8.905055
ASP:CA	1501	9.137084	8.248379	7.607099	7.671897
ASP:CB	1501	8.488145	8.885290	7.500623	7.808815
ASP:CG	1501	10.413145	9.659227	9.005392	8.974238
ASP:H	1501	1.274673	1.121736	1.187776	1.102947
ASP:HA	1501	0.949282	0.973002	0.852739	0.852344
ASP:HB2	1501	1.066147	1.045691	0.791623	0.764804
ASP:HB3	1501	1.040010	0.974573	0.988592	0.850657
ASP:N	1501	13.489851	14.438009	12.879963	13.389546
ASP:O	1501	36.692712	33.971316	31.368730	33.459258
ASP:OD1	1501	43.167720	35.791283	31.618812	31.999516
ASP:OD2	1501	38.943076	36.295361	31.980669	31.431118
GLN:C	1501	10.620852	9.663207	8.444237	8.585783
GLN:CA	1501	8.442477	8.422502	7.940282	8.037806
GLN:CB	1501	8.862644	9.535420	8.364463	9.030004
GLN:CD	1501	12.171812	11.732480	11.062254	10.452396
GLN:CG	1501	9.110130	8.571948	8.514030	8.192673
GLN:H	1501	1.337532	1.267182	1.248151	1.135579
GLN:HA	1501	1.067498	1.135415	1.027805	1.010433
GLN:HB2	1501	1.174635	0.990957	0.964693	0.923737
GLN:HB3	1501	0.965910	0.943710	0.881779	0.848623
GLN:HE21	1501	2.925023	1.964173	2.136742	2.417592
GLN:HE22	1501	2.510481	2.138370	2.218892	2.281246
GLN:HG2	1501	1.024944	1.061743	1.017275	0.938101
GLN:HG3	1501	0.984486	1.035069	1.063975	0.916919
GLN:N	1501	13.678971	14.053989	13.235499	13.172696
GLN:NE2	1501	36.205379	30.991642	26.995560	19.066628
GLN:O	1501	30.866445	29.202444	32.915914	27.462502
GLN:OE1	1501	46.547122	37.643130	40.035019	38.649419
GLY:C	4503	10.228234	9.826303	8.937968	10.172296
GLY:CA	4503	8.000604	7.902340	7.378489	7.872921
GLY:H	4503	1.431649	1.180134	1.128449	1.130901
GLY:HA2	4503	1.105774	1.215020	1.188581	1.097711
GLY:HA3	4503	1.215615	1.136995	1.155925	1.012079
GLY:N	4503	14.634049	15.294243	14.686254	14.192812
GLY:O	4503	36.765624	36.559688	34.232601	33.080154
ILE:C	1501	9.843348	8.995373	9.144979	8.869622
ILE:CA	1501	9.203096	9.053846	7.713747	8.251537
ILE:CB	1501	10.855060	10.139756	10.197397	9.658916
ILE:CD1	1501	8.686240	8.518728	7.406504	7.696214
ILE:CG1	1501	11.975623	11.334042	10.296385	9.674607
ILE:CG2	1501	9.119787	8.452888	7.732282	8.126015
ILE:H	1501	1.222823	1.380507	1.102133	1.172707
ILE:HA	1501	1.110582	1.267302	0.968271	1.076585
ILE:HB	1501	1.076485	1.057302	1.142040	1.110851
ILE:HD11	1501	0.915825	0.925857	0.891702	0.784986
ILE:HD12	1501	0.925529	0.902999	0.901410	0.786299
ILE:HD13	1501	0.914367	0.885428	0.918037	0.783897
ILE:HG12	1501	1.163726	1.104441	1.017089	1.007498
ILE:HG13	1501	1.125914	1.205259	1.055415	0.961099
ILE:HG21	1501	1.045297	0.857402	0.965938	0.821349
ILE:HG22	1501	1.137362	1.018381	1.079503	0.861371
ILE:HG23	1501	0.976037	0.955113	0.994204	0.841570
ILE:N	1501	14.226025	13.858981	13.098157	13.154877
ILE:O	1501	29.362114	28.616001	28.300128	27.207023

IUPAC	<i>n</i>	Structure	AIMNet	MOPAC	Joint
LEU:C	3002	9.630233	8.830254	8.683743	8.931021
LEU:CA	3002	8.914934	8.342829	7.428403	8.313272
LEU:CB	3002	9.767281	10.092468	8.841812	9.046332
LEU:CD1	3002	9.927851	9.646052	9.556184	8.999587
LEU:CD2	3002	9.618385	9.465763	9.169304	9.030209
LEU:CG	3002	9.824275	9.901261	9.315809	9.446951
LEU:H	3002	1.209538	1.342799	1.067325	1.110368
LEU:HA	3002	1.080909	1.100325	0.893039	0.995924
LEU:HB2	3002	0.969311	0.999899	0.985068	0.838357
LEU:HB3	3002	0.973376	0.950542	0.972173	0.859580
LEU:HD11	3002	0.961940	1.001405	0.922944	0.790959
LEU:HD12	3002	0.978836	0.919266	0.912790	0.805518
LEU:HD13	3002	1.045285	0.975674	0.935644	0.795546
LEU:HD21	3002	1.052692	0.964282	0.921968	0.807876
LEU:HD22	3002	1.059553	0.996318	0.888210	0.827459
LEU:HD23	3002	1.073328	1.002750	0.966850	0.845311
LEU:HG	3002	1.123915	1.068170	1.016371	0.967786
LEU:N	3002	15.324342	14.968173	13.673754	14.017496
LEU:O	3002	30.588288	28.447505	28.135570	28.450364
LYS:C	1501	9.793034	9.496343	8.011134	9.202213
LYS:CA	1501	8.604939	8.444185	7.679747	7.695400
LYS:CB	1501	8.408305	8.799655	8.173270	8.160840
LYS:CD	1501	9.319172	9.155491	8.224921	8.194394
LYS:CE	1501	11.417272	11.033260	9.339484	9.196897
LYS:CG	1501	9.939858	9.057356	8.086901	8.324112
LYS:H	1501	1.217431	1.043425	0.979381	0.976953
LYS:HA	1501	1.006197	1.038797	0.940574	0.956693
LYS:HB2	1501	1.042962	0.919698	0.904913	0.831158
LYS:HB3	1501	1.022967	0.921381	0.854165	0.807631
LYS:HD2	1501	1.018092	0.937363	0.918705	0.835346
LYS:HD3	1501	1.017614	0.952552	0.934635	0.835873
LYS:HE2	1501	1.159137	1.117088	0.978740	0.893460
LYS:HE3	1501	1.046788	1.077807	0.948245	0.865128
LYS:HG2	1501	1.051480	0.956976	0.941212	0.814726
LYS:HG3	1501	1.025616	1.004208	0.949367	0.859611
LYS:HZ1	1501	1.137861	1.344168	1.179388	1.108418
LYS:HZ2	1501	1.085521	1.446468	1.187716	1.167000
LYS:HZ3	1501	1.123879	1.342804	1.269736	1.465632
LYS:N	1501	13.591357	13.393352	12.320195	13.356096
LYS:NZ	1501	12.887918	12.805523	11.966046	12.382521
LYS:O	1501	31.121619	31.395872	28.799280	27.512125
PRO:C	6004	10.263919	9.370819	8.271469	9.241335
PRO:CA	6004	9.976238	9.783959	9.726415	9.243647
PRO:CB	6004	10.560367	10.151336	10.130338	9.947029
PRO:CD	6004	11.391124	11.948601	10.923415	10.334032
PRO:CG	6004	10.313858	10.279984	10.146291	9.658107
PRO:HA	6004	1.232948	1.265132	1.144716	1.165523
PRO:HB2	6004	1.066032	0.959717	0.948776	0.847329
PRO:HB3	6004	1.096878	1.094895	1.022303	0.948070
PRO:HD2	6004	1.248451	1.123739	1.048566	0.996206
PRO:HD3	6004	1.291393	1.172934	1.101271	1.007834
PRO:HG2	6004	1.042569	0.991397	0.954367	0.890630
PRO:HG3	6004	1.051545	0.973591	0.962739	0.851096
PRO:N	6004	17.914410	18.533211	18.203013	18.360275
PRO:O	6004	37.123221	39.561359	33.668506	32.887345
SER:C	4503	11.686019	11.505688	11.999441	10.439574
SER:CA	4503	13.039300	10.977733	10.760964	11.968778
SER:CB	4503	12.730741	11.143821	11.656433	11.755633
SER:H	4503	1.362048	1.264031	1.221031	1.129797
SER:HA	4503	1.248689	1.212603	1.135725	1.011306
SER:HB2	4503	1.425659	1.274868	1.234589	1.166101
SER:HB3	4503	1.459918	1.421991	1.231543	1.253266
SER:HG	4503	2.379121	2.754133	2.604761	2.617245
SER:N	4503	15.061403	15.559881	13.942331	14.364908

IUPAC	<i>n</i>	Structure	AIMNet	MOPAC	Joint
SER:O	4503	40.819099	46.898804	36.892602	36.375739
SER:OG	4503	35.944548	38.873350	36.808756	40.306701
TRP:C	1501	9.724147	9.714112	8.077065	8.223766
TRP:CA	1501	8.618765	8.366030	7.424135	8.026908
TRP:CB	1501	9.511186	8.287321	8.186421	8.829340
TRP:CD1	1501	17.514258	16.568068	22.667144	15.692098
TRP:CD2	1501	13.741171	14.224046	13.265702	15.367796
TRP:CE2	1501	18.951713	17.060379	18.799975	16.568567
TRP:CE3	1501	14.631517	12.861924	14.752137	12.119753
TRP:CG	1501	15.729913	14.742544	15.160148	14.281068
TRP:CH2	1501	12.811923	17.274811	14.708956	12.853765
TRP:CZ2	1501	12.692859	13.511120	12.191729	12.646411
TRP:CZ3	1501	14.116600	14.983334	15.251881	13.343910
TRP:H	1501	1.342178	1.218757	1.113814	1.095761
TRP:HA	1501	1.107619	1.170097	1.037885	0.991209
TRP:HB2	1501	1.020373	0.926500	1.122239	0.931535
TRP:HB3	1501	1.256370	1.025806	1.133729	1.100295
TRP:HD1	1501	1.358301	1.262090	1.423429	1.360104
TRP:HE1	1501	2.379018	1.334524	1.719803	2.056400
TRP:HE3	1501	1.295370	1.109649	1.207084	1.142047
TRP:HH2	1501	1.112252	0.983933	1.085896	1.168481
TRP:HZ2	1501	1.105399	1.105729	1.072032	1.212702
TRP:HZ3	1501	1.469443	1.052290	1.176443	1.097238
TRP:N	1501	14.860496	14.406134	12.427849	13.028359
TRP:NE1	1501	20.479337	21.753070	19.983944	20.002455
TRP:O	1501	33.477722	29.246812	29.915426	38.349936
TYR:C	1501	9.557005	9.292534	7.990388	8.455885
TYR:CA	1501	8.215582	7.814495	7.921159	7.725633
TYR:CB	1501	9.000708	8.973159	8.178440	8.275156
TYR:CD1	1501	12.142456	12.212597	12.657911	10.568827
TYR:CD2	1501	12.267356	12.675931	13.265710	10.728538
TYR:CE1	1501	13.624567	14.199846	14.718806	14.435862
TYR:CE2	1501	13.354988	13.336643	15.483203	14.642321
TYR:CG	1501	12.732988	13.003059	12.208153	11.217300
TYR:CZ	1501	16.861755	13.858704	14.699557	14.000474
TYR:H	1501	1.133315	1.281223	1.119292	1.047163
TYR:HA	1501	1.125359	1.095716	1.106746	0.999421
TYR:HB2	1501	1.106172	0.960168	0.945248	0.985366
TYR:HB3	1501	1.181264	0.962704	0.879288	1.079993
TYR:HD1	1501	1.039353	1.129063	1.011200	1.071501
TYR:HD2	1501	1.072793	1.103482	1.038653	0.990384
TYR:HE1	1501	0.947444	0.924079	1.022788	1.003379
TYR:HE2	1501	1.203725	1.082245	1.196993	1.313031
TYR:HH	1501	1.660179	1.873192	1.571137	1.719278
TYR:N	1501	13.704804	14.620276	13.217809	12.984771
TYR:O	1501	32.692719	30.678959	30.071214	32.435314
TYR:OH	1501	34.525085	29.734591	30.545701	32.178497

6.4 CLN025 (5AWL): IUPAC support

IUPAC	Element	Atoms	Status	<i>n</i>
ASP:C	6	1	scored	5001
ASP:CA	6	1	scored	5001
ASP:CB	6	1	scored	5001
ASP:CG	6	1	scored	5001
ASP:H	1	1	scored	5001
ASP:HA	1	1	scored	5001
ASP:HB2	1	1	scored	5001
ASP:HB3	1	1	scored	5001
ASP:N	7	1	scored	5001
ASP:O	8	1	scored	5001
ASP:OD1	8	1	scored	5001

IUPAC	Element	Atoms	Status	<i>n</i>
ASP:OD2	8	1	scored	5001
GLU:C	6	1	scored	5001
GLU:CA	6	1	scored	5001
GLU:CB	6	1	scored	5001
GLU:CD	6	1	scored	5001
GLU:CG	6	1	scored	5001
GLU:H	1	1	scored	5001
GLU:HA	1	1	scored	5001
GLU:HB2	1	1	scored	5001
GLU:HB3	1	1	scored	5001
GLU:HG2	1	1	scored	5001
GLU:HG3	1	1	scored	5001
GLU:N	7	1	scored	5001
GLU:O	8	1	scored	5001
GLU:OE1	8	1	scored	5001
GLU:OE2	8	1	scored	5001
GLY:C	6	1	scored	5001
GLY:CA	6	1	scored	5001
GLY:H	1	1	scored	5001
GLY:HA2	1	1	scored	5001
GLY:HA3	1	1	scored	5001
GLY:N	7	1	scored	5001
GLY:O	8	1	scored	5001
PRO:C	6	1	scored	5001
PRO:CA	6	1	scored	5001
PRO:CB	6	1	scored	5001
PRO:CD	6	1	scored	5001
PRO:CG	6	1	scored	5001
PRO:HA	1	1	scored	5001
PRO:HB2	1	1	scored	5001
PRO:HB3	1	1	scored	5001
PRO:HD2	1	1	scored	5001
PRO:HD3	1	1	scored	5001
PRO:HG2	1	1	scored	5001
PRO:HG3	1	1	scored	5001
PRO:N	7	1	scored	5001
PRO:O	8	1	scored	5001
THR:C	6	2	scored	10002
THR:CA	6	2	scored	10002
THR:CB	6	2	scored	10002
THR:CG2	6	2	scored	10002
THR:H	1	2	scored	10002
THR:HA	1	2	scored	10002
THR:HB	1	2	scored	10002
THR:HG1	1	2	scored	10002
THR:HG21	1	2	scored	10002
THR:HG22	1	2	scored	10002
THR:HG23	1	2	scored	10002
THR:N	7	2	scored	10002
THR:O	8	2	scored	10002
THR:OG1	8	2	scored	10002
TRP:C	6	1	scored	5001
TRP:CA	6	1	scored	5001
TRP:CB	6	1	scored	5001
TRP:CD1	6	1	scored	5001
TRP:CD2	6	1	scored	5001
TRP:CE2	6	1	scored	5001
TRP:CE3	6	1	scored	5001
TRP:CG	6	1	scored	5001
TRP:CH2	6	1	scored	5001
TRP:CZ2	6	1	scored	5001
TRP:CZ3	6	1	scored	5001
TRP:H	1	1	scored	5001
TRP:HA	1	1	scored	5001

IUPAC	Element	Atoms	Status	n
TRP:HB2	1	1	scored	5001
TRP:HB3	1	1	scored	5001
TRP:HD1	1	1	scored	5001
TRP:HE1	1	1	scored	5001
TRP:HE3	1	1	scored	5001
TRP:HH2	1	1	scored	5001
TRP:HZ2	1	1	scored	5001
TRP:HZ3	1	1	scored	5001
TRP:N	7	1	scored	5001
TRP:NE1	7	1	scored	5001
TRP:O	8	1	scored	5001
TYR:C	6	3	scored	15003
TYR:CA	6	3	scored	15003
TYR:CB	6	3	scored	15003
TYR:CD1	6	3	scored	15003
TYR:CD2	6	3	scored	15003
TYR:CE1	6	3	scored	15003
TYR:CE2	6	3	scored	15003
TYR:CG	6	3	scored	15003
TYR:CZ	6	3	scored	15003
TYR:H	1	2	scored	10002
TYR:H1	1	1	preparation excluded	0
TYR:H2	1	1	preparation excluded	0
TYR:H3	1	1	preparation excluded	0
TYR:HA	1	3	scored	15003
TYR:HB2	1	3	scored	15003
TYR:HB3	1	3	scored	15003
TYR:HD1	1	3	scored	15003
TYR:HD2	1	3	scored	15003
TYR:HE1	1	3	scored	15003
TYR:HE2	1	3	scored	15003
TYR:HH	1	3	scored	15003
TYR:N	7	3	scored	15003
TYR:O	8	3	scored	15003
TYR:OH	8	3	scored	15003
TYR:OXT	8	1	preparation excluded	0

6.5 CLN025 (5AWL): T0 RMSE (ppm)

IUPAC	n	Structure	AIMNet	MOPAC	Joint
ASP:C	5001	4.829227	4.631145	4.758738	4.639970
ASP:CA	5001	3.883152	3.465679	4.247605	3.774594
ASP:CB	5001	2.826440	2.826704	2.855287	2.804512
ASP:CG	5001	4.814429	4.836137	4.772838	4.799220
ASP:H	5001	0.370864	0.380524	0.406098	0.384553
ASP:HA	5001	0.286451	0.259702	0.262455	0.279379
ASP:HB2	5001	0.265887	0.279807	0.259916	0.244298
ASP:HB3	5001	0.322088	0.266281	0.274412	0.231948
ASP:N	5001	7.019555	6.097099	5.930187	6.217575
ASP:O	5001	15.821979	13.304448	14.460386	18.009779
ASP:OD1	5001	15.753230	21.458205	14.590424	15.187778
ASP:OD2	5001	15.902226	16.466913	14.761050	14.656384
GLU:C	5001	4.535189	4.412087	4.470744	4.612701
GLU:CA	5001	3.734252	3.588972	4.183229	3.477179
GLU:CB	5001	2.956695	2.982888	3.068979	2.785244
GLU:CD	5001	4.560996	4.902823	4.694456	4.745780
GLU:CG	5001	2.942219	2.763220	3.149781	2.518557
GLU:H	5001	0.447655	0.386503	0.400298	0.407058
GLU:HA	5001	0.275078	0.244944	0.228199	0.276315
GLU:HB2	5001	0.336741	0.269399	0.272585	0.238750
GLU:HB3	5001	0.295175	0.237894	0.276644	0.215527
GLU:HG2	5001	0.247779	0.237495	0.258347	0.239284

IUPAC	<i>n</i>	Structure	AIMNet	MOPAC	Joint
GLU:HG3	5001	0.242413	0.253941	0.236344	0.222315
GLU:N	5001	6.174396	5.838333	5.779432	5.831799
GLU:O	5001	14.691786	15.020881	13.396182	14.837625
GLU:OE1	5001	17.168384	14.817020	15.627824	13.566079
GLU:OE2	5001	19.184823	14.543027	15.849711	14.477071
GLY:C	5001	4.506609	4.378627	4.379267	4.509371
GLY:CA	5001	3.106927	2.973501	2.905223	2.741157
GLY:H	5001	0.396146	0.437660	0.409676	0.421134
GLY:HA2	5001	0.203444	0.243978	0.294032	0.244481
GLY:HA3	5001	0.237742	0.191032	0.219696	0.188342
GLY:N	5001	6.467490	5.848325	7.637411	5.996497
GLY:O	5001	13.672379	12.832500	15.206443	13.580024
PRO:C	5001	4.504766	4.344453	4.376593	4.448888
PRO:CA	5001	4.068287	4.106194	4.004503	3.732275
PRO:CB	5001	3.059137	2.944559	3.058933	2.942465
PRO:CD	5001	3.871036	3.786054	3.845110	3.703442
PRO:CG	5001	2.900715	2.825183	2.814777	2.632108
PRO:HA	5001	0.239121	0.238057	0.237076	0.220882
PRO:HB2	5001	0.224007	0.217680	0.270824	0.216708
PRO:HB3	5001	0.230228	0.228686	0.238692	0.218450
PRO:HD2	5001	0.270166	0.301242	0.290980	0.261364
PRO:HD3	5001	0.294300	0.292997	0.324966	0.282894
PRO:HG2	5001	0.209511	0.219110	0.205035	0.206410
PRO:HG3	5001	0.206185	0.189874	0.203725	0.197044
PRO:N	5001	8.352538	8.196047	7.706246	7.688412
PRO:O	5001	12.850118	13.075751	13.592696	13.628458
THR:C	10002	4.595436	4.329738	4.389250	4.356620
THR:CA	10002	4.800468	4.391780	4.089669	3.767742
THR:CB	10002	3.529605	3.728945	3.641144	3.794498
THR:CG2	10002	2.807894	2.356372	2.772818	2.143164
THR:H	10002	0.434316	0.399137	0.414380	0.399588
THR:HA	10002	0.241422	0.230119	0.227034	0.221271
THR:HB	10002	0.245158	0.266242	0.259366	0.274608
THR:HG1	10002	1.084965	0.990892	0.565721	0.517461
THR:HG21	10002	0.269414	0.236349	0.276513	0.233281
THR:HG22	10002	0.261317	0.246630	0.308577	0.245393
THR:HG23	10002	0.298253	0.232958	0.266907	0.253944
THR:N	10002	7.676130	6.018248	7.625764	6.541074
THR:O	10002	13.516870	12.971621	13.412259	13.593216
THR:OG1	10002	9.117866	10.233013	8.836564	9.045670
TRP:C	5001	4.486046	4.393515	4.247476	4.514039
TRP:CA	5001	3.966587	3.560473	3.729106	3.602027
TRP:CB	5001	2.493307	2.564488	2.984379	2.663777
TRP:CD1	5001	3.951893	4.238284	4.005938	4.014613
TRP:CD2	5001	4.445618	4.243385	4.108889	3.869603
TRP:CE2	5001	4.019693	4.075028	3.870598	3.903638
TRP:CE3	5001	3.871631	4.225543	3.645455	3.864675
TRP:CG	5001	4.242773	4.420439	4.171130	4.418572
TRP:CH2	5001	3.931624	4.662275	4.129244	3.790404
TRP:CZ2	5001	3.755159	3.808823	4.005002	3.502783
TRP:CZ3	5001	5.173209	4.972575	4.187963	4.754979
TRP:H	5001	0.324354	0.326388	0.295170	0.308226
TRP:HA	5001	0.301930	0.238245	0.250081	0.242587
TRP:HB2	5001	0.345584	0.244576	0.257905	0.240640
TRP:HB3	5001	0.276396	0.302756	0.287434	0.388749
TRP:HD1	5001	0.722256	0.487844	0.666125	0.329043
TRP:HE1	5001	0.371310	0.302630	0.374454	0.300677
TRP:HE3	5001	0.708776	0.621014	0.427641	0.622674
TRP:HH2	5001	0.362674	0.359254	0.260971	0.271530
TRP:HZ2	5001	0.397051	0.310448	0.311310	0.242320
TRP:HZ3	5001	0.388202	0.417584	0.354389	0.322770
TRP:N	5001	5.800014	6.003757	5.466854	5.888640
TRP:NE1	5001	5.883357	7.489894	7.680306	7.094701
TRP:O	5001	13.133415	12.533829	13.178171	15.161152

IUPAC	<i>n</i>	Structure	AIMNet	MOPAC	Joint
TYR:C	15003	4.367709	4.346069	4.314476	4.389709
TYR:CA	15003	4.041217	3.948452	4.314849	3.800031
TYR:CB	15003	3.264273	3.368132	2.861551	3.026922
TYR:CD1	15003	4.294876	4.137949	4.067130	3.798549
TYR:CD2	15003	4.043603	4.081941	3.945404	3.787725
TYR:CE1	15003	4.622621	4.375275	4.332138	4.293133
TYR:CE2	15003	5.873108	6.143674	5.386315	5.278467
TYR:CG	15003	4.583130	4.393104	4.742997	4.201851
TYR:CZ	15003	4.398499	4.086850	4.525552	4.183825
TYR:H	10002	0.370850	0.428726	0.431418	0.522554
TYR:HA	15003	0.285094	0.343005	0.295819	0.312055
TYR:HB2	15003	0.272525	0.289373	0.397912	0.300582
TYR:HB3	15003	0.375901	0.339249	0.286626	0.362921
TYR:HD1	15003	0.337085	0.297684	0.290987	0.306013
TYR:HD2	15003	0.331612	0.316692	0.381903	0.358199
TYR:HE1	15003	0.342658	0.360152	0.307813	0.380018
TYR:HE2	15003	0.287757	0.296220	0.286721	0.285694
TYR:HH	15003	0.456245	0.512600	0.463323	0.469662
TYR:N	15003	10.170256	10.836421	8.128804	7.086674
TYR:O	15003	16.546989	14.018771	12.963449	13.933257
TYR:OH	15003	8.268175	9.550001	8.271072	7.769458

6.6 CLN025 (5AWL): T2 vector RMSE (ppm)

IUPAC	<i>n</i>	Structure	AIMNet	MOPAC	Joint
ASP:C	5001	11.168742	9.815969	9.106253	9.019455
ASP:CA	5001	8.482725	8.342820	7.552707	7.707433
ASP:CB	5001	9.514026	8.256500	8.682112	8.324163
ASP:CG	5001	12.446830	10.522842	9.978972	9.984240
ASP:H	5001	1.274118	1.260239	1.336125	1.121363
ASP:HA	5001	1.041558	1.201826	1.317109	1.046828
ASP:HB2	5001	1.401948	1.161399	1.170540	0.864679
ASP:HB3	5001	1.274912	1.025265	1.324716	1.227623
ASP:N	5001	13.759834	14.504934	12.975767	13.069801
ASP:O	5001	43.550076	37.310563	37.910763	35.486586
ASP:OD1	5001	41.060468	37.839167	34.771728	35.935446
ASP:OD2	5001	41.471293	41.325760	33.505812	35.350632
GLU:C	5001	10.340028	9.653054	9.180443	9.657076
GLU:CA	5001	9.123673	8.993077	8.909025	8.891541
GLU:CB	5001	9.158805	9.934373	9.330745	9.077148
GLU:CD	5001	12.862025	11.252314	9.911313	10.592422
GLU:CG	5001	9.904291	8.146081	8.124089	8.177356
GLU:H	5001	1.486030	1.830263	1.541561	1.215886
GLU:HA	5001	1.194634	1.172895	1.239139	1.086177
GLU:HB2	5001	1.160918	1.212514	1.308470	1.150639
GLU:HB3	5001	1.169876	1.043876	1.158934	0.988645
GLU:HG2	5001	1.249904	1.002846	1.080417	0.898803
GLU:HG3	5001	1.161495	1.108661	1.215830	0.905272
GLU:N	5001	15.096254	14.763599	14.311726	13.525723
GLU:O	5001	32.966039	32.496740	41.193411	36.154575
GLU:OE1	5001	46.720557	41.993850	36.614003	38.585362
GLU:OE2	5001	55.543452	45.039552	46.152612	41.364817
GLY:C	5001	9.630649	9.695794	8.806004	9.324245
GLY:CA	5001	7.629218	7.619403	6.903381	7.913664
GLY:H	5001	1.258088	1.223637	1.390018	1.093073
GLY:HA2	5001	0.869591	0.900767	0.966352	0.910875
GLY:HA3	5001	1.036021	1.011631	1.128135	1.100055
GLY:N	5001	14.084679	14.473323	13.829781	13.141018
GLY:O	5001	33.650002	37.815316	31.144818	29.402827
PRO:C	5001	9.972661	9.409404	9.904196	8.791403
PRO:CA	5001	9.818658	9.666851	10.161987	9.126656
PRO:CB	5001	10.141756	9.972537	9.852757	9.954013

IUPAC	<i>n</i>	Structure	AIMNet	MOPAC	Joint
PRO:CD	5001	11.610256	12.978621	12.157309	11.348720
PRO:CG	5001	10.237323	10.135362	10.219102	9.843813
PRO:HA	5001	1.155547	1.193332	1.146737	1.087744
PRO:HB2	5001	1.138157	1.009513	1.020424	0.926778
PRO:HB3	5001	1.043530	0.983180	0.922154	0.833195
PRO:HD2	5001	1.098896	1.138719	1.026494	1.000573
PRO:HD3	5001	1.229755	1.192935	1.267397	1.098979
PRO:HG2	5001	0.901295	0.957565	1.098673	0.859913
PRO:HG3	5001	0.917605	0.919662	0.989420	0.826455
PRO:N	5001	18.659527	18.974777	19.051889	19.864258
PRO:O	5001	32.832474	35.990067	31.724040	31.568168
THR:C	10002	10.047178	9.662526	8.136360	8.406369
THR:CA	10002	10.775917	10.300628	9.496558	10.345223
THR:CB	10002	16.268528	16.867153	15.997665	12.269751
THR:CG2	10002	9.858446	9.542477	9.326832	9.389074
THR:H	10002	1.534133	1.576014	1.635928	1.360806
THR:HA	10002	1.048125	0.986620	1.034577	0.954248
THR:HB	10002	1.461253	1.465302	1.326645	1.216236
THR:HG1	10002	2.262570	2.973507	2.580731	2.458535
THR:HG21	10002	1.267965	1.041035	1.166926	1.008507
THR:HG22	10002	1.311266	1.189661	1.188162	0.999910
THR:HG23	10002	1.555688	1.133621	1.179792	1.152854
THR:N	10002	15.148512	15.351028	14.313190	13.952038
THR:O	10002	35.443378	34.903729	34.952179	31.399640
THR:OG1	10002	33.686337	33.089823	31.751256	32.219117
TRP:C	5001	12.470200	10.127776	10.072598	9.038493
TRP:CA	5001	10.294125	9.035012	8.214443	9.386050
TRP:CB	5001	9.811281	8.204229	7.849733	8.885565
TRP:CD1	5001	19.716040	16.671943	23.935737	15.940665
TRP:CD2	5001	13.999192	16.155434	14.391220	15.810193
TRP:CE2	5001	19.135703	18.381009	22.639230	16.839862
TRP:CE3	5001	18.509482	17.335371	15.475239	15.699954
TRP:CG	5001	14.589820	15.532650	15.779372	15.556032
TRP:CH2	5001	14.796501	14.465498	14.675557	15.931532
TRP:CZ2	5001	13.616619	13.815250	13.318121	12.664800
TRP:CZ3	5001	15.307316	15.441139	16.005980	13.220470
TRP:H	5001	1.142863	1.070461	1.051648	0.964989
TRP:HA	5001	1.218074	1.116128	1.180202	1.162756
TRP:HB2	5001	1.172760	1.336974	0.855904	1.042693
TRP:HB3	5001	1.271356	1.150236	1.037090	0.981833
TRP:HD1	5001	1.366133	1.402980	1.715969	1.324360
TRP:HE1	5001	1.397151	1.036691	1.903397	1.355777
TRP:HE3	5001	1.676492	1.307492	1.743373	1.516503
TRP:HH2	5001	1.157167	1.039603	1.229788	1.093568
TRP:HZ2	5001	1.015362	0.961305	1.188825	1.099693
TRP:HZ3	5001	1.463455	1.071483	1.187744	1.081225
TRP:N	5001	14.386958	14.988976	14.462036	14.137377
TRP:NE1	5001	18.610939	17.762989	18.980056	19.967257
TRP:O	5001	42.455702	36.916437	32.479154	29.306355
TYR:C	15003	12.457680	12.056736	10.429094	11.227990
TYR:CA	15003	10.888171	11.137363	9.622838	9.516209
TYR:CB	15003	9.614636	9.698750	8.826912	8.738211
TYR:CD1	15003	14.739615	14.558945	16.144197	12.798025
TYR:CD2	15003	15.103529	15.772316	14.731248	13.877158
TYR:CE1	15003	14.574962	17.177428	17.223299	16.175645
TYR:CE2	15003	14.439364	15.330210	16.500922	16.089975
TYR:CG	15003	14.444501	14.498347	14.848256	13.015169
TYR:CZ	15003	21.703940	18.061143	21.177358	16.668200
TYR:H	10002	1.395215	1.152406	1.311501	1.457266
TYR:HA	15003	1.284315	1.145817	1.037647	1.096442
TYR:HB2	15003	1.146051	1.054249	1.211302	1.089625
TYR:HB3	15003	1.372052	1.203895	1.190976	1.253739
TYR:HD1	15003	1.234860	1.140864	1.123008	1.077811
TYR:HD2	15003	1.303810	1.115729	1.134486	1.113919

IUPAC	<i>n</i>	Structure	AIMNet	MOPAC	Joint
TYR:HE1	15003	1.129002	1.060732	1.190021	1.103282
TYR:HE2	15003	1.260184	1.122393	1.354319	1.311139
TYR:HH	15003	2.035443	2.509380	2.024664	2.238715
TYR:N	15003	17.008773	18.851902	16.657377	17.673016
TYR:O	15003	41.418863	41.765634	38.314349	33.949052
TYR:OH	15003	39.268031	30.895641	33.572062	35.421231

7 Shift designation vocabulary

7.1 Shift IUPAC support; exact-assignment rows

IUPAC	Static present	MD present	Static fit 151	Static sel. 22	Static all 173	Static 386	Static 1,742	MD fit 151	MD sel. 22	MD all 173	Matched 45
ALA:C	yes	yes	585	94	679	2375	8064	585	94	679	134
ALA:CA	yes	yes	833	135	968	3192	11288	833	135	968	174
ALA:CB	yes	yes	805	131	936	2968	10010	805	131	936	164
ALA:H	yes	yes	847	158	1005	3295	13994	847	158	1005	184
ALA:H1	yes	yes	0	0	0	0	0	0	0	0	0
ALA:H2	yes	yes	0	0	0	0	0	0	0	0	0
ALA:H3	yes	yes	0	0	0	0	0	0	0	0	0
ALA:HA	yes	yes	853	160	1013	3098	10461	853	160	1013	182
ALA:HB1	yes	yes	0	0	0	0	0	0	0	0	0
ALA:HB2	yes	yes	0	0	0	0	0	0	0	0	0
ALA:HB3	yes	yes	0	0	0	0	0	0	0	0	0
ALA:N	yes	yes	803	132	935	3072	12046	803	132	935	171
ARG:C	yes	yes	402	74	476	1580	5059	402	74	476	140
ARG:CA	yes	yes	623	97	720	2071	7048	623	97	720	184
ARG:CB	yes	yes	592	94	686	1893	6118	592	94	686	172
ARG:CD	yes	yes	559	89	648	1617	4201	559	89	648	173
ARG:CG	yes	yes	536	88	624	1571	4121	536	88	624	167
ARG:CZ	yes	yes	2	4	6	6	84	2	4	6	0
ARG:H	yes	yes	647	106	753	2150	8923	647	106	753	191
ARG:H1	yes	yes	0	0	0	0	0	0	0	0	0
ARG:H2	yes	yes	0	0	0	0	0	0	0	0	0
ARG:H3	yes	yes	0	0	0	0	0	0	0	0	0
ARG:HA	yes	yes	647	107	754	2003	6874	647	107	754	187
ARG:HB2	yes	yes	254	41	295	680	2462	254	41	295	45
ARG:HB3	yes	yes	234	41	275	631	2274	234	41	275	45
ARG:HD2	yes	yes	285	58	343	830	2850	285	58	343	75
ARG:HD3	yes	yes	274	56	330	799	2613	274	56	330	74
ARG:HE	yes	yes	168	47	215	522	1932	168	47	215	53
ARG:HG2	yes	yes	211	40	251	654	2482	211	40	251	55
ARG:HG3	yes	yes	203	39	242	612	2290	203	39	242	55
ARG:HH11	yes	yes	13	0	13	8	85	13	0	13	0
ARG:HH12	yes	yes	13	0	13	8	77	13	0	13	0
ARG:HH21	yes	yes	9	0	9	10	73	9	0	9	0
ARG:HH22	yes	yes	9	0	9	8	66	9	0	9	0
ARG:N	yes	yes	618	99	717	1960	7420	618	99	717	169
ARG:NE	yes	yes	149	32	181	438	1157	149	32	181	44
ARG:NH1	yes	yes	7	0	7	3	20	7	0	7	0
ARG:NH2	yes	yes	7	0	7	3	14	7	0	7	0

IUPAC	Static present	MD present	Static fit 151	Static sel. 22	Static all 173	Static 386	Static 1,742	MD fit 151	MD sel. 22	MD all 173	Matched 45
ASN:C	yes	yes	265	39	304	1268	4356	265	39	304	72
ASN:CA	yes	yes	436	71	507	1712	6175	436	71	507	114
ASN:CB	yes	yes	416	67	483	1608	5514	416	67	483	109
ASN:CG	yes	yes	34	8	42	95	360	34	8	42	2
ASN:H	yes	yes	452	78	530	1787	7823	452	78	530	119
ASN:H1	yes	yes	0	0	0	0	0	0	0	0	0
ASN:H2	yes	yes	0	0	0	0	0	0	0	0	0
ASN:H3	yes	yes	0	0	0	0	0	0	0	0	0
ASN:HA	yes	yes	454	79	533	1686	6058	454	79	533	120
ASN:HB2	yes	yes	171	32	203	504	1875	171	32	203	22
ASN:HB3	yes	yes	165	32	197	477	1739	165	32	197	22
ASN:HD21	yes	yes	114	23	137	350	1273	114	23	137	16
ASN:HD22	yes	yes	114	23	137	349	1261	114	23	137	16
ASN:N	yes	yes	430	66	496	1657	6482	430	66	496	111
ASN:ND2	yes	yes	357	68	425	1217	3414	357	68	425	103
ASP:C	yes	yes	466	53	519	1855	6354	466	53	519	123
ASP:CA	yes	yes	711	89	800	2459	8884	711	89	800	171
ASP:CB	yes	yes	671	83	754	2292	7923	671	83	754	162
ASP:CG	yes	yes	3	0	3	25	149	3	0	3	0
ASP:H	yes	yes	740	98	838	2549	11056	740	98	838	182
ASP:H1	yes	yes	0	0	0	0	0	0	0	0	0
ASP:H2	yes	yes	0	0	0	0	0	0	0	0	0
ASP:H3	yes	yes	0	0	0	0	0	0	0	0	0
ASP:HA	yes	yes	736	97	833	2405	8356	736	97	833	177
ASP:HB2	yes	yes	283	28	311	684	2605	283	28	311	38
ASP:HB3	yes	yes	271	28	299	660	2433	271	28	299	38
ASP:HD2	no	yes	0	0	0	0	0	0	0	0	0
ASP:N	yes	yes	709	91	800	2388	9546	709	91	800	166
CYS:C	yes	yes	64	7	71	500	1757	64	7	71	15
CYS:CA	yes	yes	119	14	133	673	2412	119	14	133	27
CYS:CB	yes	yes	113	12	125	610	2115	113	12	125	26
CYS:H	yes	yes	146	20	166	750	4314	146	20	166	27
CYS:H1	yes	no	0	0	0	0	0	0	0	0	0
CYS:H2	yes	no	0	0	0	0	0	0	0	0	0
CYS:H3	yes	no	0	0	0	0	0	0	0	0	0
CYS:HA	yes	yes	134	20	154	710	3755	134	20	154	27
CYS:HB2	yes	yes	53	9	62	242	1291	53	9	62	10
CYS:HB3	yes	yes	52	9	61	239	1261	52	9	61	10
CYS:HG	yes	yes	0	0	0	2	35	0	0	0	0
CYS:N	yes	yes	132	14	146	643	2832	132	14	146	24
GLN:C	yes	yes	322	61	383	1381	4299	322	61	383	131
GLN:CA	yes	yes	523	83	606	1854	6040	523	83	606	170

IUPAC	Static present	MD present	Static fit 151	Static sel. 22	Static all 173	Static 386	Static 1,742	MD fit 151	MD sel. 22	MD all 173	Matched 45
GLN:CB	yes	yes	508	80	588	1730	5391	508	80	588	166
GLN:CD	yes	yes	22	19	41	96	344	22	19	41	0
GLN:CG	yes	yes	466	83	549	1532	3874	466	83	549	160
GLN:H	yes	yes	546	87	633	1940	7484	546	87	633	181
GLN:H1	yes	yes	0	0	0	0	0	0	0	0	0
GLN:H2	yes	yes	0	0	0	0	0	0	0	0	0
GLN:H3	yes	yes	0	0	0	0	0	0	0	0	0
GLN:HA	yes	yes	539	89	628	1874	5877	539	89	628	172
GLN:HB2	yes	yes	232	34	266	642	2115	232	34	266	51
GLN:HB3	yes	yes	223	34	257	597	1929	223	34	257	51
GLN:HE21	yes	yes	123	23	146	337	1144	123	23	146	26
GLN:HE22	yes	yes	124	23	147	336	1138	124	23	147	26
GLN:HG2	yes	yes	243	29	272	708	2277	243	29	272	64
GLN:HG3	yes	yes	230	28	258	676	2083	230	28	258	63
GLN:N	yes	yes	522	86	608	1769	6351	522	86	608	165
GLN:NE2	yes	yes	405	78	483	1242	3213	405	78	483	143
GLU:C	yes	yes	610	108	718	2678	8429	610	108	718	165
GLU:CA	yes	yes	925	161	1086	3444	11487	925	161	1086	207
GLU:CB	yes	yes	888	157	1045	3177	10148	888	157	1045	196
GLU:CD	yes	yes	5	0	5	23	159	5	0	5	1
GLU:CG	yes	yes	810	160	970	2782	7179	810	160	970	194
GLU:H	yes	yes	992	170	1162	3583	14010	992	170	1162	232
GLU:H1	yes	yes	0	0	0	0	0	0	0	0	0
GLU:H2	yes	yes	0	0	0	0	0	0	0	0	0
GLU:H3	yes	yes	0	0	0	0	0	0	0	0	0
GLU:HA	yes	yes	969	171	1140	3360	10647	969	171	1140	222
GLU:HB2	yes	yes	387	77	464	1156	3898	387	77	464	60
GLU:HB3	yes	yes	359	77	436	1101	3591	359	77	436	60
GLU:HE2	no	yes	0	0	0	0	0	0	0	0	0
GLU:HG2	yes	yes	345	55	400	1178	4053	345	55	400	60
GLU:HG3	yes	yes	313	50	363	1135	3746	313	50	363	59
GLU:N	yes	yes	936	163	1099	3322	12177	936	163	1099	215
GLY:C	yes	yes	523	71	594	2217	7570	523	71	594	177
GLY:CA	yes	yes	829	100	929	2990	10679	829	100	929	238
GLY:H	yes	yes	884	120	1004	3207	13687	884	120	1004	260
GLY:H1	yes	yes	0	0	0	0	0	0	0	0	0
GLY:H2	yes	yes	0	0	0	0	0	0	0	0	0
GLY:H3	yes	yes	0	0	0	0	0	0	0	0	0
GLY:HA2	yes	yes	284	46	330	959	3560	284	46	330	75
GLY:HA3	yes	yes	268	44	312	921	3340	268	44	312	75
GLY:N	yes	yes	806	97	903	2805	11293	806	97	903	225
HIS:C	yes	yes	143	19	162	702	2332	143	19	162	39

IUPAC	Static present	MD present	Static fit 151	Static sel. 22	Static all 173	Static 386	Static 1,742	MD fit 151	MD sel. 22	MD all 173	Matched 45
HIS:CA	yes	yes	231	25	256	910	3301	231	25	256	58
HIS:CB	yes	yes	221	25	246	866	2926	221	25	246	55
HIS:CD2	yes	yes	103	20	123	493	1204	103	20	123	38
HIS:CE1	yes	yes	59	13	72	381	908	59	13	72	22
HIS:CG	yes	yes	1	0	1	2	24	1	0	1	0
HIS:H	yes	yes	243	26	269	925	4058	243	26	269	57
HIS:H1	yes	no	0	0	0	0	0	0	0	0	0
HIS:H2	yes	no	0	0	0	0	0	0	0	0	0
HIS:H3	yes	no	0	0	0	0	0	0	0	0	0
HIS:HA	yes	yes	243	27	270	894	3136	243	27	270	59
HIS:HB2	yes	yes	98	7	105	263	1006	98	7	105	12
HIS:HB3	yes	yes	91	7	98	259	970	91	7	98	12
HIS:HD1	no	yes	0	0	0	0	0	3	0	3	0
HIS:HD2	yes	yes	141	25	166	610	1896	141	25	166	45
HIS:HE1	yes	yes	89	17	106	463	1512	89	17	106	24
HIS:HE2	yes	yes	2	1	3	6	60	2	1	3	2
HIS:N	yes	yes	219	23	242	832	3315	219	23	242	53
HIS:ND1	yes	yes	4	0	4	15	118	4	0	4	1
HIS:NE2	yes	yes	3	1	4	18	123	3	1	4	0
ILE:C	yes	yes	400	62	462	1767	5817	400	62	462	116
ILE:CA	yes	yes	613	83	696	2289	8113	613	83	696	157
ILE:CB	yes	yes	595	80	675	2124	7186	595	80	675	153
ILE:CD1	yes	yes	559	83	642	1957	5439	559	83	642	146
ILE:CG1	yes	yes	524	83	607	1743	4646	524	83	607	142
ILE:CG2	yes	yes	558	84	642	1913	5165	558	84	642	144
ILE:H	yes	yes	652	88	740	2409	9927	652	88	740	179
ILE:H1	yes	no	0	0	0	0	0	0	0	0	0
ILE:H2	yes	no	0	0	0	0	0	0	0	0	0
ILE:H3	yes	no	0	0	0	0	0	0	0	0	0
ILE:HA	yes	yes	629	88	717	2256	7513	629	88	717	165
ILE:HB	yes	yes	626	88	714	2151	6911	626	88	714	164
ILE:HD11	yes	yes	0	0	0	0	0	0	0	0	0
ILE:HD12	yes	yes	0	0	0	0	0	0	0	0	0
ILE:HD13	yes	yes	0	0	0	0	0	0	0	0	0
ILE:HG12	yes	yes	199	30	229	617	2380	199	30	229	22
ILE:HG13	yes	yes	193	29	222	586	2172	193	29	222	22
ILE:HG21	yes	yes	0	0	0	0	0	0	0	0	0
ILE:HG22	yes	yes	0	0	0	0	0	0	0	0	0
ILE:HG23	yes	yes	0	0	0	0	0	0	0	0	0
ILE:N	yes	yes	615	87	702	2244	8566	615	87	702	163
LEU:C	yes	yes	714	109	823	3026	9551	714	109	823	225
LEU:CA	yes	yes	1051	150	1201	3961	13124	1051	150	1201	296

IUPAC	Static present	MD present	Static fit 151	Static sel. 22	Static all 173	Static 386	Static 1,742	MD fit 151	MD sel. 22	MD all 173	Matched 45
LEU:CB	yes	yes	1016	145	1161	3712	11616	1016	145	1161	285
LEU:CD1	yes	yes	0	0	0	0	0	0	0	0	0
LEU:CD2	yes	yes	0	0	0	0	0	0	0	0	0
LEU:CG	yes	yes	927	154	1081	3058	7800	927	154	1081	269
LEU:H	yes	yes	1124	164	1288	4153	16076	1124	164	1288	327
LEU:H1	yes	yes	0	0	0	0	0	0	0	0	0
LEU:H2	yes	yes	0	0	0	0	0	0	0	0	0
LEU:H3	yes	yes	0	0	0	0	0	0	0	0	0
LEU:HA	yes	yes	1099	165	1264	3887	12074	1099	165	1264	308
LEU:HB2	yes	yes	450	55	505	1148	4036	450	55	505	66
LEU:HB3	yes	yes	436	53	489	1103	3809	436	53	489	66
LEU:HD11	yes	yes	0	0	0	0	0	0	0	0	0
LEU:HD12	yes	yes	0	0	0	0	0	0	0	0	0
LEU:HD13	yes	yes	0	0	0	0	0	0	0	0	0
LEU:HD21	yes	yes	0	0	0	0	0	0	0	0	0
LEU:HD22	yes	yes	0	0	0	0	0	0	0	0	0
LEU:HD23	yes	yes	0	0	0	0	0	0	0	0	0
LEU:HG	yes	yes	988	161	1149	3327	9753	988	161	1149	301
LEU:N	yes	yes	1056	149	1205	3846	13752	1056	149	1205	302
LYS:C	yes	yes	584	111	695	2334	7848	584	111	695	163
LYS:CA	yes	yes	895	157	1052	3055	10882	895	157	1052	216
LYS:CB	yes	yes	837	154	991	2818	9566	837	154	991	200
LYS:CD	yes	yes	745	143	888	2302	6105	745	143	888	197
LYS:CE	yes	yes	730	138	868	2269	5909	730	138	868	198
LYS:CG	yes	yes	782	154	936	2413	6434	782	154	936	200
LYS:H	yes	yes	971	174	1145	3184	13689	971	174	1145	254
LYS:H1	yes	yes	0	0	0	0	0	0	0	0	0
LYS:H2	yes	yes	0	0	0	0	0	0	0	0	0
LYS:H3	yes	yes	0	0	0	0	0	0	0	0	0
LYS:HA	yes	yes	946	173	1119	3006	10429	946	173	1119	237
LYS:HB2	yes	yes	428	80	508	973	3862	428	80	508	75
LYS:HB3	yes	yes	404	78	482	930	3628	404	78	482	75
LYS:HD2	yes	yes	471	104	575	1344	4504	471	104	575	106
LYS:HD3	yes	yes	433	102	535	1288	4145	433	102	535	104
LYS:HE2	yes	yes	502	104	606	1502	4819	502	104	606	122
LYS:HE3	yes	yes	452	101	553	1437	4372	452	101	553	121
LYS:HG2	yes	yes	329	63	392	929	3657	329	63	392	58
LYS:HG3	yes	yes	302	63	365	906	3418	302	63	365	57
LYS:HZ1	yes	yes	0	0	0	0	0	0	0	0	0
LYS:HZ2	yes	yes	0	0	0	0	0	0	0	0	0
LYS:HZ3	yes	yes	0	0	0	0	0	0	0	0	0
LYS:N	yes	yes	904	162	1066	2971	11582	904	162	1066	221

IUPAC	Static present	MD present	Static fit 151	Static sel. 22	Static all 173	Static 386	Static 1,742	MD fit 151	MD sel. 22	MD all 173	Matched 45
LYS:NZ	yes	yes	0	0	0	3	26	0	0	0	0
MET:C	yes	yes	147	23	170	698	2194	147	23	170	35
MET:CA	yes	yes	243	36	279	925	3172	243	36	279	50
MET:CB	yes	yes	229	35	264	868	2771	229	35	264	44
MET:CE	yes	yes	134	28	162	541	1247	134	28	162	41
MET:CG	yes	yes	197	35	232	705	1777	197	35	232	40
MET:H	yes	yes	226	35	261	896	3638	226	35	261	46
MET:H1	yes	yes	0	0	0	0	0	0	0	0	0
MET:H2	yes	yes	0	0	0	0	0	0	0	0	0
MET:H3	yes	yes	0	0	0	0	0	0	0	0	0
MET:HA	yes	yes	251	38	289	896	2836	251	38	289	54
MET:HB2	yes	yes	115	15	130	290	970	115	15	130	16
MET:HB3	yes	yes	110	15	125	270	883	110	15	125	16
MET:HE1	yes	yes	0	0	0	0	0	0	0	0	0
MET:HE2	yes	yes	0	0	0	0	0	0	0	0	0
MET:HE3	yes	yes	0	0	0	0	0	0	0	0	0
MET:HG2	yes	yes	81	9	90	244	886	81	9	90	14
MET:HG3	yes	yes	78	9	87	232	821	78	9	87	14
MET:N	yes	yes	218	33	251	852	3183	218	33	251	41
PHE:C	yes	yes	275	43	318	1174	4057	275	43	318	81
PHE:CA	yes	yes	389	66	455	1539	5724	389	66	455	97
PHE:CB	yes	yes	377	63	440	1447	5131	377	63	440	93
PHE:CD1	yes	yes	196	47	243	859	1935	196	47	243	73
PHE:CD2	yes	yes	150	33	183	726	1510	150	33	183	73
PHE:CE1	yes	yes	180	37	217	778	1698	180	37	217	72
PHE:CE2	yes	yes	138	26	164	671	1325	138	26	164	72
PHE:CG	yes	yes	0	0	0	0	50	0	0	0	0
PHE:CZ	yes	yes	186	29	215	727	1540	186	29	215	77
PHE:H	yes	yes	407	72	479	1621	7024	407	72	479	105
PHE:H1	yes	yes	0	0	0	0	0	0	0	0	0
PHE:H2	yes	yes	0	0	0	0	0	0	0	0	0
PHE:H3	yes	yes	0	0	0	0	0	0	0	0	0
PHE:HA	yes	yes	403	72	475	1545	5315	403	72	475	103
PHE:HB2	yes	yes	148	24	172	399	1633	148	24	172	17
PHE:HB3	yes	yes	140	24	164	393	1589	140	24	164	17
PHE:HD1	yes	yes	275	68	343	1106	3227	275	68	343	86
PHE:HD2	yes	yes	262	62	324	1036	2908	262	62	324	87
PHE:HE1	yes	yes	255	58	313	1009	2918	255	58	313	83
PHE:HE2	yes	yes	239	52	291	935	2635	239	52	291	84
PHE:HZ	yes	yes	251	48	299	924	2595	251	48	299	87
PHE:N	yes	yes	388	67	455	1497	6029	388	67	455	99
PRO:C	yes	yes	285	46	331	1382	4289	285	46	331	98

IUPAC	Static present	MD present	Static fit 151	Static sel. 22	Static all 173	Static 386	Static 1,742	MD fit 151	MD sel. 22	MD all 173	Matched 45
PRO:CA	yes	yes	466	75	541	1973	6202	466	75	541	139
PRO:CB	yes	yes	454	69	523	1814	5485	454	69	523	135
PRO:CD	yes	yes	451	79	530	1736	4122	451	79	530	149
PRO:CG	yes	yes	444	74	518	1698	4048	444	74	518	147
PRO:H2	yes	yes	0	0	0	0	0	0	0	0	0
PRO:H3	yes	yes	0	0	0	0	0	0	0	0	0
PRO:HA	yes	yes	482	89	571	1998	5993	482	89	571	158
PRO:HB2	yes	yes	123	29	152	458	1698	123	29	152	21
PRO:HB3	yes	yes	121	29	150	434	1568	121	29	150	21
PRO:HD2	yes	yes	126	32	158	523	1669	126	32	158	35
PRO:HD3	yes	yes	119	31	150	520	1635	119	31	150	34
PRO:HG2	yes	yes	167	39	206	677	2144	167	39	206	44
PRO:HG3	yes	yes	151	39	190	650	1952	151	39	190	43
PRO:N	yes	yes	8	0	8	29	125	8	0	8	0
SER:C	yes	yes	447	76	523	2272	6734	447	76	523	143
SER:CA	yes	yes	706	115	821	2936	9323	706	115	821	190
SER:CB	yes	yes	670	98	768	2669	8101	670	98	768	182
SER:H	yes	yes	722	126	848	3114	11684	722	126	848	198
SER:H1	yes	yes	0	0	0	0	0	0	0	0	0
SER:H2	yes	yes	0	0	0	0	0	0	0	0	0
SER:H3	yes	yes	0	0	0	0	0	0	0	0	0
SER:HA	yes	yes	752	129	881	3044	9305	752	129	881	223
SER:HB2	yes	yes	368	61	429	1273	3670	368	61	429	85
SER:HB3	yes	yes	338	52	390	1212	3406	338	52	390	85
SER:HG	yes	yes	10	0	10	47	147	10	0	10	3
SER:N	yes	yes	651	112	763	2717	9527	651	112	763	157
THR:C	yes	yes	383	47	430	1754	5782	383	47	430	116
THR:CA	yes	yes	628	76	704	2400	8275	628	76	704	166
THR:CB	yes	yes	599	73	672	2225	7252	599	73	672	160
THR:CG2	yes	yes	569	78	647	2036	5399	569	78	647	155
THR:H	yes	yes	653	82	735	2538	10560	653	82	735	172
THR:H1	yes	yes	0	0	0	0	0	0	0	0	0
THR:H2	yes	yes	0	0	0	0	0	0	0	0	0
THR:H3	yes	yes	0	0	0	0	0	0	0	0	0
THR:HA	yes	yes	645	83	728	2417	8146	645	83	728	166
THR:HB	yes	yes	629	83	712	2285	7398	629	83	712	166
THR:HG1	yes	yes	19	4	23	48	204	19	4	23	2
THR:HG21	yes	yes	0	0	0	0	0	0	0	0	0
THR:HG22	yes	yes	0	0	0	0	0	0	0	0	0
THR:HG23	yes	yes	0	0	0	0	0	0	0	0	0
THR:N	yes	yes	615	72	687	2323	8859	615	72	687	157
TRP:C	yes	yes	86	14	100	439	1408	86	14	100	37

IUPAC	Static present	MD present	Static fit 151	Static sel. 22	Static all 173	Static 386	Static 1,742	MD fit 151	MD sel. 22	MD all 173	Matched 45
TRP:CA	yes	yes	128	19	147	560	1908	128	19	147	48
TRP:CB	yes	yes	126	19	145	520	1696	126	19	145	48
TRP:CD1	yes	yes	93	14	107	399	882	93	14	107	43
TRP:CD2	yes	yes	2	0	2	1	9	2	0	2	2
TRP:CE2	yes	yes	0	0	0	1	19	0	0	0	0
TRP:CE3	yes	yes	90	14	104	370	775	90	14	104	43
TRP:CG	yes	yes	1	0	1	1	18	1	0	1	1
TRP:CH2	yes	yes	92	14	106	391	849	92	14	106	43
TRP:CZ2	yes	yes	93	14	107	392	853	93	14	107	43
TRP:CZ3	yes	yes	89	14	103	372	773	89	14	103	43
TRP:H	yes	yes	140	22	162	590	2490	140	22	162	50
TRP:HA	yes	yes	138	22	160	569	1891	138	22	160	50
TRP:HB2	yes	yes	41	4	45	158	573	41	4	45	12
TRP:HB3	yes	yes	39	4	43	149	552	39	4	43	12
TRP:HD1	yes	yes	127	19	146	495	1541	127	19	146	49
TRP:HE1	yes	yes	127	19	146	503	1621	127	19	146	49
TRP:HE3	yes	yes	117	19	136	455	1335	117	19	136	49
TRP:HH2	yes	yes	119	20	139	463	1381	119	20	139	49
TRP:HZ2	yes	yes	118	19	137	476	1412	118	19	137	48
TRP:HZ3	yes	yes	104	15	119	429	1270	104	15	119	44
TRP:N	yes	yes	133	21	154	540	2044	133	21	154	47
TRP:NE1	yes	yes	127	20	147	483	1348	127	20	147	50
TYR:C	yes	yes	207	34	241	992	3360	207	34	241	70
TYR:CA	yes	yes	353	54	407	1302	4738	353	54	407	91
TYR:CB	yes	yes	335	51	386	1226	4147	335	51	386	88
TYR:CD1	yes	yes	186	47	233	746	1680	186	47	233	61
TYR:CD2	yes	yes	135	35	170	621	1284	135	35	170	60
TYR:CE1	yes	yes	184	48	232	748	1663	184	48	232	60
TYR:CE2	yes	yes	135	35	170	625	1277	135	35	170	59
TYR:CG	yes	yes	4	0	4	7	58	4	0	4	2
TYR:CZ	yes	yes	0	0	0	6	35	0	0	0	0
TYR:H	yes	yes	374	63	437	1363	6133	374	63	437	92
TYR:H1	yes	no	0	0	0	0	0	0	0	0	0
TYR:H2	yes	no	0	0	0	0	0	0	0	0	0
TYR:H3	yes	no	0	0	0	0	0	0	0	0	0
TYR:HA	yes	yes	366	64	430	1302	4660	366	64	430	91
TYR:HB2	yes	yes	113	20	133	343	1484	113	20	133	18
TYR:HB3	yes	yes	108	20	128	330	1448	108	20	128	18
TYR:HD1	yes	yes	268	59	327	927	2973	268	59	327	79
TYR:HD2	yes	yes	246	56	302	860	2673	246	56	302	75
TYR:HE1	yes	yes	267	58	325	906	2869	267	58	325	79
TYR:HE2	yes	yes	244	56	300	830	2561	244	56	300	76

IUPAC	Static present	MD present	Static fit 151	Static sel. 22	Static all 173	Static 386	Static 1,742	MD fit 151	MD sel. 22	MD all 173	Matched 45
TYR:HH	yes	yes	5	0	5	21	78	5	0	5	2
TYR:N	yes	yes	360	55	415	1264	5051	360	55	415	87
VAL:C	yes	yes	522	64	586	2444	7662	522	64	586	127
VAL:CA	yes	yes	803	100	903	3189	10715	803	100	903	169
VAL:CB	yes	yes	776	96	872	2967	9469	776	96	872	160
VAL:CG1	yes	yes	0	0	0	0	0	0	0	0	0
VAL:CG2	yes	yes	0	0	0	0	0	0	0	0	0
VAL:H	yes	yes	848	107	955	3341	13227	848	107	955	182
VAL:H1	yes	yes	0	0	0	0	0	0	0	0	0
VAL:H2	yes	yes	0	0	0	0	0	0	0	0	0
VAL:H3	yes	yes	0	0	0	0	0	0	0	0	0
VAL:HA	yes	yes	848	108	956	3153	9999	848	108	956	174
VAL:HB	yes	yes	839	108	947	3034	9282	839	108	947	174
VAL:HG11	yes	yes	0	0	0	0	0	0	0	0	0
VAL:HG12	yes	yes	0	0	0	0	0	0	0	0	0
VAL:HG13	yes	yes	0	0	0	0	0	0	0	0	0
VAL:HG21	yes	yes	0	0	0	0	0	0	0	0	0
VAL:HG22	yes	yes	0	0	0	0	0	0	0	0	0
VAL:HG23	yes	yes	0	0	0	0	0	0	0	0	0
VAL:N	yes	yes	796	104	900	3118	11458	796	104	900	165

8 Static-shift IUPAC results

8.1 Development fitting 151: exact-assignment RMSE (ppm)

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ALA:C	585	101	1.093017	1.140381	1.147060	1.132345	1.169335	1.131893
ALA:CA	833	142	0.668241	0.613787	0.735428	0.679366	0.689829	0.606075
ALA:CB	805	142	0.670268	0.610646	0.702920	0.608780	0.716708	0.614389
ALA:H	847	150	0.318026	0.294875	0.324042	0.327777	0.329922	0.272669
ALA:HA	853	147	0.217522	0.176298	0.189313	0.185470	0.193795	0.169599
ALA:N	803	147	1.886528	1.703686	2.350541	1.795463	1.878220	1.717799
ARG:C	402	94	0.754230	0.696224	0.790741	0.737319	0.807969	0.682654
ARG:CA	623	137	1.003233	1.050065	1.062384	1.037625	1.048452	1.043739
ARG:CB	592	135	0.615230	0.610525	0.698432	0.622286	0.692313	0.598492
ARG:CD	559	124	0.398362	0.383479	0.391837	0.386206	0.422334	0.410329
ARG:CG	536	122	0.718823	0.703495	0.709436	0.678071	0.742747	0.699957
ARG:CZ	2	1	0.122975	0.140839	0.088424	0.112861	0.167543	0.225964
ARG:H	647	143	0.283437	0.274790	0.297675	0.307024	0.313450	0.239705
ARG:HA	647	140	0.184477	0.172222	0.164011	0.170362	0.174652	0.153369

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ARG:HB2	254	87	0.184356	0.202120	0.185793	0.192906	0.191562	0.188845
ARG:HB3	234	84	0.190968	0.187547	0.198929	0.198492	0.194089	0.178007
ARG:HD2	285	93	0.132765	0.139435	0.137533	0.137216	0.140390	0.137025
ARG:HD3	274	92	0.188966	0.186894	0.175519	0.194488	0.171788	0.147545
ARG:HE	168	68	0.683060	0.744830	0.634912	0.701475	0.729028	0.713728
ARG:HG2	211	81	0.179101	0.186182	0.185668	0.178608	0.174419	0.171161
ARG:HG3	203	78	0.198807	0.194415	0.188929	0.184765	0.191733	0.181890
ARG:HH11	13	5	0.217534	0.224633	0.232436	0.210776	0.235456	0.222582
ARG:HH12	13	5	0.222600	0.228974	0.211258	0.185053	0.213061	0.226987
ARG:HH21	9	4	0.145844	0.198719	0.170793	0.198422	0.180337	0.162778
ARG:HH22	9	4	0.168289	0.146042	0.151296	0.178840	0.188036	0.141584
ARG:N	618	142	1.831709	1.585311	2.308033	2.138713	2.395131	1.821461
ARG:NE	149	57	8.029557	8.638616	12.326036	7.794780	9.267148	10.718075
ARG:NH1	7	3	2.900352	1.836768	9.969579	2.984750	6.556022	9.583739
ARG:NH2	7	3	1.528459	0.893587	9.938040	2.522832	7.748833	9.222591
ASN:C	265	93	0.852394	0.834779	0.889706	0.854993	0.921811	0.834569
ASN:CA	436	131	0.711803	0.685884	0.729332	0.729357	0.749095	0.679346
ASN:CB	416	130	0.625342	0.722708	0.783293	0.710459	0.740539	0.682443
ASN:CG	34	12	0.847220	0.848880	0.838359	1.162874	1.055370	1.141146
ASN:H	452	139	0.433959	0.382341	0.388910	0.383892	0.402379	0.331864
ASN:HA	454	138	0.172730	0.152929	0.162012	0.156853	0.172903	0.147670
ASN:HB2	171	68	0.265718	0.231358	0.260647	0.238990	0.232998	0.211190
ASN:HB3	165	66	0.210479	0.175764	0.206328	0.206919	0.211856	0.176254
ASN:HD21	114	42	0.470074	0.484808	0.475603	0.471615	0.454034	0.446608
ASN:HD22	114	42	0.350446	0.379493	0.354204	0.379699	0.358862	0.363849
ASN:N	430	134	2.113296	1.780645	2.563303	2.120164	2.059158	1.895921
ASN:ND2	357	113	1.644062	1.506445	1.776458	1.340163	1.693403	1.595330
ASP:C	466	98	0.885486	0.846585	0.934868	0.867631	0.908432	0.860032
ASP:CA	711	139	0.724188	0.701374	0.750864	0.728944	0.758980	0.688034
ASP:CB	671	138	0.870114	0.892817	0.969774	0.899873	0.989214	0.880339
ASP:CG	3	1	0.185503	0.269719	0.292816	0.327271	0.345668	0.442012
ASP:H	740	148	0.343355	0.361253	0.365366	0.380630	0.379780	0.307481
ASP:HA	736	145	0.162592	0.136644	0.147186	0.150805	0.155553	0.131723
ASP:HB2	283	86	0.216441	0.195016	0.217330	0.199739	0.200902	0.170743
ASP:HB3	271	82	0.194666	0.191268	0.200471	0.199743	0.196773	0.188269
ASP:N	709	145	3.547793	3.361546	3.962023	3.375901	3.562787	3.396777
CYS:C	64	23	0.818802	1.019087	1.038823	0.814934	1.531809	0.721847
CYS:CA	119	40	1.379743	1.351359	1.567481	1.511557	1.541774	1.194529
CYS:CB	113	39	2.498189	2.788523	3.099452	2.654504	3.774222	2.729847
CYS:H	146	44	0.297080	0.286542	0.287448	0.343782	0.495372	0.274656
CYS:HA	134	42	0.193959	0.173807	0.212356	0.170597	0.277674	0.169962
CYS:HB2	53	18	0.341822	0.242028	0.291205	0.265244	0.380973	0.203028

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
CYS:HB3	52	17	0.234957	0.279062	0.208717	0.269446	0.395741	0.258097
CYS:N	132	42	2.188737	1.867927	2.896552	1.841141	3.000171	1.659192
GLN:C	322	90	0.733637	0.747940	0.822303	0.784217	0.818657	0.748582
GLN:CA	523	131	0.634321	0.631161	0.704928	0.682633	0.701835	0.645623
GLN:CB	508	131	0.763617	0.736919	0.835235	0.752411	0.815595	0.721638
GLN:CD	22	7	0.717004	0.736412	0.736338	0.729664	0.788195	0.655128
GLN:CG	466	120	0.725270	0.725651	0.755585	0.720755	0.746813	0.683599
GLN:H	546	138	0.272283	0.293828	0.275964	0.284181	0.292442	0.231539
GLN:HA	539	135	0.184104	0.182903	0.167487	0.175577	0.182498	0.168377
GLN:HB2	232	79	0.191868	0.154681	0.167026	0.142909	0.153047	0.147724
GLN:HB3	223	77	0.185199	0.182224	0.179196	0.169518	0.180776	0.179176
GLN:HE21	123	34	0.486312	0.503635	0.470325	0.484669	0.486371	0.485759
GLN:HE22	124	34	0.372279	0.374448	0.363210	0.375679	0.364414	0.370731
GLN:HG2	243	91	0.175589	0.168782	0.171301	0.158986	0.160601	0.160466
GLN:HG3	230	88	0.183787	0.169478	0.165275	0.168256	0.168233	0.170335
GLN:N	522	136	1.724463	1.463626	2.279504	1.596862	1.719698	1.683582
GLN:NE2	405	109	1.388951	1.237263	1.818249	1.442377	1.594106	1.483597
GLU:C	610	102	0.833545	0.817637	0.851715	0.851873	0.902772	0.835520
GLU:CA	925	143	0.663252	0.676346	0.731310	0.687777	0.703003	0.643806
GLU:CB	888	142	0.680277	0.633932	0.697066	0.659982	0.713879	0.634622
GLU:CD	5	3	17.240544	16.709828	16.544568	17.287405	17.329497	17.186398
GLU:CG	810	130	0.743604	0.772184	0.767506	0.764803	0.777271	0.765578
GLU:H	992	150	0.302232	0.298295	0.321469	0.335032	0.335732	0.253708
GLU:HA	969	147	0.166259	0.148933	0.147782	0.152368	0.158228	0.132231
GLU:HB2	387	103	0.150785	0.148435	0.155083	0.143483	0.149531	0.145459
GLU:HB3	359	101	0.156570	0.150875	0.144766	0.147626	0.153882	0.146124
GLU:HG2	345	94	0.145666	0.136433	0.138729	0.140857	0.141449	0.131854
GLU:HG3	313	91	0.137008	0.142841	0.131574	0.137237	0.138932	0.139874
GLU:N	936	149	1.625671	1.473821	2.105558	1.615566	1.744005	1.642030
GLY:C	523	95	1.264317	1.262944	1.257296	1.197706	1.271504	1.219316
GLY:CA	829	140	0.579815	0.529549	0.609884	0.558745	0.628475	0.558432
GLY:H	884	149	0.359076	0.322951	0.379278	0.360870	0.363830	0.260988
GLY:HA2	284	94	0.227540	0.260382	0.215761	0.236130	0.229258	0.218457
GLY:HA3	268	91	0.236993	0.282668	0.243111	0.235776	0.238849	0.275097
GLY:N	806	144	2.666085	2.620967	3.098178	2.559327	2.755848	2.626751
HIS:C	143	65	0.717812	0.669110	0.790184	0.802425	0.814839	0.689752
HIS:CA	231	99	0.868019	0.856005	0.985812	0.918514	1.040651	0.878504
HIS:CB	221	97	0.918055	0.983309	1.085151	0.859739	1.062458	0.991737
HIS:CD2	103	52	1.572005	1.517506	1.575573	1.172471	1.591278	1.400331
HIS:CE1	59	36	1.243389	1.321813	1.428544	1.475757	1.507051	1.178362
HIS:CG	1	1	0.229462	0.042221	0.150909	0.143326	0.026184	0.290848
HIS:H	243	107	0.378658	0.371622	0.384877	0.379580	0.414810	0.349419

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
HIS:HA	243	105	0.212382	0.191715	0.203953	0.206564	0.202013	0.181628
HIS:HB2	98	49	0.334943	0.311476	0.342769	0.340019	0.330793	0.308681
HIS:HB3	91	46	0.296898	0.303250	0.281222	0.275036	0.286776	0.290346
HIS:HD2	141	73	0.267193	0.246668	0.262171	0.259897	0.255212	0.240102
HIS:HE1	89	53	0.510319	0.503896	0.500465	0.486372	0.505487	0.473433
HIS:HE2	2	1	0.008894	0.038364	0.019604	0.016477	0.011458	0.013094
HIS:N	219	100	2.157995	2.209865	2.711850	2.105848	2.344172	2.331091
HIS:ND1	4	3	50.136301	45.283893	50.337082	47.214714	47.614861	46.680188
HIS:NE2	3	2	2.927232	2.422508	3.414006	2.201943	2.909197	1.239565
ILE:C	400	96	0.843403	0.749502	0.877622	0.807685	0.915947	0.805872
ILE:CA	613	136	0.800418	0.793179	0.834978	0.841698	0.928910	0.829463
ILE:CB	595	135	0.860020	0.831089	0.970534	0.864421	0.930457	0.848132
ILE:CD1	559	129	2.129546	2.134272	2.098824	1.996622	2.132802	2.121161
ILE:CG1	524	122	1.388522	1.366403	1.451973	1.252875	1.440852	1.371013
ILE:CG2	558	126	1.966186	1.887622	1.970462	1.855908	1.942225	1.894552
ILE:H	652	143	0.325203	0.316726	0.336451	0.328322	0.356222	0.278505
ILE:HA	629	140	0.238352	0.200753	0.207202	0.199243	0.205265	0.180737
ILE:HB	626	140	0.211426	0.166471	0.214059	0.186471	0.188738	0.162170
ILE:HG12	199	60	0.411805	0.389091	0.370136	0.383133	0.409205	0.383401
ILE:HG13	193	60	0.392273	0.398677	0.392945	0.402438	0.373477	0.386756
ILE:N	615	140	2.093611	1.983246	2.609288	2.061683	2.157280	2.097116
LEU:C	714	100	0.854451	0.778614	0.936409	0.884240	0.894730	0.799768
LEU:CA	1051	140	0.681871	0.655410	0.760920	0.734160	0.726633	0.661393
LEU:CB	1016	140	0.725338	0.667232	0.863243	0.767466	0.835055	0.657441
LEU:CG	927	130	0.716089	0.688514	0.755246	0.732724	0.714111	0.689471
LEU:H	1124	149	0.329269	0.330973	0.343844	0.333765	0.343488	0.299692
LEU:HA	1099	146	0.186667	0.158175	0.170123	0.166322	0.173591	0.147229
LEU:HB2	450	92	0.292425	0.296658	0.287228	0.288177	0.272586	0.276011
LEU:HB3	436	92	0.298744	0.302790	0.288513	0.280645	0.301043	0.295205
LEU:HG	988	142	0.267597	0.253562	0.275958	0.264793	0.256385	0.247170
LEU:N	1056	146	1.818163	1.713644	2.290113	1.899165	1.867323	1.780127
LYS:C	584	101	0.791559	0.760178	0.826592	0.778556	0.849674	0.776212
LYS:CA	895	141	0.751077	0.725217	0.835108	0.775846	0.773741	0.716886
LYS:CB	837	140	0.822381	0.844080	0.929055	0.848974	0.923540	0.837373
LYS:CD	745	129	0.986596	0.993497	0.969418	0.987656	0.989552	0.992446
LYS:CE	730	127	0.615636	0.618994	0.620376	0.613929	0.621098	0.616304
LYS:CG	782	129	0.814121	0.803754	0.833538	0.793241	0.840266	0.807271
LYS:H	971	149	0.310800	0.294023	0.331589	0.330613	0.310364	0.261242
LYS:HA	946	146	0.196604	0.169125	0.171481	0.167039	0.182712	0.155476
LYS:HB2	428	94	0.181851	0.153985	0.170232	0.152010	0.148172	0.143011
LYS:HB3	404	92	0.164149	0.151046	0.153675	0.151756	0.154862	0.147221
LYS:HD2	471	104	0.161257	0.164599	0.166232	0.165180	0.160687	0.160641

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
LYS:HD3	433	101	0.159469	0.158128	0.160464	0.158222	0.156954	0.153785
LYS:HE2	502	107	0.111351	0.109358	0.107798	0.108420	0.107411	0.104766
LYS:HE3	452	101	0.124625	0.115568	0.119732	0.116288	0.114805	0.111319
LYS:HG2	329	89	0.165171	0.153870	0.153511	0.160858	0.157158	0.146526
LYS:HG3	302	87	0.170499	0.156037	0.157899	0.152932	0.151758	0.150365
LYS:N	904	147	1.878619	1.636303	2.208108	1.744782	1.817096	1.788681
MET:C	147	72	0.944709	0.940933	1.043255	0.999411	1.090463	0.975246
MET:CA	243	106	1.116376	1.076025	1.156313	1.175144	1.175750	1.089249
MET:CB	229	104	1.051207	1.125831	1.182560	1.061494	1.147132	1.101738
MET:CE	134	66	7.947748	7.945837	7.969550	7.979432	7.967386	7.958784
MET:CG	197	88	1.097788	1.077499	1.166377	1.077347	1.139917	1.070881
MET:H	226	105	0.306190	0.282213	0.307940	0.293014	0.314885	0.261570
MET:HA	251	111	0.213353	0.176201	0.182037	0.180364	0.197953	0.158499
MET:HB2	115	55	0.211800	0.195127	0.209900	0.204022	0.193228	0.189424
MET:HB3	110	54	0.296659	0.267804	0.288820	0.285183	0.273372	0.257615
MET:HG2	81	44	0.312769	0.308875	0.312604	0.309580	0.307683	0.294194
MET:HG3	78	42	0.291749	0.334237	0.321716	0.322009	0.301768	0.320074
MET:N	218	99	1.695013	1.408167	1.914651	1.512786	1.431021	1.547756
PHE:C	275	92	0.742011	0.730378	0.862514	0.822674	0.813453	0.771753
PHE:CA	389	131	0.892320	0.882391	0.996498	0.942779	0.986757	0.826134
PHE:CB	377	130	0.876540	0.914972	0.896589	0.798623	0.986016	0.858937
PHE:CD1	196	70	1.701795	1.668994	1.720266	1.617035	1.684478	1.657802
PHE:CD2	150	52	1.846058	1.835627	1.874805	1.873017	1.854286	1.861127
PHE:CE1	180	69	2.189221	2.248585	2.246636	2.223498	2.205411	2.207585
PHE:CE2	138	52	2.378065	2.392705	2.426935	2.404243	2.470598	2.401815
PHE:CZ	186	78	6.155184	6.175521	6.201176	6.097674	6.159934	6.160444
PHE:H	407	139	0.340141	0.336020	0.359162	0.330695	0.355027	0.292185
PHE:HA	403	136	0.255250	0.223094	0.209391	0.213502	0.235805	0.204152
PHE:HB2	148	62	0.265220	0.241083	0.265509	0.241067	0.256183	0.226070
PHE:HB3	140	62	0.317922	0.297456	0.290181	0.291096	0.279666	0.277338
PHE:HD1	275	99	0.205895	0.192140	0.192795	0.200504	0.195712	0.186018
PHE:HD2	262	94	0.225165	0.187128	0.206525	0.200384	0.198178	0.180246
PHE:HE1	255	97	0.233073	0.195013	0.216149	0.203934	0.208158	0.186868
PHE:HE2	239	90	0.228759	0.200219	0.207083	0.204580	0.194304	0.192826
PHE:HZ	251	105	0.240199	0.208495	0.222649	0.219451	0.206273	0.188374
PHE:N	388	134	1.891588	1.732483	2.396664	1.836345	1.979255	1.880638
PRO:C	285	91	0.762294	0.737433	0.817452	0.734955	0.787555	0.744325
PRO:CA	466	131	0.701659	0.706295	0.697896	0.678684	0.693322	0.672557
PRO:CB	454	130	0.541535	0.536476	0.621909	0.582220	0.646057	0.549861
PRO:CD	451	124	1.422019	1.380691	1.405193	1.393739	1.417535	1.367166
PRO:CG	444	121	0.509278	0.527350	0.522827	0.544520	0.536976	0.502489
PRO:HA	482	133	0.195403	0.158309	0.171644	0.161523	0.181842	0.152463

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
PRO:HB2	123	44	0.265005	0.281300	0.297407	0.286889	0.265661	0.257277
PRO:HB3	121	44	0.319048	0.276983	0.292910	0.294380	0.290131	0.296153
PRO:HD2	126	59	0.216966	0.173302	0.188271	0.171611	0.177900	0.166267
PRO:HD3	119	58	0.178176	0.167976	0.192992	0.165836	0.190465	0.175325
PRO:HG2	167	73	0.176138	0.174463	0.178188	0.177241	0.170071	0.168543
PRO:HG3	151	69	0.145538	0.176599	0.141784	0.150028	0.167580	0.167765
PRO:N	8	2	1.265963	0.907809	2.110181	0.887674	1.053150	1.076745
SER:C	447	97	0.792082	0.719518	0.933997	0.841512	0.843094	0.742551
SER:CA	706	139	0.763263	0.749079	0.787386	0.781761	0.803196	0.736678
SER:CB	670	138	0.611309	0.604445	0.703234	0.651783	0.680196	0.580324
SER:H	722	147	0.339112	0.318978	0.324981	0.330641	0.335015	0.274994
SER:HA	752	145	0.184833	0.160353	0.165011	0.167561	0.165706	0.152889
SER:HB2	368	103	0.139834	0.125374	0.130382	0.129704	0.129103	0.117001
SER:HB3	338	99	0.152282	0.148163	0.149363	0.151322	0.155370	0.143180
SER:HG	10	7	0.365148	0.536259	0.261945	0.574833	0.386089	0.502901
SER:N	651	144	1.913817	1.630768	2.412044	1.795837	1.894348	1.828632
THR:C	383	95	0.767772	0.745769	0.855234	0.766841	0.822853	0.734836
THR:CA	628	136	0.763857	0.781654	0.799626	0.769414	0.801274	0.787217
THR:CB	599	136	0.568405	0.538969	0.676307	0.561949	0.688239	0.502486
THR:CG2	569	127	0.716803	0.609458	0.724552	0.662330	0.679453	0.605874
THR:H	653	143	0.330796	0.296576	0.329793	0.318224	0.341081	0.257308
THR:HA	645	140	0.213385	0.174523	0.184959	0.182554	0.186576	0.165709
THR:HB	629	140	0.222756	0.186757	0.214682	0.199821	0.207174	0.181336
THR:HG1	19	16	0.462658	0.513593	0.464894	0.446481	0.549924	0.472392
THR:N	615	141	2.989856	2.825129	3.324662	2.839821	2.969594	2.908237
TRP:C	86	55	0.842767	0.804041	0.946659	0.806440	0.952430	0.684759
TRP:CA	128	77	1.113252	0.945494	1.130303	1.044535	1.100864	0.948022
TRP:CB	126	76	0.681012	0.621522	0.779150	0.632592	0.751783	0.603075
TRP:CD1	93	58	1.012994	0.829848	0.979946	0.684292	1.022354	0.777553
TRP:CD2	2	2	1.715101	1.918856	1.715899	1.841071	1.922753	1.773292
TRP:CE3	90	55	0.952696	0.723813	0.844283	0.786590	0.952990	0.728356
TRP:CG	1	1	0.072304	0.131691	0.232841	0.123924	0.293503	0.016060
TRP:CH2	92	57	0.833560	0.789432	0.875500	0.724769	0.869154	0.772457
TRP:CZ2	93	58	0.601623	0.557430	0.639922	0.484816	0.586460	0.505014
TRP:CZ3	89	55	0.815472	0.780761	0.835969	0.686177	0.905727	0.745859
TRP:H	140	83	0.477889	0.421475	0.409850	0.476331	0.422199	0.335343
TRP:HA	138	83	0.253250	0.179514	0.191971	0.173691	0.193086	0.152060
TRP:HB2	41	30	0.233569	0.232235	0.206201	0.236792	0.223574	0.222132
TRP:HB3	39	28	0.240115	0.246486	0.216246	0.229850	0.210389	0.231389
TRP:HD1	127	77	0.351001	0.270803	0.313766	0.272070	0.290287	0.247243
TRP:HE1	127	77	0.447390	0.397285	0.456252	0.379936	0.444419	0.389789
TRP:HE3	117	71	0.311698	0.228246	0.247333	0.226115	0.224549	0.209022

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
TRP:HH2	119	75	0.210947	0.185730	0.176846	0.177464	0.201966	0.163622
TRP:HZ2	118	72	0.213582	0.157242	0.159178	0.157669	0.160644	0.152913
TRP:HZ3	104	64	0.264866	0.248699	0.232824	0.250844	0.256739	0.249518
TRP:N	133	81	2.519253	2.435078	3.040885	2.287771	2.689919	2.619815
TRP:NE1	127	78	1.414461	1.233852	1.497529	1.149234	1.424250	1.459010
TYR:C	207	87	0.716872	0.676309	0.740380	0.673248	0.748084	0.676307
TYR:CA	353	126	0.764179	0.767559	0.856557	0.786142	0.862443	0.734817
TYR:CB	335	125	0.730591	0.687073	0.839670	0.711826	0.858614	0.635326
TYR:CD1	186	70	0.814417	0.821687	0.800549	0.779663	0.840521	0.765150
TYR:CD2	135	55	0.734721	0.694673	0.790122	0.728183	0.719447	0.744779
TYR:CE1	184	67	0.618400	0.606530	0.647183	0.594014	0.631108	0.599706
TYR:CE2	135	54	0.581016	0.582593	0.588306	0.555178	0.598184	0.574785
TYR:CG	4	2	4.874199	4.802346	4.877777	4.580066	4.734711	4.539660
TYR:H	374	134	0.313814	0.304585	0.330122	0.321430	0.327388	0.262219
TYR:HA	366	130	0.232957	0.202207	0.215743	0.217142	0.225622	0.183565
TYR:HB2	113	60	0.278182	0.300985	0.285321	0.265162	0.295822	0.292061
TYR:HB3	108	58	0.363616	0.358465	0.352659	0.357091	0.346010	0.351573
TYR:HD1	268	96	0.208300	0.179661	0.179364	0.179300	0.180928	0.165486
TYR:HD2	246	89	0.262525	0.226925	0.246909	0.234821	0.240938	0.214784
TYR:HE1	267	96	0.169022	0.149093	0.151413	0.147573	0.146947	0.139717
TYR:HE2	244	89	0.169668	0.157414	0.158926	0.161461	0.157729	0.152690
TYR:HH	5	5	2.067757	2.107582	2.027234	2.052275	2.122731	2.084075
TYR:N	360	129	1.922492	1.491576	2.484994	1.732147	1.867322	1.707745
VAL:C	522	97	0.848263	0.807653	0.942669	1.015346	0.918054	0.852908
VAL:CA	803	137	0.769664	0.749229	1.041591	0.954177	0.827300	0.754979
VAL:CB	776	137	0.617063	0.546402	0.842671	0.683127	0.631270	0.557969
VAL:H	848	145	0.336180	0.325827	0.337574	0.351813	0.356141	0.294714
VAL:HA	848	142	0.298697	0.273132	0.349216	0.348850	0.284206	0.246712
VAL:HB	839	142	0.226223	0.217627	0.251387	0.250760	0.218615	0.201963
VAL:N	796	141	2.065957	1.906962	2.801011	2.499936	2.089053	2.049166

8.2 Development selector 22: exact-assignment RMSE (ppm)

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ALA:C	94	15	1.768177	1.712882	1.756260	1.704722	1.761564	1.682697
ALA:CA	135	20	0.668807	0.475441	0.647987	0.570485	0.577893	0.453527
ALA:CB	131	20	0.806114	0.678963	0.804978	0.910135	0.803365	0.638621
ALA:H	158	22	0.437898	0.377897	0.438105	0.405240	0.411627	0.381554
ALA:HA	160	22	0.193617	0.174442	0.185932	0.203449	0.187146	0.171411
ALA:N	132	21	1.917246	1.853050	2.228179	1.840771	1.889255	1.616179
ARG:C	74	15	0.988281	0.800707	0.861422	0.800230	0.900993	0.829083

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ARG:CA	97	19	1.005036	0.759507	0.982316	0.859948	0.933340	0.768393
ARG:CB	94	19	0.973757	0.976112	1.095355	1.063079	1.112111	1.001804
ARG:CD	89	19	0.458148	0.447187	0.485635	0.458124	0.443789	0.448766
ARG:CG	88	19	0.927759	0.922762	0.904296	0.899134	0.909834	0.929779
ARG:CZ	4	1	1.071540	0.667675	0.688363	0.680952	0.528814	0.645445
ARG:H	106	21	0.423763	0.339319	0.375612	0.338160	0.341332	0.343230
ARG:HA	107	21	0.232772	0.177274	0.196178	0.193642	0.191743	0.164701
ARG:HB2	41	16	0.239820	0.182944	0.226865	0.211164	0.201652	0.201435
ARG:HB3	41	16	0.159779	0.179094	0.157951	0.147259	0.155235	0.165963
ARG:HD2	58	19	0.117584	0.095878	0.115576	0.099659	0.105015	0.102033
ARG:HD3	56	19	0.102563	0.084568	0.091636	0.092924	0.110599	0.082847
ARG:HE	47	13	0.492225	0.405327	0.509565	0.534282	0.361524	0.379835
ARG:HG2	40	15	0.147353	0.120318	0.114539	0.114743	0.103806	0.109168
ARG:HG3	39	15	0.153823	0.090823	0.116011	0.092877	0.112983	0.090572
ARG:N	99	20	2.128980	1.852778	2.334459	2.717788	2.009116	1.896651
ARG:NE	32	8	9.812448	10.476686	11.730918	11.385831	11.234558	11.376978
ASN:C	39	14	0.816033	0.737569	0.770121	0.819972	0.915065	0.809453
ASN:CA	71	20	0.598771	0.606954	0.699134	0.574790	0.677697	0.633153
ASN:CB	67	20	0.830221	0.747077	0.868223	0.900599	0.812013	0.761016
ASN:CG	8	2	1.713201	1.009308	1.612182	0.973041	1.297710	0.769978
ASN:H	78	22	0.555933	0.548476	0.522500	0.541515	0.555690	0.436470
ASN:HA	79	22	0.250290	0.148933	0.185975	0.185628	0.175799	0.144287
ASN:HB2	32	14	0.194045	0.289921	0.260096	0.316602	0.256208	0.288888
ASN:HB3	32	14	0.285533	0.249205	0.252727	0.260941	0.247142	0.237425
ASN:HD21	23	5	0.618260	0.616619	0.579376	0.602629	0.592910	0.602892
ASN:HD22	23	5	0.679011	0.718853	0.696299	0.726362	0.659405	0.680999
ASN:N	66	20	2.241435	1.884099	2.486257	2.187542	2.182965	1.973375
ASN:ND2	68	19	2.537831	1.953636	2.201824	2.180781	2.127126	1.971281
ASP:C	53	15	0.925943	0.796250	0.756427	0.852041	0.803394	0.735671
ASP:CA	89	20	0.800276	0.714011	0.749214	0.811563	0.757415	0.701594
ASP:CB	83	20	0.772224	0.778362	0.791318	0.790971	0.855905	0.776213
ASP:H	98	22	0.455769	0.379417	0.452703	0.437620	0.431865	0.371766
ASP:HA	97	22	0.159568	0.140504	0.166466	0.148501	0.169641	0.126794
ASP:HB2	28	12	0.178061	0.144865	0.156837	0.160388	0.152422	0.130451
ASP:HB3	28	12	0.177185	0.170667	0.149255	0.168518	0.189808	0.179339
ASP:N	91	21	2.042940	1.848722	2.431770	2.258369	2.114182	1.842501
CYS:C	7	3	2.680669	2.775514	2.351173	2.500672	1.818513	2.422274
CYS:CA	14	7	1.440515	1.228656	1.232640	1.425039	1.455197	0.849953
CYS:CB	12	5	4.612917	4.882185	4.952599	5.022846	5.087742	4.550103
CYS:H	20	8	0.513467	0.460745	0.493880	0.490885	0.537896	0.488566
CYS:HA	20	8	0.419107	0.332924	0.385665	0.360865	0.735431	0.512553
CYS:HB2	9	5	0.228476	0.266032	0.240720	0.285695	0.293733	0.274600

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
CYS:HB3	9	5	0.253200	0.340179	0.257434	0.332510	0.243000	0.230371
CYS:N	14	7	2.728011	2.220205	3.616075	2.912659	2.232969	2.436446
GLN:C	61	15	0.699432	0.684267	0.827114	0.744069	0.740429	0.653824
GLN:CA	83	20	0.648548	0.657554	0.684696	0.695064	0.649902	0.623531
GLN:CB	80	20	0.938506	0.759290	0.770983	0.814371	0.761418	0.742895
GLN:CD	19	3	1.371684	1.358009	1.328188	1.646005	1.330769	1.372801
GLN:CG	83	20	0.539807	0.513705	0.565039	0.601040	0.545821	0.493075
GLN:H	87	22	0.336527	0.336957	0.351272	0.364533	0.345793	0.348589
GLN:HA	89	22	0.180820	0.156844	0.159867	0.159201	0.152183	0.142866
GLN:HB2	34	14	0.180885	0.134043	0.150319	0.152822	0.161554	0.118698
GLN:HB3	34	14	0.225316	0.183425	0.217011	0.213534	0.234240	0.170769
GLN:HE21	23	6	0.443550	0.495091	0.483119	0.491621	0.439440	0.462022
GLN:HE22	23	6	0.456723	0.441003	0.451538	0.443352	0.426499	0.436013
GLN:HG2	29	14	0.160363	0.171283	0.155677	0.152370	0.118314	0.123470
GLN:HG3	28	14	0.116799	0.134093	0.126762	0.131082	0.119520	0.162966
GLN:N	86	21	1.920776	1.509313	1.915286	1.732829	1.673221	1.806625
GLN:NE2	78	20	1.576274	1.287385	1.408233	1.297847	1.357834	1.350666
GLU:C	108	15	0.949051	0.939454	1.029928	1.039050	1.005481	0.952181
GLU:CA	161	20	0.689126	0.648322	0.723780	0.770500	0.716399	0.665834
GLU:CB	157	20	0.682772	0.690044	0.750703	0.779106	0.707095	0.660891
GLU:CG	160	20	0.620536	0.606469	0.601501	0.603739	0.617783	0.597075
GLU:H	170	22	0.349781	0.287418	0.356221	0.331311	0.359750	0.291550
GLU:HA	171	22	0.200923	0.133822	0.179928	0.160516	0.147889	0.127354
GLU:HB2	77	17	0.204615	0.127510	0.168130	0.129151	0.157676	0.132318
GLU:HB3	77	17	0.154965	0.121714	0.150008	0.136524	0.145543	0.113181
GLU:HG2	55	17	0.166075	0.136808	0.162578	0.132201	0.139215	0.130335
GLU:HG3	50	16	0.132519	0.119253	0.129858	0.120265	0.112172	0.116401
GLU:N	163	21	1.942535	1.609927	2.088628	1.866979	1.768630	1.650034
GLY:C	71	15	1.711024	1.673992	1.543797	1.615343	1.765755	1.664519
GLY:CA	100	20	0.689219	0.672994	0.640296	0.652903	0.717112	0.699047
GLY:H	120	22	0.387280	0.332392	0.349579	0.306820	0.375355	0.338913
GLY:HA2	46	16	0.228043	0.292517	0.200891	0.219754	0.235095	0.262715
GLY:HA3	44	16	0.208092	0.288416	0.230593	0.224329	0.175938	0.281571
GLY:N	97	21	2.246639	1.881195	2.364441	2.223032	2.076903	1.818978
HIS:C	19	10	1.321883	0.972809	1.315430	1.407919	1.598470	1.195040
HIS:CA	25	13	1.041913	0.946139	1.086065	1.099052	0.891972	0.945590
HIS:CB	25	13	1.429249	1.246331	1.723315	1.768631	1.437546	1.296084
HIS:CD2	20	10	4.598562	4.585063	4.664949	4.697199	4.670863	4.419878
HIS:CE1	13	9	1.443364	1.211721	1.364568	1.208794	1.253126	1.017927
HIS:H	26	14	0.678203	0.378003	0.383541	0.367144	0.400381	0.280710
HIS:HA	27	14	0.315197	0.210263	0.303704	0.201687	0.231492	0.194445
HIS:HB2	7	4	0.226066	0.283186	0.179438	0.266584	0.216093	0.262068

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
HIS:HB3	7	4	0.407680	0.195256	0.395695	0.353851	0.347638	0.190471
HIS:HD2	25	12	0.522212	0.543104	0.566281	0.527284	0.552118	0.524205
HIS:HE1	17	11	0.412874	0.359746	0.386513	0.377431	0.353692	0.386483
HIS:HE2	1	1	3.629529	3.276733	3.343632	3.346913	3.433147	3.228609
HIS:N	23	12	2.001540	1.269649	2.052976	2.115946	1.825477	1.714532
HIS:NE2	1	1	11.708725	10.939865	11.729996	9.257278	11.498886	13.791168
ILE:C	62	15	1.369315	1.390545	1.387969	1.424106	1.325959	1.326033
ILE:CA	83	20	0.929812	0.800302	0.848005	0.822530	0.941124	0.768738
ILE:CB	80	20	0.924238	0.797330	0.807341	0.916160	0.859117	0.865298
ILE:CD1	83	20	1.039750	0.935109	0.963052	1.008830	0.955098	0.894425
ILE:CG1	83	19	0.874482	0.763190	0.850447	0.919604	0.849702	0.814313
ILE:CG2	84	20	0.798048	0.759859	0.753369	0.723124	0.767392	0.764151
ILE:H	88	21	0.356503	0.288598	0.345927	0.290768	0.329350	0.276802
ILE:HA	88	21	0.301701	0.187938	0.221505	0.220190	0.212267	0.194134
ILE:HB	88	21	0.229692	0.171192	0.218094	0.199105	0.208838	0.170003
ILE:HG12	30	11	0.361273	0.489875	0.444168	0.486151	0.373172	0.494422
ILE:HG13	29	11	0.419245	0.522876	0.462784	0.506273	0.420781	0.487596
ILE:N	87	21	2.300808	2.068944	2.536488	2.208103	2.182165	2.037946
LEU:C	109	15	1.314878	1.079025	1.227958	1.263946	1.225493	1.089805
LEU:CA	150	20	0.743779	0.612469	0.753121	0.753290	0.749970	0.630113
LEU:CB	145	20	0.887314	0.697003	0.985865	0.968585	0.747816	0.734239
LEU:CG	154	20	0.636270	0.580973	0.637670	0.672066	0.604869	0.599588
LEU:H	164	22	0.415859	0.347654	0.382421	0.378374	0.405449	0.358384
LEU:HA	165	22	0.231896	0.158279	0.213818	0.190634	0.179823	0.154209
LEU:HB2	55	17	0.274419	0.219496	0.245440	0.199849	0.233646	0.226319
LEU:HB3	53	16	0.254726	0.234710	0.245609	0.235504	0.209179	0.244646
LEU:HG	161	22	0.238892	0.216984	0.256653	0.255633	0.226778	0.218146
LEU:N	149	21	2.120975	2.007880	2.477706	2.499609	2.324315	2.252900
LYS:C	111	15	0.765081	0.765724	0.820885	0.830209	0.808476	0.703937
LYS:CA	157	20	0.854777	0.833376	0.979850	0.899910	0.910257	0.862025
LYS:CB	154	19	0.602389	0.583807	0.642058	0.655573	0.604355	0.559040
LYS:CD	143	20	0.432116	0.455624	0.414478	0.491611	0.456721	0.461314
LYS:CE	138	19	0.293531	0.296695	0.285643	0.264869	0.284591	0.269603
LYS:CG	154	20	0.632303	0.629564	0.643995	0.641670	0.620209	0.600865
LYS:H	174	22	0.371572	0.303105	0.368092	0.360702	0.351146	0.334188
LYS:HA	173	22	0.202749	0.155659	0.187346	0.175285	0.161829	0.140181
LYS:HB2	80	18	0.135142	0.093752	0.116756	0.115772	0.106242	0.095978
LYS:HB3	78	17	0.122427	0.090320	0.106858	0.111350	0.104587	0.079978
LYS:HD2	104	20	0.080293	0.076488	0.077615	0.076153	0.081874	0.076130
LYS:HD3	102	20	0.078031	0.078701	0.074485	0.076723	0.092099	0.073040
LYS:HE2	104	19	0.093317	0.082802	0.088431	0.082392	0.085294	0.082427
LYS:HE3	101	18	0.098312	0.095933	0.084338	0.092369	0.094448	0.099761

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
LYS:HG2	63	17	0.134569	0.096228	0.124542	0.120024	0.120246	0.101184
LYS:HG3	63	17	0.137832	0.118174	0.127046	0.110451	0.117660	0.112382
LYS:N	162	21	1.844884	1.661035	2.079512	1.986170	1.666258	1.762289
MET:C	23	13	0.957048	1.014724	1.156409	1.238886	1.051511	0.999244
MET:CA	36	17	0.794556	0.945084	0.735926	0.922487	0.935070	0.947933
MET:CB	35	17	0.931700	0.767092	1.105487	0.977116	0.891942	0.802914
MET:CE	28	14	0.915681	0.861645	0.812562	0.920132	0.993027	0.861639
MET:CG	35	17	0.929191	0.770017	0.975452	0.953261	0.977576	0.862398
MET:H	35	17	0.352742	0.284967	0.292456	0.336336	0.350082	0.297552
MET:HA	38	18	0.272773	0.162842	0.257830	0.210393	0.221239	0.172827
MET:HB2	15	9	0.240614	0.236689	0.226735	0.229051	0.245032	0.241813
MET:HB3	15	9	0.189101	0.174503	0.205195	0.193114	0.215615	0.182687
MET:HG2	9	7	0.131380	0.206164	0.138010	0.181132	0.134381	0.213880
MET:HG3	9	7	0.161086	0.248714	0.158477	0.200071	0.135853	0.222939
MET:N	33	17	2.420483	1.553207	2.193387	2.066441	1.980311	1.894978
PHE:C	43	14	1.247586	1.199831	1.367607	1.450745	1.283231	1.154686
PHE:CA	66	19	0.919458	0.913990	0.992195	1.167596	1.087658	0.857320
PHE:CB	63	19	0.939344	0.804706	0.958185	0.962277	0.851740	0.746949
PHE:CD1	47	15	0.821473	0.877482	0.913417	0.890769	0.830482	0.869629
PHE:CD2	33	11	0.861438	0.833287	0.934034	0.905282	0.873390	0.941393
PHE:CE1	37	14	0.851930	0.778183	0.850330	0.849470	0.732730	0.750921
PHE:CE2	26	11	0.991140	0.775726	0.898072	0.788610	0.775329	0.766803
PHE:CZ	29	12	1.123055	0.993444	1.100417	1.072251	0.987700	0.939143
PHE:H	72	21	0.525386	0.444414	0.467325	0.492813	0.487676	0.456532
PHE:HA	72	21	0.302775	0.207896	0.222927	0.227046	0.247613	0.234632
PHE:HB2	24	11	0.296282	0.182405	0.253754	0.240714	0.229467	0.224489
PHE:HB3	24	11	0.304688	0.160631	0.229674	0.278321	0.258962	0.148599
PHE:HD1	68	19	0.314798	0.270958	0.310300	0.299130	0.274130	0.259218
PHE:HD2	62	17	0.296312	0.219364	0.243790	0.230348	0.242245	0.206243
PHE:HE1	58	18	0.363521	0.264467	0.334366	0.271742	0.284536	0.270686
PHE:HE2	52	16	0.305822	0.240435	0.284644	0.235827	0.274748	0.235145
PHE:HZ	48	18	0.411511	0.357710	0.384115	0.364622	0.358562	0.342526
PHE:N	67	20	2.417063	2.138204	3.092135	2.583056	2.417294	1.855308
PRO:C	46	15	1.126779	1.046017	1.105236	1.138345	1.139276	1.078136
PRO:CA	75	19	0.723469	0.668716	0.703306	0.715551	0.701704	0.653223
PRO:CB	69	18	0.550765	0.519274	0.532114	0.638769	0.641421	0.566901
PRO:CD	79	18	0.648739	0.574438	0.597862	0.651254	0.621076	0.546483
PRO:CG	74	18	0.593380	0.585843	0.579580	0.620687	0.571607	0.544635
PRO:HA	89	21	0.236751	0.193156	0.281697	0.206920	0.224162	0.227867
PRO:HB2	29	10	0.339639	0.287634	0.350157	0.312631	0.312115	0.288012
PRO:HB3	29	10	0.330900	0.283135	0.309323	0.315896	0.310299	0.301598
PRO:HD2	32	11	0.358084	0.309032	0.309834	0.310595	0.325231	0.329349

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
PRO:HD3	31	11	0.311739	0.347245	0.295289	0.314150	0.249622	0.337523
PRO:HG2	39	16	0.149126	0.190947	0.148103	0.189185	0.191697	0.172067
PRO:HG3	39	16	0.132157	0.227979	0.186831	0.209434	0.182255	0.209028
SER:C	76	14	1.587768	1.374853	1.350146	1.392306	1.461543	1.395592
SER:CA	115	18	0.790792	0.731115	0.736248	0.824557	0.814631	0.774178
SER:CB	98	18	0.702634	0.607960	0.654108	0.668078	0.643099	0.517664
SER:H	126	20	0.424441	0.364082	0.400157	0.372630	0.370657	0.374283
SER:HA	129	20	0.177752	0.145907	0.159431	0.146105	0.135879	0.160093
SER:HB2	61	15	0.137661	0.112432	0.129993	0.125716	0.151338	0.109957
SER:HB3	52	15	0.115083	0.136587	0.130866	0.142776	0.140608	0.151954
SER:N	112	19	2.210267	2.054268	2.631995	2.140467	2.213466	2.119804
THR:C	47	14	0.693894	0.702013	0.887617	0.786766	0.770813	0.712221
THR:CA	76	19	0.910059	1.001243	0.974282	1.057902	0.843981	1.053063
THR:CB	73	19	0.754725	0.752150	0.902723	0.778475	0.795261	0.757907
THR:CG2	78	19	1.344755	1.314234	1.378204	1.359830	1.343793	1.284695
THR:H	82	21	0.439160	0.374049	0.418505	0.406556	0.397183	0.350081
THR:HA	83	21	0.203650	0.213129	0.216018	0.225530	0.208847	0.188408
THR:HB	83	21	0.222998	0.160328	0.187471	0.177702	0.158861	0.164604
THR:HG1	4	1	1.815076	1.797324	1.801959	1.793504	1.750781	1.854461
THR:N	72	20	2.794043	2.552047	3.365009	3.168874	2.906841	2.617878
TRP:C	14	9	0.981774	0.591325	1.257755	1.037997	0.701679	0.904225
TRP:CA	19	13	0.949403	1.057019	1.153632	1.304282	1.424892	1.058790
TRP:CB	19	13	1.829054	1.450037	1.521742	1.479170	1.611324	1.574149
TRP:CD1	14	9	1.097969	0.920002	1.276548	1.034684	1.009538	0.777983
TRP:CE3	14	9	0.823038	0.689412	0.877888	0.837633	0.906864	0.980839
TRP:CH2	14	9	0.716300	0.561418	0.799942	0.476646	0.558419	0.515351
TRP:CZ2	14	9	0.970749	0.882795	1.178490	0.982094	0.892728	0.869198
TRP:CZ3	14	9	0.504598	0.481961	0.396268	0.602203	0.513272	0.475587
TRP:H	22	14	0.382815	0.362474	0.351670	0.348020	0.401415	0.355137
TRP:HA	22	14	0.439659	0.530975	0.514081	0.518342	0.436311	0.450735
TRP:HB2	4	3	0.290898	0.118599	0.169272	0.151313	0.376822	0.121045
TRP:HB3	4	3	0.217942	0.045134	0.149140	0.166873	0.084795	0.041164
TRP:HD1	19	13	0.263426	0.184066	0.163982	0.168208	0.219549	0.171897
TRP:HE1	19	12	0.401123	0.406019	0.373845	0.397580	0.405787	0.405777
TRP:HE3	19	12	0.476626	0.268496	0.303189	0.275283	0.279464	0.242059
TRP:HH2	20	13	0.321121	0.144195	0.169159	0.136428	0.134787	0.152128
TRP:HZ2	19	12	0.227732	0.146140	0.175402	0.193828	0.133806	0.140698
TRP:HZ3	15	10	0.231409	0.216167	0.207855	0.253970	0.246176	0.247731
TRP:N	21	14	2.153193	1.658130	2.105665	1.778527	1.551762	1.407203
TRP:NE1	20	13	2.301706	2.521317	2.230254	2.369626	2.168145	2.235766
TYR:C	34	13	0.901077	0.752300	0.827237	0.989933	0.878481	0.728669
TYR:CA	54	18	0.951964	0.976799	1.068501	0.951262	0.998546	0.830053

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
TYR:CB	51	18	0.670316	0.585288	0.696700	0.749001	0.778403	0.602914
TYR:CD1	47	15	0.635804	0.513586	0.571301	0.569167	0.626258	0.504966
TYR:CD2	35	11	0.535121	0.548873	0.552989	0.585777	0.526844	0.525748
TYR:CE1	48	15	0.649907	0.539465	0.568525	0.554912	0.492531	0.560405
TYR:CE2	35	11	0.484049	0.381326	0.438578	0.447253	0.428053	0.399140
TYR:H	63	20	0.470519	0.359796	0.453924	0.488075	0.418731	0.402400
TYR:HA	64	20	0.247105	0.179183	0.197791	0.221839	0.215410	0.273444
TYR:HB2	20	11	0.386361	0.163836	0.322668	0.273494	0.313197	0.167969
TYR:HB3	20	11	0.355772	0.214966	0.333684	0.347468	0.274156	0.236127
TYR:HD1	59	19	0.315083	0.272099	0.314497	0.292495	0.265419	0.249543
TYR:HD2	56	17	0.385864	0.258876	0.313793	0.275250	0.297178	0.244838
TYR:HE1	58	19	0.165774	0.146450	0.163752	0.146499	0.172988	0.140582
TYR:HE2	56	17	0.175681	0.135837	0.146189	0.124801	0.135550	0.139500
TYR:N	55	19	2.054075	1.794686	2.315633	2.259166	1.798533	1.576639
VAL:C	64	15	1.090181	1.054851	1.196491	1.262611	1.073116	0.938172
VAL:CA	100	20	0.851930	0.806022	0.999800	0.954583	0.907821	0.762397
VAL:CB	96	20	0.624575	0.637710	0.679859	0.610454	0.658996	0.614724
VAL:H	107	21	0.433877	0.366209	0.387332	0.417166	0.395997	0.373985
VAL:HA	108	21	0.295834	0.233851	0.289413	0.279076	0.229847	0.234103
VAL:HB	108	21	0.201895	0.166236	0.227057	0.218839	0.173732	0.176709
VAL:N	104	21	2.384953	2.112452	2.707395	2.855269	2.436983	2.147825

8.3 Development all 173: exact-assignment RMSE (ppm)

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ALA:C	679	116	1.209181	1.235560	1.249241	1.227604	1.267928	1.223030
ALA:CA	968	162	0.668320	0.596421	0.723867	0.665251	0.675333	0.587183
ALA:CB	936	162	0.690890	0.620660	0.718077	0.659299	0.729456	0.617838
ALA:H	1005	172	0.339686	0.309406	0.344486	0.341123	0.344055	0.292485
ALA:HA	1013	169	0.213924	0.176006	0.188783	0.188424	0.192760	0.169887
ALA:N	935	168	1.890895	1.725556	2.333655	1.801928	1.879781	1.703820
ARG:C	476	109	0.795152	0.713472	0.802138	0.747447	0.823122	0.707411
ARG:CA	720	156	1.003476	1.015776	1.051952	1.015502	1.033691	1.011024
ARG:CB	686	154	0.675700	0.672476	0.765094	0.699311	0.763606	0.668306
ARG:CD	648	143	0.407093	0.392841	0.406006	0.396857	0.425345	0.415818
ARG:CG	624	141	0.751813	0.738372	0.740031	0.713407	0.768514	0.736723
ARG:CZ	6	2	0.877785	0.551185	0.564360	0.559801	0.442477	0.542912
ARG:H	753	164	0.307093	0.284759	0.309834	0.311596	0.317523	0.256814
ARG:HA	754	161	0.192071	0.172948	0.168949	0.173856	0.177178	0.155028
ARG:HB2	295	103	0.193020	0.199565	0.192027	0.195545	0.192996	0.190645
ARG:HB3	275	100	0.186648	0.186311	0.193371	0.191724	0.188804	0.176263

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ARG:HD2	343	112	0.130322	0.133075	0.134073	0.131620	0.135060	0.131762
ARG:HD3	330	111	0.177295	0.173826	0.164329	0.181306	0.163031	0.138709
ARG:HE	215	81	0.646174	0.685136	0.609715	0.668507	0.666235	0.655430
ARG:HG2	251	96	0.174429	0.177332	0.176266	0.170045	0.165200	0.162870
ARG:HG3	242	93	0.192270	0.181756	0.179195	0.173282	0.181368	0.170512
ARG:HH11	13	5	0.217534	0.224633	0.232436	0.210776	0.235456	0.222582
ARG:HH12	13	5	0.222600	0.228974	0.211258	0.185053	0.213061	0.226987
ARG:HH21	9	4	0.145844	0.198719	0.170793	0.198422	0.180337	0.162778
ARG:HH22	9	4	0.168289	0.146042	0.151296	0.178840	0.188036	0.141584
ARG:N	717	162	1.875561	1.624864	2.311700	2.227644	2.345615	1.832026
ARG:NE	181	65	8.372438	8.990965	12.222931	8.540258	9.644228	10.837481
ARG:NH1	7	3	2.900352	1.836768	9.969579	2.984750	6.556022	9.583739
ARG:NH2	7	3	1.528459	0.893587	9.938040	2.522832	7.748833	9.222591
ASN:C	304	107	0.847816	0.822950	0.875278	0.850581	0.920949	0.831389
ASN:CA	507	151	0.697078	0.675387	0.725179	0.709741	0.739511	0.673068
ASN:CB	483	150	0.657587	0.726138	0.795616	0.739759	0.750860	0.693874
ASN:CG	42	14	1.067764	0.881691	1.031524	1.129178	1.105633	1.080324
ASN:H	530	161	0.453972	0.411027	0.411303	0.410902	0.428399	0.349231
ASN:HA	533	160	0.186275	0.152343	0.165783	0.161442	0.173335	0.147173
ASN:HB2	203	82	0.255757	0.241534	0.260560	0.252811	0.236807	0.225225
ASN:HB3	197	80	0.224385	0.189638	0.214549	0.216613	0.217977	0.187553
ASN:HD21	137	47	0.498041	0.509325	0.494548	0.496033	0.480163	0.476440
ASN:HD22	137	47	0.423790	0.454523	0.431037	0.456665	0.424453	0.433611
ASN:N	496	154	2.130791	1.794755	2.553185	2.129253	2.076058	1.906409
ASN:ND2	425	132	1.816856	1.586489	1.851097	1.506519	1.769955	1.661210
ASP:C	519	113	0.889701	0.841583	0.918237	0.866052	0.898269	0.848169
ASP:CA	800	159	0.733044	0.702792	0.750681	0.738593	0.758806	0.689555
ASP:CB	754	158	0.859884	0.880947	0.951770	0.888539	0.975432	0.869488
ASP:CG	3	1	0.185503	0.269719	0.292816	0.327271	0.345668	0.442012
ASP:H	838	170	0.358327	0.363424	0.376627	0.387728	0.386234	0.315675
ASP:HA	833	167	0.162243	0.137099	0.149559	0.150539	0.157258	0.131159
ASP:HB2	311	98	0.213268	0.191041	0.212590	0.196519	0.197026	0.167513
ASP:HB3	299	94	0.193096	0.189434	0.196243	0.197029	0.196132	0.187450
ASP:N	800	166	3.410253	3.225428	3.818990	3.268100	3.428995	3.257575
CYS:C	71	26	1.145781	1.302169	1.231979	1.102347	1.562415	1.023800
CYS:CA	133	47	1.386266	1.338972	1.535677	1.502684	1.532891	1.163075
CYS:CB	125	44	2.772111	3.052476	3.322511	2.965117	3.919467	2.953675
CYS:H	166	52	0.330739	0.312715	0.319467	0.364663	0.500687	0.308393
CYS:HA	154	50	0.235683	0.201693	0.241982	0.205514	0.370582	0.243421
CYS:HB2	62	23	0.327809	0.245658	0.284433	0.268309	0.369589	0.214902
CYS:HB3	61	22	0.237737	0.288894	0.216595	0.279646	0.377116	0.254197
CYS:N	146	49	2.246068	1.904534	2.973105	1.969326	2.935309	1.748763

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
GLN:C	383	105	0.728297	0.738167	0.823071	0.777961	0.806706	0.734309
GLN:CA	606	151	0.636288	0.634841	0.702191	0.684349	0.694952	0.642642
GLN:CB	588	151	0.789691	0.740002	0.826787	0.761137	0.808438	0.724567
GLN:CD	41	10	1.071344	1.070334	1.052822	1.241462	1.074261	1.050544
GLN:CG	549	140	0.700389	0.697751	0.729976	0.703964	0.720036	0.658342
GLN:H	633	160	0.281982	0.300123	0.287486	0.296518	0.300337	0.250884
GLN:HA	628	157	0.183642	0.179441	0.166428	0.173350	0.178516	0.165001
GLN:HB2	266	93	0.190499	0.152200	0.164985	0.144214	0.154161	0.144340
GLN:HB3	257	91	0.190991	0.182383	0.184643	0.175974	0.188720	0.178086
GLN:HE21	146	40	0.479829	0.502298	0.472364	0.485771	0.479283	0.482097
GLN:HE22	147	40	0.386710	0.385620	0.378393	0.387049	0.374807	0.381683
GLN:HG2	272	105	0.174030	0.169050	0.169704	0.158294	0.156637	0.156937
GLN:HG3	258	102	0.177743	0.166003	0.161540	0.164628	0.163649	0.169550
GLN:N	608	157	1.753566	1.470175	2.231599	1.616788	1.713201	1.701527
GLN:NE2	483	129	1.420875	1.245494	1.758520	1.420033	1.558378	1.462948
GLU:C	718	117	0.851921	0.837094	0.880828	0.882568	0.918955	0.854086
GLU:CA	1086	163	0.667152	0.672265	0.730199	0.700658	0.705005	0.647119
GLU:CB	1045	162	0.680652	0.642675	0.705385	0.679214	0.712864	0.638638
GLU:CD	5	3	17.240544	16.709828	16.544568	17.287405	17.329497	17.186398
GLU:CG	970	150	0.724745	0.747384	0.742684	0.740652	0.753293	0.740430
GLU:H	1162	172	0.309645	0.296728	0.326784	0.334491	0.339352	0.259589
GLU:HA	1140	169	0.171905	0.146766	0.153035	0.153618	0.156721	0.131511
GLU:HB2	464	120	0.160969	0.145171	0.157323	0.141205	0.150913	0.143361
GLU:HB3	436	118	0.156287	0.146148	0.145706	0.145727	0.152443	0.140868
GLU:HG2	400	111	0.148639	0.136484	0.142245	0.139699	0.141144	0.131646
GLU:HG3	363	107	0.136398	0.139829	0.131339	0.135026	0.135560	0.136880
GLU:N	1099	170	1.676454	1.494791	2.103056	1.655268	1.747679	1.643220
GLY:C	594	110	1.325656	1.318835	1.294881	1.254960	1.340207	1.280700
GLY:CA	929	160	0.592563	0.546800	0.613230	0.569629	0.638607	0.575221
GLY:H	1004	171	0.362563	0.324094	0.375851	0.354843	0.365227	0.271482
GLY:HA2	330	110	0.227610	0.265095	0.213750	0.233916	0.230080	0.225148
GLY:HA3	312	107	0.233134	0.283485	0.241385	0.234195	0.231017	0.276019
GLY:N	903	165	2.624244	2.551803	3.027896	2.525350	2.691140	2.552266
HIS:C	162	75	0.812258	0.711472	0.868391	0.894904	0.941150	0.766456
HIS:CA	256	112	0.886505	0.865221	0.996047	0.937677	1.027080	0.885280
HIS:CB	246	110	0.982226	1.013161	1.166059	0.990921	1.106396	1.026793
HIS:CD2	123	62	2.346887	2.312298	2.370080	2.176867	2.380736	2.195120
HIS:CE1	72	45	1.281806	1.302624	1.417207	1.431244	1.464464	1.151050
HIS:CG	1	1	0.229462	0.042221	0.150909	0.143326	0.026184	0.290848
HIS:H	269	121	0.417110	0.372243	0.384748	0.378396	0.413438	0.343378
HIS:HA	270	119	0.224790	0.193650	0.216011	0.206081	0.205152	0.182951
HIS:HB2	105	53	0.328808	0.309670	0.334372	0.335624	0.324411	0.305795

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
HIS:HB3	98	50	0.306144	0.296842	0.290897	0.281399	0.291545	0.284378
HIS:HD2	166	85	0.318920	0.310006	0.326613	0.315033	0.318171	0.300584
HIS:HE1	106	64	0.495982	0.483679	0.483999	0.470601	0.484357	0.460595
HIS:HE2	3	2	2.095522	1.892082	1.930513	1.932388	1.982151	1.864069
HIS:N	242	112	2.143616	2.138359	2.656268	2.106810	2.299910	2.279674
HIS:ND1	4	3	50.136301	45.283893	50.337082	47.214714	47.614861	46.680188
HIS:NE2	4	3	6.379661	5.858462	6.568088	5.006068	6.277234	6.978644
ILE:C	462	111	0.931394	0.863630	0.961970	0.914864	0.980977	0.893447
ILE:CA	696	156	0.816926	0.794031	0.836542	0.839435	0.930375	0.822457
ILE:CB	675	155	0.867880	0.827160	0.952654	0.870713	0.922290	0.850185
ILE:CD1	642	149	2.021988	2.019719	1.988836	1.898074	2.019576	2.005258
ILE:CG1	607	141	1.330011	1.300540	1.385225	1.212722	1.375101	1.308940
ILE:CG2	642	146	1.855641	1.781141	1.857138	1.749898	1.831864	1.787763
ILE:H	740	164	0.329081	0.313513	0.337592	0.324085	0.353133	0.278303
ILE:HA	717	161	0.247003	0.199224	0.209010	0.201930	0.206137	0.182434
ILE:HB	714	161	0.213761	0.167060	0.214560	0.188074	0.191330	0.163156
ILE:HG12	229	71	0.405544	0.403729	0.380655	0.398149	0.404667	0.399704
ILE:HG13	222	71	0.395901	0.417007	0.402756	0.417471	0.379991	0.401370
ILE:N	702	161	2.120389	1.994066	2.600376	2.080388	2.160379	2.089873
LEU:C	823	115	0.928640	0.824712	0.980018	0.943351	0.945210	0.843927
LEU:CA	1201	160	0.689907	0.650202	0.759950	0.736577	0.729588	0.657567
LEU:CB	1161	160	0.747488	0.671022	0.879493	0.795368	0.824664	0.667515
LEU:CG	1081	150	0.705270	0.674242	0.739639	0.724393	0.699591	0.677395
LEU:H	1288	171	0.341516	0.333143	0.348993	0.339770	0.351984	0.307788
LEU:HA	1264	168	0.193173	0.158189	0.176442	0.169694	0.174417	0.148159
LEU:HB2	505	109	0.290518	0.289255	0.282977	0.279913	0.268619	0.271042
LEU:HB3	489	108	0.294292	0.296168	0.284176	0.276110	0.292483	0.290151
LEU:HG	1149	164	0.263763	0.248761	0.273335	0.263528	0.252446	0.243311
LEU:N	1205	167	1.858281	1.752705	2.314134	1.983285	1.929704	1.845161
LYS:C	695	116	0.787390	0.761067	0.825683	0.787033	0.843229	0.765127
LYS:CA	1052	161	0.767443	0.742360	0.858260	0.795590	0.795603	0.740355
LYS:CB	991	159	0.792213	0.809147	0.890546	0.821911	0.881556	0.800497
LYS:CD	888	149	0.920159	0.928180	0.903382	0.925904	0.924724	0.927689
LYS:CE	868	146	0.576584	0.579856	0.580216	0.572834	0.580783	0.575325
LYS:CG	936	149	0.787098	0.777781	0.805424	0.770355	0.808189	0.777088
LYS:H	1145	171	0.320778	0.295421	0.337391	0.335359	0.316900	0.273583
LYS:HA	1119	168	0.197566	0.167114	0.174028	0.168340	0.179643	0.153211
LYS:HB2	508	112	0.175323	0.146156	0.162979	0.146898	0.142390	0.136683
LYS:HB3	482	109	0.158145	0.142979	0.147113	0.145978	0.147890	0.138570
LYS:HD2	575	124	0.149888	0.152482	0.154028	0.152965	0.149541	0.148951
LYS:HD3	535	121	0.147454	0.146349	0.147978	0.146232	0.146816	0.141979
LYS:HE2	606	126	0.108469	0.105278	0.104729	0.104416	0.103950	0.101283

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
LYS:HE3	553	119	0.120250	0.112239	0.114090	0.112300	0.111365	0.109300
LYS:HG2	392	106	0.160646	0.146147	0.149235	0.155022	0.151833	0.140231
LYS:HG3	365	104	0.165322	0.150185	0.153018	0.146482	0.146441	0.144524
LYS:N	1066	168	1.873532	1.640085	2.189052	1.783572	1.794990	1.784695
MET:C	170	85	0.946388	0.951251	1.059272	1.035057	1.085274	0.978527
MET:CA	279	123	1.080252	1.060039	1.111044	1.145679	1.147534	1.072062
MET:CB	264	121	1.036156	1.085112	1.172633	1.050697	1.116658	1.066944
MET:CE	162	80	7.238364	7.235481	7.256043	7.267238	7.257958	7.247242
MET:CG	232	105	1.074050	1.036968	1.139624	1.059558	1.116939	1.042104
MET:H	261	122	0.312835	0.282584	0.305909	0.299188	0.319830	0.266677
MET:HA	289	129	0.222076	0.174503	0.193703	0.184592	0.201168	0.160456
MET:HB2	130	64	0.215321	0.200363	0.211911	0.207065	0.199892	0.196184
MET:HB3	125	63	0.285896	0.258392	0.280106	0.275763	0.267101	0.249813
MET:HG2	90	51	0.299614	0.300190	0.299756	0.299227	0.294971	0.287175
MET:HG3	87	49	0.281063	0.326431	0.308856	0.311615	0.289055	0.311433
MET:N	251	116	1.807099	1.428077	1.953569	1.596577	1.514655	1.597721
PHE:C	318	106	0.828611	0.809926	0.946702	0.932668	0.891568	0.833879
PHE:CA	455	150	0.896307	0.887044	0.995875	0.978598	1.002024	0.830730
PHE:CB	440	149	0.885806	0.900013	0.905666	0.824051	0.967934	0.843814
PHE:CD1	243	85	1.570501	1.547804	1.596342	1.504170	1.556295	1.537210
PHE:CD2	183	63	1.710906	1.699153	1.743095	1.738779	1.719272	1.731758
PHE:CE1	217	83	2.024661	2.072989	2.076066	2.055238	2.031271	2.034361
PHE:CE2	164	63	2.216839	2.216485	2.254794	2.227684	2.287241	2.224271
PHE:CZ	215	90	5.739876	5.755529	5.781957	5.685201	5.740927	5.740301
PHE:H	479	160	0.373893	0.354436	0.377405	0.359759	0.377951	0.322285
PHE:HA	475	157	0.263006	0.220858	0.211498	0.215609	0.237632	0.209058
PHE:HB2	172	73	0.269769	0.233782	0.263900	0.241018	0.252625	0.225850
PHE:HB3	164	73	0.316020	0.281616	0.282138	0.289262	0.276733	0.262473
PHE:HD1	343	118	0.231591	0.210129	0.221110	0.223542	0.213559	0.202642
PHE:HD2	324	111	0.240414	0.193712	0.214159	0.206455	0.207337	0.185502
PHE:HE1	313	115	0.262191	0.209628	0.242446	0.218096	0.224284	0.205003
PHE:HE2	291	106	0.244320	0.207977	0.222932	0.210504	0.210942	0.201043
PHE:HZ	299	123	0.274988	0.238816	0.255539	0.248537	0.237397	0.220506
PHE:N	455	154	1.977751	1.797984	2.511197	1.964204	2.049644	1.876930
PRO:C	331	106	0.822666	0.787585	0.863205	0.803229	0.845237	0.799102
PRO:CA	541	150	0.704723	0.701206	0.698648	0.683913	0.694490	0.669910
PRO:CB	523	148	0.542762	0.534238	0.610819	0.589991	0.645447	0.552139
PRO:CD	530	142	1.335461	1.292805	1.316634	1.310032	1.329431	1.278690
PRO:CG	518	139	0.522122	0.536097	0.531306	0.556040	0.542059	0.508724
PRO:HA	571	154	0.202404	0.164227	0.192972	0.169401	0.189062	0.166478
PRO:HB2	152	54	0.280780	0.282520	0.308169	0.291976	0.275130	0.263418
PRO:HB3	150	54	0.321374	0.278183	0.296154	0.298660	0.294138	0.297214

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
PRO:HD2	158	70	0.252011	0.208069	0.218425	0.207422	0.216012	0.209797
PRO:HD3	150	69	0.212767	0.217495	0.218103	0.205460	0.204102	0.218929
PRO:HG2	206	89	0.171351	0.177701	0.172895	0.179564	0.174371	0.169215
PRO:HG3	190	85	0.142894	0.188293	0.152123	0.163987	0.170695	0.177021
PRO:N	8	2	1.265963	0.907809	2.110181	0.887674	1.053150	1.076745
SER:C	523	111	0.950034	0.846850	1.005226	0.941772	0.958084	0.868496
SER:CA	821	157	0.767179	0.746589	0.780425	0.787896	0.804808	0.742045
SER:CB	768	156	0.623707	0.604895	0.697158	0.653885	0.675575	0.572710
SER:H	848	167	0.353097	0.326074	0.337213	0.337211	0.340547	0.291892
SER:HA	881	165	0.183813	0.158320	0.164206	0.164594	0.161683	0.153965
SER:HB2	429	118	0.139527	0.123616	0.130326	0.129145	0.132492	0.116025
SER:HB3	390	114	0.147864	0.146672	0.147031	0.150211	0.153483	0.144380
SER:HG	10	7	0.365148	0.536259	0.261945	0.574833	0.386089	0.502901
SER:N	763	163	1.960143	1.699555	2.445569	1.850449	1.944473	1.874208
THR:C	430	109	0.760046	0.741112	0.858833	0.769044	0.817327	0.732398
THR:CA	704	155	0.780959	0.808237	0.820274	0.805548	0.805993	0.820077
THR:CB	672	155	0.591494	0.566028	0.704435	0.589335	0.700657	0.536157
THR:CG2	647	146	0.818458	0.731359	0.831070	0.780205	0.789746	0.722357
THR:H	735	164	0.344579	0.306192	0.340837	0.329255	0.347789	0.269247
THR:HA	728	161	0.212298	0.179344	0.188758	0.187951	0.189247	0.168452
THR:HB	712	161	0.222784	0.183872	0.211690	0.197370	0.202138	0.179466
THR:HG1	23	17	0.865900	0.883010	0.862117	0.850938	0.884820	0.884554
THR:N	687	161	2.969940	2.797759	3.328913	2.876074	2.963079	2.879181
TRP:C	100	64	0.863576	0.777771	0.996079	0.842697	0.921442	0.719526
TRP:CA	147	90	1.093457	0.960638	1.133346	1.081623	1.147904	0.963056
TRP:CB	145	89	0.917265	0.781784	0.911572	0.796512	0.911775	0.800462
TRP:CD1	107	67	1.024513	0.842193	1.023652	0.739637	1.020686	0.777609
TRP:CD2	2	2	1.715101	1.918856	1.715899	1.841071	1.922753	1.773292
TRP:CE3	104	64	0.936288	0.719278	0.848884	0.793653	0.946912	0.767199
TRP:CG	1	1	0.072304	0.131691	0.232841	0.123924	0.293503	0.016060
TRP:CH2	106	66	0.819035	0.763231	0.865899	0.697079	0.834769	0.743612
TRP:CZ2	107	67	0.661733	0.609951	0.733238	0.574883	0.634988	0.566145
TRP:CZ3	103	64	0.780522	0.747197	0.790695	0.675377	0.862929	0.715147
TRP:H	162	97	0.466117	0.413956	0.402443	0.461007	0.419437	0.338099
TRP:HA	160	97	0.286174	0.257993	0.261005	0.250926	0.241519	0.218810
TRP:HB2	45	33	0.239222	0.224475	0.203191	0.230481	0.241172	0.215079
TRP:HB3	43	31	0.238139	0.235145	0.210906	0.224737	0.202027	0.220722
TRP:HD1	146	90	0.340880	0.261151	0.298558	0.260905	0.282087	0.238787
TRP:HE1	146	89	0.441643	0.398433	0.446390	0.382278	0.439584	0.391907
TRP:HE3	136	83	0.339588	0.234285	0.255870	0.233607	0.233000	0.213944
TRP:HH2	139	88	0.230072	0.180344	0.175760	0.172163	0.193740	0.162018
TRP:HZ2	137	84	0.215600	0.155750	0.161525	0.163163	0.157196	0.151278

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
TRP:HZ3	119	74	0.260885	0.244836	0.229826	0.251240	0.255431	0.249294
TRP:N	154	95	2.472529	2.344342	2.930980	2.225201	2.564633	2.489486
TRP:NE1	147	91	1.565021	1.476539	1.616854	1.380221	1.546633	1.587192
TYR:C	241	100	0.745622	0.687539	0.753241	0.726340	0.767823	0.683937
TYR:CA	407	144	0.791661	0.798482	0.887594	0.809989	0.881710	0.748151
TYR:CB	386	143	0.722915	0.674506	0.822207	0.716848	0.848451	0.631139
TYR:CD1	233	85	0.781681	0.769535	0.759895	0.742026	0.801923	0.720274
TYR:CD2	170	66	0.698307	0.667265	0.747476	0.701232	0.684240	0.705268
TYR:CE1	232	82	0.625049	0.593277	0.631713	0.586138	0.605046	0.591789
TYR:CE2	170	65	0.562421	0.547241	0.560757	0.534742	0.567343	0.543285
TYR:CG	4	2	4.874199	4.802346	4.877777	4.580066	4.734711	4.539660
TYR:H	437	154	0.340879	0.313146	0.350677	0.350378	0.342065	0.286688
TYR:HA	430	150	0.235116	0.198949	0.213167	0.217848	0.224132	0.199523
TYR:HB2	133	71	0.296978	0.284615	0.291243	0.266432	0.298500	0.276975
TYR:HB3	128	69	0.362402	0.340058	0.349762	0.355605	0.335798	0.336158
TYR:HD1	327	115	0.231242	0.199531	0.210268	0.204412	0.198845	0.183521
TYR:HD2	302	106	0.289393	0.233180	0.260611	0.242827	0.252315	0.220667
TYR:HE1	325	115	0.168447	0.148625	0.153688	0.147382	0.151922	0.139871
TYR:HE2	300	106	0.170807	0.153617	0.156627	0.155276	0.153832	0.150315
TYR:HH	5	5	2.067757	2.107582	2.027234	2.052275	2.122731	2.084075
TYR:N	415	148	1.940444	1.535191	2.463218	1.810831	1.858352	1.690954
VAL:C	586	112	0.877932	0.838204	0.973614	1.045200	0.936239	0.862630
VAL:CA	903	157	0.779202	0.755728	1.037046	0.954222	0.836599	0.755804
VAL:CB	872	157	0.617894	0.557188	0.826320	0.675510	0.634381	0.564497
VAL:H	955	166	0.348491	0.330597	0.343507	0.359727	0.360825	0.304624
VAL:HA	956	163	0.298375	0.268982	0.342983	0.341682	0.278597	0.245320
VAL:HB	947	163	0.223582	0.212395	0.248732	0.247328	0.213973	0.199244
VAL:N	900	162	2.105290	1.931824	2.790354	2.543534	2.132162	2.060808

8.4 Like-for-like 386: exact-assignment RMSE (ppm)

IUPAC	Role	<i>n</i>	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ALA:C	C_carbonyl	2375	1.211478	1.150197	1.250659	1.284426	1.222584	1.148983
ALA:CA	C_alpha	3192	0.773944	0.656800	0.785801	0.768731	0.769709	0.667987
ALA:CB	C_beta	2968	0.961160	0.869182	0.983170	1.020313	0.970256	0.893486
ALA:H	H_amide	3295	0.430701	0.364372	0.397209	0.397503	0.412289	0.369196
ALA:HA	H_alpha	3098	0.252658	0.200166	0.226276	0.219180	0.223317	0.189993
ALA:N	N_backbone	3072	2.283020	2.155122	2.516885	2.404640	2.350763	2.148147
ARG:C	C_carbonyl	1580	1.094872	0.930874	1.076587	1.043841	1.087555	0.964407
ARG:CA	C_alpha	2071	0.967908	0.894439	0.986273	0.965883	0.949062	0.863315
ARG:CB	C_beta	1893	1.320596	1.280246	1.336299	1.350819	1.313219	1.290735

IUPAC	Role	<i>n</i>	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ARG:CD	C_other	1617	0.573970	0.583395	0.604760	0.615842	0.570848	0.576555
ARG:CG	C_other	1571	0.925730	0.919005	0.883940	0.940953	0.926197	0.922938
ARG:CZ	C_other	6	0.376819	0.334175	0.557327	0.248421	0.383538	0.248884
ARG:H	H_amide	2150	0.469492	0.390307	0.446043	0.415352	0.437565	0.405279
ARG:HA	H_alpha	2003	0.288369	0.220036	0.246457	0.231623	0.241378	0.213641
ARG:HB2	H_other	680	0.245145	0.245088	0.236739	0.237047	0.238363	0.235845
ARG:HB3	H_other	631	0.232497	0.223913	0.215816	0.220414	0.212336	0.206089
ARG:HD2	H_other	830	0.207305	0.187703	0.192997	0.187908	0.189460	0.184736
ARG:HD3	H_other	799	0.237057	0.230791	0.232557	0.224366	0.222850	0.229495
ARG:HE	H_other	522	0.659505	0.601600	0.646035	0.639757	0.629004	0.604043
ARG:HG2	H_other	654	0.234195	0.242657	0.229531	0.225251	0.214962	0.228656
ARG:HG3	H_other	612	0.241550	0.238290	0.224565	0.232224	0.219928	0.228445
ARG:HH11	H_other	8	1.023249	0.999576	1.112018	0.984490	0.961527	0.966735
ARG:HH12	H_other	8	1.008312	1.071490	1.097313	0.971314	0.962341	0.970406
ARG:HH21	H_other	10	1.188766	1.271811	1.285479	1.270234	1.253404	1.265986
ARG:HH22	H_other	8	1.349638	1.309504	1.369967	1.314464	1.362365	1.354005
ARG:N	N_backbone	1960	2.482793	2.139698	2.632101	2.922789	2.538878	2.157743
ARG:NE	N_sidechain	438	16.107848	16.070981	15.613937	15.397743	15.548651	15.725274
ARG:NH1	N_sidechain	3	7.175502	11.136740	7.128940	7.059639	7.301988	7.161165
ARG:NH2	N_sidechain	3	8.845539	7.729202	7.007652	6.947942	7.119233	6.856263
ASN:C	C_carbonyl	1268	1.043904	0.905133	1.053383	1.047315	1.055361	0.902194
ASN:CA	C_alpha	1712	0.957985	0.879063	0.948251	0.955924	0.963457	0.865951
ASN:CB	C_beta	1608	1.024320	1.006712	1.074348	1.063732	1.059415	1.030642
ASN:CG	C_other	95	1.050907	1.042331	1.070469	1.197358	1.178180	1.210763
ASN:H	H_amide	1787	0.514014	0.443563	0.471621	0.459705	0.476795	0.440062
ASN:HA	H_alpha	1686	0.247817	0.208031	0.234262	0.227779	0.231887	0.213962
ASN:HB2	H_other	504	0.334110	0.296983	0.305326	0.305517	0.298047	0.275334
ASN:HB3	H_other	477	0.337121	0.325799	0.333402	0.325816	0.320064	0.317259
ASN:HD21	H_other	350	0.435048	0.414753	0.434669	0.427405	0.406819	0.413459
ASN:HD22	H_other	349	0.525518	0.523789	0.514157	0.524741	0.510415	0.516874
ASN:N	N_backbone	1657	2.720020	2.212504	2.913357	2.562663	2.591605	2.220776
ASN:ND2	N_sidechain	1217	3.484847	3.370400	3.373051	3.413256	3.407192	3.340641
ASP:C	C_carbonyl	1855	1.029834	0.902505	1.034206	1.022965	0.991871	0.934029
ASP:CA	C_alpha	2459	0.862053	0.797504	0.862842	0.869178	0.841803	0.773895
ASP:CB	C_beta	2292	0.961423	0.876257	0.977471	0.960824	0.986459	0.879197
ASP:CG	C_other	25	30.376837	30.468126	30.347500	30.420274	30.376143	30.476044
ASP:H	H_amide	2549	0.475530	0.403615	0.436714	0.429311	0.444134	0.408608
ASP:HA	H_alpha	2405	0.223843	0.185635	0.207116	0.197331	0.210143	0.184842
ASP:HB2	H_other	684	0.274681	0.250314	0.266002	0.248500	0.252460	0.248429
ASP:HB3	H_other	660	0.241799	0.225979	0.232134	0.243094	0.230613	0.222716
ASP:N	N_backbone	2388	2.441928	2.012460	2.625440	2.386320	2.345528	2.023150
CYS:C	C_carbonyl	500	1.301632	1.144871	1.394946	1.331854	1.336645	1.207747

IUPAC	Role	<i>n</i>	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
CYS:CA	C_alpha	673	1.651367	1.580808	1.654131	1.716126	1.682147	1.682142
CYS:CB	C_beta	610	3.548757	3.430853	3.395478	3.415099	3.599199	3.240988
CYS:H	H_amide	750	0.539399	0.471733	0.554645	0.537861	0.547142	0.500690
CYS:HA	H_alpha	710	0.320001	0.252768	0.306295	0.274624	0.354569	0.313104
CYS:HB2	H_other	242	0.431553	0.353306	0.419902	0.402003	0.414789	0.360348
CYS:HB3	H_other	239	0.502051	0.459662	0.464973	0.474648	0.536187	0.484134
CYS:HG	H_other	2	1.639369	1.541968	1.677437	1.463087	1.339616	1.470923
CYS:N	N_backbone	643	3.063960	2.667335	3.182959	3.247618	3.423863	3.391098
GLN:C	C_carbonyl	1381	0.971787	0.824518	0.973984	0.980826	0.958884	0.839721
GLN:CA	C_alpha	1854	0.847943	0.793261	0.868993	0.825624	0.836452	0.771211
GLN:CB	C_beta	1730	0.995471	0.896910	0.975100	0.977586	0.935649	0.906085
GLN:CD	C_other	96	1.013871	1.005312	0.943047	1.173397	1.000949	1.043422
GLN:CG	C_other	1532	0.772973	0.725293	0.773414	0.754060	0.782849	0.727200
GLN:H	H_amide	1940	0.433221	0.362410	0.404115	0.400928	0.398156	0.367618
GLN:HA	H_alpha	1874	0.252063	0.200629	0.217528	0.213012	0.216470	0.201171
GLN:HB2	H_other	642	0.230445	0.202622	0.216453	0.196870	0.195955	0.185022
GLN:HB3	H_other	597	0.216858	0.197939	0.204865	0.194571	0.204205	0.200056
GLN:HE21	H_other	337	0.423767	0.442858	0.447383	0.461444	0.425266	0.449479
GLN:HE22	H_other	336	0.402997	0.401710	0.401191	0.407533	0.405982	0.399981
GLN:HG2	H_other	708	0.193183	0.174677	0.184333	0.175983	0.173636	0.173244
GLN:HG3	H_other	676	0.192759	0.197043	0.191621	0.195929	0.191514	0.186022
GLN:N	N_backbone	1769	2.401331	1.983633	2.582569	2.300208	2.292452	2.223827
GLN:NE2	N_sidechain	1242	2.970483	2.865812	2.873951	2.940234	2.840280	2.848275
GLU:C	C_carbonyl	2678	0.952292	0.907670	0.990990	1.003886	0.949269	0.933562
GLU:CA	C_alpha	3444	0.811713	0.753279	0.834757	0.838324	0.797215	0.749418
GLU:CB	C_beta	3177	1.045014	0.981275	1.034349	1.047003	1.007075	0.980052
GLU:CD	C_other	23	1.511565	1.292966	1.320248	1.610872	1.552367	1.625753
GLU:CG	C_other	2782	0.871199	0.841273	0.852246	0.863777	0.841734	0.822869
GLU:H	H_amide	3583	0.421897	0.372595	0.407051	0.401428	0.409527	0.371150
GLU:HA	H_alpha	3360	0.236821	0.187835	0.202832	0.201400	0.206620	0.183007
GLU:HB2	H_other	1156	0.181062	0.166931	0.173673	0.168096	0.168316	0.161798
GLU:HB3	H_other	1101	0.169583	0.165081	0.161183	0.166153	0.164999	0.155872
GLU:HG2	H_other	1178	0.160740	0.153908	0.158307	0.154934	0.146558	0.150445
GLU:HG3	H_other	1135	0.161853	0.165678	0.156066	0.160657	0.156160	0.156034
GLU:N	N_backbone	3322	2.269944	1.923386	2.457603	2.260481	2.210958	1.975152
GLY:C	C_carbonyl	2217	3.114101	3.099087	3.108495	3.129588	3.103852	3.090446
GLY:CA	C_alpha	2990	0.749747	0.631490	0.715857	0.734837	0.744412	0.659361
GLY:H	H_amide	3207	0.473879	0.379896	0.412334	0.403199	0.423751	0.376115
GLY:HA2	H_alpha	959	0.329847	0.349013	0.295073	0.307028	0.321507	0.321812
GLY:HA3	H_alpha	921	0.340459	0.372051	0.346397	0.343034	0.329180	0.360019
GLY:N	N_backbone	2805	2.840675	2.506203	2.915121	2.779242	2.817146	2.502991
HIS:C	C_carbonyl	702	1.224892	1.144569	1.239305	1.255515	1.274763	1.159853

IUPAC	Role	<i>n</i>	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
HIS:CA	C_alpha	910	1.366291	1.202759	1.354494	1.330390	1.336766	1.216105
HIS:CB	C_beta	866	1.391786	1.308711	1.341037	1.393941	1.341851	1.306613
HIS:CD2	C_other	493	4.871564	4.876685	4.882136	4.973433	4.879869	4.854986
HIS:CE1	C_other	381	1.959496	1.844235	1.845941	1.906688	1.842946	1.880008
HIS:CG	C_other	2	1.543532	2.073049	1.351456	1.420235	1.374642	2.160029
HIS:H	H_amide	925	0.491506	0.417136	0.487940	0.456329	0.466471	0.457553
HIS:HA	H_alpha	894	0.294218	0.239469	0.265574	0.257498	0.263867	0.237583
HIS:HB2	H_other	263	0.311464	0.249616	0.276100	0.258921	0.267437	0.250038
HIS:HB3	H_other	259	0.311699	0.301164	0.301914	0.294504	0.287253	0.289500
HIS:HD2	H_other	610	0.335697	0.293395	0.320645	0.302591	0.298367	0.293674
HIS:HE1	H_other	463	0.448710	0.458951	0.458857	0.449542	0.441891	0.438312
HIS:HE2	H_other	6	2.778356	2.820725	2.781906	2.798389	2.859973	2.783686
HIS:N	N_backbone	832	2.749974	2.271249	2.913868	2.706470	2.635963	2.271964
HIS:ND1	N_sidechain	15	36.307643	36.043259	35.582986	34.887740	34.828890	34.833256
HIS:NE2	N_sidechain	18	11.302768	11.182209	11.294687	11.000364	11.314219	11.152708
ILE:C	C_carbonyl	1767	1.027087	0.828736	0.991614	0.989163	0.983059	0.874713
ILE:CA	C_alpha	2289	1.135541	0.996008	1.090918	1.126318	1.181599	0.998561
ILE:CB	C_beta	2124	1.127731	0.952108	1.078956	1.137807	1.049969	0.996168
ILE:CD1	C_other	1957	1.317464	1.187521	1.200547	1.289478	1.232714	1.131083
ILE:CG1	C_other	1743	1.118827	1.083696	1.126564	1.192561	1.105023	1.049907
ILE:CG2	C_other	1913	1.088268	0.922389	1.072685	1.040382	1.056869	0.920993
ILE:H	H_amide	2409	0.449284	0.375018	0.408226	0.394601	0.412214	0.379810
ILE:HA	H_alpha	2256	0.306552	0.216154	0.247976	0.246900	0.245682	0.217026
ILE:HB	H_other	2151	0.817807	0.800650	0.813785	0.807322	0.807211	0.800049
ILE:HG12	H_other	617	0.441875	0.458778	0.449841	0.450725	0.432680	0.448018
ILE:HG13	H_other	586	0.451013	0.481244	0.470412	0.490490	0.435564	0.457903
ILE:N	N_backbone	2244	2.429101	2.255279	2.762716	2.572878	2.588623	2.354048
LEU:C	C_carbonyl	3026	1.067444	0.935413	1.069888	1.105559	1.037977	0.936164
LEU:CA	C_alpha	3961	0.778985	0.719107	0.810637	0.838857	0.811195	0.728876
LEU:CB	C_beta	3712	1.013809	0.887127	1.095708	1.107443	0.983713	0.893262
LEU:CG	C_other	3058	0.928260	0.892329	0.938077	0.938691	0.904877	0.883232
LEU:H	H_amide	4153	0.474532	0.397352	0.424413	0.418699	0.438759	0.393657
LEU:HA	H_alpha	3887	0.243084	0.180278	0.219680	0.214842	0.205376	0.180662
LEU:HB2	H_other	1148	0.299136	0.288623	0.287930	0.287031	0.275572	0.278755
LEU:HB3	H_other	1103	0.323762	0.305509	0.305620	0.310931	0.306702	0.298881
LEU:HG	H_other	3327	0.303891	0.263555	0.302195	0.293583	0.274144	0.268340
LEU:N	N_backbone	3846	2.241051	2.070776	2.535051	2.540561	2.303737	2.159905
LYS:C	C_carbonyl	2334	0.987169	0.879148	0.975685	0.983340	0.988483	0.893217
LYS:CA	C_alpha	3055	0.900960	0.839205	0.941256	0.938717	0.872782	0.842287
LYS:CB	C_beta	2818	0.893917	0.855817	0.921821	0.936289	0.898319	0.856088
LYS:CD	C_other	2302	0.819436	0.817214	0.810516	0.832309	0.806576	0.819309
LYS:CE	C_other	2269	0.484135	0.470347	0.475530	0.473141	0.474935	0.468691

IUPAC	Role	<i>n</i>	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
LYS:CG	C_other	2413	0.770634	0.739939	0.762899	0.796548	0.764106	0.738742
LYS:H	H_amide	3184	0.446376	0.370693	0.434022	0.411161	0.412456	0.377214
LYS:HA	H_alpha	3006	0.250466	0.199383	0.221740	0.213127	0.219211	0.201356
LYS:HB2	H_other	973	0.196230	0.182439	0.190013	0.185847	0.166386	0.168719
LYS:HB3	H_other	930	0.209122	0.185289	0.196209	0.187535	0.187085	0.189000
LYS:HD2	H_other	1344	3.220162	3.221440	3.221781	3.221195	3.220153	3.221366
LYS:HD3	H_other	1288	0.154300	0.138258	0.143971	0.138890	0.142802	0.139011
LYS:HE2	H_other	1502	0.171553	0.153797	0.167091	0.164931	0.150779	0.154441
LYS:HE3	H_other	1437	0.156970	0.147888	0.159708	0.157206	0.145857	0.147178
LYS:HG2	H_other	929	0.202256	0.202266	0.194376	0.198226	0.203080	0.199743
LYS:HG3	H_other	906	0.223085	0.198730	0.206711	0.196554	0.198960	0.207034
LYS:N	N_backbone	2971	2.424341	2.087055	2.602039	2.472134	2.364197	2.322824
LYS:NZ	N_sidechain	3	44.898711	43.889796	43.951603	43.751063	43.851682	43.746612
MET:C	C_carbonyl	698	1.164583	1.051077	1.184658	1.128960	1.127690	1.065611
MET:CA	C_alpha	925	0.957369	0.973825	0.989100	1.082694	1.021140	0.960027
MET:CB	C_beta	868	1.489778	1.399220	1.475329	1.521245	1.481540	1.361034
MET:CE	C_other	541	1.547216	1.525327	1.593642	1.552940	1.560114	1.511162
MET:CG	C_other	705	1.527812	1.457417	1.535424	1.514174	1.507885	1.462986
MET:H	H_amide	896	0.486581	0.451718	0.483826	0.483806	0.484179	0.442341
MET:HA	H_alpha	896	0.279876	0.222852	0.250474	0.246326	0.247414	0.224203
MET:HB2	H_other	290	0.248149	0.239442	0.244927	0.242345	0.236960	0.234605
MET:HB3	H_other	270	0.267564	0.238615	0.261212	0.252940	0.257124	0.248098
MET:HG2	H_other	244	0.300354	0.263324	0.286570	0.263485	0.269339	0.262635
MET:HG3	H_other	232	0.303327	0.286019	0.300745	0.297426	0.279399	0.277640
MET:N	N_backbone	852	2.639010	2.104238	2.712903	2.384978	2.404745	2.208004
PHE:C	C_carbonyl	1174	1.134762	0.949102	1.158372	1.166158	1.080261	0.952281
PHE:CA	C_alpha	1539	1.096662	1.010412	1.142048	1.177063	1.152377	1.029532
PHE:CB	C_beta	1447	1.105639	0.942961	1.141336	1.127404	1.067264	0.952175
PHE:CD1	C_other	859	3.988145	3.967412	3.995005	3.995747	3.991633	3.965336
PHE:CD2	C_other	726	4.267717	4.256842	4.269083	4.288563	4.272724	4.263897
PHE:CE1	C_other	778	0.921371	0.866465	0.908922	0.915691	0.865101	0.897788
PHE:CE2	C_other	671	0.809104	0.732762	0.787523	0.719946	0.693980	0.716616
PHE:CZ	C_other	727	1.219030	1.150895	1.166307	1.198106	1.164204	1.136693
PHE:H	H_amide	1621	0.491344	0.414445	0.464343	0.439693	0.471657	0.411015
PHE:HA	H_alpha	1545	0.323806	0.248445	0.272672	0.263230	0.267189	0.244231
PHE:HB2	H_other	399	0.354930	0.288022	0.343214	0.319984	0.331301	0.278233
PHE:HB3	H_other	393	0.357336	0.308489	0.342051	0.332604	0.310943	0.304938
PHE:HD1	H_other	1106	0.270472	0.227860	0.246094	0.239105	0.229603	0.217827
PHE:HD2	H_other	1036	0.276094	0.238748	0.266788	0.255320	0.251019	0.231740
PHE:HE1	H_other	1009	0.276765	0.236767	0.260042	0.247936	0.237877	0.231414
PHE:HE2	H_other	935	0.283478	0.251820	0.264307	0.260398	0.257242	0.243630
PHE:HZ	H_other	924	0.306323	0.258989	0.282356	0.272560	0.259812	0.244709

IUPAC	Role	<i>n</i>	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
PHE:N	N_backbone	1497	2.588122	2.240743	2.902261	2.580636	2.517033	2.262676
PRO:C	C_carbonyl	1382	1.028532	0.916408	1.012789	0.992778	0.977397	0.921493
PRO:CA	C_alpha	1973	0.781406	0.733645	0.767870	0.778646	0.765034	0.724596
PRO:CB	C_beta	1814	1.315440	1.261203	1.335127	1.328228	1.314510	1.262796
PRO:CD	C_other	1736	1.154424	1.113301	1.148547	1.164490	1.155455	1.125043
PRO:CG	C_other	1698	0.688603	0.653099	0.686755	0.720308	0.685357	0.649413
PRO:HA	H_alpha	1998	0.281147	0.203915	0.245577	0.217032	0.230145	0.205315
PRO:HB2	H_other	458	0.322166	0.307593	0.334121	0.339896	0.321635	0.299430
PRO:HB3	H_other	434	0.353058	0.332447	0.332861	0.343384	0.340104	0.335920
PRO:HD2	H_other	523	0.266815	0.231787	0.245111	0.226920	0.223168	0.230459
PRO:HD3	H_other	520	0.280748	0.257385	0.288766	0.274965	0.261607	0.260889
PRO:HG2	H_other	677	0.207365	0.176796	0.193724	0.175238	0.183475	0.167151
PRO:HG3	H_other	650	0.193995	0.178738	0.178801	0.176749	0.176760	0.172113
PRO:N	N_backbone	29	20.944604	21.353028	20.705112	21.414174	21.330980	21.248141
SER:C	C_carbonyl	2272	1.038987	0.909626	1.083890	1.054069	1.050913	0.940331
SER:CA	C_alpha	2936	0.949045	0.906538	0.968208	0.984964	0.969808	0.910447
SER:CB	C_beta	2669	0.844038	0.800865	0.862695	0.870576	0.846277	0.791360
SER:H	H_amide	3114	0.448904	0.374926	0.418092	0.398326	0.413308	0.371997
SER:HA	H_alpha	3044	0.253533	0.202791	0.221770	0.228666	0.223061	0.207620
SER:HB2	H_other	1273	0.216928	0.204064	0.210114	0.208547	0.203912	0.198433
SER:HB3	H_other	1212	0.199506	0.191209	0.198164	0.198124	0.198494	0.192315
SER:HG	H_other	47	0.821843	0.729433	0.804493	0.790706	0.782527	0.728943
SER:N	N_backbone	2717	2.507194	2.080676	2.751431	2.570273	2.443435	2.208748
THR:C	C_carbonyl	1754	2.838159	2.793891	2.835560	2.832302	2.824875	2.799157
THR:CA	C_alpha	2400	1.038983	1.049388	1.080786	1.121269	1.090506	1.023150
THR:CB	C_beta	2225	1.120644	1.083444	1.168336	1.150249	1.125833	1.083485
THR:CG2	C_other	2036	0.857487	0.702743	0.811754	0.818661	0.787479	0.727737
THR:H	H_amide	2538	0.499426	0.414169	0.452659	0.451432	0.463844	0.409254
THR:HA	H_alpha	2417	0.279459	0.199511	0.230616	0.221342	0.220504	0.210209
THR:HB	H_other	2285	0.247818	0.201769	0.244772	0.228721	0.228735	0.208197
THR:HG1	H_other	48	0.800816	0.713679	0.736994	0.720208	0.757887	0.776520
THR:N	N_backbone	2323	2.823462	2.484476	3.195396	2.917795	2.856499	2.610259
TRP:C	C_carbonyl	439	1.156256	1.010676	1.168682	1.167874	1.129131	0.951254
TRP:CA	C_alpha	560	1.397359	1.149263	1.346755	1.390083	1.389882	1.154186
TRP:CB	C_beta	520	1.138010	0.957789	1.064149	1.192394	1.079390	0.985949
TRP:CD1	C_other	399	1.713287	1.504402	1.621765	1.618260	1.648279	1.530593
TRP:CD2	C_other	1	4.733183	4.504805	5.069853	4.297918	4.571112	4.184874
TRP:CE2	C_other	1	15.062133	14.925461	14.672393	14.661010	14.924919	14.887764
TRP:CE3	C_other	370	1.534441	1.364424	1.449164	1.465403	1.482581	1.327297
TRP:CG	C_other	1	2.026495	1.984198	1.744925	2.002348	1.548491	2.098951
TRP:CH2	C_other	391	1.229895	1.138047	1.169630	1.272986	1.250145	1.149564
TRP:CZ2	C_other	392	1.257358	1.152686	1.233590	1.256567	1.242512	1.158630

IUPAC	Role	<i>n</i>	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
TRP:CZ3	C_other	372	1.243134	1.177025	1.220657	1.233392	1.261870	1.145382
TRP:H	H_amide	590	0.520759	0.509596	0.527394	0.516854	0.539600	0.489361
TRP:HA	H_alpha	569	0.364725	0.253678	0.290813	0.290190	0.293087	0.254884
TRP:HB2	H_other	158	0.308581	0.321182	0.321375	0.353642	0.310842	0.294066
TRP:HB3	H_other	149	0.319961	0.326486	0.299906	0.302945	0.277134	0.307569
TRP:HD1	H_other	495	0.312224	0.233988	0.275813	0.238070	0.263322	0.232737
TRP:HE1	H_other	503	0.931774	0.848889	0.957172	0.847233	0.931062	0.869226
TRP:HE3	H_other	455	0.387464	0.309096	0.333531	0.312890	0.331870	0.297169
TRP:HH2	H_other	463	0.306811	0.226304	0.255563	0.248112	0.268721	0.220218
TRP:HZ2	H_other	476	0.315082	0.237719	0.291689	0.265983	0.250003	0.226654
TRP:HZ3	H_other	429	0.335835	0.291870	0.304069	0.295283	0.306203	0.291411
TRP:N	N_backbone	540	2.645187	2.186767	2.833678	2.504017	2.705712	2.172649
TRP:NE1	N_sidechain	483	3.509214	3.336444	3.481926	3.408526	3.416184	3.371541
TYR:C	C_carbonyl	992	1.223815	0.978127	1.142926	1.157928	1.091140	1.000262
TYR:CA	C_alpha	1302	1.059845	0.993763	1.075805	1.158458	1.091343	0.943999
TYR:CB	C_beta	1226	1.130549	0.960233	1.146264	1.148302	1.053877	0.961297
TYR:CD1	C_other	746	5.931486	5.905875	5.914645	5.923038	5.916544	5.914392
TYR:CD2	C_other	621	6.521969	6.515593	6.526534	6.520912	6.517037	6.528285
TYR:CE1	C_other	748	3.084832	3.078868	3.075621	3.079028	3.077023	3.078853
TYR:CE2	C_other	625	3.241975	3.231955	3.233826	3.227204	3.229576	3.228921
TYR:CG	C_other	7	33.130993	33.247778	33.284794	33.338803	33.186759	33.390570
TYR:CZ	C_other	6	27.248612	27.433184	27.316358	27.362892	27.002031	26.883278
TYR:H	H_amide	1363	0.513398	0.417901	0.450619	0.438232	0.464781	0.417524
TYR:HA	H_alpha	1302	0.321766	0.233706	0.277614	0.257555	0.278501	0.237066
TYR:HB2	H_other	343	0.341598	0.289764	0.320302	0.298588	0.327617	0.291004
TYR:HB3	H_other	330	0.383904	0.328797	0.363829	0.344739	0.335292	0.333367
TYR:HD1	H_other	927	0.249832	0.228452	0.246500	0.238496	0.241435	0.225230
TYR:HD2	H_other	860	0.263289	0.222419	0.247948	0.237378	0.241944	0.217304
TYR:HE1	H_other	906	0.221080	0.205553	0.221948	0.213740	0.213864	0.204937
TYR:HE2	H_other	830	0.208019	0.181392	0.195026	0.189761	0.187463	0.182251
TYR:HH	H_other	21	1.207602	1.210710	1.200995	1.189634	1.199179	1.212973
TYR:N	N_backbone	1264	2.533626	2.257568	2.999674	2.664407	2.723693	2.383825
VAL:C	C_carbonyl	2444	0.988715	0.866464	1.020162	1.106934	1.013598	0.897270
VAL:CA	C_alpha	3189	0.939986	0.942735	1.183849	1.126435	0.971872	0.956297
VAL:CB	C_beta	2967	0.903103	0.872263	1.103284	1.023514	0.914294	0.873299
VAL:H	H_amide	3341	0.459462	0.394973	0.426160	0.446354	0.423323	0.389598
VAL:HA	H_alpha	3153	0.340354	0.281064	0.345829	0.348649	0.301451	0.280112
VAL:HB	H_other	3034	0.278463	0.251854	0.307497	0.307914	0.256131	0.246166
VAL:N	N_backbone	3118	2.563042	2.286060	3.207000	3.254036	2.574783	2.354237

8.5 External 1,742: exact-assignment RMSE (ppm)

IUPAC	Role	<i>n</i>	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ALA:C	C_carbonyl	8064	1.367811	1.333927	1.401645	1.440341	1.409608	1.322807
ALA:CA	C_alpha	11288	0.839468	0.739833	0.853247	0.833829	0.846268	0.751258
ALA:CB	C_beta	10010	1.230613	1.158393	1.239826	1.274080	1.240374	1.204510
ALA:H	H_amide	13994	0.458892	0.389703	0.434071	0.426731	0.463823	0.413005
ALA:HA	H_alpha	10461	0.257392	0.216804	0.236865	0.231172	0.237701	0.206557
ALA:N	N_backbone	12046	2.985726	2.817758	3.157137	3.017560	3.020773	2.840512
ARG:C	C_carbonyl	5059	1.225392	1.085523	1.207017	1.187481	1.233747	1.119980
ARG:CA	C_alpha	7048	1.073757	0.996505	1.079830	1.065622	1.060973	0.984886
ARG:CB	C_beta	6118	1.227473	1.153235	1.238943	1.228071	1.204146	1.165999
ARG:CD	C_other	4201	0.759566	0.758862	0.782195	0.782962	0.769677	0.763624
ARG:CG	C_other	4121	1.323190	1.297710	1.304845	1.332362	1.330030	1.314837
ARG:CZ	C_other	84	3.969922	3.972858	4.067259	4.000213	3.951312	3.892274
ARG:H	H_amide	8923	0.503248	0.413225	0.471721	0.438584	0.470900	0.441420
ARG:HA	H_alpha	6874	0.284171	0.233614	0.256121	0.242666	0.257168	0.231959
ARG:HB2	H_other	2462	0.250325	0.237759	0.239793	0.230790	0.234781	0.230333
ARG:HB3	H_other	2274	0.249238	0.242776	0.238050	0.234213	0.232591	0.229164
ARG:HD2	H_other	2850	0.219168	0.210944	0.212991	0.209584	0.208168	0.208257
ARG:HD3	H_other	2613	0.240267	0.235920	0.241459	0.234309	0.233519	0.239870
ARG:HE	H_other	1932	0.637548	0.587368	0.632331	0.628948	0.622378	0.583686
ARG:HG2	H_other	2482	0.238561	0.250857	0.236666	0.233840	0.228096	0.235432
ARG:HG3	H_other	2290	0.252625	0.250993	0.239080	0.251377	0.238775	0.243115
ARG:HH11	H_other	85	0.556774	0.565781	0.556345	0.549519	0.553765	0.556293
ARG:HH12	H_other	77	0.689808	0.670258	0.729865	0.703026	0.675975	0.657517
ARG:HH21	H_other	73	0.627116	0.606336	0.635025	0.622768	0.610910	0.605830
ARG:HH22	H_other	66	0.700755	0.660533	0.740480	0.689913	0.707642	0.683780
ARG:N	N_backbone	7420	2.752749	2.390130	2.826399	3.013130	2.792526	2.480719
ARG:NE	N_sidechain	1157	16.186353	16.070737	15.970861	15.501371	15.835976	15.883063
ARG:NH1	N_sidechain	20	15.901554	16.602434	15.473200	15.445274	15.883920	15.706076
ARG:NH2	N_sidechain	14	12.060161	10.972410	11.186035	11.113821	10.891457	10.855511
ASN:C	C_carbonyl	4356	1.252943	1.092411	1.222676	1.224575	1.243192	1.113829
ASN:CA	C_alpha	6175	1.034408	0.941062	1.021099	1.012802	1.046122	0.935889
ASN:CB	C_beta	5514	1.095849	1.068102	1.134232	1.126123	1.149592	1.099001
ASN:CG	C_other	360	3.691678	3.695987	3.692692	3.723394	3.690540	3.748146
ASN:H	H_amide	7823	0.527439	0.462202	0.491336	0.474190	0.503829	0.472800
ASN:HA	H_alpha	6058	0.264413	0.222746	0.245922	0.243385	0.250116	0.229273
ASN:HB2	H_other	1875	0.331287	0.316235	0.314827	0.321711	0.307262	0.303347
ASN:HB3	H_other	1739	0.313302	0.315648	0.304302	0.300935	0.296505	0.297349
ASN:HD21	H_other	1273	0.553384	0.542231	0.548356	0.548593	0.530231	0.530802
ASN:HD22	H_other	1261	0.542171	0.543979	0.531196	0.546951	0.534650	0.534415
ASN:N	N_backbone	6482	2.883691	2.309918	3.085203	2.717594	2.762982	2.396310

IUPAC	Role	<i>n</i>	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ASN:ND2	N_sidechain	3414	2.857010	2.704911	2.774117	2.814751	2.780072	2.710760
ASP:C	C_carbonyl	6354	1.212688	1.082484	1.206943	1.188496	1.179329	1.115287
ASP:CA	C_alpha	8884	0.972355	0.892327	0.966315	0.967315	0.961753	0.894015
ASP:CB	C_beta	7923	1.557112	1.506461	1.590063	1.580484	1.595239	1.517622
ASP:CG	C_other	149	12.552037	12.564092	12.542677	12.545840	12.557484	12.556538
ASP:H	H_amide	11056	0.513580	0.440549	0.475257	0.470890	0.489056	0.453625
ASP:HA	H_alpha	8356	0.233513	0.193165	0.215405	0.202525	0.216093	0.193216
ASP:HB2	H_other	2605	0.290542	0.273602	0.276655	0.269308	0.263265	0.261899
ASP:HB3	H_other	2433	0.275954	0.260006	0.267243	0.273294	0.265023	0.255746
ASP:N	N_backbone	9546	2.752865	2.294941	3.001019	2.657027	2.698710	2.338094
CYS:C	C_carbonyl	1757	1.680092	1.469412	1.741427	1.631420	1.734959	1.510098
CYS:CA	C_alpha	2412	1.780985	1.709334	1.766126	1.808699	1.795848	1.845720
CYS:CB	C_beta	2115	3.806458	3.582684	3.748024	3.693651	3.850278	3.478946
CYS:H	H_amide	4314	0.607043	0.511448	0.588414	0.564459	0.644916	0.588213
CYS:HA	H_alpha	3755	0.608850	0.562750	0.592292	0.574564	0.625805	0.605507
CYS:HB2	H_other	1291	0.438140	0.413655	0.425780	0.420150	0.462756	0.424664
CYS:HB3	H_other	1261	0.464820	0.441133	0.434304	0.453466	0.488254	0.451830
CYS:HG	H_other	35	1.874598	1.877013	1.936690	1.828329	1.737214	1.785053
CYS:N	N_backbone	2832	3.226759	2.783078	3.397138	3.142761	3.997066	3.522785
GLN:C	C_carbonyl	4299	1.077342	0.959922	1.062240	1.077945	1.080335	0.970043
GLN:CA	C_alpha	6040	0.944782	0.877593	0.954876	0.941712	0.927053	0.874249
GLN:CB	C_beta	5391	1.200699	1.074986	1.164840	1.158066	1.147832	1.093476
GLN:CD	C_other	344	1.251039	1.255751	1.215648	1.363609	1.224947	1.254522
GLN:CG	C_other	3874	1.779408	1.759713	1.781532	1.779553	1.783228	1.760884
GLN:H	H_amide	7484	0.444500	0.389627	0.420070	0.417012	0.431469	0.394659
GLN:HA	H_alpha	5877	0.260783	0.216993	0.233229	0.227767	0.236038	0.212720
GLN:HB2	H_other	2115	0.246901	0.229127	0.237458	0.224010	0.222627	0.217693
GLN:HB3	H_other	1929	0.253135	0.240667	0.239776	0.236206	0.238210	0.237241
GLN:HE21	H_other	1144	0.476413	0.485893	0.489835	0.506849	0.468857	0.475960
GLN:HE22	H_other	1138	0.440447	0.433580	0.434888	0.438542	0.436638	0.433714
GLN:HG2	H_other	2277	0.228439	0.212894	0.220645	0.214375	0.215340	0.208768
GLN:HG3	H_other	2083	0.222090	0.230336	0.219774	0.228857	0.219891	0.222285
GLN:N	N_backbone	6351	2.520103	2.104101	2.674577	2.449355	2.453352	2.320577
GLN:NE2	N_sidechain	3213	2.980356	2.888948	2.906277	2.953332	2.883415	2.874538
GLU:C	C_carbonyl	8429	2.084761	2.064371	2.108820	2.110122	2.092699	2.081262
GLU:CA	C_alpha	11487	0.937298	0.881865	0.960699	0.958671	0.955921	0.886944
GLU:CB	C_beta	10148	1.092164	1.036335	1.108038	1.108003	1.086749	1.043733
GLU:CD	C_other	159	1.762701	1.657496	1.730555	1.861257	1.867793	1.863807
GLU:CG	C_other	7179	1.034526	1.008445	1.026961	1.030412	1.012890	1.003812
GLU:H	H_amide	14010	0.455194	0.392674	0.436407	0.423334	0.442053	0.406626
GLU:HA	H_alpha	10647	0.236110	0.199165	0.211687	0.207953	0.216826	0.193195
GLU:HB2	H_other	3898	0.208751	0.189373	0.201760	0.190871	0.193192	0.184492

IUPAC	Role	<i>n</i>	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
GLU:HB3	H_other	3591	0.197880	0.193592	0.190274	0.190748	0.190613	0.187014
GLU:HG2	H_other	4053	0.199745	0.197374	0.195740	0.197072	0.191414	0.192059
GLU:HG3	H_other	3746	0.199139	0.202134	0.194751	0.199735	0.194940	0.197147
GLU:N	N_backbone	12177	2.588762	2.275656	2.766473	2.568037	2.593503	2.332369
GLY:C	C_carbonyl	7570	2.688789	2.657079	2.670713	2.710637	2.674134	2.647313
GLY:CA	C_alpha	10679	0.837722	0.725699	0.804710	0.824608	0.848807	0.768427
GLY:H	H_amide	13687	0.512888	0.439291	0.476679	0.468376	0.475947	0.443981
GLY:HA2	H_alpha	3560	0.378130	0.387040	0.343618	0.360372	0.367607	0.364591
GLY:HA3	H_alpha	3340	0.459944	0.488786	0.458228	0.448846	0.452721	0.481830
GLY:N	N_backbone	11293	3.418997	3.073118	3.476954	3.286226	3.364342	3.115260
HIS:C	C_carbonyl	2332	1.342626	1.215476	1.339820	1.349219	1.352253	1.253035
HIS:CA	C_alpha	3301	1.451442	1.305663	1.437872	1.434493	1.413044	1.355061
HIS:CB	C_beta	2926	1.699788	1.591874	1.667212	1.678971	1.665988	1.616685
HIS:CD2	C_other	1204	4.037012	4.020988	4.030418	4.043112	4.016833	3.998935
HIS:CE1	C_other	908	2.575834	2.491430	2.490875	2.506242	2.474800	2.483462
HIS:CG	C_other	24	4.198102	4.471357	4.283033	4.207707	4.163488	4.658857
HIS:H	H_amide	4058	0.575686	0.496346	0.567005	0.528025	0.549653	0.521253
HIS:HA	H_alpha	3136	0.349351	0.303011	0.325390	0.312455	0.321619	0.296108
HIS:HB2	H_other	1006	0.360857	0.320655	0.337042	0.325956	0.327179	0.319096
HIS:HB3	H_other	970	0.355088	0.326755	0.331348	0.316364	0.320050	0.329177
HIS:HD2	H_other	1896	0.647211	0.629004	0.643253	0.632718	0.634213	0.630561
HIS:HE1	H_other	1512	1.052326	1.049559	1.050880	1.049553	1.045924	1.039944
HIS:HE2	H_other	60	3.367496	3.362554	3.344856	3.332972	3.396706	3.323559
HIS:N	N_backbone	3315	3.014166	2.507438	3.059203	2.873671	2.839625	2.583536
HIS:ND1	N_sidechain	118	37.670334	37.372248	37.782203	37.454021	38.011994	37.750641
HIS:NE2	N_sidechain	123	21.380709	21.275654	21.416663	21.346588	21.261535	21.450792
ILE:C	C_carbonyl	5817	1.195318	1.014810	1.151013	1.164066	1.183743	1.046587
ILE:CA	C_alpha	8113	1.206076	1.076374	1.165113	1.197249	1.290213	1.092686
ILE:CB	C_beta	7186	1.170284	1.057303	1.161582	1.206577	1.166413	1.087576
ILE:CD1	C_other	5439	2.511524	2.441496	2.429761	2.459629	2.439495	2.402585
ILE:CG1	C_other	4646	1.537938	1.508904	1.540816	1.595517	1.526110	1.491271
ILE:CG2	C_other	5165	1.460545	1.344493	1.457510	1.411842	1.447130	1.352299
ILE:H	H_amide	9927	0.499915	0.421904	0.448688	0.426286	0.462227	0.426194
ILE:HA	H_alpha	7513	0.312547	0.233602	0.260299	0.258605	0.260911	0.232175
ILE:HB	H_other	6911	0.672206	0.649719	0.667060	0.657956	0.656802	0.650524
ILE:HG12	H_other	2380	0.441954	0.437228	0.429060	0.431760	0.431765	0.428881
ILE:HG13	H_other	2172	0.430327	0.467363	0.458160	0.483309	0.412762	0.450835
ILE:N	N_backbone	8566	2.715766	2.508249	2.995194	2.814010	2.864254	2.628901
LEU:C	C_carbonyl	9551	1.223235	1.103371	1.205973	1.253025	1.251640	1.107938
LEU:CA	C_alpha	13124	0.861103	0.815232	0.890832	0.924391	0.917394	0.833189
LEU:CB	C_beta	11616	1.131176	1.024990	1.231293	1.231141	1.133161	1.081759
LEU:CG	C_other	7800	1.195070	1.166909	1.196657	1.202883	1.182132	1.162831

IUPAC	Role	<i>n</i>	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
LEU:H	H_amide	16076	0.499418	0.430870	0.457566	0.455631	0.475526	0.433387
LEU:HA	H_alpha	12074	0.259656	0.200405	0.234736	0.228523	0.226427	0.201965
LEU:HB2	H_other	4036	0.316016	0.305584	0.305286	0.304115	0.289805	0.296328
LEU:HB3	H_other	3809	0.348002	0.328847	0.332696	0.336244	0.334048	0.315667
LEU:HG	H_other	9753	0.312394	0.276514	0.308640	0.302998	0.288416	0.282226
LEU:N	N_backbone	13752	2.387344	2.176420	2.659315	2.644691	2.454284	2.301027
LYS:C	C_carbonyl	7848	1.636077	1.563067	1.622100	1.639417	1.634520	1.580050
LYS:CA	C_alpha	10882	0.947318	0.891667	0.989659	0.980962	0.937347	0.899576
LYS:CB	C_beta	9566	1.190282	1.153364	1.222487	1.219435	1.196056	1.165224
LYS:CD	C_other	6105	1.061144	1.059599	1.059196	1.079459	1.061870	1.064057
LYS:CE	C_other	5909	0.774132	0.768047	0.769247	0.771093	0.768917	0.767629
LYS:CG	C_other	6434	0.950552	0.922184	0.943344	0.967950	0.945521	0.919744
LYS:H	H_amide	13689	0.464700	0.391791	0.446378	0.424988	0.448159	0.403765
LYS:HA	H_alpha	10429	0.264376	0.218761	0.237591	0.230090	0.239643	0.221978
LYS:HB2	H_other	3862	0.221767	0.200841	0.210256	0.201507	0.197590	0.193766
LYS:HB3	H_other	3628	0.221328	0.199508	0.209780	0.202032	0.203850	0.201585
LYS:HD2	H_other	4504	1.814451	1.814570	1.814818	1.814474	1.814387	1.814815
LYS:HD3	H_other	4145	0.468158	0.464030	0.465716	0.464532	0.465390	0.464968
LYS:HE2	H_other	4819	0.172639	0.163561	0.166068	0.167771	0.161495	0.161369
LYS:HE3	H_other	4372	0.164412	0.159949	0.164130	0.163355	0.159443	0.159463
LYS:HG2	H_other	3657	0.214180	0.211080	0.211498	0.208747	0.211920	0.205884
LYS:HG3	H_other	3418	0.227593	0.212133	0.220355	0.212412	0.211962	0.216623
LYS:N	N_backbone	11582	2.562455	2.208583	2.744574	2.552715	2.538661	2.426978
LYS:NZ	N_sidechain	26	80.477181	79.578402	78.807930	79.274133	78.566826	78.743544
MET:C	C_carbonyl	2194	1.318057	1.243710	1.334114	1.303938	1.327166	1.265172
MET:CA	C_alpha	3172	1.068354	1.063489	1.115192	1.166487	1.147673	1.068474
MET:CB	C_beta	2771	1.570365	1.488838	1.569623	1.612948	1.567140	1.475395
MET:CE	C_other	1247	2.028707	2.001048	2.077164	2.021756	2.037969	1.994376
MET:CG	C_other	1777	1.320537	1.279885	1.332065	1.330215	1.302034	1.289621
MET:H	H_amide	3638	0.493188	0.443500	0.474860	0.477207	0.486166	0.440883
MET:HA	H_alpha	2836	0.298682	0.246419	0.268980	0.262186	0.267674	0.241624
MET:HB2	H_other	970	0.452412	0.437966	0.442102	0.437418	0.432628	0.435179
MET:HB3	H_other	883	0.460672	0.440903	0.455180	0.446275	0.449081	0.442531
MET:HG2	H_other	886	1.096580	1.090550	1.091232	1.088101	1.083823	1.083718
MET:HG3	H_other	821	1.145133	1.140480	1.144855	1.141170	1.136850	1.136032
MET:N	N_backbone	3183	3.152485	2.640443	3.166746	2.911665	2.905870	2.754214
PHE:C	C_carbonyl	4057	1.366869	1.167283	1.321491	1.357907	1.297214	1.186695
PHE:CA	C_alpha	5724	1.244892	1.184923	1.295269	1.328631	1.355941	1.233774
PHE:CB	C_beta	5131	1.220665	1.064195	1.247998	1.254989	1.188761	1.090857
PHE:CD1	C_other	1935	3.006509	2.973078	3.010207	3.006713	2.999528	2.972918
PHE:CD2	C_other	1510	3.295359	3.294015	3.306726	3.321176	3.302510	3.300156
PHE:CE1	C_other	1698	1.779712	1.744476	1.770461	1.775014	1.740017	1.762293

IUPAC	Role	<i>n</i>	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
PHE:CE2	C_other	1325	1.734328	1.725117	1.733734	1.720071	1.693754	1.726345
PHE:CG	C_other	50	7.926830	8.325244	8.136091	7.888266	7.932986	8.483329
PHE:CZ	C_other	1540	1.886574	1.828410	1.856517	1.872348	1.832405	1.820092
PHE:H	H_amide	7024	0.529451	0.460390	0.505149	0.480968	0.522938	0.467205
PHE:HA	H_alpha	5315	0.339734	0.263204	0.285464	0.280092	0.291655	0.257739
PHE:HB2	H_other	1633	0.382857	0.328048	0.368962	0.350216	0.346465	0.324034
PHE:HB3	H_other	1589	0.433844	0.392526	0.420580	0.416873	0.403967	0.390412
PHE:HD1	H_other	3227	0.300554	0.260183	0.277731	0.270647	0.267147	0.255551
PHE:HD2	H_other	2908	0.309700	0.270153	0.298527	0.283648	0.285896	0.266802
PHE:HE1	H_other	2918	0.347066	0.310980	0.328674	0.320217	0.317481	0.307452
PHE:HE2	H_other	2635	0.289792	0.258793	0.274494	0.267319	0.266053	0.253343
PHE:HZ	H_other	2595	0.510980	0.462730	0.490142	0.475013	0.472613	0.458631
PHE:N	N_backbone	6029	2.665875	2.288522	3.000062	2.726193	2.703004	2.382826
PRO:C	C_carbonyl	4289	1.204109	1.081009	1.191970	1.166498	1.187884	1.085364
PRO:CA	C_alpha	6202	0.905235	0.854254	0.882091	0.892782	0.882145	0.855671
PRO:CB	C_beta	5485	1.081401	1.013527	1.101646	1.100990	1.092046	1.024186
PRO:CD	C_other	4122	1.308113	1.265395	1.295178	1.310853	1.305403	1.284862
PRO:CG	C_other	4048	0.804728	0.773906	0.804916	0.822516	0.791175	0.766363
PRO:HA	H_alpha	5993	0.293018	0.229393	0.268572	0.245998	0.256701	0.237467
PRO:HB2	H_other	1698	0.338323	0.326022	0.344709	0.359924	0.325988	0.321464
PRO:HB3	H_other	1568	0.345479	0.326317	0.329861	0.341513	0.338702	0.338143
PRO:HD2	H_other	1669	0.321557	0.291121	0.294102	0.287115	0.286621	0.293755
PRO:HD3	H_other	1635	0.321169	0.329159	0.347152	0.338430	0.323535	0.334145
PRO:HG2	H_other	2144	0.255073	0.234657	0.246913	0.235884	0.241587	0.235579
PRO:HG3	H_other	1952	0.237799	0.230203	0.228819	0.234731	0.232740	0.231802
PRO:N	N_backbone	125	16.579518	16.758143	16.391096	16.678688	16.581846	16.450607
SER:C	C_carbonyl	6734	1.291690	1.125089	1.289939	1.256174	1.267416	1.149526
SER:CA	C_alpha	9323	1.052045	1.001702	1.076521	1.081943	1.086998	1.006595
SER:CB	C_beta	8101	1.055711	0.997992	1.072595	1.081177	1.062233	1.018769
SER:H	H_amide	11684	0.495644	0.420903	0.459281	0.444136	0.471759	0.428855
SER:HA	H_alpha	9305	0.279218	0.228005	0.248442	0.246574	0.247815	0.230834
SER:HB2	H_other	3670	0.233599	0.225750	0.230299	0.224315	0.219433	0.218244
SER:HB3	H_other	3406	0.229567	0.211322	0.219265	0.214459	0.219190	0.210938
SER:HG	H_other	147	1.179074	1.160989	1.161392	1.203933	1.134092	1.153150
SER:N	N_backbone	9527	2.697241	2.250692	2.955181	2.671964	2.663922	2.395244
THR:C	C_carbonyl	5782	1.951204	1.878632	1.942532	1.943935	1.933810	1.888311
THR:CA	C_alpha	8275	1.173249	1.148450	1.180791	1.191941	1.249614	1.143940
THR:CB	C_beta	7252	2.094358	2.049451	2.105100	2.103090	2.094652	2.053375
THR:CG2	C_other	5399	1.193613	1.094870	1.174450	1.174202	1.152860	1.102546
THR:H	H_amide	10560	0.531523	0.436693	0.477057	0.463778	0.502040	0.447180
THR:HA	H_alpha	8146	0.296581	0.230839	0.258395	0.253381	0.257037	0.240876
THR:HB	H_other	7398	0.791892	0.780285	0.790304	0.784571	0.785207	0.783329

IUPAC	Role	<i>n</i>	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
THR:HG1	H_other	204	1.356284	1.291619	1.345874	1.295223	1.326955	1.338738
THR:N	N_backbone	8859	3.161532	2.821463	3.488789	3.200566	3.254241	2.946715
TRP:C	C_carbonyl	1408	1.275828	1.139073	1.338515	1.338049	1.271273	1.139857
TRP:CA	C_alpha	1908	1.555411	1.334484	1.470708	1.509563	1.515996	1.297978
TRP:CB	C_beta	1696	1.382985	1.198481	1.324028	1.404055	1.330428	1.191384
TRP:CD1	C_other	882	1.888688	1.621548	1.848433	1.797239	1.825654	1.659974
TRP:CD2	C_other	9	3.460478	3.898157	3.586850	3.534282	3.492114	3.455916
TRP:CE2	C_other	19	17.844921	18.250122	17.413967	17.487403	17.760683	17.860558
TRP:CE3	C_other	775	1.654079	1.545847	1.607824	1.630854	1.637360	1.508369
TRP:CG	C_other	18	3.203316	3.045501	3.030958	2.975079	3.003874	2.966813
TRP:CH2	C_other	849	2.205216	2.160819	2.197533	2.229998	2.226914	2.189237
TRP:CZ2	C_other	853	2.703227	2.678039	2.682007	2.678830	2.694120	2.684414
TRP:CZ3	C_other	773	1.464679	1.424552	1.428365	1.445929	1.483628	1.408487
TRP:H	H_amide	2490	0.539997	0.523924	0.545773	0.548038	0.575015	0.516461
TRP:HA	H_alpha	1891	0.376235	0.280591	0.311553	0.301339	0.314855	0.285628
TRP:HB2	H_other	573	0.369576	0.349215	0.363125	0.376989	0.362916	0.329366
TRP:HB3	H_other	552	0.350028	0.339164	0.322861	0.333045	0.308934	0.322164
TRP:HD1	H_other	1541	0.333392	0.245342	0.298322	0.260372	0.275966	0.245277
TRP:HE1	H_other	1621	0.727087	0.666632	0.747004	0.673552	0.721007	0.679337
TRP:HE3	H_other	1335	0.383462	0.324182	0.340741	0.336903	0.351990	0.321030
TRP:HH2	H_other	1381	0.339473	0.277160	0.296459	0.293777	0.307485	0.272797
TRP:HZ2	H_other	1412	0.313240	0.245763	0.285812	0.259250	0.277151	0.239284
TRP:HZ3	H_other	1270	0.348059	0.314249	0.319510	0.323685	0.330567	0.313490
TRP:N	N_backbone	2044	3.616259	3.260177	3.763901	3.582518	3.683722	3.286392
TRP:NE1	N_sidechain	1348	2.887390	2.728108	2.782342	2.746558	2.740419	2.687513
TYR:C	C_carbonyl	3360	1.333372	1.111339	1.277822	1.288547	1.257870	1.140898
TYR:CA	C_alpha	4738	1.227827	1.174459	1.246808	1.299565	1.266328	1.182224
TYR:CB	C_beta	4147	1.321119	1.132542	1.320225	1.330263	1.309549	1.171225
TYR:CD1	C_other	1680	4.412756	4.388171	4.398038	4.401636	4.397431	4.398247
TYR:CD2	C_other	1284	5.065249	5.067378	5.075401	5.069062	5.065630	5.077188
TYR:CE1	C_other	1663	2.787794	2.779212	2.781300	2.780705	2.781718	2.776513
TYR:CE2	C_other	1277	3.037540	3.018327	3.026078	3.019929	3.021910	3.009840
TYR:CG	C_other	58	14.382666	14.652523	14.388898	14.278843	14.391483	14.767364
TYR:CZ	C_other	35	27.156607	27.027882	27.153606	26.918709	26.632337	26.433103
TYR:H	H_amide	6133	0.557922	0.465986	0.505039	0.489622	0.530094	0.482008
TYR:HA	H_alpha	4660	0.343730	0.276088	0.307447	0.287943	0.321287	0.282158
TYR:HB2	H_other	1484	0.345299	0.313517	0.333518	0.323839	0.333311	0.308638
TYR:HB3	H_other	1448	0.356377	0.312680	0.337622	0.330977	0.321637	0.314993
TYR:HD1	H_other	2973	0.320237	0.297218	0.307793	0.301117	0.305362	0.295058
TYR:HD2	H_other	2673	0.276406	0.240477	0.261639	0.251556	0.256825	0.238073
TYR:HE1	H_other	2869	0.225146	0.206385	0.219146	0.213557	0.214466	0.205392
TYR:HE2	H_other	2561	0.213192	0.188948	0.204121	0.197997	0.196777	0.191267

IUPAC	Role	<i>n</i>	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
TYR:HH	H_other	78	1.668794	1.665184	1.670690	1.654188	1.650930	1.655440
TYR:N	N_backbone	5051	2.715468	2.358452	3.043610	2.762994	2.983969	2.513725
VAL:C	C_carbonyl	7662	1.174965	1.049877	1.192706	1.263994	1.195868	1.080568
VAL:CA	C_alpha	10715	1.114380	1.099358	1.301860	1.233032	1.159912	1.132212
VAL:CB	C_beta	9469	1.012270	0.979119	1.199746	1.131815	1.038053	1.025031
VAL:H	H_amide	13227	0.496117	0.424306	0.472461	0.480780	0.468795	0.430371
VAL:HA	H_alpha	9999	0.340869	0.293268	0.353658	0.357951	0.312819	0.293417
VAL:HB	H_other	9282	0.661650	0.649008	0.669576	0.671260	0.650054	0.647896
VAL:N	N_backbone	11458	2.817617	2.518426	3.314749	3.365848	2.885509	2.629757

9 MD-shift IUPAC results

9.1 Single Frame at 150 ps; Fitting 151: exact-assignment RMSE (ppm)

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ALA:C	585	101	1.182349	1.162520	1.217898	1.190677	1.145892	1.191034
ALA:CA	833	142	0.854211	0.787962	0.801142	0.787701	0.802764	0.783379
ALA:CB	805	142	0.793493	0.807135	0.801611	0.811318	0.779730	0.839064
ALA:H	847	150	0.396884	0.366027	0.369859	0.367998	0.390278	0.399403
ALA:HA	853	147	0.231325	0.206470	0.254078	0.212354	0.225123	0.219281
ALA:N	803	147	2.423232	2.178374	2.274491	2.326211	2.190283	2.092692
ARG:C	402	94	0.927858	0.878847	0.920436	0.887337	0.916961	0.904832
ARG:CA	623	137	1.211212	1.155827	1.205886	1.177784	1.230672	1.176749
ARG:CB	592	135	0.823019	0.820640	0.789582	0.794580	0.819776	0.791306
ARG:CD	559	124	0.462425	0.428985	0.440313	0.442015	0.439983	0.413476
ARG:CG	536	122	0.870863	0.788319	0.835628	0.792007	0.816499	0.758709
ARG:CZ	2	1	0.103531	0.199027	0.168634	0.050492	0.264309	0.109616
ARG:H	647	143	0.344378	0.337452	0.317446	0.343664	0.361322	0.379893
ARG:HA	647	140	0.221919	0.208583	0.217301	0.209711	0.217686	0.230503
ARG:HB2	254	87	0.203592	0.195574	0.200285	0.198455	0.195059	0.198040
ARG:HB3	234	84	0.216686	0.202561	0.224766	0.198642	0.211648	0.210555
ARG:HD2	285	93	0.142167	0.142475	0.145966	0.145761	0.142032	0.147300
ARG:HD3	274	92	0.238399	0.230761	0.252437	0.227978	0.208139	0.251891
ARG:HE	168	68	0.757713	0.738887	0.696330	0.731663	0.800658	0.774196
ARG:HG2	211	81	0.179386	0.180498	0.200041	0.182109	0.184098	0.185928
ARG:HG3	203	78	0.193616	0.189668	0.189641	0.195070	0.200747	0.203913
ARG:HH11	13	5	0.189441	0.241115	0.237265	0.233046	0.233141	0.256995
ARG:HH12	13	5	0.173656	0.194446	0.247717	0.226232	0.228086	0.213631
ARG:HH21	9	4	0.169917	0.169202	0.134751	0.181379	0.181347	0.177163
ARG:HH22	9	4	0.150742	0.155843	0.136063	0.158047	0.183818	0.158428

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ARG:N	618	142	2.376309	2.227141	2.249674	2.388977	2.248038	2.198865
ARG:NE	149	57	12.165347	12.213909	10.333658	12.781418	12.228013	11.790104
ARG:NH1	7	3	9.692566	9.695289	4.395391	10.107438	9.747075	9.181860
ARG:NH2	7	3	10.229211	9.506734	2.363230	10.071538	9.366596	9.057843
ASN:C	265	93	1.019427	0.974089	0.958684	0.969536	0.947348	0.927713
ASN:CA	436	131	0.935984	0.867307	0.909646	0.868304	0.882843	0.862666
ASN:CB	416	130	0.920199	0.913362	0.889201	0.865717	0.950576	0.915348
ASN:CG	34	12	0.934712	1.045741	0.898914	1.048190	1.129230	1.091482
ASN:H	452	139	0.455046	0.426407	0.410945	0.417996	0.456192	0.446754
ASN:HA	454	138	0.229941	0.203337	0.225626	0.202689	0.216834	0.215993
ASN:HB2	171	68	0.256029	0.247719	0.270765	0.251578	0.239117	0.243381
ASN:HB3	165	66	0.218229	0.236043	0.237260	0.235273	0.230132	0.231705
ASN:HD21	114	42	0.444418	0.477638	0.450070	0.473196	0.462624	0.478772
ASN:HD22	114	42	0.369378	0.368587	0.371464	0.359891	0.374714	0.353775
ASN:N	430	134	2.655938	2.501912	2.379184	2.632104	2.400850	2.374977
ASN:ND2	357	113	1.874910	1.812046	1.629802	1.991149	1.827240	1.806952
ASP:C	466	98	0.977104	0.906196	0.975281	0.946433	0.971618	0.931247
ASP:CA	711	139	0.993304	0.913039	0.965229	0.930749	0.932642	0.953231
ASP:CB	671	138	1.029419	1.025867	1.006631	1.030759	1.019727	1.075070
ASP:CG	3	1	0.385831	0.276663	0.255981	0.255350	0.340748	0.379823
ASP:H	740	148	0.394811	0.391484	0.369280	0.393552	0.417331	0.416850
ASP:HA	736	145	0.172727	0.162527	0.175611	0.172452	0.178879	0.177622
ASP:HB2	283	86	0.233257	0.209320	0.240352	0.209114	0.211469	0.207203
ASP:HB3	271	82	0.210934	0.213187	0.219288	0.212093	0.216303	0.216006
ASP:N	709	145	3.881584	3.732085	3.792865	3.925316	3.828568	3.740027
CYS:C	64	23	1.121833	0.928684	1.252819	1.016269	1.109942	0.951686
CYS:CA	119	40	1.599813	1.543202	1.578688	1.594924	1.471567	1.671521
CYS:CB	113	39	2.805406	2.254568	2.330620	2.445002	3.433356	3.303390
CYS:H	146	44	0.352027	0.346656	0.364869	0.351092	0.487127	0.468507
CYS:HA	134	42	0.216907	0.211612	0.247446	0.200802	0.292667	0.263669
CYS:HB2	53	18	0.323424	0.312342	0.333331	0.336679	0.404991	0.381958
CYS:HB3	52	17	0.296521	0.424003	0.365731	0.392537	0.383544	0.391995
CYS:N	132	42	2.750031	2.584597	2.462456	2.824610	2.550964	2.685365
GLN:C	322	90	0.910304	0.877274	0.901966	0.833929	0.907029	0.874275
GLN:CA	523	131	0.832054	0.783685	0.791866	0.785804	0.830508	0.796198
GLN:CB	508	131	0.901649	0.894325	0.873938	0.920781	0.902541	0.898827
GLN:CD	22	7	0.793710	0.712286	0.690102	0.732218	0.836256	0.808112
GLN:CG	466	120	0.791570	0.753732	0.726995	0.728447	0.735975	0.724918
GLN:H	546	138	0.320141	0.323329	0.334348	0.334812	0.372462	0.367274
GLN:HA	539	135	0.191856	0.201468	0.206430	0.213733	0.213696	0.213256
GLN:HB2	232	79	0.193244	0.194729	0.215132	0.194143	0.180436	0.189747
GLN:HB3	223	77	0.199837	0.214231	0.208688	0.223575	0.223338	0.228075

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
GLN:HE21	123	34	0.510168	0.502845	0.494910	0.475424	0.468206	0.485702
GLN:HE22	124	34	0.374304	0.368223	0.348862	0.364263	0.376556	0.368691
GLN:HG2	243	91	0.178577	0.192169	0.190369	0.193873	0.183421	0.195121
GLN:HG3	230	88	0.185803	0.185328	0.190530	0.179743	0.189851	0.183968
GLN:N	522	136	2.332330	2.244988	2.065567	2.378457	2.119174	2.174916
GLN:NE2	405	109	1.636469	1.775418	1.625916	1.854602	1.784632	1.722813
GLU:C	610	102	0.953523	0.916296	0.964941	0.941920	0.928888	0.951817
GLU:CA	925	143	0.911114	0.842843	0.858348	0.845505	0.845208	0.850787
GLU:CB	888	142	0.764919	0.775502	0.773679	0.829203	0.766514	0.808003
GLU:CD	5	3	17.408647	17.226873	16.789629	17.080014	17.276308	17.489411
GLU:CG	810	130	0.796027	0.788583	0.773284	0.794348	0.783563	0.782634
GLU:H	992	150	0.362320	0.365660	0.374135	0.368677	0.409355	0.410404
GLU:HA	969	147	0.189125	0.168022	0.192587	0.180013	0.182611	0.176912
GLU:HB2	387	103	0.157680	0.149584	0.164198	0.156316	0.144928	0.155638
GLU:HB3	359	101	0.168084	0.160085	0.171204	0.162430	0.175925	0.166334
GLU:HG2	345	94	0.134826	0.140800	0.154237	0.143794	0.133692	0.138991
GLU:HG3	313	91	0.140161	0.139714	0.145946	0.142460	0.154171	0.149420
GLU:N	936	149	2.254731	2.109631	2.009967	2.263851	2.152578	2.068293
GLY:C	523	95	1.342727	1.308750	1.305585	1.310124	1.266223	1.248755
GLY:CA	829	140	0.815121	0.759229	0.768322	0.693155	0.761866	0.736224
GLY:H	884	149	0.432214	0.390016	0.440710	0.427783	0.466800	0.456617
GLY:HA2	284	94	0.253882	0.246691	0.264841	0.262834	0.249988	0.277586
GLY:HA3	268	91	0.251973	0.248038	0.245000	0.262252	0.262916	0.303225
GLY:N	806	144	3.107769	3.013682	2.904932	3.174992	2.988143	2.960941
HIS:C	143	65	0.892291	0.737399	0.888766	0.755667	0.718798	0.792961
HIS:CA	231	99	1.160839	1.055153	1.107394	1.059064	1.125566	1.082595
HIS:CB	221	97	1.100084	1.035865	1.044534	1.016650	1.087364	0.969080
HIS:CD2	103	52	1.496380	1.550042	1.461387	1.523135	1.512319	1.527614
HIS:CE1	59	36	1.420689	1.403318	1.141040	1.534770	1.381678	1.197008
HIS:CG	1	1	0.073578	0.116287	0.240219	0.002487	0.024475	0.013794
HIS:H	243	107	0.370671	0.340547	0.364814	0.348318	0.408941	0.419210
HIS:HA	243	105	0.246123	0.216771	0.245747	0.228955	0.231286	0.235273
HIS:HB2	98	49	0.343101	0.354897	0.351464	0.346286	0.333335	0.350768
HIS:HB3	91	46	0.257234	0.322983	0.308613	0.315528	0.305471	0.323029
HIS:HD1	3	3	1.000486	1.051212	1.032231	1.061796	1.060752	1.031500
HIS:HD2	141	73	0.240329	0.245983	0.281247	0.257573	0.241418	0.261934
HIS:HE1	89	53	0.450191	0.469487	0.480728	0.466292	0.471128	0.473286
HIS:HE2	2	1	0.019214	0.042521	0.019460	0.067235	0.044168	0.048241
HIS:N	219	100	2.626210	2.717313	2.505556	2.787457	2.584284	2.699531
HIS:ND1	4	3	17.663068	18.226519	18.751141	18.236902	17.869972	18.364150
HIS:NE2	3	2	3.343385	3.331377	1.309219	3.815576	3.900027	3.466264
ILE:C	400	96	0.998851	0.903507	0.963793	0.928376	0.952999	0.938924

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ILE:CA	613	136	1.056084	0.921875	0.995402	0.961352	1.036546	0.950886
ILE:CB	595	135	1.118917	1.082775	1.117669	1.090486	1.100531	1.095629
ILE:CD1	559	129	2.112517	2.125362	2.135679	2.140600	2.152395	2.081029
ILE:CG1	524	122	1.530943	1.446204	1.451640	1.448402	1.447706	1.447370
ILE:CG2	558	126	1.968056	1.952618	1.970916	1.906803	1.915979	1.875826
ILE:H	652	143	0.367951	0.384565	0.373678	0.365719	0.398216	0.414336
ILE:HA	629	140	0.227368	0.225268	0.237266	0.234075	0.242464	0.244262
ILE:HB	626	140	0.235955	0.198255	0.252185	0.201488	0.218638	0.204721
ILE:HG12	199	60	0.415044	0.397596	0.419600	0.393607	0.393576	0.398954
ILE:HG13	193	60	0.411400	0.405700	0.389436	0.389821	0.407033	0.396111
ILE:N	615	140	2.571687	2.488068	2.475284	2.586333	2.493867	2.491364
LEU:C	714	100	1.041331	0.894085	1.019203	0.905584	0.967037	0.926589
LEU:CA	1051	140	0.887164	0.835040	0.859109	0.826907	0.855576	0.830499
LEU:CB	1016	140	1.053026	0.927284	0.940766	0.907065	0.944598	0.922901
LEU:CG	927	130	0.752805	0.721486	0.731101	0.727680	0.737496	0.715758
LEU:H	1124	149	0.391488	0.385494	0.373597	0.382918	0.410992	0.412095
LEU:HA	1099	146	0.216678	0.192890	0.217932	0.205294	0.216027	0.207994
LEU:HB2	450	92	0.295851	0.285153	0.311014	0.300343	0.282799	0.294106
LEU:HB3	436	92	0.326244	0.323071	0.320887	0.323435	0.320773	0.312740
LEU:HG	988	142	0.277335	0.255627	0.289223	0.257189	0.258451	0.261682
LEU:N	1056	146	2.212813	2.110058	2.107973	2.284274	2.146206	2.103869
LYS:C	584	101	0.945556	0.849212	0.921155	0.854816	0.932237	0.908640
LYS:CA	895	141	0.986294	0.914605	0.952705	0.917060	0.931973	0.918830
LYS:CB	837	140	1.006668	0.997031	0.975952	0.992762	0.996734	1.030910
LYS:CD	745	129	1.009733	0.997178	0.988691	0.999121	1.010453	0.988050
LYS:CE	730	127	0.618689	0.617299	0.623172	0.616276	0.617250	0.622863
LYS:CG	782	129	0.854642	0.845197	0.851138	0.845937	0.844064	0.840542
LYS:H	971	149	0.325112	0.336793	0.340499	0.347643	0.358927	0.379471
LYS:HA	946	146	0.198410	0.192340	0.207020	0.202408	0.220925	0.224176
LYS:HB2	428	94	0.179945	0.175259	0.189858	0.173299	0.184462	0.173966
LYS:HB3	404	92	0.171878	0.166923	0.183486	0.171891	0.175113	0.181751
LYS:HD2	471	104	0.159133	0.168704	0.166937	0.172868	0.164993	0.182529
LYS:HD3	433	101	0.163193	0.167846	0.166146	0.170943	0.169724	0.179627
LYS:HE2	502	107	0.112840	0.116332	0.116195	0.119024	0.119555	0.120204
LYS:HE3	452	101	0.121408	0.128953	0.126699	0.133301	0.132072	0.132928
LYS:HG2	329	89	0.162460	0.170806	0.163935	0.171909	0.170780	0.173699
LYS:HG3	302	87	0.168042	0.168264	0.178848	0.163775	0.169728	0.175470
LYS:N	904	147	2.225761	2.155798	2.106883	2.277639	2.176445	2.123867
MET:C	147	72	1.143827	1.129561	1.106759	1.039539	1.184520	1.110491
MET:CA	243	106	1.314265	1.310658	1.300722	1.281339	1.335851	1.329424
MET:CB	229	104	1.227636	1.178268	1.231958	1.239178	1.228722	1.238853
MET:CE	134	66	7.974303	7.945569	7.915224	7.920727	7.987335	7.935661

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
MET:CG	197	88	1.186006	1.158257	1.151890	1.140066	1.129442	1.153019
MET:H	226	105	0.328777	0.329814	0.318260	0.329460	0.360471	0.369528
MET:HA	251	111	0.240341	0.205350	0.236891	0.212901	0.228175	0.240865
MET:HB2	115	55	0.235965	0.237108	0.253982	0.243652	0.245245	0.243115
MET:HB3	110	54	0.311369	0.274977	0.312209	0.287171	0.289985	0.297627
MET:HG2	81	44	0.306578	0.316676	0.324102	0.296713	0.304027	0.302117
MET:HG3	78	42	0.306877	0.293081	0.300784	0.285028	0.308506	0.286575
MET:N	218	99	2.064587	1.922786	1.880490	2.006977	1.750292	1.903540
PHE:C	275	92	1.000374	0.893285	1.020159	0.897249	0.955451	0.955464
PHE:CA	389	131	1.183185	1.109219	1.159583	1.119786	1.115925	1.142787
PHE:CB	377	130	1.008444	1.018171	1.076398	1.026769	1.050615	1.053665
PHE:CD1	196	70	1.668734	1.638937	1.706341	1.629001	1.674172	1.645391
PHE:CD2	150	52	1.893217	1.879767	1.877877	1.853776	1.782748	1.910151
PHE:CE1	180	69	2.207589	2.228315	2.211861	2.212611	2.196138	2.202805
PHE:CE2	138	52	2.500070	2.510335	2.471108	2.491816	2.435350	2.493844
PHE:CZ	186	78	6.186107	6.177095	6.128562	6.152802	6.153851	6.196076
PHE:H	407	139	0.387994	0.382713	0.402812	0.410448	0.436850	0.444137
PHE:HA	403	136	0.274210	0.256617	0.276703	0.252116	0.270371	0.273934
PHE:HB2	148	62	0.320347	0.283905	0.346306	0.295572	0.292738	0.289316
PHE:HB3	140	62	0.321738	0.307426	0.339905	0.297887	0.314957	0.305646
PHE:HD1	275	99	0.225869	0.207312	0.238928	0.216814	0.210085	0.218336
PHE:HD2	262	94	0.228319	0.226484	0.258827	0.216543	0.232743	0.241715
PHE:HE1	255	97	0.246989	0.210521	0.254476	0.201852	0.208215	0.211672
PHE:HE2	239	90	0.235353	0.220591	0.248364	0.218507	0.219609	0.230718
PHE:HZ	251	105	0.272138	0.256411	0.296868	0.248212	0.238357	0.250977
PHE:N	388	134	2.484270	2.408126	2.296955	2.450050	2.309865	2.325305
PRO:C	285	91	0.878868	0.858963	0.871161	0.805446	0.886730	0.841027
PRO:CA	466	131	0.751287	0.741136	0.725494	0.772438	0.742494	0.743211
PRO:CB	454	130	0.719293	0.635904	0.705029	0.649706	0.711057	0.707748
PRO:CD	451	124	1.417440	1.394534	1.412268	1.389841	1.398452	1.402600
PRO:CG	444	121	0.727910	0.627460	0.686324	0.629159	0.715526	0.654675
PRO:HA	482	133	0.230651	0.180133	0.216292	0.191395	0.215327	0.201838
PRO:HB2	123	44	0.266277	0.274262	0.288508	0.263042	0.271279	0.260480
PRO:HB3	121	44	0.337329	0.302550	0.315000	0.261341	0.330550	0.302348
PRO:HD2	126	59	0.206126	0.168888	0.237545	0.184734	0.180522	0.163265
PRO:HD3	119	58	0.187397	0.175359	0.196273	0.180843	0.199200	0.179558
PRO:HG2	167	73	0.176679	0.172588	0.191048	0.170032	0.165931	0.185845
PRO:HG3	151	69	0.169786	0.181679	0.180015	0.191728	0.182348	0.197593
PRO:N	8	2	2.104284	1.428825	1.750662	2.198689	1.492035	1.660741
SER:C	447	97	0.994306	0.914280	0.964202	0.887589	0.948660	0.941261
SER:CA	706	139	0.995992	0.964408	0.950019	0.974640	0.949305	0.993403
SER:CB	670	138	0.781031	0.779364	0.738169	0.730955	0.754915	0.755853

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
SER:H	722	147	0.390654	0.367141	0.370205	0.366316	0.394242	0.406606
SER:HA	752	145	0.207751	0.194635	0.202527	0.205830	0.206412	0.211561
SER:HB2	368	103	0.156563	0.151439	0.159522	0.150723	0.149079	0.151882
SER:HB3	338	99	0.160573	0.158775	0.157354	0.163506	0.163677	0.162391
SER:HG	10	7	0.453756	0.484993	0.466124	0.554800	0.560494	0.530671
SER:N	651	144	2.559757	2.277518	2.290043	2.452889	2.332500	2.234158
THR:C	383	95	0.940194	0.885512	0.947283	0.897105	0.903803	0.883204
THR:CA	628	136	1.046510	1.040195	1.057470	0.993544	1.013584	0.997919
THR:CB	599	136	0.771840	0.725282	0.734119	0.711414	0.759217	0.724439
THR:CG2	569	127	0.737067	0.695348	0.721937	0.685928	0.726499	0.668436
THR:H	653	143	0.392964	0.361223	0.376909	0.366946	0.388903	0.391165
THR:HA	645	140	0.254456	0.214580	0.227721	0.217253	0.221714	0.222369
THR:HB	629	140	0.228144	0.215496	0.251691	0.222309	0.239604	0.237427
THR:HG1	19	16	0.494991	0.510352	0.510794	0.472015	0.508814	0.544812
THR:N	615	141	3.449890	3.432217	3.359266	3.534512	3.231007	3.348618
TRP:C	86	55	1.093200	1.054351	1.110908	0.913844	1.015106	0.963976
TRP:CA	128	77	1.398114	1.048955	1.290123	1.183941	1.235663	1.263043
TRP:CB	126	76	0.993198	0.880172	0.812158	0.857227	0.904727	0.891400
TRP:CD1	93	58	1.129000	1.220951	1.141229	1.093287	1.191789	1.077441
TRP:CD2	2	2	1.851049	1.756969	1.229683	1.704428	1.923961	1.975412
TRP:CE3	90	55	0.948851	1.018119	0.859309	0.972388	0.982665	0.983665
TRP:CG	1	1	0.031708	0.093781	0.019897	0.055275	0.049232	0.067215
TRP:CH2	92	57	0.906674	0.932979	0.806011	0.844330	0.835497	0.866526
TRP:CZ2	93	58	0.638682	0.614664	0.563333	0.589354	0.595080	0.540522
TRP:CZ3	89	55	0.851243	0.846573	0.803671	0.786892	0.839845	0.815939
TRP:H	140	83	0.445391	0.420942	0.411465	0.443780	0.481901	0.451665
TRP:HA	138	83	0.224625	0.201987	0.240305	0.215096	0.244822	0.231246
TRP:HB2	41	30	0.267779	0.258829	0.265131	0.235905	0.248853	0.227712
TRP:HB3	39	28	0.233645	0.241229	0.234333	0.205610	0.251007	0.226923
TRP:HD1	127	77	0.386730	0.333245	0.363580	0.322463	0.334973	0.322640
TRP:HE1	127	77	0.478063	0.479379	0.481217	0.484534	0.467598	0.472836
TRP:HE3	117	71	0.365141	0.307449	0.364164	0.283769	0.315409	0.333724
TRP:HH2	119	75	0.259496	0.227334	0.247163	0.235773	0.222375	0.262530
TRP:HZ2	118	72	0.251721	0.227442	0.249261	0.246154	0.247825	0.260875
TRP:HZ3	104	64	0.244289	0.248123	0.267016	0.250157	0.251166	0.274845
TRP:N	133	81	2.950843	2.903649	2.974331	3.134805	2.809368	2.985381
TRP:NE1	127	78	1.568687	1.526183	1.251359	1.753186	1.497246	1.433185
TYR:C	207	87	0.980721	0.872673	0.979382	0.908118	0.925415	0.883519
TYR:CA	353	126	1.110675	1.021614	1.100860	1.009888	1.071914	1.041760
TYR:CB	335	125	0.994545	0.855195	0.954365	0.901745	0.915765	0.844205
TYR:CD1	186	70	0.791783	0.824267	0.797854	0.819072	0.860040	0.818632
TYR:CD2	135	55	0.765974	0.745386	0.762257	0.771705	0.744485	0.779310

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
TYR:CE1	184	67	0.648765	0.642933	0.593505	0.626883	0.632502	0.630232
TYR:CE2	135	54	0.589105	0.600117	0.557610	0.588156	0.613345	0.609447
TYR:CG	4	2	5.087579	4.930415	4.986530	4.899112	4.733970	4.815636
TYR:H	374	134	0.378887	0.366812	0.368591	0.356578	0.404070	0.394238
TYR:HA	366	130	0.254215	0.215314	0.262641	0.235775	0.244381	0.235205
TYR:HB2	113	60	0.305236	0.304950	0.322954	0.320519	0.320974	0.325430
TYR:HB3	108	58	0.357493	0.331151	0.369274	0.341130	0.357794	0.339544
TYR:HD1	268	96	0.228617	0.199159	0.243326	0.203421	0.196356	0.216545
TYR:HD2	246	89	0.269036	0.236775	0.280009	0.237901	0.253333	0.244039
TYR:HE1	267	96	0.174234	0.160088	0.187015	0.158953	0.161962	0.167600
TYR:HE2	244	89	0.179395	0.170625	0.180963	0.169291	0.171146	0.173402
TYR:HH	5	5	2.088911	2.117736	2.107725	2.127243	2.132523	2.131353
TYR:N	360	129	2.484828	2.462082	2.403406	2.495908	2.461390	2.221865
VAL:C	522	97	1.103310	0.974011	0.991114	0.961245	1.015635	0.998867
VAL:CA	803	137	1.076328	0.986244	1.041138	0.990923	0.981636	0.999061
VAL:CB	776	137	0.793917	0.798100	0.826034	0.772308	0.819364	0.821439
VAL:H	848	145	0.413453	0.402566	0.384425	0.389257	0.418280	0.424420
VAL:HA	848	142	0.296101	0.275375	0.296452	0.281857	0.297322	0.289607
VAL:HB	839	142	0.230604	0.197692	0.246775	0.197909	0.209362	0.206073
VAL:N	796	141	3.009204	2.574422	2.685837	2.833563	2.787866	2.538297

9.2 Single Frame at 150 ps; Selector 22: exact-assignment RMSE (ppm)

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ALA:C	94	15	1.848685	1.735851	1.805208	1.767093	1.784212	1.757025
ALA:CA	135	20	0.695009	0.596835	0.635699	0.642417	0.647961	0.604731
ALA:CB	131	20	0.905290	0.835818	0.940224	0.848981	0.842973	0.841874
ALA:H	158	22	0.543341	0.454085	0.503425	0.465148	0.484106	0.477497
ALA:HA	160	22	0.238190	0.211202	0.239517	0.212806	0.206631	0.197664
ALA:N	132	21	2.481536	2.375390	2.312228	2.372082	2.168117	2.169563
ARG:C	74	15	1.091860	0.973126	0.998684	0.873263	0.958101	0.891830
ARG:CA	97	19	1.129806	1.101786	1.059059	0.994460	1.141796	1.078940
ARG:CB	94	19	1.261290	1.074533	1.076155	1.092311	1.160316	1.070523
ARG:CD	89	19	0.513060	0.453485	0.513525	0.448915	0.486673	0.418835
ARG:CG	88	19	0.957406	0.916463	0.982672	0.913490	0.952713	0.878229
ARG:CZ	4	1	0.504003	0.715771	0.668885	0.630587	0.547250	0.796067
ARG:H	106	21	0.437709	0.378145	0.426681	0.369253	0.389024	0.401086
ARG:HA	107	21	0.237907	0.199965	0.240008	0.196996	0.239772	0.211600
ARG:HB2	41	16	0.291396	0.147551	0.253338	0.143470	0.186493	0.149326
ARG:HB3	41	16	0.192948	0.171215	0.186858	0.168819	0.180842	0.162620
ARG:HD2	58	19	0.133478	0.095903	0.116118	0.102265	0.108865	0.095721

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ARG:HD3	56	19	0.128637	0.097538	0.159830	0.099571	0.137393	0.100950
ARG:HE	47	13	0.344435	0.325824	0.477522	0.351336	0.310801	0.324853
ARG:HG2	40	15	0.161327	0.098388	0.156071	0.102520	0.102799	0.111460
ARG:HG3	39	15	0.141889	0.104462	0.140495	0.144504	0.124278	0.116477
ARG:N	99	20	2.597594	2.142251	2.619861	1.964943	2.120417	1.943343
ARG:NE	32	8	11.916768	11.740550	11.338311	11.796368	11.902841	11.919416
ASN:C	39	14	0.988246	0.860363	1.035358	0.835959	1.029208	0.867889
ASN:CA	71	20	0.863668	0.776695	0.905110	0.737673	0.857466	0.803718
ASN:CB	67	20	1.038712	0.929000	0.965493	0.888409	1.005036	0.938838
ASN:CG	8	2	1.572818	1.220869	1.592039	1.429157	1.521859	1.153192
ASN:H	78	22	0.642255	0.572361	0.624972	0.530101	0.591140	0.549550
ASN:HA	79	22	0.236974	0.182061	0.278241	0.185416	0.225013	0.183349
ASN:HB2	32	14	0.313187	0.290618	0.316209	0.268651	0.306075	0.303174
ASN:HB3	32	14	0.251226	0.259767	0.247881	0.278552	0.246857	0.244751
ASN:HD21	23	5	0.546678	0.617357	0.571343	0.575356	0.546574	0.554931
ASN:HD22	23	5	0.700855	0.728552	0.712870	0.682593	0.701989	0.670675
ASN:N	66	20	2.783744	2.133552	2.407747	2.120613	2.380832	2.152589
ASN:ND2	68	19	2.388634	2.376552	2.564518	2.432976	2.484668	2.527147
ASP:C	53	15	1.034741	0.977586	0.964537	0.938443	0.954482	0.901039
ASP:CA	89	20	1.023307	0.861868	1.035892	0.918821	0.949162	0.902449
ASP:CB	83	20	0.963306	1.066282	0.947858	0.990950	0.959371	0.978835
ASP:H	98	22	0.505085	0.439159	0.501793	0.465566	0.511883	0.455649
ASP:HA	97	22	0.186631	0.176080	0.189954	0.180474	0.188709	0.170684
ASP:HB2	28	12	0.275010	0.243282	0.261778	0.258434	0.244551	0.244271
ASP:HB3	28	12	0.302020	0.275889	0.281195	0.275531	0.282979	0.280136
ASP:N	91	21	2.427796	2.162339	2.473283	2.448875	2.284914	2.141754
CYS:C	7	3	2.239714	2.164844	2.164028	2.163440	2.017930	1.997698
CYS:CA	14	7	1.776287	1.516229	1.674820	1.487703	1.709963	1.440005
CYS:CB	12	5	5.034325	4.973651	4.905019	4.975270	5.410329	5.300742
CYS:H	20	8	0.635289	0.532822	0.464869	0.576911	0.473381	0.578793
CYS:HA	20	8	0.622231	0.389706	0.516055	0.433813	0.558221	0.633477
CYS:HB2	9	5	0.279429	0.216386	0.234170	0.227731	0.293276	0.255949
CYS:HB3	9	5	0.211877	0.426344	0.333683	0.384049	0.233527	0.254108
CYS:N	14	7	3.132099	2.254834	3.024344	2.010199	3.305492	2.502373
GLN:C	61	15	0.894147	0.875179	0.927397	0.863387	0.928454	0.840466
GLN:CA	83	20	0.759275	0.673262	0.695393	0.635359	0.796551	0.706414
GLN:CB	80	20	0.781736	0.772222	0.897295	0.824793	0.735964	0.759068
GLN:CD	19	3	1.370427	1.378869	1.389376	1.280230	1.388201	1.335144
GLN:CG	83	20	0.639749	0.554689	0.617112	0.570633	0.576670	0.550306
GLN:H	87	22	0.384189	0.378346	0.399224	0.360991	0.406077	0.362801
GLN:HA	89	22	0.195605	0.162164	0.192330	0.180601	0.183000	0.159696
GLN:HB2	34	14	0.229791	0.228705	0.257797	0.224509	0.218995	0.203061

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
GLN:HB3	34	14	0.227861	0.192755	0.264362	0.199176	0.193944	0.207547
GLN:HE21	23	6	0.452276	0.509123	0.446196	0.469396	0.436663	0.444880
GLN:HE22	23	6	0.455300	0.407827	0.406443	0.420560	0.450994	0.411009
GLN:HG2	29	14	0.131930	0.147055	0.195617	0.149014	0.120603	0.139862
GLN:HG3	28	14	0.159687	0.116263	0.174711	0.116322	0.129617	0.106224
GLN:N	86	21	2.081844	1.946101	1.930716	2.155258	1.953566	1.905680
GLN:NE2	78	20	1.471249	1.454848	1.502929	1.489858	1.476752	1.484998
GLU:C	108	15	1.047991	1.027224	0.972537	1.082826	1.028159	0.993746
GLU:CA	161	20	0.966322	0.830684	0.879699	0.843023	0.856620	0.804303
GLU:CB	157	20	0.805146	0.810035	0.832839	0.814418	0.804508	0.814366
GLU:CG	160	20	0.698884	0.659270	0.691773	0.669434	0.672258	0.661885
GLU:H	170	22	0.458482	0.356179	0.416418	0.382060	0.381625	0.386163
GLU:HA	171	22	0.283561	0.219760	0.269480	0.227731	0.236159	0.229503
GLU:HB2	77	17	0.232080	0.221796	0.238714	0.215088	0.214646	0.219245
GLU:HB3	77	17	0.160871	0.155296	0.154073	0.151528	0.165277	0.154151
GLU:HG2	55	17	0.176019	0.156124	0.166661	0.150314	0.134764	0.142889
GLU:HG3	50	16	0.178257	0.165845	0.178129	0.159967	0.159376	0.154647
GLU:N	163	21	2.230122	1.944382	2.265317	1.992943	2.151395	1.936312
GLY:C	71	15	1.718237	1.659494	1.612830	1.787774	1.670213	1.671348
GLY:CA	100	20	0.834594	0.775180	0.777225	0.734940	0.735735	0.767740
GLY:H	120	22	0.436334	0.363308	0.415623	0.376813	0.377965	0.392249
GLY:HA2	46	16	0.207993	0.209921	0.227211	0.310430	0.226850	0.285146
GLY:HA3	44	16	0.279893	0.299434	0.237156	0.330970	0.278931	0.335653
GLY:N	97	21	2.527985	2.307639	2.682672	2.310870	2.734905	2.216433
HIS:C	19	10	1.627414	1.321375	1.591912	1.395906	1.519733	1.519380
HIS:CA	25	13	1.197937	1.115002	1.227980	0.940003	1.219177	1.101619
HIS:CB	25	13	1.698674	1.600630	1.438733	1.454576	1.492672	1.429492
HIS:CD2	20	10	4.552452	4.653975	4.551429	4.546991	4.639762	4.576445
HIS:CE1	13	9	1.423867	1.196106	1.071060	1.320098	1.268192	1.300967
HIS:H	26	14	0.575887	0.446107	0.517090	0.392951	0.414143	0.366188
HIS:HA	27	14	0.359810	0.285685	0.440622	0.323198	0.342338	0.316213
HIS:HB2	7	4	0.237348	0.186657	0.177341	0.191336	0.190512	0.169287
HIS:HB3	7	4	0.443061	0.477398	0.429567	0.466533	0.432853	0.430152
HIS:HD2	25	12	0.676874	0.650170	0.664307	0.649370	0.687402	0.647198
HIS:HE1	17	11	0.430651	0.422207	0.435463	0.411579	0.423429	0.410651
HIS:HE2	1	1	3.251718	3.294281	3.412486	3.205225	3.296535	3.179738
HIS:N	23	12	2.736534	2.293629	2.173073	1.937629	2.467175	2.156301
HIS:NE2	1	1	11.971893	13.685425	12.341492	11.703415	11.387421	12.094986
ILE:C	62	15	1.396322	1.311626	1.398167	1.303923	1.308861	1.303662
ILE:CA	83	20	1.053949	1.024964	1.011220	0.923364	1.072945	0.984062
ILE:CB	80	20	1.163533	1.075261	1.136217	1.068204	1.048074	1.124437
ILE:CD1	83	20	1.040042	1.031933	1.023967	0.974575	1.044639	0.997984

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ILE:CG1	83	19	1.054314	0.886580	0.973602	0.958356	0.907164	0.885542
ILE:CG2	84	20	0.774314	0.746582	0.824120	0.675653	0.807951	0.745475
ILE:H	88	21	0.432752	0.365305	0.374516	0.338451	0.385768	0.382256
ILE:HA	88	21	0.295062	0.250538	0.318308	0.233715	0.262418	0.231775
ILE:HB	88	21	0.252387	0.176716	0.247312	0.172425	0.182341	0.177358
ILE:HG12	30	11	0.388758	0.359470	0.381510	0.334960	0.333204	0.344143
ILE:HG13	29	11	0.458051	0.417377	0.438018	0.385442	0.400232	0.384645
ILE:N	87	21	2.737943	2.271888	2.396738	2.278351	2.308751	2.363321
LEU:C	109	15	1.383101	1.290822	1.322795	1.236364	1.323555	1.293770
LEU:CA	150	20	0.908094	0.851065	0.849644	0.780573	0.907686	0.808387
LEU:CB	145	20	0.985520	0.876292	1.127838	1.004720	0.930756	1.001905
LEU:CG	154	20	0.623621	0.613264	0.646760	0.608814	0.619001	0.618887
LEU:H	164	22	0.480777	0.411432	0.438673	0.416700	0.462348	0.425948
LEU:HA	165	22	0.241494	0.183428	0.242412	0.203281	0.210964	0.201796
LEU:HB2	55	17	0.296390	0.269864	0.299361	0.313310	0.267763	0.267337
LEU:HB3	53	16	0.295764	0.269220	0.305689	0.236169	0.278113	0.254875
LEU:HG	161	22	0.269784	0.226264	0.264595	0.229337	0.238322	0.228363
LEU:N	149	21	2.633141	2.340504	2.615117	2.329274	2.663621	2.401654
LYS:C	111	15	1.051809	0.968594	0.993348	0.867339	0.980745	0.916421
LYS:CA	157	20	1.119191	0.995477	1.030497	0.925035	0.995794	0.982015
LYS:CB	154	19	0.775091	0.768129	0.777990	0.781965	0.703417	0.771047
LYS:CD	143	20	0.518281	0.526690	0.539506	0.530577	0.525135	0.516993
LYS:CE	138	19	0.284822	0.295722	0.276599	0.302622	0.272399	0.284404
LYS:CG	154	20	0.719025	0.660995	0.684319	0.656079	0.665853	0.660418
LYS:H	174	22	0.491473	0.420618	0.433014	0.418182	0.442328	0.433832
LYS:HA	173	22	0.228305	0.184215	0.227171	0.173992	0.187435	0.181964
LYS:HB2	80	18	0.115413	0.103904	0.163582	0.089548	0.111171	0.093658
LYS:HB3	78	17	0.140580	0.128223	0.141837	0.126077	0.134618	0.133741
LYS:HD2	104	20	0.089633	0.076529	0.087228	0.086644	0.087050	0.102648
LYS:HD3	102	20	0.104105	0.085272	0.099240	0.090137	0.109589	0.098081
LYS:HE2	104	19	0.100200	0.094438	0.098747	0.109901	0.100433	0.129628
LYS:HE3	101	18	0.102344	0.111867	0.096320	0.111237	0.094305	0.128937
LYS:HG2	63	17	0.129506	0.125876	0.120932	0.127479	0.125260	0.130964
LYS:HG3	63	17	0.144327	0.114139	0.137400	0.127690	0.129623	0.128414
LYS:N	162	21	2.394687	2.025228	2.181757	2.085753	2.251934	1.952182
MET:C	23	13	1.303267	1.185238	1.217737	1.172449	1.296901	1.185514
MET:CA	36	17	1.092187	1.118935	1.058387	1.034413	1.226659	1.042799
MET:CB	35	17	1.248105	1.150335	1.197249	1.070487	1.137320	1.072822
MET:CE	28	14	0.945287	1.020037	0.941361	0.963141	0.950698	1.033192
MET:CG	35	17	1.230519	1.116196	1.130884	1.020177	1.131868	1.056972
MET:H	35	17	0.468672	0.406262	0.447410	0.405025	0.411393	0.407733
MET:HA	38	18	0.360320	0.219725	0.399871	0.226975	0.295767	0.240822

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
MET:HB2	15	9	0.311136	0.259141	0.342965	0.268567	0.255821	0.259871
MET:HB3	15	9	0.260067	0.247721	0.313455	0.225554	0.233067	0.255041
MET:HG2	9	7	0.194100	0.171045	0.241214	0.147670	0.128038	0.182519
MET:HG3	9	7	0.304398	0.217413	0.236706	0.254835	0.245357	0.248218
MET:N	33	17	2.518156	2.324284	2.886790	2.385402	2.482014	2.237325
PHE:C	43	14	1.249906	1.366842	1.294764	1.385727	1.164994	1.191883
PHE:CA	66	19	1.189364	1.076591	1.198894	1.108991	1.166817	1.079724
PHE:CB	63	19	1.068988	0.996550	1.009965	1.002108	0.990132	1.090209
PHE:CD1	47	15	0.882102	0.956827	0.935568	0.841427	0.802375	0.943745
PHE:CD2	33	11	0.879336	0.864677	0.947675	0.915723	0.814673	0.978107
PHE:CE1	37	14	0.885471	0.739721	0.840423	0.788788	0.840557	0.810935
PHE:CE2	26	11	0.889658	0.703937	0.834642	0.722497	0.941808	0.795937
PHE:CZ	29	12	1.088317	1.029557	1.132024	0.964849	1.139599	0.953658
PHE:H	72	21	0.605801	0.536158	0.576261	0.556742	0.601124	0.545861
PHE:HA	72	21	0.344356	0.259134	0.353516	0.269269	0.290144	0.232350
PHE:HB2	24	11	0.402787	0.329260	0.383955	0.332511	0.307920	0.297854
PHE:HB3	24	11	0.277057	0.284795	0.292994	0.306638	0.319446	0.299212
PHE:HD1	68	19	0.352295	0.316290	0.350470	0.307634	0.328356	0.321170
PHE:HD2	62	17	0.345474	0.269021	0.347795	0.262020	0.246341	0.273981
PHE:HE1	58	18	0.373676	0.295967	0.378686	0.297377	0.297996	0.315863
PHE:HE2	52	16	0.379185	0.275192	0.365329	0.275280	0.280514	0.297616
PHE:HZ	48	18	0.466300	0.400492	0.484366	0.397459	0.391940	0.398619
PHE:N	67	20	2.839874	2.620650	2.841590	2.433932	2.538339	2.509445
PRO:C	46	15	1.236154	1.078346	1.139597	1.031405	1.105329	0.941913
PRO:CA	75	19	0.709090	0.693580	0.729452	0.736756	0.756379	0.736455
PRO:CB	69	18	0.705281	0.585913	0.695636	0.672407	0.651594	0.670725
PRO:CD	79	18	0.660234	0.588478	0.633672	0.564117	0.599840	0.586307
PRO:CG	74	18	0.645649	0.550946	0.635967	0.592397	0.642564	0.552658
PRO:HA	89	21	0.206510	0.221852	0.209471	0.243281	0.214982	0.215151
PRO:HB2	29	10	0.419368	0.325432	0.451562	0.250630	0.362902	0.282797
PRO:HB3	29	10	0.370020	0.349272	0.375032	0.275206	0.342595	0.319976
PRO:HD2	32	11	0.371040	0.411936	0.384309	0.394347	0.408011	0.425456
PRO:HD3	31	11	0.252806	0.402956	0.291826	0.427033	0.419205	0.452379
PRO:HG2	39	16	0.190050	0.175328	0.175879	0.206428	0.166546	0.197252
PRO:HG3	39	16	0.203317	0.201086	0.180070	0.246988	0.181771	0.237741
SER:C	76	14	1.533306	1.518952	1.435670	1.482457	1.410132	1.514467
SER:CA	115	18	0.864007	0.898409	0.885187	0.861215	0.924811	0.874669
SER:CB	98	18	0.790437	0.808196	0.817343	0.772427	0.735358	0.778827
SER:H	126	20	0.468419	0.453977	0.514112	0.449368	0.424946	0.448545
SER:HA	129	20	0.237410	0.159930	0.209786	0.178293	0.189916	0.184364
SER:HB2	61	15	0.159214	0.122760	0.136826	0.133119	0.122565	0.129347
SER:HB3	52	15	0.194144	0.173952	0.127740	0.154579	0.174335	0.139203

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
SER:N	112	19	2.566830	2.399535	2.178791	2.177433	2.528837	2.205615
THR:C	47	14	0.857094	0.661867	0.853657	0.751074	0.785863	0.723210
THR:CA	76	19	1.146366	1.083973	0.997130	1.084115	1.048861	1.108306
THR:CB	73	19	0.876288	0.825387	0.885705	0.850127	0.849847	0.848539
THR:CG2	78	19	1.350397	1.298266	1.369044	1.279037	1.351501	1.308637
THR:H	82	21	0.495793	0.404210	0.452022	0.413570	0.418185	0.389837
THR:HA	83	21	0.237802	0.210959	0.226865	0.208440	0.213004	0.181184
THR:HB	83	21	0.228546	0.206894	0.246733	0.195847	0.209430	0.208610
THR:HG1	4	1	1.701670	1.722745	1.717931	1.886477	1.883505	1.876282
THR:N	72	20	3.106188	2.839669	3.356071	2.826380	2.880301	2.681276
TRP:C	14	9	0.891103	0.768771	0.729204	0.771066	0.742488	0.671501
TRP:CA	19	13	1.633311	1.277396	1.495321	1.417841	1.287248	1.646118
TRP:CB	19	13	1.877058	1.756924	1.916646	1.873610	1.882249	1.950691
TRP:CD1	14	9	1.123821	1.110817	1.054211	0.839458	1.051467	0.909289
TRP:CE3	14	9	0.835433	0.919910	0.775283	0.899860	1.132964	0.997069
TRP:CH2	14	9	0.845576	0.653908	0.919748	0.552680	0.614336	0.641164
TRP:CZ2	14	9	1.052640	1.013490	1.097033	0.986047	1.052416	1.060629
TRP:CZ3	14	9	0.560257	0.400489	0.575316	0.462730	0.326728	0.452347
TRP:H	22	14	0.609010	0.565680	0.544848	0.550702	0.567838	0.558600
TRP:HA	22	14	0.413471	0.441672	0.380068	0.402498	0.362942	0.425573
TRP:HB2	4	3	0.168565	0.324138	0.098794	0.219298	0.223121	0.259761
TRP:HB3	4	3	0.341628	0.240828	0.227773	0.236641	0.172416	0.175925
TRP:HD1	19	13	0.250392	0.310673	0.271168	0.254745	0.230309	0.254567
TRP:HE1	19	12	0.374797	0.387952	0.401328	0.330484	0.362727	0.361477
TRP:HE3	19	12	0.685503	0.343553	0.540163	0.304610	0.405049	0.393647
TRP:HH2	20	13	0.296176	0.171261	0.266076	0.117295	0.171803	0.125730
TRP:HZ2	19	12	0.252878	0.219161	0.259168	0.171003	0.191765	0.200266
TRP:HZ3	15	10	0.273157	0.205787	0.284402	0.180913	0.187986	0.226744
TRP:N	21	14	2.798866	2.369690	1.835273	2.609329	2.296323	2.200975
TRP:NE1	20	13	2.184621	2.159782	2.086205	1.998921	2.076879	2.096422
TYR:C	34	13	0.983523	0.863793	0.964568	0.849873	0.966987	0.890601
TYR:CA	54	18	1.123195	1.233426	1.086936	1.050274	1.105184	1.111390
TYR:CB	51	18	1.027982	0.959648	0.954547	1.075322	0.780627	0.963129
TYR:CD1	47	15	0.651065	0.627685	0.627719	0.595122	0.623215	0.560778
TYR:CD2	35	11	0.682090	0.663331	0.650676	0.586766	0.542583	0.579516
TYR:CE1	48	15	0.513682	0.501317	0.615018	0.500472	0.537151	0.487383
TYR:CE2	35	11	0.459566	0.478108	0.513263	0.452087	0.485863	0.434604
TYR:H	63	20	0.632080	0.554226	0.610458	0.562492	0.599980	0.581875
TYR:HA	64	20	0.394658	0.268521	0.426407	0.240309	0.303406	0.260559
TYR:HB2	20	11	0.318689	0.283374	0.358810	0.276301	0.283087	0.279095
TYR:HB3	20	11	0.378546	0.412743	0.368842	0.366171	0.358473	0.371743
TYR:HD1	59	19	0.387561	0.281516	0.426233	0.259765	0.284493	0.278577

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
TYR:HD2	56	17	0.399881	0.293869	0.440840	0.279413	0.331241	0.291746
TYR:HE1	58	19	0.199574	0.167686	0.221813	0.152073	0.194109	0.168690
TYR:HE2	56	17	0.199958	0.168133	0.217574	0.156176	0.158490	0.157485
TYR:N	55	19	2.159258	1.868218	2.425184	2.116733	2.288271	2.037551
VAL:C	64	15	1.359912	1.133433	1.278917	1.295440	1.268785	1.176434
VAL:CA	100	20	1.172476	1.009132	1.061060	0.981574	1.083644	1.016767
VAL:CB	96	20	0.813285	0.699492	0.788927	0.689796	0.758484	0.769715
VAL:H	107	21	0.506099	0.468547	0.514972	0.457454	0.450667	0.455733
VAL:HA	108	21	0.295356	0.258126	0.304264	0.255631	0.282900	0.275490
VAL:HB	108	21	0.243052	0.212110	0.234680	0.216673	0.217178	0.209613
VAL:N	104	21	3.438163	2.914067	3.246383	2.993545	3.294126	2.827003

9.3 Single Frame at 150 ps; All 173: exact-assignment RMSE (ppm)

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ALA:C	679	116	1.295204	1.257577	1.314943	1.285977	1.253793	1.284351
ALA:CA	968	162	0.833834	0.764180	0.780177	0.769088	0.783013	0.760985
ALA:CB	936	162	0.810069	0.811210	0.822418	0.816693	0.788887	0.839458
ALA:H	1005	172	0.423280	0.381221	0.393870	0.384899	0.406467	0.412661
ALA:HA	1013	169	0.232423	0.207225	0.251834	0.212425	0.222305	0.216011
ALA:N	935	168	2.431548	2.207255	2.279856	2.332742	2.187167	2.103715
ARG:C	476	109	0.955204	0.894157	0.933032	0.885164	0.923477	0.902823
ARG:CA	720	156	1.200566	1.148694	1.187164	1.154784	1.219076	1.164051
ARG:CB	686	154	0.895842	0.859873	0.834688	0.841628	0.874317	0.835104
ARG:CD	648	143	0.469703	0.432432	0.451073	0.442969	0.446684	0.414216
ARG:CG	624	141	0.883581	0.807623	0.857893	0.810244	0.837052	0.776679
ARG:CZ	6	2	0.415835	0.595614	0.554753	0.515697	0.472167	0.653060
ARG:H	753	164	0.358986	0.343472	0.334984	0.347380	0.365349	0.382947
ARG:HA	754	161	0.224257	0.207382	0.220666	0.207954	0.220955	0.227916
ARG:HB2	295	103	0.217923	0.189629	0.208468	0.191758	0.193891	0.192011
ARG:HB3	275	100	0.213314	0.198203	0.219530	0.194486	0.207346	0.204123
ARG:HD2	343	112	0.140736	0.135727	0.141362	0.139363	0.136989	0.139920
ARG:HD3	330	111	0.223601	0.214076	0.239261	0.211746	0.197923	0.233262
ARG:HE	215	81	0.688880	0.670681	0.654773	0.667299	0.722518	0.701015
ARG:HG2	251	96	0.176631	0.170089	0.193703	0.171912	0.173710	0.176181
ARG:HG3	242	93	0.186254	0.178704	0.182617	0.187843	0.190510	0.192525
ARG:HH11	13	5	0.189441	0.241115	0.237265	0.233046	0.233141	0.256995
ARG:HH12	13	5	0.173656	0.194446	0.247717	0.226232	0.228086	0.213631
ARG:HH21	9	4	0.169917	0.169202	0.134751	0.181379	0.181347	0.177163
ARG:HH22	9	4	0.150742	0.155843	0.136063	0.158047	0.183818	0.158428
ARG:N	717	162	2.408073	2.215613	2.304329	2.335015	2.230851	2.165379

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ARG:NE	181	65	12.121771	12.131565	10.518262	12.612865	12.171157	11.813069
ARG:NH1	7	3	9.692566	9.695289	4.395391	10.107438	9.747075	9.181860
ARG:NH2	7	3	10.229211	9.506734	2.363230	10.071538	9.366596	9.057843
ASN:C	304	107	1.015480	0.960253	0.968860	0.953446	0.958241	0.920256
ASN:CA	507	151	0.926197	0.855196	0.909012	0.851218	0.879333	0.854656
ASN:CB	483	150	0.937534	0.915547	0.900170	0.868900	0.958315	0.918642
ASN:CG	42	14	1.085570	1.081288	1.066261	1.130696	1.213847	1.103502
ASN:H	530	161	0.487134	0.450862	0.448893	0.436306	0.478447	0.463316
ASN:HA	533	160	0.230997	0.200326	0.234172	0.200223	0.218065	0.211473
ASN:HB2	203	82	0.265856	0.254961	0.278421	0.254346	0.250861	0.253744
ASN:HB3	197	80	0.223920	0.240056	0.239018	0.242829	0.232930	0.233874
ASN:HD21	137	47	0.463165	0.503808	0.472609	0.491831	0.477749	0.492381
ASN:HD22	137	47	0.442716	0.449621	0.447365	0.431277	0.446732	0.423863
ASN:N	496	154	2.673297	2.456085	2.383005	2.569921	2.398196	2.346601
ASN:ND2	425	132	1.966147	1.913591	1.812052	2.068194	1.947401	1.940231
ASP:C	519	113	0.983144	0.913742	0.974189	0.945620	0.969882	0.928207
ASP:CA	800	159	0.996686	0.907489	0.973344	0.929429	0.934494	0.947716
ASP:CB	754	158	1.022351	1.030394	1.000330	1.026452	1.013260	1.064902
ASP:CG	3	1	0.385831	0.276663	0.255981	0.255350	0.340748	0.379823
ASP:H	838	170	0.409244	0.397355	0.387126	0.402639	0.429465	0.421572
ASP:HA	833	167	0.174403	0.164163	0.177341	0.173405	0.180051	0.176828
ASP:HB2	311	98	0.237317	0.212600	0.242359	0.214021	0.214656	0.210808
ASP:HB3	299	94	0.221063	0.219819	0.225807	0.218815	0.223393	0.222796
ASP:N	800	166	3.744773	3.588309	3.666778	3.786502	3.685711	3.594227
CYS:C	71	26	1.276322	1.113318	1.369860	1.180014	1.229625	1.099941
CYS:CA	133	47	1.619295	1.540385	1.589081	1.583980	1.498449	1.648683
CYS:CB	125	44	3.089954	2.640052	2.687010	2.789348	3.669655	3.544318
CYS:H	166	52	0.397011	0.374028	0.378320	0.385375	0.485492	0.483130
CYS:HA	154	50	0.302027	0.242255	0.296418	0.243979	0.339115	0.335572
CYS:HB2	62	23	0.317416	0.300321	0.320844	0.323151	0.390761	0.366365
CYS:HB3	61	22	0.285615	0.424350	0.361181	0.391296	0.365306	0.374855
CYS:N	146	49	2.788938	2.554822	2.521769	2.756964	2.632707	2.668362
GLN:C	383	105	0.907750	0.876941	0.906064	0.838690	0.910475	0.868978
GLN:CA	606	151	0.822467	0.769498	0.779358	0.766945	0.825939	0.784508
GLN:CB	588	151	0.886288	0.878710	0.877152	0.908318	0.881729	0.881116
GLN:CD	41	10	1.099255	1.073927	1.072429	1.023336	1.126186	1.084667
GLN:CG	549	140	0.770539	0.727145	0.711472	0.706852	0.714175	0.701314
GLN:H	633	160	0.329682	0.331432	0.343990	0.338530	0.377260	0.366663
GLN:HA	628	157	0.192391	0.196376	0.204491	0.209356	0.209620	0.206512
GLN:HB2	266	93	0.198292	0.199395	0.221045	0.198284	0.185811	0.191501
GLN:HB3	257	91	0.203766	0.211515	0.216875	0.220502	0.219675	0.225466
GLN:HE21	146	40	0.501492	0.503839	0.487559	0.474480	0.463379	0.479502

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
GLN:HE22	147	40	0.388094	0.374696	0.358482	0.373632	0.389144	0.375627
GLN:HG2	272	105	0.174200	0.187875	0.190936	0.189597	0.177784	0.189996
GLN:HG3	258	102	0.183149	0.179126	0.188878	0.173983	0.184269	0.177189
GLN:N	608	157	2.298558	2.205173	2.047032	2.348174	2.096544	2.138892
GLN:NE2	483	129	1.610935	1.727681	1.606692	1.800708	1.738608	1.686680
GLU:C	718	117	0.968321	0.933824	0.966087	0.964431	0.944487	0.958241
GLU:CA	1086	163	0.919507	0.841051	0.861547	0.845137	0.846909	0.844058
GLU:CB	1045	162	0.771097	0.780788	0.782853	0.826999	0.772342	0.808962
GLU:CD	5	3	17.408647	17.226873	16.789629	17.080014	17.276308	17.489411
GLU:CG	970	150	0.780836	0.768753	0.760440	0.775131	0.766317	0.764032
GLU:H	1162	172	0.377920	0.364288	0.380614	0.370665	0.405417	0.406948
GLU:HA	1140	169	0.206068	0.176751	0.205959	0.187945	0.191600	0.185752
GLU:HB2	464	120	0.172265	0.163786	0.178727	0.167502	0.158632	0.167870
GLU:HB3	436	118	0.166833	0.159249	0.168306	0.160558	0.174092	0.164248
GLU:HG2	400	111	0.141204	0.143005	0.156004	0.144708	0.133840	0.139534
GLU:HG3	363	107	0.146000	0.143596	0.150787	0.144997	0.154898	0.150151
GLU:N	1099	170	2.251098	2.085949	2.049850	2.225755	2.152403	2.049255
GLY:C	594	110	1.392948	1.355458	1.346005	1.375970	1.321028	1.306480
GLY:CA	929	160	0.817239	0.760962	0.769285	0.697773	0.759097	0.739681
GLY:H	1004	171	0.432708	0.386920	0.437788	0.422015	0.457092	0.449408
GLY:HA2	330	110	0.247995	0.241901	0.259922	0.269972	0.246893	0.278652
GLY:HA3	312	107	0.256095	0.255912	0.243909	0.272993	0.265233	0.308005
GLY:N	903	165	3.050776	2.945962	2.881878	3.093761	2.961978	2.890175
HIS:C	162	75	1.006691	0.827505	0.997237	0.855917	0.852614	0.908731
HIS:CA	256	112	1.164514	1.061146	1.119742	1.048033	1.135047	1.084468
HIS:CB	246	110	1.174921	1.106498	1.091116	1.069373	1.135179	1.025351
HIS:CD2	123	62	2.290186	2.352409	2.270851	2.303155	2.327147	2.315094
HIS:CE1	72	45	1.421263	1.368228	1.128726	1.498287	1.361887	1.216436
HIS:CG	1	1	0.073578	0.116287	0.240219	0.002487	0.024475	0.013794
HIS:H	269	121	0.395186	0.352134	0.382190	0.352879	0.409446	0.414382
HIS:HA	270	119	0.259741	0.224616	0.271601	0.240050	0.244670	0.244575
HIS:HB2	105	53	0.337085	0.346233	0.342620	0.338172	0.325768	0.341681
HIS:HB3	98	50	0.274708	0.336371	0.318778	0.328624	0.316275	0.331830
HIS:HD1	3	3	1.000486	1.051212	1.032231	1.061796	1.060752	1.031500
HIS:HD2	166	85	0.343598	0.339202	0.365580	0.346206	0.347373	0.348366
HIS:HE1	106	64	0.447115	0.462230	0.473760	0.457957	0.463809	0.463811
HIS:HE2	3	2	1.877446	1.902271	1.970264	1.851352	1.903597	1.836245
HIS:N	242	112	2.636894	2.679928	2.475877	2.718138	2.573383	2.652690
HIS:ND1	4	3	17.663068	18.226519	18.751141	18.236902	17.869972	18.364150
HIS:NE2	4	3	6.649453	7.426053	6.274045	6.720226	6.620120	6.751549
ILE:C	462	111	1.060878	0.968321	1.032755	0.987110	1.008080	0.995664
ILE:CA	696	156	1.055829	0.934766	0.997301	0.956901	1.040954	0.954903

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ILE:CB	675	155	1.124297	1.081887	1.119883	1.087869	1.094445	1.099083
ILE:CD1	642	149	2.006393	2.017632	2.026573	2.027946	2.043267	1.974730
ILE:CG1	607	141	1.474888	1.383111	1.395969	1.391617	1.386289	1.384073
ILE:CG2	642	146	1.856048	1.840322	1.861483	1.794409	1.809992	1.769475
ILE:H	740	164	0.376242	0.382325	0.373778	0.362583	0.396756	0.410652
ILE:HA	717	161	0.236721	0.228520	0.248639	0.234031	0.245000	0.242764
ILE:HB	714	161	0.238041	0.195728	0.251590	0.198137	0.214497	0.201550
ILE:HG12	229	71	0.411695	0.392812	0.414809	0.386431	0.386205	0.392210
ILE:HG13	222	71	0.417790	0.407245	0.396120	0.389252	0.406151	0.394632
ILE:N	702	161	2.592870	2.462307	2.465685	2.550184	2.471678	2.475855
LEU:C	823	115	1.092754	0.956135	1.064398	0.955991	1.021429	0.983130
LEU:CA	1201	160	0.889805	0.837058	0.857933	0.821263	0.862257	0.827770
LEU:CB	1161	160	1.044834	0.921070	0.966112	0.919828	0.942881	0.933133
LEU:CG	1081	150	0.735788	0.707081	0.719690	0.711960	0.721804	0.702774
LEU:H	1288	171	0.403955	0.388893	0.382499	0.387383	0.417882	0.413885
LEU:HA	1264	168	0.220076	0.191682	0.221281	0.205032	0.215373	0.207195
LEU:HB2	505	109	0.295910	0.283528	0.309766	0.301783	0.281200	0.291310
LEU:HB3	489	108	0.323080	0.317676	0.319274	0.315147	0.316427	0.306996
LEU:HG	1149	164	0.276290	0.251719	0.285900	0.253471	0.255726	0.257273
LEU:N	1205	167	2.269010	2.139898	2.177092	2.289886	2.216738	2.142934
LYS:C	695	116	0.963313	0.869380	0.933060	0.856828	0.940152	0.909887
LYS:CA	1052	161	1.007241	0.927123	0.964713	0.918255	0.941772	0.928533
LYS:CB	991	159	0.974301	0.965029	0.947906	0.963038	0.957071	0.994992
LYS:CD	888	149	0.947962	0.937500	0.931111	0.939587	0.949212	0.928479
LYS:CE	868	146	0.578634	0.578255	0.582036	0.577904	0.576387	0.582356
LYS:CG	936	149	0.833846	0.817747	0.826010	0.817735	0.817417	0.813652
LYS:H	1145	171	0.355446	0.350824	0.356110	0.359256	0.372805	0.388223
LYS:HA	1119	168	0.203319	0.191107	0.210262	0.198281	0.216087	0.218184
LYS:HB2	508	112	0.171402	0.166068	0.185966	0.162990	0.174969	0.163950
LYS:HB3	482	109	0.167211	0.161291	0.177410	0.165341	0.169218	0.174878
LYS:HD2	575	124	0.148984	0.156117	0.155575	0.160737	0.153849	0.170870
LYS:HD3	535	121	0.153690	0.155523	0.155625	0.158743	0.160012	0.167177
LYS:HE2	606	126	0.110774	0.112877	0.113391	0.117509	0.116496	0.121873
LYS:HE3	553	119	0.118156	0.126005	0.121718	0.129552	0.126022	0.132208
LYS:HG2	392	106	0.157629	0.164415	0.157816	0.165574	0.164317	0.167568
LYS:HG3	365	104	0.164194	0.160233	0.172407	0.158136	0.163510	0.168290
LYS:N	1066	168	2.252250	2.136469	2.118432	2.249533	2.188085	2.098681
MET:C	170	85	1.166673	1.137253	1.122416	1.058498	1.200340	1.120935
MET:CA	279	123	1.287764	1.287525	1.272049	1.252217	1.322269	1.296007
MET:CB	264	121	1.230370	1.174603	1.227413	1.218158	1.216999	1.218143
MET:CE	162	80	7.263138	7.238797	7.209397	7.214892	7.275095	7.230125
MET:CG	232	105	1.192827	1.152010	1.148746	1.122800	1.129808	1.139048

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
MET:H	261	122	0.350791	0.341062	0.338452	0.340568	0.367709	0.374877
MET:HA	289	129	0.259306	0.207297	0.264127	0.214804	0.238160	0.240860
MET:HB2	130	64	0.245815	0.239754	0.265774	0.246655	0.246488	0.245107
MET:HB3	125	63	0.305667	0.271851	0.312359	0.280493	0.283759	0.292844
MET:HG2	90	51	0.297252	0.305256	0.316791	0.285334	0.291253	0.292368
MET:HG3	87	49	0.306621	0.286183	0.294802	0.282055	0.302585	0.282849
MET:N	251	116	2.129742	1.980226	2.041315	2.060702	1.862977	1.950687
PHE:C	318	106	1.037630	0.970919	1.061453	0.977677	0.986392	0.990737
PHE:CA	455	150	1.184083	1.104546	1.165368	1.118226	1.123450	1.133857
PHE:CB	440	149	1.017334	1.015104	1.067140	1.023275	1.042170	1.058975
PHE:CD1	243	85	1.548086	1.530899	1.586741	1.509082	1.544428	1.534908
PHE:CD2	183	63	1.754241	1.741021	1.747129	1.722790	1.650683	1.778549
PHE:CE1	217	83	2.043570	2.052329	2.044158	2.041322	2.030057	2.033991
PHE:CE2	164	63	2.320543	2.319758	2.291011	2.303807	2.265233	2.309483
PHE:CZ	215	90	5.767665	5.757846	5.715417	5.733782	5.739078	5.773705
PHE:H	479	160	0.427873	0.409466	0.433340	0.435587	0.465262	0.460863
PHE:HA	475	157	0.285952	0.257000	0.289659	0.254790	0.273460	0.268046
PHE:HB2	172	73	0.333077	0.290659	0.351801	0.300999	0.294903	0.290523
PHE:HB3	164	73	0.315595	0.304219	0.333452	0.299184	0.315618	0.304713
PHE:HD1	343	118	0.255945	0.233003	0.264802	0.237595	0.238245	0.242218
PHE:HD2	324	111	0.254938	0.235220	0.278063	0.225954	0.235406	0.248214
PHE:HE1	313	115	0.274907	0.228776	0.281658	0.222668	0.227542	0.234500
PHE:HE2	291	106	0.266807	0.231296	0.272968	0.229684	0.231670	0.244022
PHE:HZ	299	123	0.311571	0.284501	0.334134	0.277631	0.268988	0.279975
PHE:N	455	154	2.539762	2.440583	2.384976	2.447683	2.344906	2.353325
PRO:C	331	106	0.936710	0.892683	0.913199	0.840491	0.920222	0.855759
PRO:CA	541	150	0.745580	0.734727	0.726044	0.767591	0.744435	0.742278
PRO:CB	523	148	0.717460	0.629536	0.703797	0.652746	0.703500	0.702975
PRO:CD	530	142	1.332155	1.306319	1.325541	1.300449	1.310647	1.313502
PRO:CG	518	139	0.716736	0.617110	0.679358	0.624040	0.705565	0.641095
PRO:HA	571	154	0.227057	0.187248	0.215243	0.200368	0.215274	0.203970
PRO:HB2	152	54	0.301546	0.284736	0.325975	0.260720	0.290996	0.264883
PRO:HB3	150	54	0.343892	0.312129	0.327465	0.264079	0.332913	0.305835
PRO:HD2	158	70	0.248526	0.238986	0.273701	0.242303	0.244344	0.240661
PRO:HD3	150	69	0.202653	0.240734	0.219458	0.252255	0.260381	0.260522
PRO:HG2	206	89	0.179287	0.173110	0.188270	0.177496	0.166048	0.188058
PRO:HG3	190	85	0.177187	0.185828	0.180026	0.204294	0.182229	0.206472
PRO:N	8	2	2.104284	1.428825	1.750662	2.198689	1.492035	1.660741
SER:C	523	111	1.089321	1.024554	1.045994	0.996338	1.028656	1.044282
SER:CA	821	157	0.978577	0.955438	0.941207	0.959560	0.945912	0.977641
SER:CB	768	156	0.782238	0.783102	0.748738	0.736377	0.752448	0.758823
SER:H	848	167	0.403159	0.381297	0.394918	0.379806	0.398954	0.413107

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
SER:HA	881	165	0.212353	0.189950	0.203606	0.202032	0.204080	0.207801
SER:HB2	429	118	0.156943	0.147701	0.156495	0.148347	0.145604	0.148886
SER:HB3	390	114	0.165443	0.160882	0.153736	0.162344	0.165138	0.159494
SER:HG	10	7	0.453756	0.484993	0.466124	0.554800	0.560494	0.530671
SER:N	763	163	2.560797	2.295835	2.274053	2.414424	2.362342	2.229991
THR:C	430	109	0.931472	0.863890	0.937504	0.882321	0.891671	0.867154
THR:CA	704	155	1.057744	1.045009	1.051122	1.003715	1.017451	1.010417
THR:CB	672	155	0.783861	0.736815	0.752066	0.727763	0.769579	0.738930
THR:CG2	647	146	0.835235	0.792727	0.827234	0.781664	0.827269	0.774208
THR:H	735	164	0.405729	0.366269	0.386014	0.372437	0.392278	0.391017
THR:HA	728	161	0.252613	0.214170	0.227624	0.216267	0.220739	0.218067
THR:HB	712	161	0.228191	0.214511	0.251118	0.219388	0.236285	0.234250
THR:HG1	23	17	0.840239	0.855166	0.853699	0.896087	0.911504	0.925985
THR:N	687	161	3.415492	3.375000	3.358931	3.467088	3.196058	3.285044
TRP:C	100	64	1.067213	1.019198	1.065731	0.895227	0.981508	0.928591
TRP:CA	147	90	1.430691	1.081201	1.318444	1.216706	1.242451	1.318832
TRP:CB	145	89	1.148418	1.038106	1.026902	1.048110	1.084211	1.090453
TRP:CD1	107	67	1.128324	1.207112	1.130225	1.063526	1.174383	1.056962
TRP:CD2	2	2	1.851049	1.756969	1.229683	1.704428	1.923961	1.975412
TRP:CE3	104	64	0.934385	1.005458	0.848483	0.962943	1.004208	0.985480
TRP:CG	1	1	0.031708	0.093781	0.019897	0.055275	0.049232	0.067215
TRP:CH2	106	66	0.898843	0.901088	0.821935	0.811838	0.809757	0.840233
TRP:CZ2	107	67	0.706768	0.680275	0.658246	0.655063	0.672833	0.633344
TRP:CZ3	103	64	0.817794	0.800671	0.776586	0.751093	0.789923	0.776580
TRP:H	162	97	0.470958	0.443379	0.432002	0.459762	0.494448	0.467624
TRP:HA	160	97	0.258893	0.249022	0.263948	0.249359	0.264214	0.266505
TRP:HB2	45	33	0.260495	0.265286	0.254782	0.234476	0.246674	0.230741
TRP:HB3	43	31	0.245700	0.241192	0.233731	0.208691	0.244763	0.222673
TRP:HD1	146	90	0.371828	0.330395	0.352926	0.314477	0.323276	0.314616
TRP:HE1	146	89	0.465922	0.468492	0.471587	0.467369	0.455320	0.459873
TRP:HE3	136	83	0.424678	0.312744	0.393512	0.286772	0.329401	0.342726
TRP:HH2	139	88	0.265087	0.220147	0.249973	0.222644	0.215830	0.247548
TRP:HZ2	137	84	0.251882	0.226311	0.250658	0.237158	0.240831	0.253337
TRP:HZ3	119	74	0.248113	0.243193	0.269269	0.242520	0.244105	0.269255
TRP:N	154	95	2.930583	2.836761	2.845977	3.068453	2.745059	2.890976
TRP:NE1	147	91	1.665925	1.626954	1.394631	1.788605	1.588587	1.540298
TYR:C	241	100	0.981117	0.871426	0.977306	0.900129	0.931393	0.884521
TYR:CA	407	144	1.112344	1.052173	1.099022	1.015339	1.076387	1.051263
TYR:CB	386	143	0.999027	0.869715	0.954389	0.926545	0.899075	0.860860
TYR:CD1	233	85	0.765483	0.788569	0.766581	0.779098	0.817809	0.773570
TYR:CD2	170	66	0.749472	0.729248	0.740660	0.737431	0.707642	0.742583
TYR:CE1	232	82	0.623223	0.616309	0.598019	0.602907	0.613991	0.603458

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
TYR:CE2	170	65	0.564869	0.577110	0.548773	0.562838	0.589357	0.577792
TYR:CG	4	2	5.087579	4.930415	4.986530	4.899112	4.733970	4.815636
TYR:H	437	154	0.424803	0.399295	0.412308	0.392977	0.437756	0.426413
TYR:HA	430	150	0.279622	0.224035	0.292875	0.236455	0.254036	0.239149
TYR:HB2	133	71	0.307296	0.301804	0.328596	0.314268	0.315567	0.318893
TYR:HB3	128	69	0.360864	0.345174	0.369207	0.345162	0.357901	0.344774
TYR:HD1	327	115	0.264455	0.216349	0.285139	0.214683	0.214948	0.228983
TYR:HD2	302	106	0.297674	0.248355	0.316074	0.246128	0.269486	0.253564
TYR:HE1	325	115	0.179019	0.161470	0.193684	0.157747	0.168150	0.167795
TYR:HE2	300	106	0.183409	0.170163	0.188338	0.166921	0.168855	0.170543
TYR:HH	5	5	2.088911	2.117736	2.107725	2.127243	2.132523	2.131353
TYR:N	415	148	2.444174	2.391867	2.406304	2.449033	2.439153	2.198327
VAL:C	586	112	1.134162	0.992669	1.026480	1.003174	1.046267	1.019765
VAL:CA	903	157	1.087394	0.988804	1.043363	0.989892	0.993448	1.001038
VAL:CB	872	157	0.796072	0.787848	0.822031	0.763661	0.812885	0.815905
VAL:H	955	166	0.424839	0.410487	0.401171	0.397480	0.422033	0.428042
VAL:HA	956	163	0.296017	0.273481	0.297345	0.279018	0.295728	0.288047
VAL:HB	947	163	0.232057	0.199389	0.245426	0.200138	0.210268	0.206480
VAL:N	900	162	3.061845	2.615924	2.756443	2.852509	2.850965	2.573315

9.4 100-Frame Mean; Fitting 151: exact-assignment RMSE (ppm)

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ALA:C	585	101	1.224714	1.166554	1.251765	1.189588	1.181947	1.199676
ALA:CA	833	142	0.840712	0.754738	0.800414	0.757830	0.803543	0.751855
ALA:CB	805	142	0.838619	0.795241	0.834056	0.794937	0.792471	0.826622
ALA:H	847	150	0.419072	0.365580	0.375399	0.374980	0.390056	0.386844
ALA:HA	853	147	0.240878	0.202487	0.254316	0.209210	0.217076	0.204780
ALA:N	803	147	2.435154	2.211166	2.342915	2.240280	2.267669	2.093336
ARG:C	402	94	0.995677	0.896324	1.000558	0.916772	0.968526	0.928116
ARG:CA	623	137	1.224359	1.171147	1.215711	1.169690	1.236929	1.180598
ARG:CB	592	135	0.905420	0.839364	0.881123	0.838038	0.891406	0.844113
ARG:CD	559	124	0.481158	0.438100	0.462497	0.448431	0.453801	0.433881
ARG:CG	536	122	0.910428	0.819632	0.907365	0.811340	0.845913	0.789474
ARG:CZ	2	1	0.370897	0.235511	0.209871	0.318326	0.199679	0.615452
ARG:H	647	143	0.384419	0.349389	0.346428	0.360100	0.366834	0.369532
ARG:HA	647	140	0.242797	0.204103	0.233763	0.207083	0.211355	0.205154
ARG:HB2	254	87	0.219843	0.199349	0.219708	0.197915	0.196717	0.198155
ARG:HB3	234	84	0.217328	0.191159	0.230038	0.189323	0.203200	0.193548
ARG:HD2	285	93	0.140668	0.136527	0.148998	0.135959	0.138049	0.135093
ARG:HD3	274	92	0.247233	0.230848	0.254867	0.231163	0.216612	0.252152

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ARG:HE	168	68	0.800779	0.759404	0.734477	0.741795	0.825785	0.779501
ARG:HG2	211	81	0.189526	0.179103	0.205107	0.177593	0.184183	0.185045
ARG:HG3	203	78	0.209741	0.186242	0.203820	0.188504	0.195799	0.192039
ARG:HH11	13	5	0.234481	0.221216	0.267314	0.233144	0.223425	0.224801
ARG:HH12	13	5	0.243137	0.210811	0.246962	0.219048	0.211101	0.209885
ARG:HH21	9	4	0.199861	0.201286	0.203666	0.191326	0.205539	0.190477
ARG:HH22	9	4	0.211630	0.185173	0.164366	0.178977	0.204267	0.191400
ARG:N	618	142	2.441216	2.197709	2.299620	2.305776	2.318330	2.198537
ARG:NE	149	57	12.648780	12.490763	11.817288	12.768226	12.527766	12.239456
ARG:NH1	7	3	10.519215	10.052858	10.091219	10.193455	10.343167	9.630078
ARG:NH2	7	3	10.787656	10.167272	9.366777	10.152417	10.240320	9.601335
ASN:C	265	93	1.027546	0.961102	0.998632	0.955441	0.944884	0.927805
ASN:CA	436	131	0.912991	0.839032	0.895894	0.855714	0.894383	0.883903
ASN:CB	416	130	0.974864	0.928792	0.964430	0.874026	0.979704	0.934444
ASN:CG	34	12	1.035702	1.041192	1.048079	1.079437	1.221746	1.152125
ASN:H	452	139	0.488965	0.441841	0.438728	0.437716	0.454120	0.442408
ASN:HA	454	138	0.245717	0.213629	0.254040	0.210509	0.217192	0.218416
ASN:HB2	171	68	0.269879	0.248828	0.281927	0.242420	0.238537	0.240796
ASN:HB3	165	66	0.225977	0.235452	0.239238	0.236778	0.234526	0.232838
ASN:HD21	114	42	0.466338	0.499076	0.483624	0.488464	0.476085	0.482624
ASN:HD22	114	42	0.374550	0.383320	0.376805	0.366606	0.376278	0.362090
ASN:N	430	134	2.576200	2.436165	2.436171	2.483095	2.468406	2.369515
ASN:ND2	357	113	1.974616	1.876284	1.847715	1.979683	1.934291	1.843251
ASP:C	466	98	1.008085	0.942180	0.985320	0.968720	0.976930	0.940840
ASP:CA	711	139	1.014064	0.958856	0.988661	0.987100	0.966464	0.990605
ASP:CB	671	138	1.071116	1.045360	1.070088	1.046701	1.061000	1.095363
ASP:CG	3	1	0.358363	0.335950	0.334489	0.306935	0.415057	0.494746
ASP:H	740	148	0.415573	0.410361	0.403924	0.415078	0.429634	0.425301
ASP:HA	736	145	0.192945	0.171349	0.192381	0.180908	0.179206	0.173324
ASP:HB2	283	86	0.235609	0.209110	0.238875	0.203259	0.211078	0.207754
ASP:HB3	271	82	0.217084	0.199170	0.217796	0.191082	0.206427	0.202054
ASP:N	709	145	3.945027	3.767485	3.861890	3.901582	3.923748	3.810940
CYS:C	64	23	1.296487	1.012517	1.293329	1.111338	1.256796	1.098309
CYS:CA	119	40	1.682772	1.647166	1.714925	1.635432	1.644599	1.838687
CYS:CB	113	39	3.066494	2.711595	2.843531	2.698405	3.548374	3.457224
CYS:H	146	44	0.416103	0.365327	0.409406	0.382559	0.486285	0.462858
CYS:HA	134	42	0.274642	0.222102	0.280165	0.211142	0.322065	0.307539
CYS:HB2	53	18	0.397938	0.303168	0.380173	0.313550	0.415578	0.366973
CYS:HB3	52	17	0.391017	0.445892	0.396956	0.410605	0.407611	0.411635
CYS:N	132	42	2.997534	2.794835	2.955453	2.741682	3.172533	3.088431
GLN:C	322	90	0.952126	0.902889	0.935169	0.888295	0.951431	0.913853
GLN:CA	523	131	0.843645	0.778767	0.811636	0.779479	0.849191	0.790467

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
GLN:CB	508	131	0.971694	0.943766	0.951322	0.940839	0.952196	0.970003
GLN:CD	22	7	0.791764	0.770973	0.759493	0.766226	0.842114	0.793084
GLN:CG	466	120	0.811320	0.784827	0.792075	0.775512	0.768241	0.767434
GLN:H	546	138	0.375540	0.352417	0.369628	0.355689	0.383960	0.360023
GLN:HA	539	135	0.213158	0.199084	0.217200	0.204365	0.206352	0.202751
GLN:HB2	232	79	0.200498	0.191654	0.219987	0.187216	0.180011	0.185750
GLN:HB3	223	77	0.205769	0.205055	0.210010	0.205750	0.206562	0.207207
GLN:HE21	123	34	0.517759	0.510204	0.509117	0.484882	0.473659	0.487016
GLN:HE22	124	34	0.379914	0.363249	0.359010	0.359881	0.375773	0.360818
GLN:HG2	243	91	0.179297	0.164559	0.190842	0.161944	0.162240	0.170398
GLN:HG3	230	88	0.185017	0.171327	0.194117	0.170798	0.180323	0.176367
GLN:N	522	136	2.374402	2.089785	2.146980	2.117601	2.163664	2.067807
GLN:NE2	405	109	1.783329	1.813202	1.723873	1.873100	1.853549	1.762497
GLU:C	610	102	0.996064	0.936360	0.989999	0.981165	0.977678	0.962215
GLU:CA	925	143	0.922782	0.846391	0.873265	0.843865	0.864156	0.855661
GLU:CB	888	142	0.804944	0.794939	0.814139	0.801315	0.800991	0.817730
GLU:CD	5	3	17.412869	17.373771	17.033965	17.281248	17.268515	17.484904
GLU:CG	810	130	0.815918	0.804323	0.811283	0.804845	0.804430	0.793730
GLU:H	992	150	0.405423	0.382817	0.408146	0.398514	0.412350	0.403920
GLU:HA	969	147	0.211060	0.170867	0.212469	0.179504	0.186035	0.170755
GLU:HB2	387	103	0.156472	0.146299	0.166558	0.146395	0.142801	0.146850
GLU:HB3	359	101	0.175825	0.160384	0.176873	0.155464	0.175571	0.158370
GLU:HG2	345	94	0.150057	0.139305	0.162655	0.139574	0.139057	0.139888
GLU:HG3	313	91	0.153155	0.138938	0.151569	0.139490	0.152349	0.145463
GLU:N	936	149	2.266343	2.018488	2.096754	2.133518	2.156844	2.030291
GLY:C	523	95	1.359210	1.321622	1.316126	1.331247	1.302919	1.289805
GLY:CA	829	140	0.793056	0.735964	0.747568	0.669339	0.746930	0.706672
GLY:H	884	149	0.471823	0.402120	0.451014	0.430437	0.476065	0.448488
GLY:HA2	284	94	0.258287	0.240006	0.264839	0.249455	0.246779	0.253933
GLY:HA3	268	91	0.260655	0.252826	0.273337	0.272664	0.273984	0.306589
GLY:N	806	144	3.131908	2.966306	2.952017	3.074565	3.075461	2.901065
HIS:C	143	65	0.976078	0.833429	0.939847	0.837439	0.853927	0.853998
HIS:CA	231	99	1.181008	1.081572	1.179771	1.086723	1.171594	1.091530
HIS:CB	221	97	1.169137	1.083843	1.141867	1.050987	1.138017	1.036775
HIS:CD2	103	52	1.540774	1.554456	1.543192	1.546751	1.565031	1.539419
HIS:CE1	59	36	1.503570	1.433377	1.322703	1.560443	1.418776	1.357387
HIS:CG	1	1	0.058090	0.022552	0.104248	0.093414	0.248352	0.430099
HIS:H	243	107	0.445195	0.397680	0.457910	0.409788	0.437005	0.425718
HIS:HA	243	105	0.262575	0.244604	0.274977	0.243083	0.249460	0.242586
HIS:HB2	98	49	0.351918	0.326484	0.360355	0.312995	0.318355	0.317638
HIS:HB3	91	46	0.285320	0.274712	0.297804	0.274829	0.283839	0.280000
HIS:HD1	3	3	1.033811	1.077948	1.046511	1.076510	1.054173	1.078218

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
HIS:HD2	141	73	0.276957	0.250921	0.303302	0.260033	0.258579	0.265969
HIS:HE1	89	53	0.467378	0.473982	0.484232	0.474741	0.486758	0.478402
HIS:HE2	2	1	0.052161	0.033938	0.076304	0.024004	0.040002	0.062775
HIS:N	219	100	2.758856	2.799793	2.741630	2.737406	2.724721	2.776609
HIS:ND1	4	3	17.850332	18.266581	18.786398	18.265396	18.018913	18.337478
HIS:NE2	3	2	3.611086	3.860855	3.004282	3.656194	3.906564	3.727392
ILE:C	400	96	1.050417	0.938182	1.008136	0.953118	1.010013	0.958467
ILE:CA	613	136	1.101514	0.967035	1.055417	0.978816	1.066953	0.980966
ILE:CB	595	135	1.191926	1.065520	1.179835	1.097463	1.149501	1.113447
ILE:CD1	559	129	2.186829	2.175291	2.201731	2.165306	2.197406	2.167715
ILE:CG1	524	122	1.576273	1.489942	1.519062	1.500876	1.486504	1.481361
ILE:CG2	558	126	1.976860	1.959334	1.987192	1.939455	1.942492	1.930288
ILE:H	652	143	0.414579	0.387805	0.404166	0.390389	0.402753	0.400718
ILE:HA	629	140	0.254819	0.219309	0.250279	0.229648	0.237710	0.231875
ILE:HB	626	140	0.243556	0.195282	0.255100	0.195358	0.218198	0.197665
ILE:HG12	199	60	0.421756	0.404397	0.425430	0.391989	0.388981	0.393762
ILE:HG13	193	60	0.418127	0.406253	0.402371	0.391182	0.409679	0.390660
ILE:N	615	140	2.556788	2.478874	2.483486	2.439140	2.522087	2.470872
LEU:C	714	100	1.057131	0.913031	1.027656	0.911767	0.973063	0.936759
LEU:CA	1051	140	0.865618	0.797432	0.817279	0.786882	0.844467	0.798412
LEU:CB	1016	140	1.075917	0.910925	1.010091	0.910044	0.953052	0.925899
LEU:CG	927	130	0.759444	0.733242	0.747287	0.728234	0.751903	0.724937
LEU:H	1124	149	0.410971	0.394531	0.400865	0.389913	0.413673	0.401050
LEU:HA	1099	146	0.222441	0.189036	0.223907	0.201091	0.209326	0.192131
LEU:HB2	450	92	0.300201	0.274780	0.308831	0.284208	0.276333	0.278981
LEU:HB3	436	92	0.321731	0.322613	0.328573	0.319111	0.322123	0.309768
LEU:HG	988	142	0.280308	0.250480	0.288482	0.251147	0.257341	0.250298
LEU:N	1056	146	2.257520	2.091731	2.189050	2.125911	2.193537	2.114185
LYS:C	584	101	0.983849	0.868739	0.948785	0.873348	0.947307	0.906817
LYS:CA	895	141	0.965149	0.863681	0.948694	0.855346	0.927731	0.881014
LYS:CB	837	140	1.056111	0.985077	1.044373	0.983618	1.010555	1.003834
LYS:CD	745	129	1.013091	1.001997	1.003452	1.004591	1.021321	1.001719
LYS:CE	730	127	0.624520	0.623850	0.625201	0.622477	0.623680	0.626150
LYS:CG	782	129	0.890714	0.861293	0.882048	0.853224	0.862307	0.850927
LYS:H	971	149	0.382165	0.361367	0.365654	0.364579	0.374250	0.368018
LYS:HA	946	146	0.220347	0.193608	0.223547	0.198853	0.214722	0.203079
LYS:HB2	428	94	0.200037	0.180261	0.210988	0.176602	0.189111	0.178119
LYS:HB3	404	92	0.176088	0.162914	0.181292	0.164418	0.170760	0.166412
LYS:HD2	471	104	0.170495	0.155539	0.170878	0.152243	0.157134	0.160514
LYS:HD3	433	101	0.169233	0.166409	0.174413	0.165937	0.168648	0.170561
LYS:HE2	502	107	0.116328	0.113125	0.117345	0.114491	0.120610	0.114291
LYS:HE3	452	101	0.125786	0.120452	0.129273	0.124581	0.124836	0.121933

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
LYS:HG2	329	89	0.174259	0.159728	0.174061	0.161170	0.165601	0.163841
LYS:HG3	302	87	0.176633	0.166696	0.184413	0.168325	0.170182	0.167037
LYS:N	904	147	2.332827	2.145274	2.238778	2.202771	2.260955	2.145053
MET:C	147	72	1.150797	1.150975	1.144956	1.114988	1.210735	1.119347
MET:CA	243	106	1.382761	1.320014	1.336322	1.311201	1.381061	1.344616
MET:CB	229	104	1.380324	1.273830	1.321489	1.306521	1.316488	1.306341
MET:CE	134	66	7.993543	7.972209	7.959673	7.966699	7.975823	7.960025
MET:CG	197	88	1.206341	1.166328	1.193785	1.145053	1.157596	1.161890
MET:H	226	105	0.361519	0.337976	0.341450	0.342346	0.354474	0.344894
MET:HA	251	111	0.282691	0.220036	0.277627	0.218669	0.238138	0.234774
MET:HB2	115	55	0.236873	0.210778	0.256744	0.216964	0.214608	0.207888
MET:HB3	110	54	0.301992	0.259349	0.310489	0.256725	0.272531	0.271040
MET:HG2	81	44	0.319971	0.322992	0.344104	0.308507	0.310173	0.313583
MET:HG3	78	42	0.319158	0.303536	0.331685	0.302134	0.307242	0.300459
MET:N	218	99	1.991886	1.805182	1.841460	1.915013	1.886962	1.868641
PHE:C	275	92	1.037366	0.900264	1.033607	0.907659	0.981595	0.916663
PHE:CA	389	131	1.210089	1.104192	1.194400	1.111960	1.154847	1.135214
PHE:CB	377	130	1.150228	1.066237	1.130100	1.053333	1.113090	1.079019
PHE:CD1	196	70	1.726407	1.706582	1.747925	1.712195	1.711836	1.715291
PHE:CD2	150	52	1.900575	1.866299	1.907741	1.879583	1.840192	1.890324
PHE:CE1	180	69	2.234167	2.246367	2.254266	2.245482	2.212161	2.215890
PHE:CE2	138	52	2.495658	2.499830	2.489627	2.492854	2.432295	2.475232
PHE:CZ	186	78	6.204246	6.206344	6.170875	6.183021	6.166101	6.184917
PHE:H	407	139	0.444533	0.413803	0.443641	0.431395	0.443851	0.442070
PHE:HA	403	136	0.307164	0.255291	0.304126	0.262923	0.278693	0.274329
PHE:HB2	148	62	0.332089	0.279166	0.362788	0.285816	0.279797	0.276599
PHE:HB3	140	62	0.327970	0.299074	0.336334	0.298229	0.315756	0.294482
PHE:HD1	275	99	0.245263	0.213628	0.248509	0.225112	0.216059	0.225863
PHE:HD2	262	94	0.239687	0.204949	0.259746	0.206406	0.211990	0.227041
PHE:HE1	255	97	0.261343	0.212324	0.263424	0.210591	0.212803	0.218640
PHE:HE2	239	90	0.249945	0.213385	0.257054	0.210501	0.214728	0.221291
PHE:HZ	251	105	0.300741	0.249297	0.309776	0.239227	0.236369	0.246943
PHE:N	388	134	2.487516	2.373963	2.394245	2.319448	2.390147	2.319699
PRO:C	285	91	0.933427	0.866678	0.917131	0.843402	0.911214	0.866234
PRO:CA	466	131	0.757127	0.737482	0.738219	0.747139	0.766155	0.741831
PRO:CB	454	130	0.749377	0.613627	0.751842	0.640098	0.721261	0.699256
PRO:CD	451	124	1.422105	1.399762	1.418621	1.389573	1.409401	1.407445
PRO:CG	444	121	0.730927	0.623065	0.696877	0.643989	0.724259	0.653124
PRO:HA	482	133	0.232165	0.177219	0.231521	0.187344	0.203101	0.184490
PRO:HB2	123	44	0.300962	0.282827	0.302438	0.257098	0.281809	0.274588
PRO:HB3	121	44	0.342177	0.310888	0.335143	0.263295	0.337584	0.308368
PRO:HD2	126	59	0.210648	0.171571	0.226793	0.173525	0.191384	0.170110

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
PRO:HD3	119	58	0.189216	0.156369	0.195645	0.151456	0.203516	0.162278
PRO:HG2	167	73	0.174871	0.172356	0.188627	0.173031	0.171931	0.175277
PRO:HG3	151	69	0.171471	0.166081	0.179784	0.171197	0.172738	0.176247
PRO:N	8	2	2.009589	1.364576	1.763399	1.742017	1.411103	1.464522
SER:C	447	97	1.012661	0.890715	0.992120	0.902667	0.934073	0.894263
SER:CA	706	139	0.982752	0.947839	0.966295	0.964887	0.937022	0.983879
SER:CB	670	138	0.796749	0.775446	0.795955	0.740930	0.773721	0.751879
SER:H	722	147	0.421688	0.384585	0.402198	0.389853	0.411080	0.402012
SER:HA	752	145	0.207246	0.186139	0.200091	0.193120	0.193831	0.192667
SER:HB2	368	103	0.156003	0.140628	0.162491	0.140598	0.141652	0.142453
SER:HB3	338	99	0.171061	0.159220	0.165176	0.163300	0.171681	0.158441
SER:HG	10	7	0.503063	0.483518	0.559466	0.566646	0.563777	0.548422
SER:N	651	144	2.588811	2.203734	2.322551	2.295639	2.389828	2.235677
THR:C	383	95	0.966018	0.881721	0.959556	0.896239	0.922221	0.895762
THR:CA	628	136	1.056044	1.037663	1.053380	0.967474	1.025183	0.982193
THR:CB	599	136	0.797727	0.751016	0.807733	0.718398	0.783566	0.748885
THR:CG2	569	127	0.753379	0.711450	0.736303	0.692425	0.734566	0.693080
THR:H	653	143	0.428018	0.372449	0.380071	0.382191	0.390168	0.381162
THR:HA	645	140	0.263398	0.217753	0.240127	0.220407	0.218731	0.216167
THR:HB	629	140	0.231768	0.215075	0.250434	0.215529	0.240148	0.226093
THR:HG1	19	16	0.525158	0.508151	0.561434	0.500182	0.522797	0.544718
THR:N	615	141	3.571685	3.585149	3.553546	3.567007	3.423216	3.510940
TRP:C	86	55	1.254457	1.117558	1.190222	0.949358	1.140452	1.034039
TRP:CA	128	77	1.442984	1.144944	1.362026	1.197388	1.319010	1.272262
TRP:CB	126	76	1.109565	0.954132	0.992740	0.974085	1.030441	0.985225
TRP:CD1	93	58	1.192804	1.281512	1.288712	1.205546	1.296375	1.184790
TRP:CD2	2	2	1.860927	1.800007	1.746109	1.812590	1.958595	1.877648
TRP:CE3	90	55	1.009564	1.034020	1.014414	0.983134	0.988761	0.985795
TRP:CG	1	1	0.037827	0.077042	0.046318	0.018066	0.310616	0.298630
TRP:CH2	92	57	0.979721	0.952965	0.934535	0.910041	0.897784	0.863292
TRP:CZ2	93	58	0.681879	0.620681	0.636971	0.583791	0.613303	0.576824
TRP:CZ3	89	55	0.916547	0.887682	0.903905	0.842900	0.872337	0.854419
TRP:H	140	83	0.499151	0.481253	0.501106	0.496425	0.496444	0.471063
TRP:HA	138	83	0.274543	0.204754	0.286834	0.210521	0.241581	0.220055
TRP:HB2	41	30	0.288480	0.226440	0.269213	0.223414	0.239344	0.224526
TRP:HB3	39	28	0.237327	0.215921	0.234067	0.200733	0.221249	0.209392
TRP:HD1	127	77	0.386295	0.314153	0.366505	0.306699	0.328050	0.308480
TRP:HE1	127	77	0.478299	0.481257	0.481191	0.486601	0.471363	0.473622
TRP:HE3	117	71	0.384356	0.295003	0.383568	0.279499	0.303614	0.326692
TRP:HH2	119	75	0.253447	0.202032	0.251397	0.197937	0.186070	0.221187
TRP:HZ2	118	72	0.266043	0.200100	0.251384	0.217359	0.219470	0.231630
TRP:HZ3	104	64	0.282936	0.253142	0.292866	0.259988	0.239554	0.278436

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
TRP:N	133	81	3.100058	3.000948	3.061161	3.146913	3.042490	2.977623
TRP:NE1	127	78	1.654845	1.622733	1.536789	1.796052	1.586597	1.534940
TYR:C	207	87	0.947703	0.818873	0.965644	0.829929	0.909741	0.803955
TYR:CA	353	126	1.144553	1.047966	1.125326	1.037006	1.139953	1.041274
TYR:CB	335	125	1.010440	0.890276	1.010463	0.906505	0.975969	0.890926
TYR:CD1	186	70	0.851714	0.856417	0.853229	0.852859	0.881510	0.835778
TYR:CD2	135	55	0.783071	0.765125	0.791091	0.780415	0.792960	0.777296
TYR:CE1	184	67	0.671869	0.658978	0.640740	0.648301	0.668254	0.651176
TYR:CE2	135	54	0.626951	0.607680	0.591992	0.599640	0.624044	0.611315
TYR:CG	4	2	5.015210	4.890843	4.830293	4.861692	4.870334	4.874249
TYR:H	374	134	0.406167	0.377976	0.392141	0.363434	0.390512	0.388668
TYR:HA	366	130	0.301811	0.223773	0.287040	0.230319	0.242703	0.228545
TYR:HB2	113	60	0.313216	0.304732	0.320291	0.306324	0.314764	0.307592
TYR:HB3	108	58	0.380173	0.352011	0.378533	0.356124	0.365441	0.350435
TYR:HD1	268	96	0.249436	0.191201	0.250509	0.189403	0.197694	0.204035
TYR:HD2	246	89	0.276308	0.242871	0.287049	0.241781	0.260861	0.250686
TYR:HE1	267	96	0.177348	0.154909	0.181519	0.154799	0.156171	0.161002
TYR:HE2	244	89	0.178408	0.167842	0.182354	0.167712	0.169015	0.169928
TYR:HH	5	5	2.109378	2.132069	2.120635	2.146172	2.140071	2.144969
TYR:N	360	129	2.498474	2.339068	2.459072	2.305503	2.478580	2.236431
VAL:C	522	97	1.088406	0.958393	1.018628	0.997701	1.032016	0.962330
VAL:CA	803	137	1.049179	0.976059	1.037163	0.966220	1.001589	0.967398
VAL:CB	776	137	0.793416	0.747568	0.802566	0.731917	0.824892	0.801281
VAL:H	848	145	0.441864	0.403197	0.400741	0.405032	0.422855	0.418521
VAL:HA	848	142	0.323903	0.275000	0.313599	0.277779	0.301650	0.280343
VAL:HB	839	142	0.237691	0.192504	0.252347	0.192923	0.210872	0.197532
VAL:N	796	141	2.996341	2.705541	2.792075	2.736142	2.810839	2.623774

9.5 100-Frame Mean; Selector 22: exact-assignment RMSE (ppm)

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ALA:C	94	15	1.779517	1.726003	1.803186	1.717602	1.766820	1.719115
ALA:CA	135	20	0.693919	0.606435	0.620298	0.611751	0.646619	0.591102
ALA:CB	131	20	0.854489	0.767608	0.854639	0.785237	0.824676	0.757249
ALA:H	158	22	0.503335	0.413590	0.438023	0.411283	0.451273	0.426413
ALA:HA	160	22	0.246009	0.200999	0.257400	0.198870	0.197967	0.193009
ALA:N	132	21	2.249666	2.209612	2.173894	2.097377	2.120133	2.032568
ARG:C	74	15	0.954996	0.884198	0.938574	0.849103	0.875437	0.853151
ARG:CA	97	19	1.042478	0.993192	0.957223	0.887490	1.067533	0.965843
ARG:CB	94	19	1.159259	1.038455	1.078173	1.059919	1.132873	1.018214
ARG:CD	89	19	0.490815	0.437519	0.480734	0.421424	0.475972	0.404179

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ARG:CG	88	19	0.931411	0.886303	0.924612	0.866112	0.932150	0.869957
ARG:CZ	4	1	0.589196	0.526206	0.533689	0.488030	0.521662	0.573990
ARG:H	106	21	0.388087	0.326535	0.356072	0.328142	0.354249	0.353243
ARG:HA	107	21	0.223377	0.185078	0.211962	0.179752	0.205572	0.183989
ARG:HB2	41	16	0.268074	0.150668	0.248493	0.141555	0.177437	0.132326
ARG:HB3	41	16	0.192502	0.155332	0.172813	0.154766	0.177203	0.154128
ARG:HD2	58	19	0.107326	0.096108	0.117698	0.097306	0.109257	0.094982
ARG:HD3	56	19	0.114051	0.091446	0.112648	0.095532	0.107836	0.090057
ARG:HE	47	13	0.350061	0.322061	0.446784	0.353772	0.312307	0.332372
ARG:HG2	40	15	0.142774	0.108582	0.151078	0.107284	0.122522	0.109368
ARG:HG3	39	15	0.140236	0.104996	0.138985	0.111031	0.131853	0.111367
ARG:N	99	20	2.417597	2.009066	2.415009	2.026569	2.189950	1.979306
ARG:NE	32	8	11.889645	11.806395	11.309483	11.804310	11.986940	11.881405
ASN:C	39	14	0.889067	0.720868	0.930566	0.782139	0.894835	0.767936
ASN:CA	71	20	0.823830	0.693092	0.794567	0.686422	0.797847	0.686079
ASN:CB	67	20	0.935575	0.831916	0.900396	0.776633	0.920597	0.862327
ASN:CG	8	2	1.428888	1.169342	1.601619	1.430240	1.229107	0.980014
ASN:H	78	22	0.543166	0.567990	0.522804	0.502597	0.571985	0.522589
ASN:HA	79	22	0.219858	0.160071	0.227923	0.164818	0.199333	0.170802
ASN:HB2	32	14	0.336049	0.273629	0.341218	0.246969	0.306469	0.274790
ASN:HB3	32	14	0.241186	0.245197	0.268174	0.238197	0.255819	0.201192
ASN:HD21	23	5	0.560745	0.611142	0.568656	0.578149	0.546787	0.556918
ASN:HD22	23	5	0.695733	0.710312	0.703199	0.670729	0.703502	0.663302
ASN:N	66	20	2.453100	2.001536	2.198981	1.823925	2.298508	2.021397
ASN:ND2	68	19	2.329945	2.270884	2.313957	2.385575	2.382110	2.343465
ASP:C	53	15	0.896713	0.880550	0.912742	0.904548	0.906381	0.889726
ASP:CA	89	20	0.942044	0.874816	0.942846	0.888925	0.884086	0.872318
ASP:CB	83	20	0.870501	0.934891	0.879064	0.886213	0.864485	0.933309
ASP:H	98	22	0.479231	0.450088	0.467527	0.463520	0.517024	0.463293
ASP:HA	97	22	0.174611	0.168423	0.171181	0.167366	0.176591	0.155315
ASP:HB2	28	12	0.276716	0.263050	0.271244	0.264518	0.265060	0.267569
ASP:HB3	28	12	0.296170	0.275319	0.278772	0.278159	0.293820	0.278800
ASP:N	91	21	2.217568	2.007052	2.270576	2.202461	2.106129	1.912421
CYS:C	7	3	2.138365	2.125950	2.186606	1.923951	2.283147	1.895108
CYS:CA	14	7	1.739546	1.527549	1.574875	1.458851	1.622409	1.572617
CYS:CB	12	5	5.080854	4.936714	4.975089	5.383379	5.255621	5.337580
CYS:H	20	8	0.579583	0.409685	0.451600	0.425774	0.484142	0.507680
CYS:HA	20	8	0.622756	0.439101	0.579755	0.463681	0.630119	0.602146
CYS:HB2	9	5	0.272802	0.217241	0.267079	0.202362	0.239690	0.215132
CYS:HB3	9	5	0.260828	0.367159	0.271573	0.363107	0.262793	0.239158
CYS:N	14	7	3.841978	2.497949	3.379013	3.317225	3.530205	2.757492
GLN:C	61	15	0.869503	0.777512	0.826269	0.738196	0.854198	0.753545

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
GLN:CA	83	20	0.720102	0.597541	0.641274	0.602571	0.740553	0.626097
GLN:CB	80	20	0.714944	0.759591	0.755302	0.773879	0.718820	0.757024
GLN:CD	19	3	1.383005	1.361008	1.333340	1.280574	1.371580	1.360732
GLN:CG	83	20	0.648810	0.606835	0.607497	0.602529	0.609297	0.585096
GLN:H	87	22	0.350846	0.334507	0.327448	0.330074	0.373027	0.320780
GLN:HA	89	22	0.198177	0.170986	0.192525	0.165390	0.177775	0.161743
GLN:HB2	34	14	0.240370	0.235613	0.271285	0.220829	0.210064	0.191510
GLN:HB3	34	14	0.223006	0.198811	0.251400	0.199717	0.207362	0.201160
GLN:HE21	23	6	0.439723	0.473505	0.435305	0.441541	0.426697	0.424357
GLN:HE22	23	6	0.452896	0.437127	0.436479	0.432529	0.457460	0.432089
GLN:HG2	29	14	0.152511	0.138723	0.184753	0.126846	0.125445	0.123090
GLN:HG3	28	14	0.135352	0.121034	0.163076	0.122225	0.134062	0.117812
GLN:N	86	21	1.839871	1.603917	1.723285	1.751611	1.664062	1.559134
GLN:NE2	78	20	1.514640	1.461499	1.429566	1.457399	1.455416	1.468256
GLU:C	108	15	1.013069	0.968744	0.946381	1.032104	0.985257	0.948356
GLU:CA	161	20	0.854545	0.779425	0.757805	0.750463	0.813767	0.758039
GLU:CB	157	20	0.759146	0.734765	0.785723	0.764381	0.701865	0.737597
GLU:CG	160	20	0.685475	0.662463	0.683931	0.659939	0.662457	0.653276
GLU:H	170	22	0.411669	0.330874	0.387922	0.344440	0.385313	0.359958
GLU:HA	171	22	0.250830	0.164353	0.247921	0.172532	0.195412	0.173112
GLU:HB2	77	17	0.226306	0.209986	0.248852	0.195428	0.206232	0.199969
GLU:HB3	77	17	0.152840	0.154721	0.159406	0.153375	0.177404	0.157523
GLU:HG2	55	17	0.144004	0.135659	0.173461	0.127775	0.125380	0.123311
GLU:HG3	50	16	0.171859	0.135184	0.179672	0.127912	0.148674	0.124884
GLU:N	163	21	2.230941	1.906136	2.064684	2.025266	2.105425	1.932598
GLY:C	71	15	1.694901	1.641258	1.606453	1.696501	1.617795	1.610418
GLY:CA	100	20	0.823982	0.744573	0.750701	0.680649	0.731241	0.744613
GLY:H	120	22	0.364390	0.311172	0.348208	0.321872	0.357289	0.331755
GLY:HA2	46	16	0.209821	0.208739	0.220048	0.281248	0.203119	0.247715
GLY:HA3	44	16	0.264508	0.271356	0.239686	0.298991	0.295720	0.293540
GLY:N	97	21	2.652025	2.149706	2.353175	2.153988	2.511328	2.110249
HIS:C	19	10	1.548478	1.300892	1.405131	1.299540	1.569344	1.515239
HIS:CA	25	13	0.974060	1.005835	1.108552	0.968718	1.077188	1.077362
HIS:CB	25	13	1.679121	1.604325	1.504607	1.487888	1.673599	1.486099
HIS:CD2	20	10	4.487913	4.462621	4.468004	4.417075	4.435486	4.410362
HIS:CE1	13	9	1.469107	1.280659	1.288101	1.342834	1.253633	1.416074
HIS:H	26	14	0.567829	0.419025	0.563069	0.447898	0.416061	0.360438
HIS:HA	27	14	0.320547	0.243078	0.309284	0.292690	0.285351	0.281140
HIS:HB2	7	4	0.216666	0.176859	0.170411	0.191400	0.179852	0.169399
HIS:HB3	7	4	0.416811	0.412943	0.440089	0.404622	0.406839	0.403113
HIS:HD2	25	12	0.635152	0.590805	0.647226	0.598855	0.626444	0.609549
HIS:HE1	17	11	0.429566	0.424609	0.444680	0.415311	0.418854	0.414423

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
HIS:HE2	1	1	3.098278	3.268247	3.400754	3.174133	3.272151	3.175236
HIS:N	23	12	2.038832	2.001221	2.025633	1.807274	2.266570	2.022291
HIS:NE2	1	1	11.811523	12.509460	12.273590	11.832809	11.364975	12.107712
ILE:C	62	15	1.419064	1.337045	1.411935	1.347653	1.348781	1.327845
ILE:CA	83	20	0.972844	0.838752	0.877891	0.798202	1.024903	0.861820
ILE:CB	80	20	1.131265	1.093254	1.147401	1.078234	1.104399	1.141087
ILE:CD1	83	20	1.097381	1.017540	1.036934	1.007706	1.092619	1.010553
ILE:CG1	83	19	1.007166	0.863675	0.878852	0.829947	0.871556	0.840077
ILE:CG2	84	20	0.783858	0.719851	0.796872	0.670122	0.821034	0.693805
ILE:H	88	21	0.386652	0.328047	0.383000	0.341010	0.358476	0.353999
ILE:HA	88	21	0.287489	0.212331	0.267655	0.225100	0.259751	0.213972
ILE:HB	88	21	0.257254	0.196136	0.263612	0.190038	0.211364	0.198337
ILE:HG12	30	11	0.384755	0.352841	0.376365	0.338127	0.341553	0.338857
ILE:HG13	29	11	0.422501	0.418175	0.410234	0.399659	0.429437	0.403812
ILE:N	87	21	2.751157	2.281151	2.454444	2.238393	2.393456	2.377368
LEU:C	109	15	1.382794	1.254645	1.302247	1.245487	1.321368	1.265971
LEU:CA	150	20	0.870553	0.781576	0.755983	0.764671	0.873928	0.777594
LEU:CB	145	20	0.912127	0.782445	0.956380	0.811363	0.860907	0.827488
LEU:CG	154	20	0.626224	0.613375	0.639725	0.613557	0.618026	0.613110
LEU:H	164	22	0.465374	0.420075	0.440390	0.422161	0.461858	0.423781
LEU:HA	165	22	0.228387	0.179760	0.245951	0.184698	0.194552	0.181185
LEU:HB2	55	17	0.289399	0.238850	0.284758	0.288477	0.238766	0.255403
LEU:HB3	53	16	0.296356	0.268983	0.300470	0.236779	0.271642	0.253160
LEU:HG	161	22	0.258758	0.229061	0.262312	0.227488	0.244455	0.224502
LEU:N	149	21	2.451254	2.374134	2.492299	2.297718	2.567136	2.381276
LYS:C	111	15	0.982816	0.879050	0.931594	0.879976	0.916984	0.880204
LYS:CA	157	20	1.054993	0.954304	1.001140	0.932224	1.008162	0.957411
LYS:CB	154	19	0.667162	0.619872	0.659050	0.606209	0.647593	0.610260
LYS:CD	143	20	0.487758	0.486689	0.495849	0.479847	0.497558	0.474308
LYS:CE	138	19	0.279301	0.282737	0.272974	0.279432	0.282558	0.285372
LYS:CG	154	20	0.692545	0.659485	0.681262	0.649622	0.629272	0.646071
LYS:H	174	22	0.412202	0.379242	0.392825	0.391454	0.426888	0.398517
LYS:HA	173	22	0.216389	0.165513	0.210980	0.162081	0.177197	0.159848
LYS:HB2	80	18	0.137496	0.097526	0.151164	0.098756	0.108777	0.098400
LYS:HB3	78	17	0.151549	0.127347	0.150055	0.130117	0.136927	0.137613
LYS:HD2	104	20	0.089224	0.082055	0.073880	0.087873	0.086171	0.088393
LYS:HD3	102	20	0.095956	0.083399	0.089277	0.085636	0.097110	0.087102
LYS:HE2	104	19	0.092642	0.079341	0.092169	0.080085	0.081988	0.087512
LYS:HE3	101	18	0.090984	0.092782	0.089149	0.087703	0.088797	0.105859
LYS:HG2	63	17	0.125340	0.107439	0.127222	0.114883	0.115901	0.110265
LYS:HG3	63	17	0.130494	0.109812	0.135704	0.106773	0.119352	0.100539
LYS:N	162	21	2.163509	1.796167	1.969438	1.896144	2.099431	1.767682

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
MET:C	23	13	1.303905	1.245164	1.183045	1.252620	1.314088	1.166600
MET:CA	36	17	1.015842	1.055620	0.938322	0.984241	1.245774	1.057196
MET:CB	35	17	1.098457	1.086108	1.003581	1.073013	1.048145	0.893666
MET:CE	28	14	0.955775	0.897948	0.979520	0.814988	0.947467	0.904242
MET:CG	35	17	1.152751	1.049470	1.045970	0.993888	1.061497	1.029203
MET:H	35	17	0.405805	0.364383	0.385479	0.404125	0.422052	0.389704
MET:HA	38	18	0.348099	0.245284	0.379045	0.236858	0.261310	0.246242
MET:HB2	15	9	0.262325	0.238757	0.276240	0.241021	0.247594	0.238025
MET:HB3	15	9	0.236555	0.234577	0.247522	0.221937	0.224093	0.235856
MET:HG2	9	7	0.185914	0.201401	0.214154	0.172398	0.177371	0.175475
MET:HG3	9	7	0.235053	0.218148	0.215953	0.204450	0.202274	0.215058
MET:N	33	17	2.113581	2.342481	2.303398	2.266558	2.231719	2.306723
PHE:C	43	14	1.237255	1.228486	1.275672	1.212845	1.190120	1.158243
PHE:CA	66	19	1.223656	1.061687	1.186733	1.114126	1.225366	1.132239
PHE:CB	63	19	0.922820	0.899617	0.987393	0.912393	0.997621	0.943146
PHE:CD1	47	15	0.916632	0.877302	0.946871	0.823864	0.896639	0.892200
PHE:CD2	33	11	0.897478	0.815485	0.910582	0.824310	0.860476	0.877527
PHE:CE1	37	14	0.877178	0.721206	0.829682	0.735204	0.789761	0.741271
PHE:CE2	26	11	0.847404	0.723397	0.787283	0.701948	0.808291	0.774750
PHE:CZ	29	12	1.080545	0.995296	1.106111	0.939276	1.084037	0.911837
PHE:H	72	21	0.541485	0.496458	0.526684	0.502472	0.546242	0.506516
PHE:HA	72	21	0.298563	0.222392	0.335019	0.231011	0.246884	0.221747
PHE:HB2	24	11	0.323786	0.296899	0.362934	0.279854	0.285715	0.259314
PHE:HB3	24	11	0.267602	0.278015	0.287285	0.281880	0.292733	0.277229
PHE:HD1	68	19	0.331582	0.284005	0.344600	0.270070	0.277242	0.285606
PHE:HD2	62	17	0.325694	0.268238	0.357263	0.254837	0.250098	0.270321
PHE:HE1	58	18	0.371214	0.276255	0.362715	0.280462	0.278699	0.300786
PHE:HE2	52	16	0.350590	0.261745	0.350211	0.259826	0.262079	0.280075
PHE:HZ	48	18	0.463984	0.385964	0.478200	0.373608	0.380282	0.386304
PHE:N	67	20	2.592232	2.381594	2.625271	2.322124	2.467533	2.374521
PRO:C	46	15	1.072075	1.011738	1.139492	1.015574	1.069343	0.968164
PRO:CA	75	19	0.688795	0.703619	0.674478	0.671513	0.743317	0.669452
PRO:CB	69	18	0.652797	0.540700	0.647252	0.551110	0.612087	0.683145
PRO:CD	79	18	0.654859	0.562342	0.633499	0.544372	0.601978	0.578051
PRO:CG	74	18	0.648147	0.537854	0.626234	0.559669	0.648039	0.564410
PRO:HA	89	21	0.166688	0.186518	0.180267	0.209836	0.179804	0.170424
PRO:HB2	29	10	0.411298	0.336157	0.447668	0.248010	0.370693	0.280833
PRO:HB3	29	10	0.361643	0.335029	0.353684	0.264441	0.335162	0.315765
PRO:HD2	32	11	0.403391	0.373899	0.438790	0.347900	0.418457	0.385662
PRO:HD3	31	11	0.237371	0.346195	0.230068	0.350646	0.318786	0.357574
PRO:HG2	39	16	0.176572	0.172864	0.193118	0.185956	0.166687	0.154414
PRO:HG3	39	16	0.184489	0.193032	0.155244	0.224834	0.169949	0.200325

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
SER:C	76	14	1.459398	1.447479	1.396284	1.467442	1.366580	1.439505
SER:CA	115	18	0.851381	0.871784	0.824223	0.853063	0.878936	0.879810
SER:CB	98	18	0.712803	0.692208	0.715210	0.656972	0.674688	0.688181
SER:H	126	20	0.427420	0.403716	0.408420	0.403413	0.411744	0.407286
SER:HA	129	20	0.218742	0.167222	0.210158	0.178239	0.183283	0.180978
SER:HB2	61	15	0.133327	0.119918	0.138045	0.128916	0.117906	0.123612
SER:HB3	52	15	0.167616	0.154364	0.124299	0.128184	0.159200	0.128633
SER:N	112	19	2.378601	2.018515	2.097158	2.100443	2.237132	2.046248
THR:C	47	14	0.795255	0.669589	0.784155	0.705601	0.764381	0.736178
THR:CA	76	19	0.988364	0.963460	0.963139	1.011087	0.949048	0.994560
THR:CB	73	19	0.844812	0.831719	0.897151	0.813664	0.882200	0.834006
THR:CG2	78	19	1.338643	1.309708	1.349659	1.314637	1.349271	1.316517
THR:H	82	21	0.443552	0.374424	0.392015	0.396263	0.400722	0.374682
THR:HA	83	21	0.206202	0.209391	0.215991	0.209524	0.215903	0.181242
THR:HB	83	21	0.239457	0.209587	0.256275	0.198854	0.221444	0.207478
THR:HG1	4	1	1.698566	1.607379	1.701425	1.802351	1.774408	1.795048
THR:N	72	20	2.991889	2.845668	3.132813	2.758716	2.920625	2.837850
TRP:C	14	9	0.813536	0.721113	0.692011	0.869287	0.826356	0.714190
TRP:CA	19	13	1.568724	1.143738	1.333800	1.307488	1.235532	1.354496
TRP:CB	19	13	1.892993	1.828797	1.802011	1.911742	1.879258	1.852623
TRP:CD1	14	9	1.047688	1.147555	1.076735	0.982091	1.102134	0.986783
TRP:CE3	14	9	0.864594	0.899916	0.808037	0.822483	1.038801	0.943943
TRP:CH2	14	9	0.786073	0.592020	0.728640	0.536114	0.569505	0.535483
TRP:CZ2	14	9	1.006000	0.966214	1.037029	0.962377	0.996076	0.956433
TRP:CZ3	14	9	0.456026	0.403785	0.431989	0.424622	0.310690	0.322730
TRP:H	22	14	0.521015	0.523072	0.521796	0.496526	0.567483	0.529280
TRP:HA	22	14	0.352437	0.424517	0.318321	0.407691	0.331717	0.412764
TRP:HB2	4	3	0.187229	0.164312	0.146590	0.215156	0.226204	0.227958
TRP:HB3	4	3	0.322780	0.255837	0.250538	0.212643	0.218535	0.163295
TRP:HD1	19	13	0.211070	0.221436	0.206662	0.197496	0.171445	0.190083
TRP:HE1	19	12	0.356117	0.384655	0.365802	0.360733	0.355570	0.365097
TRP:HE3	19	12	0.651065	0.343121	0.508704	0.312940	0.427030	0.408274
TRP:HH2	20	13	0.257034	0.203372	0.244812	0.171613	0.173883	0.155532
TRP:HZ2	19	12	0.217078	0.165222	0.195093	0.155126	0.153241	0.154100
TRP:HZ3	15	10	0.267212	0.255868	0.274017	0.267188	0.253054	0.255941
TRP:N	21	14	2.935305	2.485951	2.404108	2.677006	2.466485	2.366867
TRP:NE1	20	13	2.068616	2.184424	2.067470	2.027580	2.055622	2.154234
TYR:C	34	13	0.960855	0.820367	0.977228	0.821525	0.882110	0.855044
TYR:CA	54	18	1.066453	1.054139	1.034729	0.967294	1.064849	1.000331
TYR:CB	51	18	0.895344	0.764148	0.766585	0.820434	0.805511	0.737970
TYR:CD1	47	15	0.645238	0.612223	0.621478	0.577340	0.630306	0.575303
TYR:CD2	35	11	0.663689	0.617101	0.580190	0.553191	0.556228	0.532670

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
TYR:CE1	48	15	0.507932	0.470879	0.577280	0.485712	0.503100	0.453798
TYR:CE2	35	11	0.439625	0.482547	0.516179	0.455567	0.461168	0.433328
TYR:H	63	20	0.553666	0.503963	0.552384	0.513088	0.515404	0.525456
TYR:HA	64	20	0.368589	0.219673	0.417087	0.196442	0.253363	0.229341
TYR:HB2	20	11	0.346418	0.302528	0.344022	0.294879	0.302102	0.299615
TYR:HB3	20	11	0.368218	0.354497	0.373393	0.336280	0.347525	0.336203
TYR:HD1	59	19	0.387572	0.297805	0.422295	0.281542	0.292207	0.295316
TYR:HD2	56	17	0.396889	0.289807	0.437689	0.264456	0.331520	0.288067
TYR:HE1	58	19	0.195798	0.170215	0.215665	0.161497	0.173868	0.169345
TYR:HE2	56	17	0.196067	0.167354	0.211750	0.158782	0.161915	0.159759
TYR:N	55	19	2.201641	1.920878	2.291165	1.990225	2.111822	1.881779
VAL:C	64	15	1.291237	1.151644	1.149835	1.237534	1.281244	1.136146
VAL:CA	100	20	1.024809	0.881586	1.029914	0.860758	1.063236	0.935059
VAL:CB	96	20	0.779614	0.687327	0.716952	0.673463	0.793008	0.689569
VAL:H	107	21	0.468407	0.433902	0.436453	0.424732	0.449702	0.426963
VAL:HA	108	21	0.257330	0.216101	0.264220	0.205835	0.254567	0.226000
VAL:HB	108	21	0.206270	0.168925	0.219214	0.164317	0.185875	0.158765
VAL:N	104	21	2.954813	2.643937	2.800302	2.473500	3.001139	2.634112

9.6 100-Frame Mean; All 173: exact-assignment RMSE (ppm)

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ALA:C	679	116	1.315549	1.258918	1.341688	1.275786	1.278968	1.284179
ALA:CA	968	162	0.821815	0.735851	0.777801	0.739192	0.783546	0.731559
ALA:CB	936	162	0.840858	0.791432	0.836967	0.793586	0.797057	0.817267
ALA:H	1005	172	0.433406	0.373537	0.385918	0.380917	0.400301	0.393328
ALA:HA	1013	169	0.241695	0.202253	0.254806	0.207611	0.214171	0.202966
ALA:N	935	168	2.409833	2.210947	2.319800	2.220663	2.247427	2.084864
ARG:C	476	109	0.989463	0.894450	0.991177	0.906583	0.954650	0.916864
ARG:CA	720	156	1.201462	1.148781	1.184180	1.135765	1.215484	1.153998
ARG:CB	686	154	0.944245	0.869345	0.910648	0.871787	0.928214	0.870032
ARG:CD	648	143	0.482496	0.438020	0.465044	0.444819	0.456910	0.429923
ARG:CG	624	141	0.913417	0.829359	0.909817	0.819286	0.858600	0.801314
ARG:CZ	6	2	0.526583	0.450648	0.452288	0.438816	0.441261	0.588135
ARG:H	753	164	0.384937	0.346263	0.347802	0.355775	0.365089	0.367282
ARG:HA	754	161	0.240137	0.201512	0.230795	0.203428	0.210544	0.202285
ARG:HB2	295	103	0.227160	0.193318	0.223930	0.191079	0.194152	0.190373
ARG:HB3	275	100	0.213810	0.186255	0.222442	0.184582	0.199539	0.188196
ARG:HD2	343	112	0.135607	0.130574	0.144183	0.130232	0.133617	0.129189
ARG:HD3	330	111	0.230128	0.213697	0.236828	0.214283	0.202316	0.232739
ARG:HE	215	81	0.726537	0.687969	0.682031	0.676262	0.744427	0.706359

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ARG:HG2	251	96	0.182878	0.169837	0.197489	0.168367	0.175811	0.175189
ARG:HG3	242	93	0.200178	0.175707	0.194835	0.178308	0.186977	0.181478
ARG:HH11	13	5	0.234481	0.221216	0.267314	0.233144	0.223425	0.224801
ARG:HH12	13	5	0.243137	0.210811	0.246962	0.219048	0.211101	0.209885
ARG:HH21	9	4	0.199861	0.201286	0.203666	0.191326	0.205539	0.190477
ARG:HH22	9	4	0.211630	0.185173	0.164366	0.178977	0.204267	0.191400
ARG:N	717	162	2.437969	2.172637	2.315894	2.269269	2.301030	2.169585
ARG:NE	181	65	12.517919	12.372525	11.729110	12.603176	12.433862	12.176921
ARG:NH1	7	3	10.519215	10.052858	10.091219	10.193455	10.343167	9.630078
ARG:NH2	7	3	10.787656	10.167272	9.366777	10.152417	10.240320	9.601335
ASN:C	304	107	1.010842	0.933745	0.990162	0.935006	0.938612	0.908869
ASN:CA	507	151	0.901036	0.820160	0.882405	0.834078	0.881501	0.858948
ASN:CB	483	150	0.969509	0.915966	0.955803	0.861174	0.971720	0.924776
ASN:CG	42	14	1.121276	1.066789	1.173816	1.154504	1.223151	1.121380
ASN:H	530	161	0.497312	0.462571	0.452084	0.447855	0.473312	0.455095
ASN:HA	533	160	0.242058	0.206569	0.250341	0.204383	0.214639	0.212035
ASN:HB2	203	82	0.281345	0.252899	0.292073	0.243143	0.250472	0.246467
ASN:HB3	197	80	0.228517	0.237062	0.244172	0.237009	0.238114	0.227996
ASN:HD21	137	47	0.483477	0.519581	0.498912	0.504635	0.488670	0.495875
ASN:HD22	137	47	0.444970	0.454940	0.448511	0.432855	0.448222	0.427740
ASN:N	496	154	2.560161	2.382908	2.405958	2.405823	2.446479	2.326200
ASN:ND2	425	132	2.035641	1.944808	1.929898	2.050033	2.012649	1.932008
ASP:C	519	113	0.997282	0.936073	0.978156	0.962363	0.969961	0.935748
ASP:CA	800	159	1.006307	0.949874	0.983670	0.976666	0.957650	0.978153
ASP:CB	754	158	1.050910	1.033778	1.050763	1.030260	1.041186	1.078718
ASP:CG	3	1	0.358363	0.335950	0.334489	0.306935	0.415057	0.494746
ASP:H	838	170	0.423512	0.415203	0.411870	0.421031	0.440749	0.429917
ASP:HA	833	167	0.190901	0.171011	0.190034	0.179384	0.178904	0.171324
ASP:HB2	311	98	0.239599	0.214523	0.241966	0.209509	0.216491	0.213826
ASP:HB3	299	94	0.225669	0.207490	0.224211	0.200845	0.216116	0.210432
ASP:N	800	166	3.788443	3.610761	3.715393	3.747344	3.761529	3.645170
CYS:C	71	26	1.402133	1.170349	1.406832	1.215833	1.392028	1.200599
CYS:CA	133	47	1.688838	1.634987	1.700726	1.617753	1.642277	1.812520
CYS:CB	125	44	3.313443	2.997751	3.112168	3.060152	3.746184	3.679672
CYS:H	166	52	0.439036	0.370953	0.414717	0.388021	0.486027	0.468486
CYS:HA	154	50	0.340587	0.260698	0.334589	0.258289	0.376590	0.359701
CYS:HB2	62	23	0.382322	0.292267	0.365931	0.299978	0.394936	0.349054
CYS:HB3	61	22	0.374665	0.435172	0.381060	0.403949	0.389644	0.391002
CYS:N	146	49	3.088533	2.767747	2.998662	2.802000	3.208559	3.058250
GLN:C	383	105	0.939454	0.884112	0.918689	0.866132	0.936621	0.890256
GLN:CA	606	151	0.827815	0.756516	0.790476	0.757694	0.835147	0.770031
GLN:CB	588	151	0.940889	0.920876	0.927092	0.919906	0.923915	0.943855

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
GLN:CD	41	10	1.105783	1.085057	1.064601	1.036808	1.119068	1.093416
GLN:CG	549	140	0.788902	0.760595	0.767025	0.751918	0.746386	0.742745
GLN:H	633	160	0.372243	0.350010	0.364120	0.352279	0.382476	0.354886
GLN:HA	628	157	0.211100	0.195348	0.213877	0.199306	0.202548	0.197458
GLN:HB2	266	93	0.206025	0.197818	0.227190	0.191841	0.184126	0.186497
GLN:HB3	257	91	0.208132	0.204240	0.215942	0.204962	0.206668	0.206417
GLN:HE21	146	40	0.506265	0.504600	0.498215	0.478315	0.466575	0.477691
GLN:HE22	147	40	0.392230	0.375768	0.372197	0.372185	0.389686	0.372870
GLN:HG2	272	105	0.176635	0.162001	0.190202	0.158572	0.158723	0.165997
GLN:HG3	258	102	0.180290	0.166605	0.190992	0.166215	0.175892	0.170985
GLN:N	608	157	2.306328	2.028140	2.092266	2.069766	2.100226	2.003713
GLN:NE2	483	129	1.742745	1.761167	1.679840	1.812436	1.795242	1.718394
GLU:C	718	117	0.998640	0.941302	0.983561	0.988995	0.978821	0.960143
GLU:CA	1086	163	0.912988	0.836802	0.857131	0.830682	0.856873	0.841904
GLU:CB	1045	162	0.798231	0.786192	0.809934	0.795875	0.786896	0.806200
GLU:CD	5	3	17.412869	17.373771	17.033965	17.281248	17.268515	17.484904
GLU:CG	970	150	0.795875	0.782696	0.791689	0.782793	0.782787	0.772324
GLU:H	1162	172	0.406342	0.375667	0.405251	0.391071	0.408506	0.397792
GLU:HA	1140	169	0.217489	0.169906	0.218155	0.178475	0.187471	0.171111
GLU:HB2	464	120	0.170057	0.158647	0.182797	0.155605	0.155133	0.156914
GLU:HB3	436	118	0.171989	0.159399	0.173916	0.155097	0.175896	0.158221
GLU:HG2	400	111	0.149240	0.138809	0.164183	0.138011	0.137257	0.137727
GLU:HG3	363	107	0.155864	0.138427	0.155741	0.137953	0.151848	0.142805
GLU:N	1099	170	2.261127	2.002222	2.092029	2.117812	2.149296	2.016101
GLY:C	594	110	1.403566	1.363775	1.354108	1.380001	1.344441	1.332194
GLY:CA	929	160	0.796443	0.736895	0.747906	0.670565	0.745257	0.710853
GLY:H	1004	171	0.460303	0.392361	0.439992	0.418944	0.463473	0.436182
GLY:HA2	330	110	0.252091	0.235896	0.259061	0.254125	0.241167	0.253075
GLY:HA3	312	107	0.261202	0.255521	0.268846	0.276528	0.277152	0.304782
GLY:N	903	165	3.083941	2.889672	2.893637	2.989299	3.019919	2.826742
HIS:C	162	75	1.059344	0.900900	1.005624	0.903948	0.965671	0.955539
HIS:CA	256	112	1.162422	1.074411	1.173007	1.075769	1.162712	1.090155
HIS:CB	246	110	1.230650	1.147565	1.183816	1.103314	1.203377	1.090919
HIS:CD2	123	62	2.294121	2.293827	2.289157	2.275054	2.291289	2.268764
HIS:CE1	72	45	1.497406	1.407029	1.316522	1.523454	1.390410	1.368170
HIS:CG	1	1	0.058090	0.022552	0.104248	0.093414	0.248352	0.430099
HIS:H	269	121	0.458483	0.399793	0.469104	0.413624	0.435025	0.419851
HIS:HA	270	119	0.268935	0.244452	0.278598	0.248490	0.253278	0.246713
HIS:HB2	105	53	0.344557	0.318702	0.350905	0.306394	0.311046	0.309968
HIS:HB3	98	50	0.296651	0.286804	0.310140	0.286060	0.294334	0.290529
HIS:HD1	3	3	1.033811	1.077948	1.046511	1.076510	1.054173	1.078218
HIS:HD2	166	85	0.354837	0.325649	0.375799	0.333833	0.340433	0.340650

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
HIS:HE1	106	64	0.461522	0.466416	0.478109	0.465720	0.476519	0.468730
HIS:HE2	3	2	1.789299	1.887127	1.964414	1.832692	1.889460	1.833940
HIS:N	242	112	2.698698	2.733945	2.681814	2.663012	2.684542	2.713949
HIS:ND1	4	3	17.850332	18.266581	18.786398	18.265396	18.018913	18.337478
HIS:NE2	4	3	6.682663	7.092341	6.665548	6.710414	6.613365	6.860704
ILE:C	462	111	1.107044	1.000986	1.071205	1.015013	1.061774	1.015870
ILE:CA	696	156	1.086970	0.952645	1.035846	0.959065	1.062026	0.967528
ILE:CB	675	155	1.184899	1.068845	1.176038	1.095201	1.144249	1.116758
ILE:CD1	642	149	2.078376	2.062521	2.088040	2.052725	2.087744	2.055119
ILE:CG1	607	141	1.511158	1.420696	1.448321	1.427864	1.418243	1.410981
ILE:CG2	642	146	1.864684	1.845127	1.874923	1.824304	1.835150	1.816997
ILE:H	740	164	0.411357	0.381190	0.401708	0.384849	0.397746	0.395452
ILE:HA	717	161	0.259051	0.218465	0.252476	0.229095	0.240524	0.229753
ILE:HB	714	161	0.245286	0.195388	0.256164	0.194710	0.217368	0.197748
ILE:HG12	229	71	0.417095	0.398023	0.419330	0.385362	0.383102	0.387013
ILE:HG13	222	71	0.418701	0.407830	0.403407	0.392300	0.412314	0.392403
ILE:N	702	161	2.581671	2.455234	2.479905	2.415167	2.506504	2.459477
LEU:C	823	115	1.105786	0.965246	1.068087	0.962636	1.026009	0.986691
LEU:CA	1201	160	0.866236	0.795469	0.809877	0.784142	0.848203	0.795842
LEU:CB	1161	160	1.056849	0.895886	1.003540	0.898312	0.942037	0.914187
LEU:CG	1081	150	0.741928	0.717390	0.732928	0.713025	0.734323	0.710082
LEU:H	1288	171	0.418291	0.397875	0.406111	0.394166	0.420116	0.404015
LEU:HA	1264	168	0.223227	0.187851	0.226906	0.199028	0.207457	0.190738
LEU:HB2	505	109	0.299043	0.271098	0.306301	0.284676	0.272493	0.276511
LEU:HB3	489	108	0.319078	0.317239	0.325645	0.311242	0.317040	0.304142
LEU:HG	1149	164	0.277390	0.247590	0.284960	0.247968	0.255575	0.246846
LEU:N	1205	167	2.282367	2.128682	2.228784	2.147900	2.243107	2.149010
LYS:C	695	116	0.983684	0.870394	0.946060	0.874410	0.942530	0.902619
LYS:CA	1052	161	0.979081	0.877800	0.956703	0.867252	0.940171	0.892831
LYS:CB	991	159	1.005591	0.937706	0.994342	0.935020	0.963169	0.953396
LYS:CD	888	149	0.948359	0.938329	0.940404	0.940088	0.956550	0.937059
LYS:CE	868	146	0.583454	0.583114	0.583591	0.581626	0.582948	0.585387
LYS:CG	936	149	0.861249	0.831462	0.852270	0.823194	0.828483	0.820744
LYS:H	1145	171	0.386880	0.364140	0.369911	0.368789	0.382716	0.372814
LYS:HA	1119	168	0.219740	0.189537	0.221651	0.193625	0.209361	0.197016
LYS:HB2	508	112	0.191547	0.169926	0.202742	0.166771	0.178869	0.168092
LYS:HB3	482	109	0.172354	0.157704	0.176612	0.159369	0.165754	0.162099
LYS:HD2	575	124	0.158905	0.145033	0.157814	0.142767	0.146862	0.150060
LYS:HD3	535	121	0.157908	0.154073	0.161678	0.153895	0.157535	0.158086
LYS:HE2	606	126	0.112617	0.108081	0.113422	0.109359	0.114909	0.110159
LYS:HE3	553	119	0.120184	0.115893	0.122926	0.118703	0.119070	0.119159
LYS:HG2	392	106	0.167364	0.152538	0.167420	0.154668	0.158667	0.156473

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
LYS:HG3	365	104	0.169568	0.158344	0.176966	0.159407	0.162548	0.157576
LYS:N	1066	168	2.307897	2.095970	2.199973	2.158981	2.237159	2.092095
MET:C	170	85	1.172682	1.164164	1.150183	1.134586	1.225228	1.125856
MET:CA	279	123	1.341070	1.288950	1.291875	1.273737	1.364358	1.311075
MET:CB	264	121	1.346352	1.250564	1.283876	1.278019	1.284141	1.259430
MET:CE	162	80	7.280848	7.260198	7.250637	7.253500	7.264567	7.249267
MET:CG	232	105	1.198409	1.149459	1.172679	1.123552	1.143616	1.142860
MET:H	261	122	0.367767	0.341636	0.347678	0.351262	0.364265	0.351235
MET:HA	289	129	0.292129	0.223519	0.292974	0.221146	0.241312	0.236314
MET:HB2	130	64	0.239948	0.214193	0.259068	0.219875	0.218668	0.211584
MET:HB3	125	63	0.294907	0.256502	0.303623	0.252804	0.267183	0.267063
MET:HG2	90	51	0.309192	0.312966	0.333396	0.297710	0.299554	0.302621
MET:HG3	87	49	0.311513	0.295848	0.321649	0.293540	0.298102	0.292782
MET:N	251	116	2.008307	1.884589	1.908587	1.964826	1.935797	1.931917
PHE:C	318	106	1.066587	0.951291	1.069547	0.954647	1.012306	0.952917
PHE:CA	455	150	1.212066	1.098128	1.193291	1.112274	1.165341	1.134783
PHE:CB	440	149	1.120502	1.044012	1.110793	1.034332	1.097303	1.060633
PHE:CD1	243	85	1.602036	1.580500	1.624106	1.579834	1.587167	1.589690
PHE:CD2	183	63	1.762401	1.724789	1.769943	1.737324	1.705631	1.751518
PHE:CE1	217	83	2.066788	2.067474	2.081495	2.067516	2.040981	2.041236
PHE:CE2	164	63	2.314031	2.311146	2.305180	2.303745	2.254268	2.291422
PHE:CZ	215	90	5.784299	5.784184	5.753990	5.761265	5.748993	5.762431
PHE:H	479	160	0.460412	0.427250	0.457088	0.442808	0.460698	0.452344
PHE:HA	475	157	0.305876	0.250583	0.309007	0.258339	0.274109	0.267025
PHE:HB2	172	73	0.330943	0.281707	0.362809	0.284991	0.280630	0.274253
PHE:HB3	164	73	0.319849	0.296086	0.329612	0.295893	0.312493	0.292021
PHE:HD1	343	118	0.264623	0.229303	0.270288	0.234711	0.229489	0.238897
PHE:HD2	324	111	0.258370	0.218483	0.281037	0.216514	0.219794	0.235938
PHE:HE1	313	115	0.284919	0.225543	0.284451	0.225181	0.226466	0.236030
PHE:HE2	291	106	0.270690	0.222799	0.276017	0.220128	0.223925	0.232887
PHE:HZ	299	123	0.332394	0.275838	0.342442	0.265425	0.264796	0.274132
PHE:N	455	154	2.503211	2.375088	2.429644	2.319842	2.401698	2.327853
PRO:C	331	106	0.953901	0.888256	0.951149	0.869372	0.934791	0.881105
PRO:CA	541	150	0.748027	0.732881	0.729715	0.737118	0.763030	0.732225
PRO:CB	523	148	0.737360	0.604510	0.738892	0.629079	0.707823	0.697151
PRO:CD	530	142	1.335984	1.309358	1.331290	1.298949	1.320734	1.317361
PRO:CG	518	139	0.719685	0.611620	0.687230	0.632632	0.713869	0.641203
PRO:HA	571	154	0.223227	0.178700	0.224304	0.191024	0.199649	0.182369
PRO:HB2	152	54	0.324918	0.293750	0.335041	0.255389	0.300802	0.275790
PRO:HB3	150	54	0.346026	0.315700	0.338807	0.263517	0.337117	0.309812
PRO:HD2	158	70	0.261424	0.227571	0.282865	0.220286	0.254311	0.230652
PRO:HD3	150	69	0.200120	0.210160	0.203237	0.208826	0.232080	0.217522

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
PRO:HG2	206	89	0.175194	0.172452	0.189485	0.175551	0.170950	0.171522
PRO:HG3	190	85	0.174222	0.171958	0.175028	0.183490	0.172169	0.181450
PRO:N	8	2	2.009589	1.364576	1.763399	1.742017	1.411103	1.464522
SER:C	523	111	1.089019	0.991236	1.060461	1.004651	1.008508	0.992278
SER:CA	821	157	0.965428	0.937558	0.947678	0.950017	0.929104	0.969974
SER:CB	768	156	0.786536	0.765329	0.786114	0.730754	0.761801	0.744054
SER:H	848	167	0.422545	0.387488	0.403129	0.391898	0.411179	0.402800
SER:HA	881	165	0.208969	0.183491	0.201596	0.191013	0.192323	0.191000
SER:HB2	429	118	0.152983	0.137873	0.159244	0.138997	0.138524	0.139929
SER:HB3	390	114	0.170606	0.158581	0.160329	0.159067	0.170070	0.154799
SER:HG	10	7	0.503063	0.483518	0.559466	0.566646	0.563777	0.548422
SER:N	763	163	2.559036	2.177533	2.290855	2.268039	2.368030	2.208888
THR:C	430	109	0.948850	0.861082	0.941975	0.877420	0.906308	0.879729
THR:CA	704	155	1.048948	1.029910	1.044014	0.972276	1.017238	0.983536
THR:CB	672	155	0.802976	0.760198	0.817920	0.729350	0.794873	0.758595
THR:CG2	647	146	0.845688	0.807424	0.834498	0.793729	0.833075	0.794606
THR:H	735	164	0.429779	0.372670	0.381422	0.383787	0.391359	0.380445
THR:HA	728	161	0.257520	0.216816	0.237499	0.219194	0.218411	0.212475
THR:HB	712	161	0.232677	0.214443	0.251122	0.213652	0.238043	0.224002
THR:HG1	23	17	0.854160	0.814029	0.873980	0.878421	0.879405	0.897495
THR:N	687	161	3.515409	3.514955	3.511817	3.491086	3.374056	3.446570
TRP:C	100	64	1.202500	1.070927	1.133730	0.938559	1.101882	0.995466
TRP:CA	147	90	1.459846	1.144789	1.358411	1.212181	1.308520	1.283188
TRP:CB	145	89	1.240712	1.108748	1.132208	1.141670	1.177046	1.137196
TRP:CD1	107	67	1.174836	1.264792	1.263001	1.178720	1.272647	1.160805
TRP:CD2	2	2	1.860927	1.800007	1.746109	1.812590	1.958595	1.877648
TRP:CE3	104	64	0.991284	1.016998	0.989143	0.963070	0.995644	0.980265
TRP:CG	1	1	0.037827	0.077042	0.046318	0.018066	0.310616	0.298630
TRP:CH2	106	66	0.956395	0.913504	0.910015	0.869916	0.861626	0.827474
TRP:CZ2	107	67	0.732488	0.676009	0.702393	0.646065	0.675827	0.639437
TRP:CZ3	103	64	0.868413	0.838473	0.855193	0.799010	0.818937	0.803095
TRP:H	162	97	0.502176	0.487143	0.503966	0.496439	0.506676	0.479384
TRP:HA	160	97	0.286512	0.246859	0.291366	0.247142	0.255865	0.255328
TRP:HB2	45	33	0.280961	0.221624	0.260659	0.222693	0.238205	0.224833
TRP:HB3	43	31	0.246529	0.219940	0.235648	0.201870	0.220998	0.205540
TRP:HD1	146	90	0.368242	0.303694	0.349861	0.294786	0.312149	0.295767
TRP:HE1	146	89	0.464222	0.469811	0.467788	0.472124	0.457954	0.460947
TRP:HE3	136	83	0.431637	0.302187	0.403390	0.284408	0.323696	0.339270
TRP:HH2	139	88	0.253966	0.202225	0.250460	0.194369	0.184366	0.212990
TRP:HZ2	137	84	0.259804	0.195635	0.244353	0.209833	0.211527	0.222497
TRP:HZ3	119	74	0.281003	0.253487	0.290557	0.260906	0.241297	0.275702
TRP:N	154	95	3.078111	2.936045	2.980105	3.087050	2.970528	2.901917

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
TRP:NE1	147	91	1.717010	1.710031	1.619244	1.829275	1.658225	1.633059
TYR:C	241	100	0.949569	0.819084	0.967287	0.828749	0.905894	0.811358
TYR:CA	407	144	1.134500	1.048787	1.113730	1.028029	1.130276	1.035935
TYR:CB	386	143	0.995996	0.874655	0.981721	0.895607	0.955193	0.872256
TYR:CD1	233	85	0.814290	0.813085	0.811825	0.804912	0.836931	0.790179
TYR:CD2	170	66	0.760027	0.737083	0.752518	0.739364	0.750351	0.733630
TYR:CE1	232	82	0.641398	0.624725	0.628137	0.618180	0.637604	0.615553
TYR:CE2	170	65	0.593239	0.584113	0.577198	0.572947	0.594172	0.579160
TYR:CG	4	2	5.015210	4.890843	4.830293	4.861692	4.870334	4.874249
TYR:H	437	154	0.430560	0.398603	0.419040	0.388581	0.410865	0.411205
TYR:HA	430	150	0.312655	0.223167	0.309872	0.225599	0.244319	0.228664
TYR:HB2	133	71	0.318430	0.304402	0.323970	0.304630	0.312893	0.306406
TYR:HB3	128	69	0.378330	0.352401	0.377734	0.353097	0.362700	0.348249
TYR:HD1	327	115	0.279454	0.214391	0.289151	0.209052	0.217800	0.223281
TYR:HD2	302	106	0.302322	0.252235	0.320377	0.246144	0.275336	0.258027
TYR:HE1	325	115	0.180778	0.157749	0.188068	0.156016	0.159473	0.162522
TYR:HE2	300	106	0.181834	0.167751	0.188190	0.166082	0.167713	0.168077
TYR:HH	5	5	2.109378	2.132069	2.120635	2.146172	2.140071	2.144969
TYR:N	415	148	2.461193	2.288043	2.437484	2.266242	2.433153	2.192729
VAL:C	586	112	1.112359	0.981352	1.033768	1.026624	1.062084	0.982810
VAL:CA	903	157	1.046509	0.966052	1.036363	0.955114	1.008601	0.963870
VAL:CB	872	157	0.791908	0.741176	0.793593	0.725712	0.821442	0.789757
VAL:H	955	166	0.444917	0.406753	0.404899	0.407287	0.425947	0.419476
VAL:HA	956	163	0.317084	0.268993	0.308417	0.270612	0.296706	0.274743
VAL:HB	947	163	0.234321	0.189963	0.248792	0.189879	0.208173	0.193503
VAL:N	900	162	2.991572	2.698494	2.793027	2.707095	2.833483	2.624971

10 Matched preparation and motion IUPAC results

10.1 tleap; evaluation 45: exact-assignment RMSE (ppm)

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ALA:C	134	35	0.974719	0.849594	1.088533	1.067486	0.948603	0.912435
ALA:CA	174	42	0.728305	0.616849	0.799956	0.804575	0.696059	0.651893
ALA:CB	164	42	0.852614	0.667374	0.793110	0.820691	0.768582	0.716001
ALA:H	184	45	0.403196	0.335471	0.395922	0.403551	0.354015	0.350611
ALA:HA	182	43	0.223031	0.164814	0.221172	0.194007	0.196908	0.179993
ALA:N	171	44	2.558318	1.888889	2.736256	2.285650	2.224811	1.786256
ARG:C	140	33	1.101856	0.808696	1.020809	1.004982	0.896022	0.785048
ARG:CA	184	40	0.903166	0.914814	1.050355	0.999088	0.892074	0.984353

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ARG:CB	172	40	0.921690	0.784491	0.928431	0.863172	0.807824	0.790716
ARG:CD	173	37	0.536241	0.585811	0.635054	0.617686	0.560828	0.530956
ARG:CG	167	37	1.181116	0.754590	1.066859	0.898731	0.906251	0.801918
ARG:H	191	42	0.420370	0.335591	0.422565	0.367029	0.360404	0.371805
ARG:HA	187	40	0.232148	0.184946	0.230582	0.203383	0.190059	0.173893
ARG:HB2	45	18	0.220140	0.160915	0.211225	0.158724	0.185019	0.140049
ARG:HB3	45	18	0.273202	0.167228	0.221013	0.203612	0.203885	0.160109
ARG:HD2	75	28	0.174531	0.133520	0.151097	0.135520	0.139784	0.131400
ARG:HD3	74	28	0.411794	0.199050	0.346124	0.290969	0.268276	0.244309
ARG:HE	53	24	1.083585	1.142817	1.051598	1.114419	1.126197	1.107747
ARG:HG2	55	25	0.196767	0.150894	0.145322	0.136512	0.151973	0.140778
ARG:HG3	55	25	0.232862	0.125087	0.199289	0.142792	0.142282	0.148395
ARG:N	169	41	2.546932	2.176081	2.504174	2.477869	2.379376	2.195251
ARG:NE	44	19	13.728473	13.936058	14.103983	13.209621	14.314530	12.669641
ASN:C	72	30	0.982620	0.810711	0.987974	1.102671	0.901482	0.932971
ASN:CA	114	39	0.776773	0.731058	0.835085	0.910740	0.822048	0.889797
ASN:CB	109	39	0.908287	0.927387	0.880764	0.900005	0.909027	0.813514
ASN:CG	2	2	2.683333	2.778931	2.785762	2.173193	3.008759	2.862746
ASN:H	119	42	0.598324	0.419432	0.480598	0.420853	0.452841	0.449166
ASN:HA	120	41	0.251101	0.185626	0.207090	0.199894	0.205221	0.169519
ASN:HB2	22	11	0.434550	0.381901	0.443665	0.396744	0.409280	0.371529
ASN:HB3	22	11	0.214932	0.173136	0.214287	0.164279	0.241261	0.188418
ASN:HD21	16	9	0.588083	0.553494	0.502669	0.519876	0.535023	0.572227
ASN:HD22	16	9	0.333945	0.284820	0.274052	0.267509	0.275540	0.282421
ASN:N	111	39	2.980065	2.421338	3.133876	2.653334	2.684106	2.274838
ASN:ND2	103	37	1.918906	1.852737	1.851320	1.869389	1.856404	1.882595
ASP:C	123	33	1.003260	0.876159	0.887771	0.845335	0.810483	0.806491
ASP:CA	171	40	0.847900	0.935330	0.936300	1.021276	0.831234	0.976438
ASP:CB	162	40	1.254929	1.139370	1.199594	1.212041	1.265942	1.177001
ASP:H	182	43	0.500207	0.426106	0.442300	0.423633	0.443581	0.444351
ASP:HA	177	41	0.179036	0.147438	0.175306	0.167838	0.158663	0.147796
ASP:HB2	38	18	0.284566	0.249929	0.287644	0.280977	0.263172	0.224319
ASP:HB3	38	18	0.260227	0.192642	0.239487	0.227010	0.211597	0.204934
ASP:N	166	42	2.616661	2.119945	2.949106	2.383398	2.448263	1.977064
CYS:C	15	6	1.479676	1.290531	1.721833	1.703135	1.980584	1.981522
CYS:CA	27	10	2.006353	1.836325	1.749386	1.744030	1.799484	2.054769
CYS:CB	26	10	3.033661	3.483114	3.139332	3.088121	4.053248	3.732208
CYS:H	27	10	0.678571	0.392946	0.485237	0.475899	0.893428	0.703576
CYS:HA	27	10	0.404651	0.227688	0.287342	0.288388	0.385481	0.308939
CYS:HB2	10	2	0.331577	0.379476	0.307551	0.292710	0.391290	0.410909
CYS:HB3	10	2	0.403504	0.472386	0.507552	0.320632	0.579474	0.465757
CYS:N	24	9	3.993422	2.857555	3.870218	2.532401	2.551870	2.669021

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
GLN:C	131	32	1.001669	0.784505	0.995176	1.021466	1.014967	0.801116
GLN:CA	170	39	0.820046	0.702912	0.856866	0.946422	0.799540	0.767744
GLN:CB	166	39	0.861343	0.757713	0.991781	0.870124	0.806525	0.755688
GLN:CG	160	36	0.964900	0.996356	0.961358	0.988232	0.951425	0.937395
GLN:H	181	42	0.415826	0.348687	0.378572	0.379018	0.391956	0.385039
GLN:HA	172	40	0.282489	0.204636	0.250765	0.227357	0.252163	0.220136
GLN:HB2	51	22	0.206531	0.134237	0.175426	0.154336	0.160839	0.139297
GLN:HB3	51	22	0.201598	0.199656	0.199354	0.187463	0.182089	0.189199
GLN:HE21	26	6	0.436273	0.431484	0.415309	0.486606	0.451512	0.436329
GLN:HE22	26	6	0.293428	0.301359	0.302755	0.308100	0.295156	0.303533
GLN:HG2	64	28	0.229556	0.245249	0.246696	0.220119	0.257087	0.272595
GLN:HG3	63	28	0.226372	0.244344	0.243354	0.230166	0.268452	0.261404
GLN:N	165	41	2.446999	2.137584	2.586340	2.136746	2.286763	1.874958
GLN:NE2	143	36	2.317520	2.336562	2.410765	2.384648	2.398325	2.285182
GLU:C	165	35	1.009752	0.863125	0.988328	0.969052	0.951437	0.869284
GLU:CA	207	42	0.871118	0.808350	0.802338	0.898683	0.751021	0.766034
GLU:CB	196	41	0.739098	0.766602	0.904146	0.958667	0.816358	0.825082
GLU:CD	1	1	37.909952	37.731226	37.664088	36.751017	37.323038	37.607386
GLU:CG	194	39	0.890790	0.886193	0.879777	0.931243	0.867184	0.875998
GLU:H	232	45	0.575894	0.451166	0.522683	0.517999	0.459522	0.474711
GLU:HA	222	43	0.270653	0.189607	0.245358	0.210740	0.200815	0.187362
GLU:HB2	60	27	0.197459	0.145229	0.179174	0.154974	0.150357	0.146710
GLU:HB3	60	27	0.188831	0.146481	0.185191	0.142174	0.173270	0.163149
GLU:HG2	60	25	0.149239	0.121018	0.131939	0.146364	0.131419	0.116946
GLU:HG3	59	25	0.126647	0.098614	0.127308	0.111356	0.116158	0.103860
GLU:N	215	44	2.365613	2.176103	2.436121	2.257566	2.042697	2.171404
GLY:C	177	32	1.065617	0.991113	1.149071	1.153025	1.098646	1.134713
GLY:CA	238	40	0.711847	0.598263	0.695508	0.659311	0.719911	0.597855
GLY:H	260	45	0.385540	0.398089	0.375054	0.381241	0.401877	0.378759
GLY:HA2	75	30	0.170517	0.175376	0.184734	0.168173	0.190506	0.165516
GLY:HA3	75	30	0.179797	0.172468	0.198025	0.202210	0.182345	0.224420
GLY:N	225	41	3.427263	3.348215	3.447964	3.205100	3.341985	3.162584
HIS:C	39	20	1.048331	0.759750	0.853520	0.874345	0.895685	0.774312
HIS:CA	58	27	1.134312	1.245893	1.353970	1.195890	1.220154	1.076190
HIS:CB	55	26	0.962239	0.793762	1.062676	1.016536	0.859804	0.897812
HIS:CD2	38	17	1.924552	1.765864	1.933792	1.873773	1.867861	1.829756
HIS:CE1	22	15	1.622424	1.326599	1.498885	1.479603	1.460252	1.340676
HIS:H	57	29	0.439571	0.385473	0.497215	0.419279	0.500086	0.437878
HIS:HA	59	27	0.242586	0.304966	0.376386	0.290366	0.244637	0.220463
HIS:HB2	12	8	0.181275	0.125031	0.166167	0.185433	0.162086	0.158734
HIS:HB3	12	8	0.130537	0.147028	0.237949	0.167738	0.237867	0.143495
HIS:HD2	45	21	0.258305	0.232841	0.314511	0.310323	0.184717	0.207861

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
HIS:HE1	24	17	0.570127	0.429254	0.518158	0.443270	0.490101	0.401484
HIS:HE2	2	1	5.279658	5.395353	5.394707	5.326341	5.349836	5.385338
HIS:N	53	28	3.430924	3.240024	3.569981	3.363375	3.040382	2.867373
HIS:ND1	1	1	96.931982	96.788153	96.647498	94.319098	96.559027	102.666449
ILE:C	116	34	0.931571	0.772175	0.922867	0.921068	0.921544	0.821841
ILE:CA	157	41	0.842982	0.866942	0.993734	1.080637	0.875053	0.844934
ILE:CB	153	41	0.968638	0.868970	1.121936	1.107729	1.023880	0.897456
ILE:CD1	146	39	3.724945	3.751230	3.721598	3.748368	3.709670	3.695482
ILE:CG1	142	37	1.239630	1.238121	1.254028	1.285307	1.197862	1.234675
ILE:CG2	144	38	3.652206	3.639403	3.672862	3.636582	3.632882	3.644598
ILE:H	179	44	0.470265	0.370295	0.452725	0.456375	0.450392	0.376834
ILE:HA	165	42	0.269685	0.194110	0.252772	0.219038	0.235717	0.195361
ILE:HB	164	42	0.240091	0.174373	0.250700	0.221723	0.209880	0.170555
ILE:HG12	22	11	0.431175	0.563384	0.501977	0.455756	0.555149	0.559571
ILE:HG13	22	11	0.414764	0.251389	0.335599	0.331964	0.287356	0.285696
ILE:N	163	43	2.635349	2.346275	3.049833	2.720752	2.556772	2.296071
LEU:C	225	35	1.018686	0.805893	1.017160	1.134467	0.985822	0.889328
LEU:CA	296	42	0.738499	0.747191	0.740815	0.964801	0.690816	0.785078
LEU:CB	285	42	1.130428	0.878965	1.187237	1.361445	1.006676	1.040918
LEU:CG	269	39	0.635081	0.597489	0.653419	0.682562	0.596445	0.589010
LEU:H	327	45	0.449559	0.396032	0.427096	0.429781	0.404688	0.416434
LEU:HA	308	43	0.233511	0.155454	0.208615	0.210061	0.191165	0.161383
LEU:HB2	66	29	0.349365	0.254007	0.312745	0.364045	0.270352	0.211595
LEU:HB3	66	29	0.372660	0.328602	0.352331	0.316897	0.348192	0.281626
LEU:HG	301	43	0.263530	0.218771	0.256905	0.252442	0.232783	0.237136
LEU:N	302	44	2.402529	2.150250	2.586329	2.806471	2.336191	2.005160
LYS:C	163	35	0.967069	0.794323	0.967781	0.998488	1.017944	0.805473
LYS:CA	216	42	0.891249	0.870828	0.999767	0.955306	0.948028	0.889343
LYS:CB	200	42	0.888194	0.650321	0.911409	0.763641	0.870648	0.717516
LYS:CD	197	40	0.657947	0.672424	0.681456	0.678079	0.686683	0.648660
LYS:CE	198	39	0.822895	0.815306	0.820457	0.819955	0.816753	0.812959
LYS:CG	200	40	0.705040	0.689403	0.736625	0.673802	0.669045	0.615546
LYS:H	254	45	0.378003	0.328238	0.367464	0.353236	0.332616	0.321805
LYS:HA	237	43	0.239478	0.192589	0.233118	0.228840	0.216088	0.192761
LYS:HB2	75	30	0.266832	0.152037	0.235183	0.197095	0.157237	0.154271
LYS:HB3	75	30	0.244625	0.156295	0.203212	0.170315	0.174248	0.158565
LYS:HD2	106	30	0.177868	0.106950	0.151841	0.107973	0.111395	0.103589
LYS:HD3	104	30	0.161175	0.110426	0.135889	0.113843	0.116353	0.109773
LYS:HE2	122	32	0.126696	0.102488	0.113867	0.102685	0.107052	0.102825
LYS:HE3	121	32	0.123713	0.094906	0.115343	0.105458	0.114006	0.104193
LYS:HG2	58	23	0.193224	0.149842	0.177545	0.152265	0.132530	0.150192
LYS:HG3	57	23	0.174912	0.129344	0.168637	0.133028	0.134982	0.128854

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
LYS:N	221	44	2.432479	2.072353	2.501541	2.255625	2.569657	2.031250
MET:C	35	22	1.331220	1.083390	1.178058	1.243281	1.267048	1.200377
MET:CA	50	28	1.032954	0.906461	1.039388	0.959234	1.097399	1.046941
MET:CB	44	28	1.434601	1.366679	1.473475	1.356483	1.348544	1.434138
MET:CE	41	24	0.830396	0.652124	0.878766	0.819351	0.818684	0.678301
MET:CG	40	22	0.654569	0.608703	0.734917	0.705091	0.706956	0.653896
MET:H	46	26	0.399241	0.307708	0.395206	0.327490	0.340601	0.296239
MET:HA	54	30	0.260286	0.226490	0.241639	0.239825	0.225830	0.199407
MET:HB2	16	11	0.235777	0.137226	0.187010	0.146396	0.119115	0.096292
MET:HB3	16	11	0.198245	0.122844	0.204058	0.118324	0.169426	0.133890
MET:HG2	14	10	0.339562	0.312465	0.313079	0.374136	0.315038	0.319665
MET:HG3	14	10	0.321117	0.282693	0.310279	0.318059	0.289322	0.272859
MET:N	41	24	2.004882	1.939652	2.309333	2.132248	1.777702	1.942117
PHE:C	81	34	1.184084	0.884863	1.251458	1.373361	1.138771	0.936803
PHE:CA	97	40	0.993419	0.894910	1.067745	1.205966	0.963528	0.918061
PHE:CB	93	40	1.043724	0.877396	1.119925	1.040172	1.229792	0.939827
PHE:CD1	73	29	0.692202	0.737557	0.674389	0.644668	0.633448	0.660198
PHE:CD2	73	28	0.933270	0.614370	0.721700	0.699644	0.679354	0.666154
PHE:CE1	72	29	0.717679	0.737958	0.765527	0.751846	0.704452	0.718337
PHE:CE2	72	28	0.818022	0.821728	0.922562	0.857104	0.799700	0.866621
PHE:CZ	77	32	1.417168	1.539543	1.580860	1.612091	1.516776	1.523269
PHE:H	105	43	0.510774	0.337918	0.447406	0.407775	0.338519	0.397628
PHE:HA	103	41	0.306700	0.233371	0.256842	0.245160	0.273885	0.211375
PHE:HB2	17	11	0.443177	0.355875	0.372002	0.401522	0.365840	0.369330
PHE:HB3	17	11	0.345371	0.336813	0.344206	0.352967	0.364293	0.323755
PHE:HD1	86	35	0.316005	0.247158	0.297872	0.227497	0.241162	0.231386
PHE:HD2	87	35	0.308137	0.226231	0.313927	0.286541	0.253401	0.217978
PHE:HE1	83	34	0.256566	0.185612	0.254125	0.194527	0.189174	0.177962
PHE:HE2	84	34	0.241963	0.228809	0.311845	0.218514	0.221784	0.216963
PHE:HZ	87	36	0.278803	0.200543	0.299256	0.216464	0.197971	0.210946
PHE:N	99	42	2.016873	1.771619	2.138356	2.237023	2.046612	1.624863
PRO:C	98	34	0.923052	0.867357	1.075710	1.006556	1.021569	0.966151
PRO:CA	139	42	0.603252	0.692397	0.563713	0.710251	0.585342	0.578656
PRO:CB	135	42	0.856007	0.750444	0.842302	0.853987	0.811405	0.780767
PRO:CD	149	40	2.329479	2.312118	2.342563	2.313248	2.332725	2.300542
PRO:CG	147	39	0.540727	0.569735	0.663900	0.667471	0.605898	0.652607
PRO:HA	158	43	0.307870	0.187627	0.278834	0.217216	0.238443	0.204763
PRO:HB2	21	10	0.321287	0.225815	0.299263	0.230572	0.228749	0.220214
PRO:HB3	21	10	0.491897	0.331895	0.367669	0.321848	0.349777	0.383301
PRO:HD2	35	23	0.463311	0.116340	0.209644	0.132414	0.169237	0.108363
PRO:HD3	34	23	0.357341	0.166007	0.240226	0.176870	0.230433	0.194934
PRO:HG2	44	24	0.202308	0.137066	0.165932	0.120313	0.118619	0.128605

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
PRO:HG3	43	24	0.131291	0.129952	0.139464	0.117003	0.122682	0.163451
SER:C	143	33	1.011497	0.852978	1.092543	1.040171	0.902204	0.887896
SER:CA	190	41	0.874859	0.927131	1.047583	1.027696	1.004679	0.952253
SER:CB	182	41	0.743197	0.622574	0.692187	0.748407	0.637739	0.676244
SER:H	198	45	0.454003	0.430107	0.448573	0.419754	0.413253	0.394657
SER:HA	223	43	0.183371	0.163045	0.189520	0.170335	0.170913	0.160199
SER:HB2	85	30	0.110115	0.096660	0.102476	0.122131	0.114048	0.113369
SER:HB3	85	30	0.107949	0.090994	0.122089	0.104332	0.136095	0.103208
SER:HG	3	2	0.316084	0.692810	0.338741	0.286193	0.240085	0.176645
SER:N	157	43	2.611884	2.206111	2.745775	2.169402	2.360036	2.201146
THR:C	116	32	0.918649	0.782592	0.994487	0.900551	0.891189	0.822975
THR:CA	166	40	0.786076	0.728377	0.832059	0.922658	0.927451	0.781019
THR:CB	160	40	0.816109	0.848854	0.918684	0.887784	0.924165	0.743689
THR:CG2	155	38	0.602384	0.525921	0.620876	0.592992	0.604205	0.533838
THR:H	172	43	0.425453	0.359864	0.403885	0.399976	0.417038	0.352412
THR:HA	166	41	0.233414	0.183340	0.225904	0.198079	0.199630	0.204890
THR:HB	166	41	0.234094	0.152591	0.220824	0.191369	0.178645	0.159780
THR:HG1	2	2	0.620601	0.496875	0.522150	0.499549	0.555046	0.472201
THR:N	157	42	3.767317	3.295700	4.103127	3.389606	3.904704	3.241787
TRP:C	37	25	1.319369	1.018261	1.297405	1.238016	1.224352	0.939835
TRP:CA	48	28	1.274418	1.124349	1.206513	1.565996	1.220287	1.089557
TRP:CB	48	28	1.332711	0.842103	1.022597	1.177022	1.176422	0.921798
TRP:CD1	43	25	1.170753	0.970240	1.363820	1.307860	1.193195	1.035102
TRP:CD2	2	2	4.960309	5.164167	4.866852	4.933292	4.970616	6.513640
TRP:CE3	43	24	0.783145	0.754845	0.809769	0.786191	0.703857	0.846600
TRP:CG	1	1	12.111568	12.648845	12.315616	13.239383	12.723987	11.823245
TRP:CH2	43	24	1.092443	0.904602	1.034553	1.070448	0.954469	0.910506
TRP:CZ2	43	24	0.518129	0.485928	0.520897	0.654770	0.421770	0.464748
TRP:CZ3	43	24	0.871427	0.887995	0.946544	0.833907	0.740798	0.808197
TRP:H	50	28	0.553796	0.455430	0.527899	0.487117	0.481020	0.425793
TRP:HA	50	28	0.337231	0.216529	0.306531	0.294344	0.296563	0.192801
TRP:HB2	12	6	0.242776	0.192313	0.215688	0.185738	0.187966	0.212113
TRP:HB3	12	6	0.258538	0.165024	0.210685	0.142905	0.215906	0.191833
TRP:HD1	49	27	0.329698	0.224226	0.285195	0.262781	0.247559	0.227060
TRP:HE1	49	27	0.433167	0.352527	0.394646	0.375386	0.349334	0.312465
TRP:HE3	49	27	0.562105	0.271897	0.403547	0.313128	0.305262	0.284901
TRP:HH2	49	27	0.337420	0.191922	0.296013	0.205752	0.207123	0.201171
TRP:HZ2	48	26	0.203719	0.107227	0.201938	0.187513	0.138437	0.146573
TRP:HZ3	44	24	0.470913	0.307321	0.332223	0.314534	0.368815	0.359773
TRP:N	47	28	2.488474	2.132223	2.712810	2.087035	3.333582	2.284811
TRP:NE1	50	28	1.549679	1.277302	1.259613	1.218571	1.425308	1.347392
TYR:C	70	33	0.854241	0.685964	0.991475	1.008721	0.899883	0.906756

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
TYR:CA	91	41	0.965499	0.850133	1.096895	1.138357	1.053474	0.969630
TYR:CB	88	41	1.023502	0.986199	1.172353	1.023120	0.947090	0.943960
TYR:CD1	61	29	1.061915	1.080306	0.999836	1.162234	1.056898	0.962641
TYR:CD2	60	28	1.013939	0.984431	0.974728	1.041225	1.012940	0.971244
TYR:CE1	60	28	0.732652	0.668863	0.655318	0.658667	0.707862	0.629304
TYR:CE2	59	27	0.781600	0.693123	0.757713	0.669901	0.751303	0.716331
TYR:CG	2	1	9.075719	9.012942	9.118257	8.791468	8.638261	9.048630
TYR:H	92	44	0.389933	0.312609	0.312272	0.376223	0.340629	0.378725
TYR:HA	91	41	0.251460	0.210309	0.278484	0.213548	0.198342	0.206358
TYR:HB2	18	13	0.371724	0.238087	0.317620	0.265211	0.298628	0.249371
TYR:HB3	18	13	0.382372	0.322277	0.367544	0.302102	0.333035	0.330257
TYR:HD1	79	36	0.250192	0.198372	0.228640	0.214766	0.246583	0.193180
TYR:HD2	75	34	0.253149	0.195932	0.243455	0.206349	0.217364	0.186589
TYR:HE1	79	36	0.203907	0.191426	0.210203	0.198386	0.202799	0.192363
TYR:HE2	76	35	0.210766	0.185296	0.213157	0.190701	0.184276	0.175459
TYR:HH	2	2	2.850409	2.865676	2.838933	2.871453	2.883629	2.870101
TYR:N	87	42	1.987693	1.707185	2.472244	2.132568	2.502807	1.773637
VAL:C	127	34	0.986113	0.864718	0.952184	1.123518	0.900727	0.873967
VAL:CA	169	41	0.902874	0.928257	1.091022	1.081608	0.930232	0.958685
VAL:CB	160	41	0.751168	0.712876	0.808972	0.977492	0.724650	0.764020
VAL:H	182	44	0.386551	0.341861	0.354400	0.395074	0.331099	0.340919
VAL:HA	174	42	0.316405	0.221112	0.305985	0.308263	0.247272	0.238046
VAL:HB	174	42	0.212378	0.191445	0.209687	0.198202	0.190480	0.179829
VAL:N	165	42	2.542845	2.208497	3.162828	3.050088	2.375787	2.213855

10.2 GROMACS Single Frame at 150 ps; evaluation 45: exact-assignment RMSE (ppm)

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ALA:C	134	35	1.117590	1.022833	1.335734	1.018220	1.121396	1.023535
ALA:CA	174	42	0.945456	0.756138	0.876227	0.746175	0.782715	0.797492
ALA:CB	164	42	0.954063	0.970714	0.968354	0.926196	0.977905	0.905124
ALA:H	184	45	0.461246	0.391861	0.479775	0.406540	0.451566	0.419283
ALA:HA	182	43	0.266130	0.196053	0.250191	0.208080	0.217047	0.205810
ALA:N	171	44	3.066086	2.397139	3.045351	2.459463	2.908570	2.780238
ARG:C	140	33	1.156807	1.003374	1.210770	0.976344	1.149027	0.987510
ARG:CA	184	40	1.218242	1.059664	1.178603	1.051487	1.112218	1.140084
ARG:CB	172	40	0.955426	0.961620	0.957790	0.971449	0.916333	0.895886
ARG:CD	173	37	0.610037	0.598517	0.644783	0.613304	0.586949	0.530478
ARG:CG	167	37	1.170892	0.914928	1.163729	0.962567	1.043397	0.931222
ARG:H	191	42	0.418338	0.389572	0.397603	0.376975	0.347419	0.408881
ARG:HA	187	40	0.292264	0.244366	0.263049	0.250281	0.257027	0.246167

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ARG:HB2	45	18	0.251270	0.253295	0.248061	0.244021	0.233674	0.224891
ARG:HB3	45	18	0.334296	0.260054	0.296250	0.229330	0.268349	0.253774
ARG:HD2	75	28	0.259671	0.136620	0.148574	0.144240	0.115457	0.137864
ARG:HD3	74	28	0.530552	0.393201	0.446926	0.395893	0.385722	0.393495
ARG:HE	53	24	1.124923	1.069491	1.150921	1.027757	1.129825	1.061183
ARG:HG2	55	25	0.289796	0.160174	0.220310	0.163449	0.166072	0.169065
ARG:HG3	55	25	0.289750	0.155731	0.192691	0.137073	0.172500	0.165715
ARG:N	169	41	2.832241	2.284478	2.778913	2.366096	2.740661	2.690406
ARG:NE	44	19	13.494210	13.685138	14.227873	14.154228	14.180709	14.202453
ASN:C	72	30	1.223909	1.003845	1.159237	1.080787	1.107520	1.050782
ASN:CA	114	39	0.876169	0.901853	0.818347	0.809414	0.867971	0.919491
ASN:CB	109	39	1.015529	1.006538	1.013294	0.941927	0.897375	1.020781
ASN:CG	2	2	2.929027	3.391515	2.920309	3.442260	3.086660	3.040821
ASN:H	119	42	0.621937	0.542662	0.618508	0.464069	0.567322	0.539969
ASN:HA	120	41	0.247096	0.227308	0.251639	0.216570	0.257061	0.218289
ASN:HB2	22	11	0.436231	0.413199	0.445204	0.421812	0.415211	0.409333
ASN:HB3	22	11	0.239532	0.276192	0.249277	0.278209	0.253431	0.255675
ASN:HD21	16	9	0.596023	0.528723	0.545665	0.497873	0.593384	0.530054
ASN:HD22	16	9	0.325797	0.305890	0.339132	0.307425	0.317771	0.306573
ASN:N	111	39	3.075222	2.807256	3.126041	3.007431	3.238030	2.918699
ASN:ND2	103	37	2.101862	2.048522	1.931976	1.930451	1.965733	1.894865
ASP:C	123	33	1.153534	0.863555	1.031437	0.856238	1.014724	1.004315
ASP:CA	171	40	1.150177	1.082588	1.228282	1.038962	1.066625	1.224769
ASP:CB	162	40	1.342406	1.268126	1.293735	1.263071	1.266545	1.239750
ASP:H	182	43	0.447872	0.453123	0.421600	0.421540	0.479705	0.431166
ASP:HA	177	41	0.196715	0.188221	0.200913	0.202522	0.186572	0.188142
ASP:HB2	38	18	0.298021	0.215216	0.280713	0.228875	0.208731	0.206925
ASP:HB3	38	18	0.272008	0.207186	0.260509	0.201958	0.236112	0.207606
ASP:N	166	42	2.932348	2.197068	2.824047	2.736825	3.087840	2.797594
CYS:C	15	6	1.473334	1.321444	1.849673	1.386431	2.012788	2.086147
CYS:CA	27	10	1.893626	1.853931	1.901336	1.925295	1.912918	1.754824
CYS:CB	26	10	3.043944	3.165917	3.194941	3.232812	4.130101	3.887459
CYS:H	27	10	0.787183	0.630592	0.667800	0.575723	0.804463	0.738986
CYS:HA	27	10	0.416562	0.312092	0.337478	0.298495	0.444260	0.394211
CYS:HB2	10	2	0.383305	0.509662	0.258929	0.424863	0.665238	0.524384
CYS:HB3	10	2	0.445882	0.431355	0.478432	0.435702	0.512926	0.466602
CYS:N	24	9	4.052016	3.515836	3.458571	3.398204	3.693892	3.116203
GLN:C	131	32	1.172016	0.965031	1.170210	0.943848	1.047046	0.955755
GLN:CA	170	39	1.032420	0.863740	0.998727	0.961865	0.930380	0.953533
GLN:CB	166	39	1.018377	0.985370	1.038203	1.029389	1.046472	0.989028
GLN:CG	160	36	0.972773	0.924056	0.966826	0.957095	0.943212	0.924587
GLN:H	181	42	0.439104	0.397445	0.427751	0.449003	0.434430	0.416833

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
GLN:HA	172	40	0.305044	0.251131	0.297193	0.252461	0.259389	0.240204
GLN:HB2	51	22	0.217774	0.184842	0.199589	0.162977	0.153787	0.161136
GLN:HB3	51	22	0.200748	0.190238	0.193383	0.189833	0.190819	0.192179
GLN:HE21	26	6	0.449120	0.444886	0.460072	0.442324	0.432829	0.437635
GLN:HE22	26	6	0.302110	0.292893	0.286886	0.293146	0.304301	0.285449
GLN:HG2	64	28	0.235814	0.237436	0.233492	0.228921	0.258431	0.249821
GLN:HG3	63	28	0.230596	0.224780	0.241800	0.218484	0.250633	0.235541
GLN:N	165	41	2.620816	2.211812	2.724341	2.447462	2.797205	2.567115
GLN:NE2	143	36	2.453255	2.381875	2.348811	2.418261	2.425496	2.393880
GLU:C	165	35	1.054138	0.982499	1.212164	1.025774	1.032480	0.981728
GLU:CA	207	42	1.057167	0.893153	0.993999	0.923332	0.905121	0.998462
GLU:CB	196	41	1.172591	0.966371	1.092130	1.077717	1.077519	1.026027
GLU:CD	1	1	37.932062	37.991953	37.624506	37.761606	38.135874	38.046183
GLU:CG	194	39	0.906939	0.925278	0.921989	0.920791	0.933509	0.889367
GLU:H	232	45	0.518105	0.466699	0.500992	0.458345	0.490987	0.452602
GLU:HA	222	43	0.291560	0.216134	0.285119	0.222492	0.232838	0.220853
GLU:HB2	60	27	0.184568	0.166311	0.188092	0.168940	0.154738	0.168276
GLU:HB3	60	27	0.198759	0.148181	0.200719	0.173483	0.188938	0.165389
GLU:HG2	60	25	0.139088	0.125974	0.143426	0.127702	0.122295	0.121196
GLU:HG3	59	25	0.125274	0.122661	0.141872	0.118131	0.130870	0.121831
GLU:N	215	44	2.585410	2.790100	2.611654	2.465836	2.579664	2.583160
GLY:C	177	32	1.224362	1.128337	1.297717	1.094110	1.075798	1.075135
GLY:CA	238	40	0.820296	0.768348	0.792719	0.737031	0.823148	0.774400
GLY:H	260	45	0.439047	0.438144	0.475093	0.431109	0.434565	0.424674
GLY:HA2	75	30	0.191570	0.202324	0.183812	0.203118	0.166125	0.196835
GLY:HA3	75	30	0.208475	0.190555	0.214252	0.188808	0.204293	0.217459
GLY:N	225	41	3.458122	3.354172	3.374888	3.371768	3.570434	3.507792
HIS:C	39	20	1.250395	0.898134	1.158171	0.796492	0.917726	0.790718
HIS:CA	58	27	1.301145	1.211851	1.248609	1.130992	1.182720	1.209237
HIS:CB	55	26	1.140484	1.044458	1.189888	1.048702	1.042765	1.087579
HIS:CD2	38	17	1.784428	1.805103	1.768976	1.802416	1.811074	1.807169
HIS:CE1	22	15	1.478285	1.446780	1.408468	1.500602	1.378000	1.407956
HIS:H	57	29	0.541985	0.379494	0.491941	0.407143	0.443059	0.425184
HIS:HA	59	27	0.237661	0.298675	0.286735	0.256962	0.309968	0.300775
HIS:HB2	12	8	0.248711	0.156404	0.187175	0.157635	0.158006	0.162506
HIS:HB3	12	8	0.247258	0.105592	0.309655	0.138437	0.221918	0.151630
HIS:HD2	45	21	0.341934	0.219267	0.234791	0.211939	0.207163	0.214881
HIS:HE1	24	17	0.507219	0.456895	0.434355	0.407013	0.401275	0.413539
HIS:HE2	2	1	2.786784	2.794944	2.860795	2.770240	2.795539	2.764691
HIS:N	53	28	3.114389	3.727670	3.870460	3.361087	3.820928	3.430610
HIS:ND1	1	1	96.947745	97.873358	97.249472	96.885916	97.024512	96.868536
ILE:C	116	34	1.043616	0.983840	1.174017	0.953620	0.951520	0.937065

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ILE:CA	157	41	1.059424	0.961889	1.084447	0.999307	1.094461	0.942114
ILE:CB	153	41	1.335096	1.268423	1.355024	1.212984	1.264336	1.236184
ILE:CD1	146	39	3.755681	3.722790	3.697931	3.727203	3.712603	3.686486
ILE:CG1	142	37	1.358667	1.255977	1.317606	1.265664	1.269453	1.246885
ILE:CG2	144	38	3.718082	3.637423	3.676398	3.616515	3.607283	3.625939
ILE:H	179	44	0.558286	0.502796	0.522877	0.509152	0.479034	0.505685
ILE:HA	165	42	0.332463	0.256891	0.301773	0.261697	0.299673	0.268874
ILE:HB	164	42	0.260517	0.232330	0.276316	0.223682	0.243453	0.217113
ILE:HG12	22	11	0.396908	0.270356	0.456831	0.294134	0.309984	0.292068
ILE:HG13	22	11	0.458186	0.476496	0.412764	0.454133	0.449213	0.447133
ILE:N	163	43	3.283328	2.646720	2.976225	2.579256	3.194162	2.876885
LEU:C	225	35	1.186503	0.976048	1.203406	0.964279	1.048749	1.000558
LEU:CA	296	42	1.084210	0.831098	0.939324	0.865463	0.867801	0.951292
LEU:CB	285	42	1.413953	1.263675	1.410303	1.351714	1.328770	1.275519
LEU:CG	269	39	0.667527	0.623544	0.665647	0.647724	0.634791	0.621420
LEU:H	327	45	0.477278	0.443669	0.471501	0.460044	0.460540	0.453073
LEU:HA	308	43	0.300886	0.234204	0.282821	0.238998	0.250271	0.243027
LEU:HB2	66	29	0.351680	0.274128	0.351510	0.286170	0.292363	0.287279
LEU:HB3	66	29	0.412548	0.369052	0.426942	0.356335	0.376281	0.357043
LEU:HG	301	43	0.270423	0.216875	0.268812	0.217462	0.214185	0.212500
LEU:N	302	44	2.554144	2.327933	2.482637	2.451922	2.663276	2.737642
LYS:C	163	35	1.072739	0.915011	1.186596	0.889157	1.059477	0.977375
LYS:CA	216	42	1.132096	1.079383	1.097804	1.031412	1.043648	1.184101
LYS:CB	200	42	0.985277	1.033318	0.970780	0.910771	0.973071	1.005820
LYS:CD	197	40	0.701444	0.726420	0.703908	0.705353	0.696571	0.695266
LYS:CE	198	39	0.823783	0.825508	0.833939	0.817652	0.817132	0.813847
LYS:CG	200	40	0.738834	0.708416	0.733491	0.688864	0.710022	0.703971
LYS:H	254	45	0.415688	0.391771	0.411498	0.393978	0.374067	0.402218
LYS:HA	237	43	0.263890	0.232698	0.275953	0.231838	0.255987	0.253969
LYS:HB2	75	30	0.243906	0.213088	0.243413	0.212550	0.203299	0.200757
LYS:HB3	75	30	0.242771	0.223754	0.266275	0.229917	0.224556	0.213575
LYS:HD2	106	30	0.195260	0.142192	0.182420	0.146062	0.122096	0.147038
LYS:HD3	104	30	0.195341	0.152155	0.196166	0.160968	0.154365	0.157369
LYS:HE2	122	32	0.121227	0.103303	0.131494	0.110425	0.111682	0.114563
LYS:HE3	121	32	0.130614	0.111883	0.131663	0.118795	0.119774	0.114515
LYS:HG2	58	23	0.184585	0.154598	0.187093	0.163998	0.154616	0.151972
LYS:HG3	57	23	0.173992	0.158373	0.195590	0.142828	0.155536	0.155346
LYS:N	221	44	2.666085	2.386366	2.608595	2.275410	2.520704	2.480486
MET:C	35	22	1.379101	1.189082	1.370413	1.112423	1.367880	1.237091
MET:CA	50	28	1.086643	1.147357	1.041322	0.943023	1.069394	1.189616
MET:CB	44	28	1.547103	1.676007	1.550586	1.598702	1.508436	1.554137
MET:CE	41	24	0.862050	0.695588	0.802648	0.782054	0.824322	0.780225

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
MET:CG	40	22	0.714644	0.641779	0.744257	0.695204	0.685483	0.682320
MET:H	46	26	0.438378	0.381098	0.397893	0.402354	0.421870	0.410779
MET:HA	54	30	0.348811	0.262650	0.315221	0.240284	0.286769	0.270683
MET:HB2	16	11	0.219751	0.172014	0.207395	0.175583	0.156293	0.156589
MET:HB3	16	11	0.238333	0.158532	0.306972	0.153491	0.172088	0.148471
MET:HG2	14	10	0.310213	0.287354	0.362392	0.320632	0.312917	0.286931
MET:HG3	14	10	0.298757	0.333532	0.292241	0.305725	0.354789	0.320694
MET:N	41	24	2.156004	2.114611	2.462614	2.283057	2.353506	2.260348
PHE:C	81	34	1.418374	1.147993	1.586641	1.137640	1.272500	1.110048
PHE:CA	97	40	1.368023	1.158013	1.242442	1.285031	1.193048	1.300910
PHE:CB	93	40	1.379645	1.250258	1.270683	1.176899	1.243504	1.148543
PHE:CD1	73	29	0.684054	0.629474	0.677360	0.632999	0.645233	0.646337
PHE:CD2	73	28	0.758934	0.710454	0.824791	0.661737	0.719245	0.703380
PHE:CE1	72	29	0.788410	0.741230	0.840230	0.757935	0.788308	0.750484
PHE:CE2	72	28	0.872219	0.827027	0.910852	0.874224	0.832984	0.860652
PHE:CZ	77	32	1.553230	1.554315	1.603312	1.583555	1.553667	1.563041
PHE:H	105	43	0.494691	0.427224	0.481877	0.447605	0.465398	0.449736
PHE:HA	103	41	0.364788	0.297857	0.343970	0.310033	0.355506	0.311911
PHE:HB2	17	11	0.405657	0.443867	0.405326	0.433818	0.454560	0.463129
PHE:HB3	17	11	0.361927	0.357306	0.386958	0.377114	0.398315	0.391204
PHE:HD1	86	35	0.299715	0.247273	0.353744	0.261191	0.248205	0.253320
PHE:HD2	87	35	0.319615	0.306555	0.358781	0.270045	0.310572	0.285291
PHE:HE1	83	34	0.231957	0.209421	0.290460	0.189756	0.206584	0.189629
PHE:HE2	84	34	0.252314	0.226927	0.272258	0.224862	0.232948	0.222536
PHE:HZ	87	36	0.330742	0.261531	0.341810	0.249917	0.260438	0.252073
PHE:N	99	42	2.639204	2.018509	2.451215	2.223346	2.526690	2.328690
PRO:C	98	34	1.054152	1.015258	1.051735	0.979709	1.049710	1.044565
PRO:CA	139	42	0.699637	0.728908	0.702666	0.725284	0.727695	0.700017
PRO:CB	135	42	0.869009	0.740032	0.871801	0.775986	0.859713	0.806894
PRO:CD	149	40	2.325484	2.305791	2.330258	2.322158	2.328750	2.318417
PRO:CG	147	39	0.765355	0.703965	0.751086	0.719601	0.728780	0.724773
PRO:HA	158	43	0.380508	0.208133	0.334933	0.215761	0.234573	0.227601
PRO:HB2	21	10	0.479698	0.254786	0.418158	0.239933	0.250280	0.235503
PRO:HB3	21	10	0.560134	0.352436	0.434258	0.330615	0.379757	0.320469
PRO:HD2	35	23	0.504127	0.146564	0.343817	0.173995	0.196863	0.182748
PRO:HD3	34	23	0.401804	0.190584	0.346071	0.203628	0.234325	0.163382
PRO:HG2	44	24	0.296522	0.180972	0.255735	0.227835	0.221944	0.212170
PRO:HG3	43	24	0.209269	0.170330	0.194402	0.189486	0.188577	0.200774
SER:C	143	33	1.196777	1.020930	1.208071	0.992005	1.042622	1.076503
SER:CA	190	41	1.266920	1.166976	1.303514	1.124007	1.196167	1.285090
SER:CB	182	41	0.748668	0.729199	0.744876	0.695818	0.746439	0.771843
SER:H	198	45	0.492693	0.446747	0.501377	0.470756	0.453923	0.441418

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
SER:HA	223	43	0.212473	0.181172	0.185971	0.181188	0.196488	0.189129
SER:HB2	85	30	0.134369	0.117494	0.146797	0.111656	0.123334	0.118714
SER:HB3	85	30	0.122206	0.107808	0.129795	0.114038	0.124143	0.114071
SER:HG	3	2	0.358911	0.256959	0.235513	0.281287	0.165831	0.283853
SER:N	157	43	2.979606	2.546259	3.025711	2.583426	3.220084	2.770004
THR:C	116	32	0.965942	0.936167	1.075073	0.905274	0.907611	0.944912
THR:CA	166	40	1.060966	0.990359	1.027780	0.867894	1.048085	1.055805
THR:CB	160	40	1.041995	0.870932	0.915723	0.841758	0.907924	0.837863
THR:CG2	155	38	0.610057	0.561432	0.656904	0.608077	0.588405	0.581258
THR:H	172	43	0.474721	0.414897	0.465828	0.419271	0.429210	0.449448
THR:HA	166	41	0.293502	0.255396	0.259349	0.254436	0.244443	0.244113
THR:HB	166	41	0.275928	0.237484	0.276780	0.232356	0.243573	0.234037
THR:HG1	2	2	0.559190	0.430652	0.654556	0.450533	0.434789	0.552751
THR:N	157	42	4.372399	3.663791	4.174950	3.861734	4.415421	4.131440
TRP:C	37	25	1.342220	1.249237	1.723005	1.079671	1.181832	1.179141
TRP:CA	48	28	1.646964	1.448023	1.722367	1.489473	1.404806	1.384940
TRP:CB	48	28	1.113835	1.432669	1.114543	1.149026	1.218526	1.135507
TRP:CD1	43	25	1.272864	1.090871	1.289898	1.136500	1.304116	1.085518
TRP:CD2	2	2	4.593146	4.971861	4.467194	4.922924	4.516612	5.406720
TRP:CE3	43	24	0.781660	0.758904	0.743231	0.801175	0.708646	0.822284
TRP:CG	1	1	12.572253	12.901973	12.901500	12.612598	12.872409	12.335071
TRP:CH2	43	24	1.034672	0.950144	1.073238	0.846583	1.026755	0.990362
TRP:CZ2	43	24	0.594373	0.492615	0.645174	0.437391	0.651749	0.469540
TRP:CZ3	43	24	0.939376	0.838443	1.029035	0.928621	0.825322	0.862357
TRP:H	50	28	0.547964	0.475940	0.504230	0.435417	0.463688	0.535328
TRP:HA	50	28	0.332507	0.231843	0.325883	0.247488	0.292009	0.264276
TRP:HB2	12	6	0.228717	0.295804	0.245669	0.310308	0.359939	0.298795
TRP:HB3	12	6	0.249706	0.217116	0.235452	0.186968	0.232290	0.212368
TRP:HD1	49	27	0.319275	0.313135	0.310823	0.278301	0.307369	0.295658
TRP:HE1	49	27	0.396896	0.403111	0.346079	0.387280	0.367299	0.354330
TRP:HE3	49	27	0.605267	0.505586	0.518853	0.459942	0.429254	0.404747
TRP:HH2	49	27	0.362461	0.278347	0.316992	0.207368	0.273984	0.229545
TRP:HZ2	48	26	0.218016	0.208514	0.212750	0.200299	0.208566	0.174646
TRP:HZ3	44	24	0.521379	0.395145	0.377510	0.335568	0.349364	0.358590
TRP:N	47	28	2.393910	2.381305	2.544619	2.691868	2.732556	2.626561
TRP:NE1	50	28	1.395527	1.098622	1.317192	1.385654	1.342683	1.222161
TYR:C	70	33	1.167002	1.050907	1.242364	0.934229	1.164244	1.063065
TYR:CA	91	41	1.332512	1.158369	1.299521	1.182405	1.282216	1.206414
TYR:CB	88	41	1.191169	1.205572	1.054487	1.179038	1.266244	1.029097
TYR:CD1	61	29	1.073216	0.980364	1.001913	1.025292	1.091852	0.984985
TYR:CD2	60	28	1.061197	0.969765	1.027714	1.027075	1.006206	0.992602
TYR:CE1	60	28	0.706667	0.697794	0.637955	0.681974	0.751192	0.710785

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
TYR:CE2	59	27	0.678383	0.720071	0.671630	0.721281	0.718674	0.725385
TYR:CG	2	1	9.275694	8.911162	9.028251	9.004297	8.740873	9.401135
TYR:H	92	44	0.420620	0.402450	0.475589	0.450730	0.446865	0.414675
TYR:HA	91	41	0.289184	0.218994	0.254451	0.242927	0.260146	0.219800
TYR:HB2	18	13	0.372813	0.361144	0.393225	0.381566	0.353595	0.366429
TYR:HB3	18	13	0.375729	0.358435	0.380774	0.374649	0.310334	0.368699
TYR:HD1	79	36	0.251624	0.204358	0.274435	0.216804	0.216342	0.212806
TYR:HD2	75	34	0.248420	0.210109	0.251946	0.210028	0.215540	0.206513
TYR:HE1	79	36	0.215202	0.196460	0.239818	0.201450	0.216880	0.198916
TYR:HE2	76	35	0.221352	0.186679	0.199152	0.192325	0.189320	0.187981
TYR:HH	2	2	2.841346	2.819410	2.899297	2.869208	2.922735	2.841622
TYR:N	87	42	2.434208	2.323084	2.387014	2.340721	2.332535	2.387629
VAL:C	127	34	1.196103	1.022883	1.169252	0.967558	1.156361	0.972592
VAL:CA	169	41	1.345735	1.148658	1.316673	1.089508	1.252535	1.210008
VAL:CB	160	41	0.889238	0.988296	0.828612	0.902522	0.951294	0.882600
VAL:H	182	44	0.425347	0.390505	0.418534	0.413387	0.382862	0.388317
VAL:HA	174	42	0.286147	0.238701	0.278003	0.239128	0.253543	0.226024
VAL:HB	174	42	0.220491	0.186625	0.213988	0.173442	0.207510	0.187203
VAL:N	165	42	3.522488	2.962350	3.735445	3.162492	3.631902	3.432158

10.3 GROMACS 100-Frame Mean; evaluation 45: exact-assignment RMSE (ppm)

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ALA:C	134	35	1.080738	0.933745	1.324813	0.932590	0.981079	0.959683
ALA:CA	174	42	0.900502	0.691320	0.813419	0.701999	0.754687	0.770832
ALA:CB	164	42	0.927042	0.861626	0.910989	0.850427	0.863150	0.840400
ALA:H	184	45	0.421772	0.360839	0.394821	0.347617	0.373889	0.370189
ALA:HA	182	43	0.252234	0.186470	0.234243	0.192149	0.198180	0.184674
ALA:N	171	44	2.844090	2.118759	2.908144	2.393355	2.809039	2.651692
ARG:C	140	33	1.092959	0.936185	1.187895	0.937073	1.048731	0.982564
ARG:CA	184	40	1.174563	0.939393	1.089114	0.968649	1.021284	1.028646
ARG:CB	172	40	0.889905	0.850366	0.882963	0.835650	0.925009	0.781364
ARG:CD	173	37	0.602269	0.560931	0.628410	0.586756	0.570078	0.502295
ARG:CG	167	37	1.161553	0.846598	1.136504	0.930651	0.956732	0.897189
ARG:H	191	42	0.374843	0.358555	0.353892	0.336433	0.337683	0.349450
ARG:HA	187	40	0.280715	0.213481	0.252304	0.208972	0.217814	0.207131
ARG:HB2	45	18	0.237553	0.225179	0.240308	0.226279	0.224637	0.209789
ARG:HB3	45	18	0.319560	0.239041	0.299523	0.231896	0.259369	0.247646
ARG:HD2	75	28	0.239687	0.117656	0.134165	0.119094	0.107320	0.112200
ARG:HD3	74	28	0.522341	0.397814	0.430830	0.389973	0.390224	0.389783
ARG:HE	53	24	1.129776	1.103976	1.147342	1.074109	1.158955	1.089886

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ARG:HG2	55	25	0.276608	0.157922	0.163997	0.146531	0.161570	0.150623
ARG:HG3	55	25	0.280078	0.158606	0.186362	0.160333	0.175678	0.166518
ARG:N	169	41	2.532040	2.168494	2.798633	2.368334	2.696205	2.508713
ARG:NE	44	19	13.295887	13.613698	13.957220	14.059837	14.089178	14.134994
ASN:C	72	30	1.161278	0.909766	1.128770	0.954175	0.991578	0.945484
ASN:CA	114	39	0.850244	0.864327	0.841820	0.763129	0.824786	0.883075
ASN:CB	109	39	0.986803	0.941228	1.018147	0.909691	0.891206	0.936492
ASN:CG	2	2	3.094041	3.493777	3.210810	3.510381	3.166732	3.273489
ASN:H	119	42	0.575818	0.504097	0.530299	0.441136	0.535943	0.508581
ASN:HA	120	41	0.238247	0.197778	0.243153	0.187068	0.208021	0.188348
ASN:HB2	22	11	0.429291	0.415606	0.433835	0.420829	0.415127	0.414019
ASN:HB3	22	11	0.225917	0.258978	0.256355	0.252693	0.235879	0.233744
ASN:HD21	16	9	0.626615	0.604865	0.610054	0.588408	0.609900	0.601701
ASN:HD22	16	9	0.306509	0.306786	0.330505	0.309124	0.305144	0.296082
ASN:N	111	39	3.066094	2.628442	3.079706	2.928425	3.198233	2.818039
ASN:ND2	103	37	1.916072	1.850141	1.859466	1.840716	1.872600	1.894566
ASP:C	123	33	1.119082	0.850433	0.984884	0.809120	0.946237	0.918888
ASP:CA	171	40	1.133071	1.073247	1.193066	1.027820	1.037465	1.206832
ASP:CB	162	40	1.317221	1.215632	1.271022	1.236916	1.229724	1.229287
ASP:H	182	43	0.415835	0.416578	0.407458	0.401917	0.439661	0.420308
ASP:HA	177	41	0.178909	0.167940	0.167917	0.176688	0.167984	0.158060
ASP:HB2	38	18	0.299195	0.205564	0.283552	0.212708	0.200500	0.207967
ASP:HB3	38	18	0.274741	0.227720	0.269168	0.219919	0.233125	0.224301
ASP:N	166	42	2.670475	2.067511	2.713386	2.590072	3.146488	2.680167
CYS:C	15	6	1.516529	1.317192	1.706513	1.420153	1.835519	1.547583
CYS:CA	27	10	1.882414	1.709137	1.958795	1.862600	1.552703	1.695587
CYS:CB	26	10	3.165540	3.123414	2.895523	2.888591	3.842301	3.706991
CYS:H	27	10	0.776053	0.516699	0.641010	0.486151	0.738495	0.712123
CYS:HA	27	10	0.394604	0.268268	0.303321	0.290034	0.461836	0.374503
CYS:HB2	10	2	0.387361	0.424745	0.297212	0.344846	0.520010	0.426979
CYS:HB3	10	2	0.462863	0.438277	0.462813	0.425634	0.502038	0.464290
CYS:N	24	9	3.757214	3.907525	3.621736	2.863381	3.504104	3.066707
GLN:C	131	32	1.150178	0.910246	1.153758	0.863660	0.997097	0.906604
GLN:CA	170	39	1.049184	0.793260	0.989200	0.921851	0.868406	0.908925
GLN:CB	166	39	0.944172	0.885642	0.932010	0.886299	0.966804	0.883441
GLN:CG	160	36	0.974389	0.953286	0.969480	0.972033	0.964383	0.964592
GLN:H	181	42	0.402675	0.366228	0.387837	0.391343	0.392482	0.364873
GLN:HA	172	40	0.291073	0.203543	0.250324	0.217085	0.226878	0.212278
GLN:HB2	51	22	0.212394	0.161616	0.193183	0.152018	0.138697	0.155191
GLN:HB3	51	22	0.197190	0.157191	0.188915	0.162313	0.160943	0.166196
GLN:HE21	26	6	0.451261	0.438335	0.448387	0.421570	0.392728	0.431932
GLN:HE22	26	6	0.295491	0.285363	0.296206	0.291599	0.295843	0.277736

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
GLN:HG2	64	28	0.210517	0.198783	0.211605	0.195948	0.211374	0.215702
GLN:HG3	63	28	0.216341	0.203926	0.225995	0.205247	0.204845	0.214929
GLN:N	165	41	2.433818	2.079387	2.671413	2.114098	2.687091	2.206510
GLN:NE2	143	36	2.424899	2.348182	2.359107	2.409156	2.408069	2.398778
GLU:C	165	35	1.003084	0.934569	1.156579	0.979883	0.974258	0.927379
GLU:CA	207	42	1.035785	0.797854	0.955282	0.860899	0.849908	0.868035
GLU:CB	196	41	1.118836	0.879712	1.043469	1.000025	0.973329	0.990128
GLU:CD	1	1	37.889811	38.012903	37.621332	37.863383	37.896860	38.019648
GLU:CG	194	39	0.917869	0.902162	0.919017	0.901479	0.917339	0.884875
GLU:H	232	45	0.493438	0.422194	0.459082	0.433944	0.448557	0.422554
GLU:HA	222	43	0.274243	0.191858	0.262100	0.194673	0.204516	0.187823
GLU:HB2	60	27	0.191314	0.143782	0.193931	0.153811	0.148675	0.149420
GLU:HB3	60	27	0.208074	0.152584	0.209828	0.173231	0.187770	0.173315
GLU:HG2	60	25	0.142940	0.114951	0.137159	0.119826	0.114489	0.118988
GLU:HG3	59	25	0.127973	0.123536	0.137656	0.121848	0.129295	0.128546
GLU:N	215	44	2.403171	2.232042	2.525717	2.234550	2.438695	2.204657
GLY:C	177	32	1.200541	1.105710	1.285570	1.097547	1.053913	1.083503
GLY:CA	238	40	0.806273	0.727854	0.766774	0.699149	0.802294	0.745317
GLY:H	260	45	0.406411	0.379766	0.394609	0.369996	0.403121	0.392265
GLY:HA2	75	30	0.191232	0.197175	0.194577	0.192346	0.155484	0.177650
GLY:HA3	75	30	0.205336	0.188899	0.222881	0.202864	0.205962	0.210642
GLY:N	225	41	3.356045	3.230045	3.335121	3.209882	3.567642	3.296270
HIS:C	39	20	1.030120	0.773832	1.147566	0.747566	0.866437	0.768866
HIS:CA	58	27	1.251388	1.059358	1.251534	1.084984	1.201404	1.094354
HIS:CB	55	26	1.170300	0.974422	1.182866	1.034552	0.936981	1.044985
HIS:CD2	38	17	1.821964	1.800353	1.804252	1.788282	1.823877	1.816627
HIS:CE1	22	15	1.446093	1.438976	1.409576	1.506951	1.385928	1.428544
HIS:H	57	29	0.517585	0.412074	0.449119	0.420385	0.443687	0.427020
HIS:HA	59	27	0.225570	0.259723	0.272549	0.242444	0.284836	0.265822
HIS:HB2	12	8	0.202999	0.163977	0.155981	0.160673	0.177421	0.157821
HIS:HB3	12	8	0.250267	0.111320	0.292445	0.117158	0.182670	0.145068
HIS:HD2	45	21	0.311877	0.207002	0.237948	0.196093	0.195820	0.188110
HIS:HE1	24	17	0.478357	0.419921	0.399952	0.389404	0.383749	0.393476
HIS:HE2	2	1	2.799467	2.831891	2.900772	2.804954	2.827620	2.829983
HIS:N	53	28	3.388338	3.413390	3.598247	3.373263	3.638876	3.565731
HIS:ND1	1	1	97.224509	97.796255	97.275046	96.896231	97.044470	96.860663
ILE:C	116	34	1.028297	0.932390	1.155582	0.916549	0.947366	0.928097
ILE:CA	157	41	1.081248	1.042210	1.094810	1.087678	1.084131	1.011355
ILE:CB	153	41	1.293318	1.160158	1.280190	1.104335	1.147862	1.123884
ILE:CD1	146	39	3.725960	3.663703	3.700490	3.700669	3.680600	3.663269
ILE:CG1	142	37	1.295679	1.193440	1.262709	1.246660	1.212371	1.154996
ILE:CG2	144	38	3.706924	3.626957	3.674601	3.622500	3.636102	3.633267

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
ILE:H	179	44	0.527882	0.466232	0.466643	0.453301	0.459875	0.481645
ILE:HA	165	42	0.322843	0.253234	0.289118	0.263187	0.264279	0.265192
ILE:HB	164	42	0.253295	0.213137	0.255953	0.209207	0.234524	0.207487
ILE:HG12	22	11	0.381597	0.297273	0.429885	0.315629	0.321531	0.309802
ILE:HG13	22	11	0.460929	0.425793	0.439225	0.409809	0.417621	0.394596
ILE:N	163	43	3.129577	2.400186	2.971040	2.597085	3.247417	2.857972
LEU:C	225	35	1.140899	0.923836	1.211823	0.881901	0.978792	0.907766
LEU:CA	296	42	1.020854	0.723107	0.869085	0.793386	0.810037	0.816111
LEU:CB	285	42	1.388711	1.135038	1.354253	1.238093	1.240930	1.131541
LEU:CG	269	39	0.647639	0.595599	0.639452	0.614856	0.622402	0.595512
LEU:H	327	45	0.432996	0.398501	0.414355	0.404302	0.422720	0.417520
LEU:HA	308	43	0.272489	0.205604	0.241279	0.215908	0.227941	0.212316
LEU:HB2	66	29	0.346690	0.243908	0.338485	0.260728	0.265674	0.265471
LEU:HB3	66	29	0.412631	0.356195	0.416527	0.344463	0.365054	0.355521
LEU:HG	301	43	0.262771	0.204637	0.259328	0.199795	0.208225	0.200372
LEU:N	302	44	2.383371	2.217956	2.469870	2.306083	2.586519	2.418000
LYS:C	163	35	0.981976	0.856010	1.089219	0.794129	0.900743	0.853836
LYS:CA	216	42	1.094551	0.936639	1.082879	0.918434	0.944632	1.028228
LYS:CB	200	42	0.933820	0.850343	0.909652	0.768968	0.792028	0.818450
LYS:CD	197	40	0.689229	0.672022	0.673506	0.662328	0.667284	0.655468
LYS:CE	198	39	0.823206	0.821150	0.829374	0.814583	0.817670	0.816246
LYS:CG	200	40	0.721637	0.635624	0.702754	0.661901	0.677103	0.662836
LYS:H	254	45	0.401598	0.352859	0.367244	0.366983	0.354049	0.369158
LYS:HA	237	43	0.244750	0.200294	0.245205	0.195609	0.223223	0.212214
LYS:HB2	75	30	0.249578	0.185029	0.248535	0.199555	0.200666	0.190613
LYS:HB3	75	30	0.230771	0.185228	0.239442	0.190728	0.197823	0.187126
LYS:HD2	106	30	0.197195	0.123360	0.173499	0.139555	0.125946	0.131089
LYS:HD3	104	30	0.188370	0.144861	0.179490	0.157935	0.151829	0.152343
LYS:HE2	122	32	0.118301	0.097711	0.126978	0.101685	0.108898	0.106454
LYS:HE3	121	32	0.124285	0.098664	0.124339	0.099818	0.108585	0.100514
LYS:HG2	58	23	0.178452	0.139837	0.172614	0.146576	0.154382	0.147087
LYS:HG3	57	23	0.163784	0.135661	0.171796	0.145843	0.151715	0.144495
LYS:N	221	44	2.406665	2.148758	2.386832	2.213037	2.449353	2.349506
MET:C	35	22	1.188981	1.156642	1.346689	1.089339	1.245339	1.145684
MET:CA	50	28	1.005402	0.998400	0.972231	0.904268	1.014824	1.068960
MET:CB	44	28	1.465330	1.466436	1.496666	1.488538	1.402674	1.451897
MET:CE	41	24	0.840961	0.739222	0.796742	0.749559	0.825524	0.772620
MET:CG	40	22	0.685753	0.631988	0.692270	0.638416	0.657936	0.640163
MET:H	46	26	0.410938	0.330719	0.378412	0.354008	0.380423	0.344608
MET:HA	54	30	0.300298	0.261331	0.289874	0.243004	0.261787	0.272942
MET:HB2	16	11	0.182287	0.141902	0.193495	0.144575	0.154946	0.119908
MET:HB3	16	11	0.231966	0.125858	0.281182	0.135614	0.160699	0.131190

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
MET:HG2	14	10	0.298463	0.296980	0.311520	0.301558	0.302602	0.299729
MET:HG3	14	10	0.299842	0.307232	0.290422	0.301140	0.319401	0.293793
MET:N	41	24	2.193426	1.877768	2.211908	2.110747	2.349911	2.080393
PHE:C	81	34	1.336649	1.064711	1.560295	1.074302	1.166784	0.989306
PHE:CA	97	40	1.384669	1.094940	1.261513	1.270937	1.165908	1.195928
PHE:CB	93	40	1.291742	1.099556	1.209739	1.100791	1.213795	1.091246
PHE:CD1	73	29	0.685514	0.649152	0.681276	0.661831	0.667300	0.674408
PHE:CD2	73	28	0.749969	0.734768	0.798485	0.661091	0.686656	0.699494
PHE:CE1	72	29	0.783313	0.724103	0.816093	0.723632	0.757390	0.743827
PHE:CE2	72	28	0.869608	0.804143	0.892863	0.852451	0.823706	0.838928
PHE:CZ	77	32	1.541880	1.529379	1.564422	1.527559	1.494258	1.539065
PHE:H	105	43	0.461730	0.340266	0.394628	0.385347	0.382889	0.362540
PHE:HA	103	41	0.360006	0.252937	0.325843	0.258020	0.302157	0.284324
PHE:HB2	17	11	0.427890	0.420809	0.410284	0.425923	0.425753	0.435144
PHE:HB3	17	11	0.343500	0.371287	0.385264	0.391106	0.399554	0.390830
PHE:HD1	86	35	0.301227	0.251426	0.341712	0.261783	0.251557	0.261728
PHE:HD2	87	35	0.310897	0.281303	0.343366	0.228233	0.270724	0.248465
PHE:HE1	83	34	0.225383	0.200790	0.269233	0.176455	0.198603	0.181541
PHE:HE2	84	34	0.238921	0.209923	0.264896	0.188613	0.204796	0.201507
PHE:HZ	87	36	0.321737	0.251206	0.323095	0.214295	0.235972	0.221447
PHE:N	99	42	2.257907	1.774621	2.301535	1.992349	2.436916	2.109747
PRO:C	98	34	1.046297	0.965051	1.009176	0.967675	0.961253	0.945936
PRO:CA	139	42	0.664472	0.585817	0.632268	0.613520	0.675853	0.642271
PRO:CB	135	42	0.889344	0.706007	0.889346	0.735463	0.868908	0.789955
PRO:CD	149	40	2.324293	2.303112	2.322890	2.305090	2.330004	2.318905
PRO:CG	147	39	0.764624	0.642790	0.738209	0.692416	0.719685	0.704261
PRO:HA	158	43	0.369536	0.196098	0.323584	0.203645	0.210078	0.184536
PRO:HB2	21	10	0.454576	0.210222	0.381600	0.200618	0.276838	0.190969
PRO:HB3	21	10	0.556398	0.349715	0.461490	0.343123	0.406765	0.309908
PRO:HD2	35	23	0.485521	0.135429	0.297443	0.120590	0.169441	0.121609
PRO:HD3	34	23	0.389278	0.185331	0.331036	0.186215	0.235201	0.154416
PRO:HG2	44	24	0.274101	0.155094	0.207272	0.187263	0.162593	0.151425
PRO:HG3	43	24	0.195660	0.125313	0.180282	0.128848	0.150969	0.134981
SER:C	143	33	1.140933	0.931863	1.176302	0.936353	0.984540	0.954707
SER:CA	190	41	1.206641	1.084401	1.198370	1.063794	1.091577	1.180417
SER:CB	182	41	0.703949	0.644600	0.704345	0.674481	0.656924	0.663988
SER:H	198	45	0.482964	0.431782	0.466000	0.451824	0.451437	0.445264
SER:HA	223	43	0.190523	0.159542	0.165248	0.152298	0.163804	0.157252
SER:HB2	85	30	0.132185	0.107355	0.125762	0.108628	0.116832	0.106935
SER:HB3	85	30	0.119224	0.094635	0.125973	0.107903	0.120192	0.110047
SER:HG	3	2	0.430016	0.124443	0.287574	0.252279	0.098975	0.126687
SER:N	157	43	2.802889	2.307714	2.877990	2.475411	3.055540	2.611321

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
THR:C	116	32	0.933584	0.887536	1.079190	0.840734	0.897195	0.915868
THR:CA	166	40	1.059241	0.985740	1.020322	0.828940	0.985328	0.934863
THR:CB	160	40	1.017418	0.795202	0.832262	0.797433	0.850207	0.742320
THR:CG2	155	38	0.600575	0.551519	0.628931	0.559852	0.552211	0.544683
THR:H	172	43	0.459222	0.394485	0.414218	0.367891	0.396271	0.399088
THR:HA	166	41	0.254776	0.209103	0.223580	0.218725	0.211529	0.195614
THR:HB	166	41	0.268917	0.212469	0.266536	0.206266	0.224759	0.209150
THR:HG1	2	2	0.453046	0.399409	0.533925	0.530386	0.444953	0.518522
THR:N	157	42	4.300198	3.754587	4.063152	3.916745	4.290783	4.059332
TRP:C	37	25	1.358577	1.118629	1.748255	1.028642	1.088047	1.090473
TRP:CA	48	28	1.681891	1.308643	1.567807	1.338689	1.357443	1.255230
TRP:CB	48	28	1.080625	1.124904	1.061912	0.964873	1.232357	1.090961
TRP:CD1	43	25	1.294611	1.105384	1.240939	1.211493	1.343300	1.126364
TRP:CD2	2	2	4.600722	5.088537	4.604604	4.896036	4.913572	5.453656
TRP:CE3	43	24	0.778499	0.765672	0.706895	0.777626	0.699559	0.832640
TRP:CG	1	1	12.614055	12.789752	12.739230	12.611415	12.519153	12.178836
TRP:CH2	43	24	1.038286	0.950900	1.049237	0.892605	1.013863	0.993380
TRP:CZ2	43	24	0.591699	0.463349	0.639262	0.409441	0.641172	0.480135
TRP:CZ3	43	24	0.882894	0.818133	0.919293	0.772335	0.747398	0.843530
TRP:H	50	28	0.526686	0.441252	0.485871	0.426730	0.454022	0.447875
TRP:HA	50	28	0.333026	0.186242	0.304546	0.220967	0.233373	0.237390
TRP:HB2	12	6	0.250126	0.244846	0.267634	0.285525	0.321627	0.255276
TRP:HB3	12	6	0.246815	0.216464	0.237313	0.191084	0.240400	0.183079
TRP:HD1	49	27	0.307106	0.249962	0.284079	0.198107	0.248601	0.202076
TRP:HE1	49	27	0.396797	0.396360	0.354685	0.365245	0.351482	0.342564
TRP:HE3	49	27	0.586726	0.487637	0.503015	0.420516	0.391716	0.386325
TRP:HH2	49	27	0.368922	0.267813	0.289156	0.183007	0.235381	0.200190
TRP:HZ2	48	26	0.217233	0.190477	0.206430	0.165425	0.185000	0.145699
TRP:HZ3	44	24	0.481263	0.389152	0.346181	0.311541	0.302241	0.339009
TRP:N	47	28	2.428863	2.505986	2.444604	2.703743	2.549695	2.557348
TRP:NE1	50	28	1.330513	1.022621	1.301967	1.380360	1.322253	1.207872
TYR:C	70	33	1.089399	0.878586	1.168486	0.848974	1.006730	0.881320
TYR:CA	91	41	1.340984	1.243864	1.276231	1.202774	1.229058	1.177774
TYR:CB	88	41	1.060214	1.016810	0.968068	1.040749	1.121699	0.873791
TYR:CD1	61	29	1.054865	0.973169	1.021020	0.995261	1.087517	0.982205
TYR:CD2	60	28	1.041466	0.967910	1.035445	0.983195	1.029310	0.976247
TYR:CE1	60	28	0.693569	0.707217	0.653377	0.687017	0.724151	0.712181
TYR:CE2	59	27	0.677898	0.712492	0.677352	0.696027	0.708767	0.717336
TYR:CG	2	1	9.011853	8.849181	8.888676	8.990282	8.919647	9.226107
TYR:H	92	44	0.418384	0.355075	0.406879	0.389672	0.410286	0.375461
TYR:HA	91	41	0.258457	0.184970	0.230938	0.200558	0.225528	0.196040
TYR:HB2	18	13	0.384738	0.362113	0.388674	0.378478	0.350400	0.377821

IUPAC	<i>n</i>	Proteins	Structure / no WT	Structure / WT	AIMNet2 / no WT	AIMNet2 / WT	MOPAC PM7 / no WT	MOPAC PM7 / WT
TYR:HB3	18	13	0.366289	0.354046	0.373440	0.367555	0.322165	0.359311
TYR:HD1	79	36	0.237526	0.186400	0.255765	0.183947	0.195444	0.189653
TYR:HD2	75	34	0.241894	0.214301	0.242449	0.204723	0.218859	0.207339
TYR:HE1	79	36	0.198537	0.188736	0.223299	0.189703	0.199036	0.184917
TYR:HE2	76	35	0.208713	0.182183	0.195674	0.184311	0.184093	0.181777
TYR:HH	2	2	2.873591	2.855557	2.914076	2.912302	2.923725	2.860608
TYR:N	87	42	2.278261	2.045996	2.337178	2.039732	2.318649	2.055621
VAL:C	127	34	1.135991	0.933975	1.126403	0.899836	1.039483	0.913717
VAL:CA	169	41	1.259701	1.008601	1.176980	0.953655	1.089216	1.070150
VAL:CB	160	41	0.803149	0.858186	0.777268	0.832968	0.819240	0.764798
VAL:H	182	44	0.389959	0.347437	0.346150	0.363439	0.343866	0.348446
VAL:HA	174	42	0.262712	0.191637	0.240688	0.196646	0.216851	0.191703
VAL:HB	174	42	0.217519	0.177579	0.217574	0.168974	0.202932	0.172107
VAL:N	165	42	3.402142	2.502936	3.628298	3.166853	3.528458	3.295296